

# Feeding the City: How Rural Productivity Shapes Urban Structural Change\*

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*Job Market Paper* (most recent version [here](#))

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## Abstract

How does rural productivity affect cities? I use Indonesia's Transmigration Program, which resettled over two million people to newly established villages in the 1980s. The haphazard rollout of the policy generated plausibly exogenous variation in agro-climatic similarity between migrants' places of origin and destinations, with higher similarity resulting in more productive villages. Cities surrounded by more productive villages undergo sectoral employment changes without much population growth: the agricultural employment share falls by about 5.1 percentage points per standard deviation of village productivity, and modern-sector employment expands while the remaining peri-urban agriculture contracts. The cities add far fewer workers than modern-sector jobs. With the incumbent workforce fixed and net employment growth equated to arrivals, new arrivals cannot account for the shift. Repeated cross-sections do not identify individual occupational transitions. The effects concentrate in cities less connected to global markets, where the food expenditure share falls, and non-agricultural employment grows most. In a spatial general equilibrium model, I show that the observed reallocation of urban employment out of agriculture can occur with lower levels of internal migration, but not without agricultural trade. Thus, agricultural productivity gains can translate into expanding services and manufacturing in cities without substantial migration, especially when food trade reduces the subsistence constraints.

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# 1 Introduction

Many development programs by governments and inter-governmental organizations often target rural areas: raising farm productivity, expanding cultivation, and improving rural livelihoods. In agriculture-based countries, governments directed about 4% of their public spending to agriculture ([World Bank 2007](#)). These policies are typically evaluated based on the rural outcomes they achieve. Yet, when goods and labor markets are linked across space, rural productivity gains can also reshape the towns and cities nearby. Higher productivity may attract and retain workers in agriculture earning a higher income, while also reducing liquidity constraints for farmers to finance migration and leave for cities. Higher agricultural output can lower the local cost of living (especially when staple foods are locally sourced), raise rural incomes and demand for non-agricultural goods and services, and alter migration incentives and urban labor supply. The more cities rely on local food supply, the more likely rural spending is also urban policy. Most existing research estimates the effects of agricultural productivity gains *within* the region that received them, so the effect on cities cannot be separated from the effect on the villages themselves. How trade in goods or in labor spreads across space and whether these channels translate into urban growth and structural transformation is an open question.

In this paper, I study rural-to-urban spillovers by leveraging a policy that generated a large and spatially dispersed rural shock, produced plausibly exogenous differences in rural productivity across the newly settled villages, and can be linked to urban outcomes over several decades. Indonesia’s Transmigration Program provides this setting, having resettled over 2 million people from the more central islands of Java and Bali to the Outer Islands. Although the program was designed as a rural resettlement and agricultural development policy, it created sharp, localized changes in rural population and agricultural production in the hinterlands of many towns. Transmigrant households did not choose their destination villages, and farming households who ended up in destination villages more similar to their places of origin are more productive. Higher agro-climatic similarity between origin and destination raises village rice productivity by roughly 20% per standard deviation, for reasons unrelated to local economic conditions, since farming skills transfer better to places that resemble home in geography and climate ([Bazzi et al. 2016](#)). These villages trade in food with the cities around them, increasing the available food supply in a setting where households spend half of their household budget on agricultural goods.

I estimate rural-to-urban spillovers comparing cities exposed to a more or less productive rural hinterland, measured as the average agro-climatic similarity of the transmigration villages within a five-hour journey of the city. I use three decades of Indonesian census microdata and village and household surveys along with the satellite-derived urban footprints of the Global Human Settlement Layer to construct a consistent-boundary sample of urban centers on the Outer Islands. I measure long-run changes in urban outcomes, holding baseline levels of population, employment, and urban structure in cities in 1980 constant. A placebo test supports the main identifying assumption with exposure being uncorrelated with most baseline levels of city population, employment, and sectoral employment composition in 1980. Cities surrounded by more productive villages change their sectoral composition

more than their size: employment moves out of urban agriculture into manufacturing and services with only a small and insignificant effect on population growth. Per standard deviation of hinterland productivity, the agricultural employment share falls by about 5.1 percentage points, against a 2010 mean of 37.4% and about a quarter of the average city’s shift out of agriculture over the three decades, and modern-sector employment rises by about 17%. Comparing the growth in the workforce to the growth in modern-sector employment, I show using back-of-the-envelope calculations that new arrivals cannot account for the structural shift with the incumbent workforce fixed and without offsetting departures or labor-force entry and exit. Much of the fall in the agricultural share is within birthplace groups. Repeated cross-sections do not identify individual moves or occupational transitions.

Cities transform through the trade in food, as lower spending on agricultural goods among urban residents frees income for other sectors, and the response concentrates in the closed cities, those less connected to global markets, where food availability and food trade are more likely to bind. Urban residents benefit from higher food availability from more productive rural hinterland as they consume more food, measured in quantity and calories, while spending relatively less with the food expenditure share falling. Higher overall household spending is consistent with the higher demand for non-agricultural goods and services, pulling labor into the modern sector. The countryside moves the opposite way. Villages with higher agro-climatic similarity keep at least as large a share of their workers in agriculture and produce more food. The settler farmers keep farming, with no distinct migration response by their village productivity, and the migration that does occur comes from other rural areas in the same province, which now have to compete with the more productive villages, while the share of urban residents born on Java or Bali falls.

To provide further evidence on the role of food trade in urban employment composition, I compare the effects of productivity exposure in cities that are more vs. less connected to global trade networks. I conjecture that less connected (“closed”) cities must rely more on local food supply to alleviate the subsistence constraint, and thus respond more strongly to rural productivity gains. I define city-level trade openness using predetermined proximity to historical global trade routes, measuring each city’s travel time by land and sea to the Thai and Vietnamese hubs that ship most of Indonesia’s rice imports. The coefficient estimates for population and sectoral employment reallocation are larger in the closed cities, by more than double on modern-sector employment. These estimates provide further suggestive evidence on trade openness shaping the urban response to rural productivity, complementing the predictions of [Matsuyama \(1992\)](#) and the cross-country evidence in [Glaeser \(2014\)](#).

I then use a spatial general equilibrium model to quantify the contribution of agricultural productivity to urban sectoral employment change, and to decompose the trade and migration channels behind the fall in the urban agricultural employment share. I do not observe trade flows directly in any Indonesian data source at this level; in the data both trade and migration operate at the same time and only their general equilibrium effects are observed, so the model decomposes them: I shut down trade, raising trade costs, and shut down migration separately. The two sectors are agriculture, producing food, and combined manufacturing and services, the same two sectors as in the empirical analysis.

Workers choose a location, with migration costly, and within each location a sector. Preferences are Stone-Geary, with a subsistence level of food consumption that has to be satisfied, which generates the sectoral shift. With the counterfactual exercise in the model, I show that even in the absence of migration the sectoral reallocation can be achieved, whereas in the absence of trade it cannot.

I make two contributions. First, my finding that a village productivity gain transforms the sectoral composition of the cities it feeds adds a supply-side spillover to the literature on spillovers of local shocks across space, which has followed such shocks through demand and through capital, and adds the hinterland's productivity to the explanations of urbanization without growth. Local shocks have been followed through demand, from cash transfers that raise consumption in nearby non-recipient villages (Egger et al. 2022) to resource booms that raise income and employment in surrounding counties (Feyrer et al. 2017; Allcott & Keniston 2018) and new jobs in one urban sector that create jobs in others (Moretti 2010), and agro-processing plants that raise employment in the villages around them (Edwards 2024; Kraus et al. 2024), and through capital, as agricultural windfalls flow through the banking system into industrial growth elsewhere (Bustos et al. 2020). The shock here reaches cities through the supply of the food sold by the nearby villages, and in related work on India, my co-authors and I study the reverse direction of urban employment shocks reducing rural population, but increasing non-agricultural employment in surrounding villages (Asher et al. 2026). Studies of local productivity shocks should therefore account for spillovers on population and sectoral employment in the places that buy goods from affected villages. Research on urbanization and structural transformation documents cities that gain population without commensurate industrialization (Gollin et al. 2016; Jedwab & Vollrath 2015; Gollin 2023; Henderson & Turner 2020), while spatial accounts of structural change emphasize the importance of sectoral reallocation within regions (Eckert & Peters 2025). I identify a rural source of urban structural change: productivity gains in the hinterland shift city employment toward manufacturing and services, with a comparatively modest population response. More broadly, the finding speaks to the macro-development literature on agricultural productivity and structural transformation (Gollin et al. 2007; Restuccia et al. 2008; Gollin et al. 2014; Herrendorf et al. 2014; McArthur & McCord 2017; Gollin et al. 2021), which studies a country as one closed economy, in which the countryside that becomes more productive and the cities it supplies form a single aggregate. Closest to this paper, Bazzi et al. (2016) establish the productivity variation itself. Settlers assigned to agro-climatically familiar destinations farm more productively, raising village rice yields and nightlights within the settlements. I take that village-level variation as the shock and follow it out of the settlements, asking what the productivity of a city's surrounding villages does to the city itself. Complementary work in progress studies the villages' own outcomes under the same program's variation in market access, where better-connected villages shift out of agriculture in place yet their income per capita falls (Mitchell 2026), while I show that hinterland productivity moves the cities those villages feed.

Second, I show that the urban response concentrates in cities weakly connected to outside food markets. The idea that cities need food to grow is not novel, and the evidence for it is cross-country and historical (Glaeser 2014; Gollin et al. 2007; Nunn & Qian 2011); I establish it with causal evidence from an exogenous shock to productivity. It also adds to the literature on internal trade and migration a

case in which goods trade transmits the shock while labor flows stay small. Matsuyama (1992) predicts that higher agricultural productivity moves labor into manufacturing in a closed economy and into agriculture in a small open economy, where comparative advantage sets the manufacturing share; in both cases the positive productivity shock is within the same economy where outcomes are measured. Existing studies estimate what an agricultural shock does inside the locality with the source of the shock, where the sign of the non-farm response varies with which markets happen to be open (Bustos et al. 2016; Emerick 2018; Santangelo 2019; Foster & Rosenzweig 2004; Hornbeck & Keskin 2015); across Indian districts, Moscona (2019) finds that the Green Revolution’s productivity gains raised agricultural employment and land use and reduced urbanization and manufacturing employment, most in the districts more involved in trade. The empirical tests of that prediction (Bustos et al. 2016) and the within-country tests that interact an agricultural shock with regional openness (Moscona 2019; Liu et al. 2023; Marden 2016) measure the response on the shocked side. The city-and-hinterland economy of Glaeser (2014) is closer to my setting: the farms around the city become more productive and the city responds, and urbanization depends on rural prosperity when that economy is closed and on rural desperation when it is open. Matsuyama’s open and closed cases have been tested on cross-country data, and Glaeser extends the theory from the sectoral response, agricultural contraction and the growth of the modern sector, to the growth of urban areas. In his closed economy urbanization increases with farm productivity, which is mostly consistent with my result: the population estimate is larger in the closed cities, and the change in sectoral employment is much greater than the change in population. The literature on internal trade and migration (Tombe & Zhu 2019; Bryan & Morten 2019; Morten & Oliveira 2024; Lagakos et al. 2023; Budí-Ors 2026) quantifies how goods-market and labor-market integration shape aggregate income and infers integration from how people move; here goods trade transmits the shock while labor flows stay small, and in a world where migration is low, trade still accounts for the urban structural change, whereas migration without trade does not enable that transformation. The importance of food trade as the channel has implications for studying trade and migration frictions in spatial models of structural transformation (Tombe 2015; Nath 2025).

The rest of the paper is organized as follows. Section 2 describes the program and the urban setting, Section 3 the data and the construction of the exposure measure, Section 4 the framework and the empirical strategy, Section 5 the reduced-form results, Section 6 the model and its quantification; Section 7 concludes.

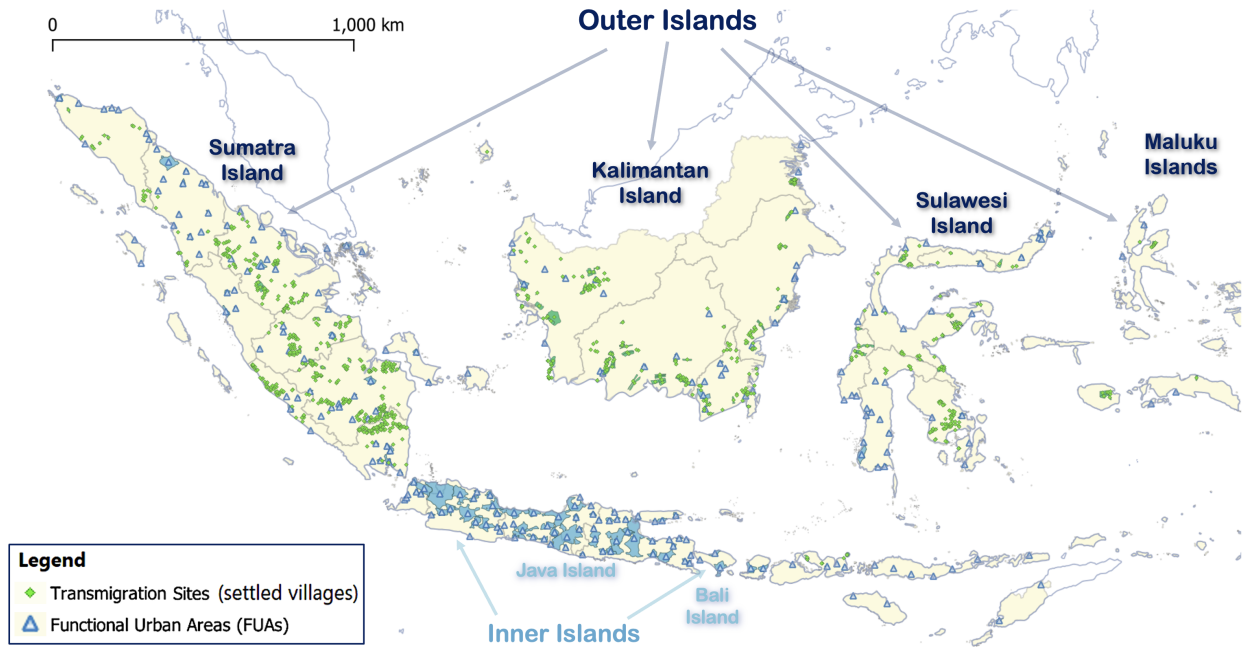
## 2 Setting

### 2.1 The Transmigration Program and Rural Productivity

The Transmigration Program aimed to alleviate population pressure on Java and Bali by resettling households in less densely populated parts of Indonesia, including Sumatra, Kalimantan, Sulawesi, and Papua. During its peak period, from 1979 to 1988, it moved more than 2,000,000 volunteer migrants to newly established villages on the Outer Islands. Participating households received a land plot, a house, and free transportation to the destination. The program expanded agricultural settlement, largely

on previously unused land, with the aim of increasing food production in the receiving regions. Its expansion in the early 1980s was followed by a sharp contraction as falling oil revenues reduced the funds available for resettlement (Bazzi et al. 2016). The settled villages were geographically dispersed across the Outer Islands (Figure 1).

Figure 1: Transmigration Villages and Urban Agglomerations



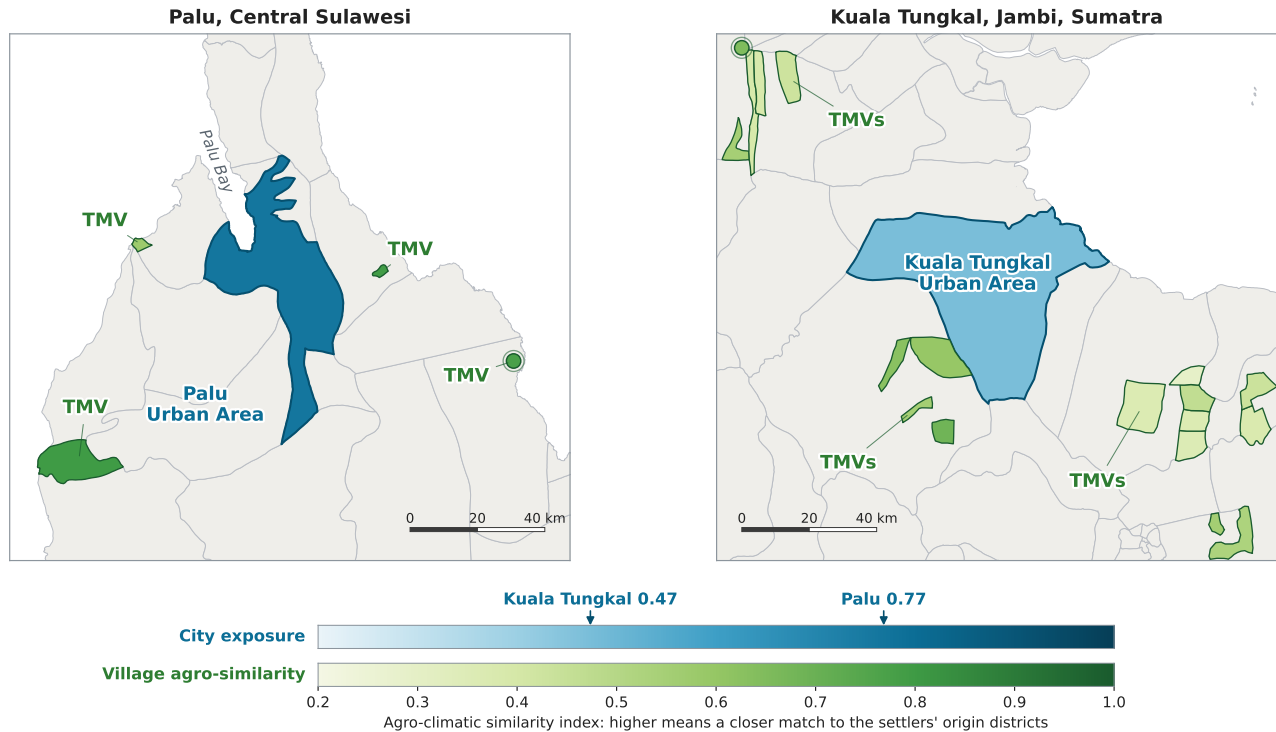
**Notes:** Diamonds mark the settled transmigration villages, from the Transmigration census (Bazzi et al. 2016). Triangles and shaded areas mark functional urban areas of the Global Human Settlement Layer (Moreno-Monroy et al. 2021). Papua is not shown: it is excluded from the analysis throughout (Section 2.2.1).

The government selected settlement sites, while households could not choose their destination villages. Assignment depended on the timing of a household’s arrival at the resettlement center and the availability of newly opened settlements. Trains and ships departed for different destinations from one day to the next, and officials did not systematically match households to destinations on the basis of their farming skills or preferences. Conditional on where the government opened a settlement, the composition of settlers arriving there was therefore plausibly as good as random. Consistent with this assignment process, Bazzi et al. (2016) find that the agro-climatic similarity between settlers’ origins and their destinations is uncorrelated with predetermined village characteristics, including potential crop yields and development in the surrounding area.

Agro-climatic similarity measures how closely a destination’s growing conditions resemble those in the migrants’ regions of origin, providing a proxy for the transferability of their farming skills. Bazzi et al. (2016) estimate that a one-standard-deviation increase in similarity raises village rice productivity by roughly 20%. Rice accounts for more than 25% of urban household expenditures on agricultural goods (Badan Pusat Statistik 2011). The replication data used here cover 814 settled villages where roughly 1,500,000 transmigrants were initially placed. Section 3.3 defines the similarity index.

I construct a city-level measure of exposure to rural productivity by estimating travel times between cities and transmigration villages along land and sea routes. For each city, I then average the agro-climatic similarity of the villages within 300 travel minutes.<sup>1</sup> In Figure 2, the villages around Palu in Central Sulawesi have higher average similarity than those around Kuala Tungkal in Sumatra. Appendix Figure A1a shows the joint variation in agro-climatic similarity and settler exposure across cities.

Figure 2: Sample Areas of Urban Agglomerations and Nearby Transmigration Villages



*Notes:* Transmigration villages are shaded by agro-climatic similarity between settlers' origins and destinations. Urban subdistricts are shaded by their city's mean similarity across villages within 300 travel minutes, used as a proxy for rural productivity exposure. Palu, Central Sulawesi (left), and Kuala Tungkal, Jambi, Sumatra (right), are drawn at a common scale. Mean similarity, on the 0 to 1 index, is 0.77 across 9 villages for Palu and 0.47 across 14 for Kuala Tungkal. The map frames differ from the travel-time catchments: the left panel shows 4 settler villages, of which 3 enter Palu's exposure; the right shows 18, of which 8 enter Kuala Tungkal's. A village whose boundary is absent from the village polygon layer is drawn as a circle at its recorded location.

To compare resettlement with the size of the receiving countryside, I define a city's rural hinterland as rural subdistricts outside every urban center and within 135 km of the city. This straight-line radius gives the closest approximation, among the distances examined, to the set of city-village pairs within the 300-minute travel-time window.<sup>2</sup> Across the 134 cities, settlers placed in these hinterlands amount to about 7% of the hinterlands' 1980 population, with roughly a seventh of cities above 15%. The 1980 census precedes the settlement of 84% of the program's villages, so the denominator includes some early arrivals.

<sup>1</sup>I augment an inherited land-travel friction surface with maritime connections and calculate the pairwise distances used here; Appendix B.1 describes the spatial construction and source limitations.

<sup>2</sup>The matching criterion penalizes both pairs omitted by the distance cutoff and pairs included by it that lie beyond the travel-time window.

## 2.2 Urbanization and Structural Transformation on the Outer Islands

The Outer Islands entered the study period less urban and more agricultural than Java and Bali. In 1980, agriculture accounted for 67% of employment on the Outer Islands, compared with 51% on Java and Bali. The urban population shares were 18% and 25%, respectively.<sup>3</sup> By 2010, the Outer Islands' urban share had risen to 37%, and their agricultural employment share had fallen to 52%. The shift toward non-agricultural employment was concentrated in services, whose share rose from 23% to 35%, while manufacturing's fell from 6% to 5%. On Java and Bali, manufacturing's employment share rose from 11% to 15%.

### 2.2.1 The Urban Sample

The analysis follows urban centers on the Outer Islands. Java and Bali do not have settled transmigration villages in the program sample. I exclude Papua because the 1980 census measures needed for baseline controls are unavailable and early spatial coverage is sparse. Indonesia's administrative hierarchy consists of provinces, districts, subdistricts, and villages. I aggregate subdistricts into cities to match the geographic detail available consistently across the data sources.

I define the satellite-based cities using the urban-center polygons of the Global Human Settlement (GHS) Urban Center Database. The analysis uses the 2020 footprints, with city membership held fixed across census years. I overlay these polygons with subdistrict boundaries and aggregate the overlapping subdistricts into urban centers, including the rural parts of those subdistricts. The functional urban areas shown in [Figure 1](#) are a separate GHS layer that includes commuting zones around urban centers ([Moreno-Monroy et al. 2021](#)). Administrative urban status is a third measure: the baseline urban population share records the fraction of each city's population living in villages classified as urban by BPS ([Section 3.4](#)).<sup>4</sup>

I complement the satellite-defined sample with areas identified from the 1980 census, because incomplete historical satellite coverage limits the measurement of Indonesia's earlier urban development. A subdistrict is added to the urban sample when its 1980 population density exceeds 500 persons per square kilometer and its agricultural employment share is below one half. These additions use baseline observables, while the satellite-defined component uses the later footprints described above. [Appendix Figure A2a](#) reports the diagnostics for this rule, and [Appendix A.1](#) documents the added and re-attached centers.

The main estimation sample retains centers whose nearest transmigration village lies within 300 travel minutes and whose 1980 baseline is directly observed in the census. [Section 3.3](#) describes the

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<sup>3</sup>These aggregates use the harmonized census files described in [Section 3](#) and exclude Papua in both years ([Section 2.2.1](#)). Badan Pusat Statistik (BPS), Indonesia's national statistical agency, classifies villages as urban or rural using population density, the share of farm households, and access to urban facilities.

<sup>4</sup>[Appendix A.1](#) details the polygon-to-city mapping. [Section 3.2](#) explains outcome aggregation, and [Section 4.3](#) discusses the implications of using endline geography.

exposure and sample restrictions. [Table 1](#) summarizes these cities, and Appendix [Table A5](#) compares them with the other outer-island urban centers.<sup>5</sup>

Table 1: Summary Statistics: The Main City Sample

| <b>Panel A: Levels (counts and the travel time to the nearest transmigration village)</b> |         |         |         |        |           |     |
|-------------------------------------------------------------------------------------------|---------|---------|---------|--------|-----------|-----|
|                                                                                           | Mean    | SD      | Median  | Min    | Max       | N   |
| Total population 1980                                                                     | 118,652 | 220,180 | 67,979  | 5,785  | 2,078,951 | 134 |
| Total population 2010                                                                     | 229,366 | 404,467 | 115,447 | 28,284 | 3,803,645 | 134 |
| Total employed 1980                                                                       | 32,841  | 57,395  | 18,874  | 2,226  | 537,303   | 134 |
| Total employed 2010                                                                       | 90,259  | 152,082 | 48,181  | 10,952 | 1,419,054 | 134 |
| Agricultural employment 1980                                                              | 17,376  | 21,328  | 10,622  | 261    | 148,091   | 134 |
| Agricultural employment 2010                                                              | 27,455  | 29,925  | 18,530  | 1,298  | 161,274   | 134 |
| Modern-sector employment 2010                                                             | 53,037  | 110,425 | 23,658  | 1,533  | 1,061,804 | 134 |
| Minutes to nearest transmigration village                                                 | 109     | 86      | 78      | 1      | 300       | 134 |

| <b>Panel B: Shares, the unemployment rate, and the exposure measures</b> |       |       |        |        |       |     |
|--------------------------------------------------------------------------|-------|-------|--------|--------|-------|-----|
|                                                                          | Mean  | SD    | Median | Min    | Max   | N   |
| Urban population share 1980                                              | 0.260 | 0.228 | 0.239  | 0.000  | 0.807 | 134 |
| Agricultural employment share 1980                                       | 0.560 | 0.219 | 0.579  | 0.045  | 0.981 | 134 |
| Agricultural employment share 2010                                       | 0.374 | 0.183 | 0.362  | 0.050  | 0.899 | 134 |
| Modern-sector employment share 2010                                      | 0.516 | 0.156 | 0.523  | 0.094  | 0.827 | 134 |
| Unemployment rate 2010                                                   | 0.122 | 0.040 | 0.116  | 0.038  | 0.223 | 134 |
| Settler count (thousands)                                                | 0.466 | 0.684 | 0.130  | 0.001  | 4.125 | 134 |
| Agro-similarity                                                          | 0.008 | 0.996 | -0.094 | -2.930 | 2.692 | 134 |

*Notes:* The main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline); every row is measured on the whole sample. The rows are the seven main 2010 outcomes, the exposure measures, the four 1980 baselines of the main specification, and the 1980 agricultural employment share. The settler count sums, over the transmigration villages within 300 travel minutes, the number of settlers placed in each divided by its travel minutes to the city, in thousands, so a village of 1,000 settlers 100 minutes away adds 0.010; agro-similarity is the standardized mean agro-climatic similarity of those villages. The modern sector is manufacturing plus services. Sources: 1980 and 2010 population censuses and the Transmigration census. Comparison with the other outer-island urban centers: Appendix [Table A5](#).

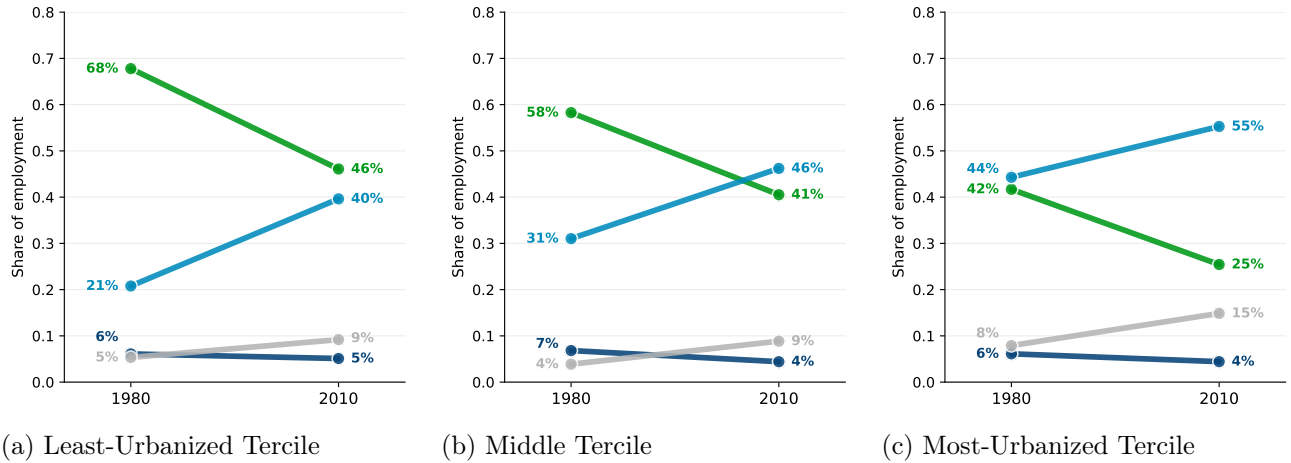
Agricultural employment shares fell and service shares rose in each tercile of cities ranked by their 1980 administrative urban population share ([Figure 3](#), panels (a)-(c)). Even in the most urbanized tercile, agriculture accounted for 42% of employment in 1980 and 25% in 2010. The decline extended across more than 90% of cities, with a median of 19 percentage points (panel (d)). The measured extent of urban agriculture also depends on how much rural territory these city boundaries include.

## 2.2.2 Urban Agriculture

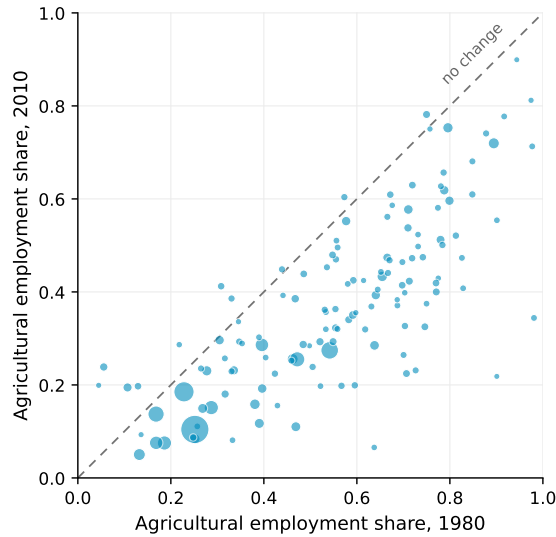
I use urban agriculture to denote agricultural employment among city residents. The census records where workers live, so these measures do not identify the location of their jobs or farm plots. When whole member subdistricts are counted, the mean agricultural employment share in the estimation sample fell from 56.0% in 1980 to 37.4% in 2010. Restricting the calculation to villages that touch the satellite footprint gives a 2010 mean of 18%; requiring those villages also to be classified as urban by BPS gives 14% (Appendix [Table D3](#)). These alternatives count all workers in each selected village, and the set of cities observed varies across definitions.

<sup>5</sup> Appendix [Table A3](#) summarizes the program villages, and Appendix [Table A4](#) compares cities with rural subdistricts hosting a transmigration village and rural subdistricts without one.

Figure 3: Sectoral Employment Change Across Urban Centers, 1980 to 2010



● Agriculture   
 ● Manufacturing   
 ● Services   
 ● Other (constr. + resid.)



(d) Agricultural Share of Each Center, 1980 and 2010

*Notes:* The sample is the main city sample. Panels (a)-(c) split the urban centers into terciles by 1980 urban population share (cutoffs 11% and 37%), held constant across both years; each line is the mean employment share of a sector across the centers of a tercile from 1980 to 2010, and “Other” combines construction with mining, utilities, and undefined industry codes. Panel (d): each center’s 1980 (horizontal) and 2010 (vertical) agricultural employment share, marker area proportional to 2010 census population; the dashed line is the 45-degree line. Source: 1980 and 2010 population censuses.

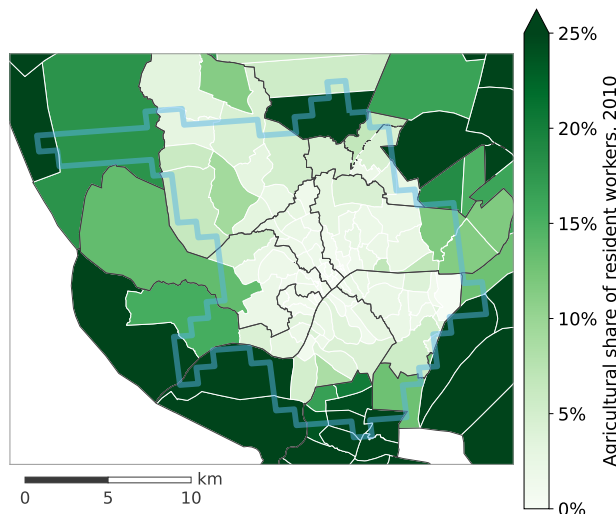
To measure variation within these city boundaries, I classify mapped villages by whether their interior points fall inside the footprint. Each selected village is counted in full, including any territory extending beyond the footprint. For each city, I pool workers separately across the two sets of villages.

Across the 120 cities with a measured share for the inside group, the median city-level agricultural employment share is 8% for that group and 49% for the outside group.<sup>6</sup>

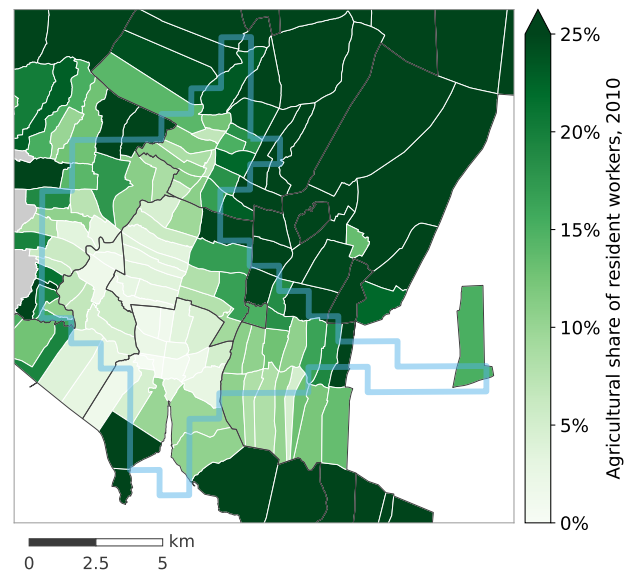
Figure 4 shows the village-level variation in Palembang, Sumatra, and Gorontalo, Sulawesi. Among mapped villages whose interior points lie inside the footprints, agriculture accounts for 3.4% of employed residents in Palembang and 7.2% in Gorontalo; the corresponding shares for the outside villages are 50.4% and 29.0%. The maps show higher shares toward the edges of both footprints, with considerable variation among the inside villages: 6 of 100 in Palembang and 33 of 79 in Gorontalo have agricultural employment shares above 10%.

Figure 4: Agricultural Employment Among Village Residents in Palembang and Gorontalo

(a) Palembang, Sumatra



(b) Gorontalo, Sulawesi



Notes: Agricultural workers as a share of employed residents in each mapped 2010 village (*desa* or *kelurahan*) of the cities' member subdistricts, from the 2010 population census. The wide outline is the Global Human Settlement urban-center footprint; thin dark lines separate subdistricts. Both maps use a common color scale from 0 to 25%; higher shares receive the darkest shade. The inside group consists of whole villages whose interior points lie inside the footprint, including any parts extending beyond it. Of these villages, 1 in Palembang and 3 in Gorontalo exceed the color limit. Gray denotes villages without measured employment; white areas fall outside the mapped member villages. The maps identify workers' residential villages; workplace and farm-plot locations are unobserved.

When city outcomes are recalculated from less agricultural member subdistricts, the estimated agricultural-share response to exposure remains negative and statistically significant at the 1% level (Appendix Table D4). These regressions use smaller city samples, and the modern-employment estimate becomes imprecise under the stricter restriction.

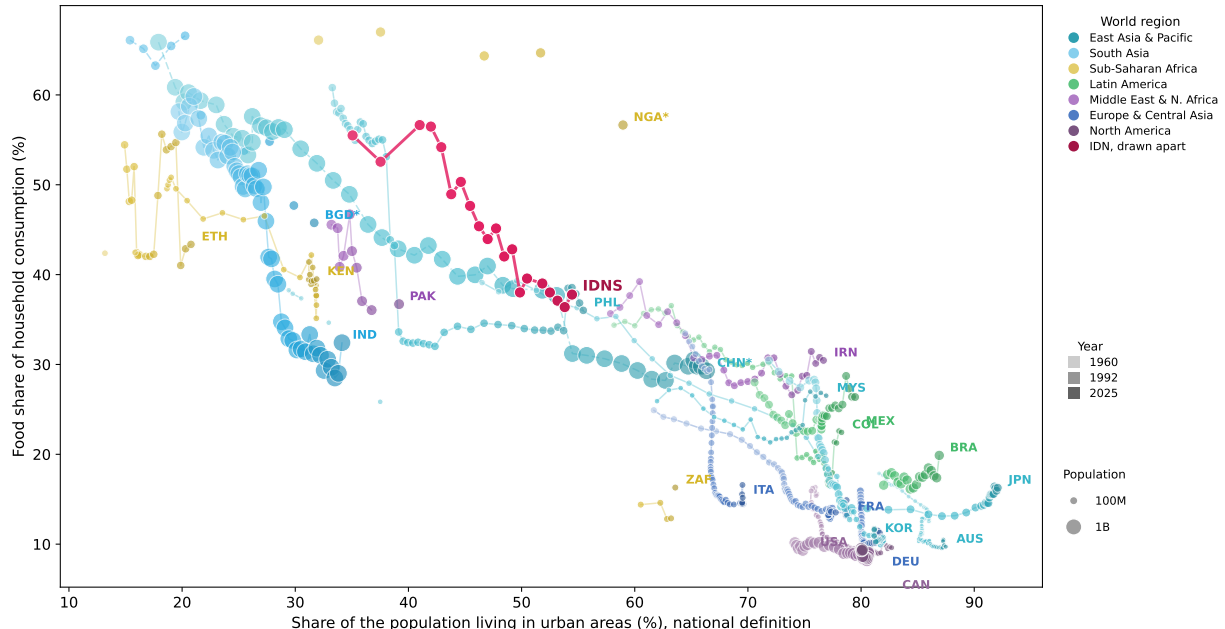
Urban agriculture is also documented outside Indonesia. In Nairobi, 20% of households grew crops within city limits in the 1984-1985 survey (Foeken & Mwangi 1998). More recently, Gallaher

<sup>6</sup>McGee (1991) describes Southeast Asian areas combining rice cultivation and non-agricultural activity at urban densities as *desakota*, from the Indonesian words for village and city. Across countries, agricultural employment within cities declines with development and city size (Gollin et al. 2016; Jedwab et al. 2025).

et al. (2015) document vegetable cultivation in sacks in densely populated Kibera, where access to land is limited.<sup>7</sup>

Limited access to affordable food is one possible reason that urban residents continue farming. Indonesia’s household food expenditure share fell as its urban share rose, a pattern also observed in the large countries shown in Figure 5. This aggregate association does not establish that food scarcity explains agricultural employment in the sample cities.

Figure 5: Food Share of Household Consumption vs. Urban Population Share, Largest Countries



Notes: Country paths plot the food share of household consumption against the urban population share under national definitions. Marker size indicates population; darker shades indicate later years. Indonesia (IDNS) uses the adjusted Susenas series. Dashed paths and asterisks identify broader food definitions whose levels are not directly comparable. Appendix A.3 documents country coverage, data sources, definition adjustments, and interpolation.

### 3 Data and Outcomes

I combine population censuses, transmigration records, household and village surveys, and spatial data to measure city outcomes and exposure to nearby settlements. The urban boundaries and subdistrict membership are defined in Section 2.2.1.

#### 3.1 Data Sources

The population censuses provide individual records for a 5% sample in 1980 and the full population in 2000 and 2010. I use the 1980 records, expanded by census weights, for baseline controls and the 2010 records for city population, employment, and demographic composition. The 2000 census supplies village agricultural-employment and settler-retention measures. I link all waves to year-2000

<sup>7</sup>The Nairobi percentage measures household participation in crop cultivation; the Indonesian shares measure agricultural employment among residents.

subdistricts using the historical BPS code links in [Olken \(2009\)](#), supplemented by census codes and village administrative histories. The Online Supplement documents coverage and comparability.

The Transmigration Census records settlement locations, establishment years, and the number of individuals initially placed in each village. I obtain these records and the village agro-climatic similarity index from the replication data of [Bazzi et al. \(2016\)](#) and [Bazzi et al. \(2019\)](#). The National Socioeconomic Survey (Susenas) supplies household consumption and earnings; the Village Potential Statistics (PODES) supply rice production. [Table 2](#) summarizes their years, geographic detail, and coverage.

Spatial data serve two purposes. The Global Human Settlement Layer (GHS) provides urban boundaries and the larger functional urban areas used in descriptive maps. I augment a general land-travel friction surface with shipping connections and calculate travel times between the paper’s geographic locations ([Appendix B.1](#)).

### 3.2 Urban Outcomes

I aggregate the constituent subdistricts into a city area that approximates an urban labor market. The common geography permits comparison with the 1980 baseline, whose finest observed location is the subdistrict; it does not establish commuting integration. Population and employment are sums across residents. Survey outcomes pool individuals or households within these areas using their survey weights. The available cities vary by outcome and wave, as summarized in [Table 2](#) and [Appendix A.1](#).

**Population and employment.** The headline employment count and the sector counts use the different definitions in [Table 2](#). Sector totals therefore need not add to headline employment. Modern employment comprises manufacturing and services; agriculture is reported separately. Construction, mining, utilities, and other industry codes remain in the sector-share denominator. The sectoral analysis also separates service activities, including trade and transportation. I report the unemployment rate and use  $\ln(1 + x)$  for log outcomes.

**Birthplace and recent migration.** The 2010 census records birthplace and residence five years earlier at the district level. Birthplace shares describe residents’ origins; five-year residence identifies recent moves across district or province boundaries. Both distinguish Java and Bali from other origins. These measures do not locate moves within districts or identify former transmigrants among city residents. The censuses do not follow individuals between waves. Worker-specific decompositions restrict the denominator to workers and are labeled accordingly. The 1980 birthplace measures identify origins at the province level.

**Household spending and consumption.** Susenas provides consumption per person, household spending, and the food share, household food spending divided by total spending. The 1996 wave supplies the main food-spending, calorie, staple-quantity, and agricultural-price measures before the 2004 rice import ban. It covers more of the main city sample than the 2002 wave. The latter provides the

Table 2: Outcome Definitions and Source Coverage

| Outcome and source                            | Location detail                                    | Definition                                                                                           | Coverage                                          |
|-----------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Population, 2010 census                       | Village or urban neighborhood                      | Resident count                                                                                       | Main city sample                                  |
| Total employment, 2010 census                 | Village or urban neighborhood                      | Worked or ran a business in the preceding week                                                       | Main city sample                                  |
| Sector employment, 2010 census                | Village or urban neighborhood                      | Counts by reported industry; shares divide by everyone with a valid industry code                    | Main city sample                                  |
| Food outcomes, Susenas 1996, 2002, and 2011   | Household, located by subdistrict                  | Weighted means of spending and quantities; price indices from item unit values                       | Surveyed cities, varying by wave and outcome      |
| Wages and rents, Susenas 2011                 | Individuals and households, located by subdistrict | Weighted median monthly earnings among positive earners; median rent relative to the national median | Cities with the required survey measure           |
| Rice production, PODES 2003                   | Village                                            | Kilograms of paddy per person, using population in villages with observed output                     | Cities with observed rice production              |
| Rice net imports, Susenas 2002 and PODES 2003 | Subdistrict                                        | Rice consumption minus production, in tons of milled rice                                            | Shared subdistrict coverage, aggregated to cities |

*Notes:* The main specifications apply the city sample’s exposure and baseline-control requirements. Survey and production outcomes additionally require the stated source coverage; their observation counts are reported with the estimates. Source geography is mapped to year-2000 subdistricts and then to cities. Household statistics use survey weights. BPS denotes Indonesia’s statistical office; Susenas is the National Socioeconomic Survey and PODES the Village Potential Statistics. Appendix [Table A6](#) reports survey coverage by wave. Sources: the 2010 population census, Susenas, and PODES.

consumption quantities paired with 2003 production. The 2011 wave supplies later spending measures, wages, and rents. These outcomes lack comparable 1980 measures, so the regressions compare their levels across cities.

**Prices and wages.** Agricultural price indices use survey-weighted medians of household item unit values. In 1996, these divide consumption value by quantity, including own production valued by BPS. Each index compares an item’s survey-weighted median unit value over the city’s households with its median over all urban-center households and aggregates the items under the expenditure weights of all urban centers, the city’s own weights, or equal weights; the rice price is the value of rice consumed over the kilograms consumed across the city’s households. Later waves distinguish purchases, permitting purchase unit values. The indices can reflect both prices and the quality and composition of consumption. I deflate 2011 monthly earnings by local price indices to measure real wages. The housing-rent index divides the survey-weighted city median monthly rent, including actual and imputed rents, by the national weighted median; it serves as a proxy for non-tradable prices.

**Rice production and net imports.** The production measure and the net-import calculation use different coverage rules. Production per person uses all villages with observed rice output and their recorded population. The net-import calculation sums 2002 consumption and 2003 production over

subdistricts observed in both sources, expressing both in tons of milled rice. Rice surplus is production minus consumption; the net-import proxy in [Figure 11](#) reverses that sign, so a positive value indicates a local shortfall. This accounting measure does not observe shipments between a city and its hinterland. The Online Supplement gives the detailed outcome definitions.

### 3.3 Exposure: Agro-Climatic Similarity, Settler Counts, and Cost Distance

I measure exposure to rural agricultural productivity using the agro-climatic similarity index of [Bazzi et al. \(2016\)](#). The index measures the match between settlers’ origin environments and their destination village, which predicts agricultural productivity through the transferability of farming skills.

**Village agro-climatic similarity.** Let  $x_o$  and  $x_v$  denote agro-climatic characteristics of origin district  $o$  and destination village  $v$ , and let  $\omega_{ov}$  be the share of settlers in  $v$  born in  $o$ . The underlying similarity index is

$$AS_v^{\text{raw}} = - \sum_{o=1}^I \omega_{ov} D(x_o, x_v), \quad (1)$$

where  $D(\cdot)$  sums absolute differences across standardized characteristics. The negative sign makes a closer match a higher value. I use the min-max rescaling stored in the replication data, denoted  $AS_v$ , which ranges from 0 to 1 across the 814 program villages. It is a proxy for agricultural productivity.

**Travel time and city exposure.** I start with an inherited land-travel friction surface, add shipping connections, and calculate least-cost travel times for the paper’s subdistricts, urban areas, and transmigration villages. I align these geographic layers, locate urban endpoints using the mapped urban footprints, and aggregate the pairwise calculations to the city definitions used here. These measures determine village catchments, travel-time weights, and nearest-village eligibility. Maritime connections matter because straight-line distance does not reflect the routes linking Indonesian islands. [Appendix B.1](#) documents the construction, distance adjustments, and source-date limitations.

I average village agro-similarity over the settlements within 300 travel minutes of city  $c$ , giving each village equal weight:

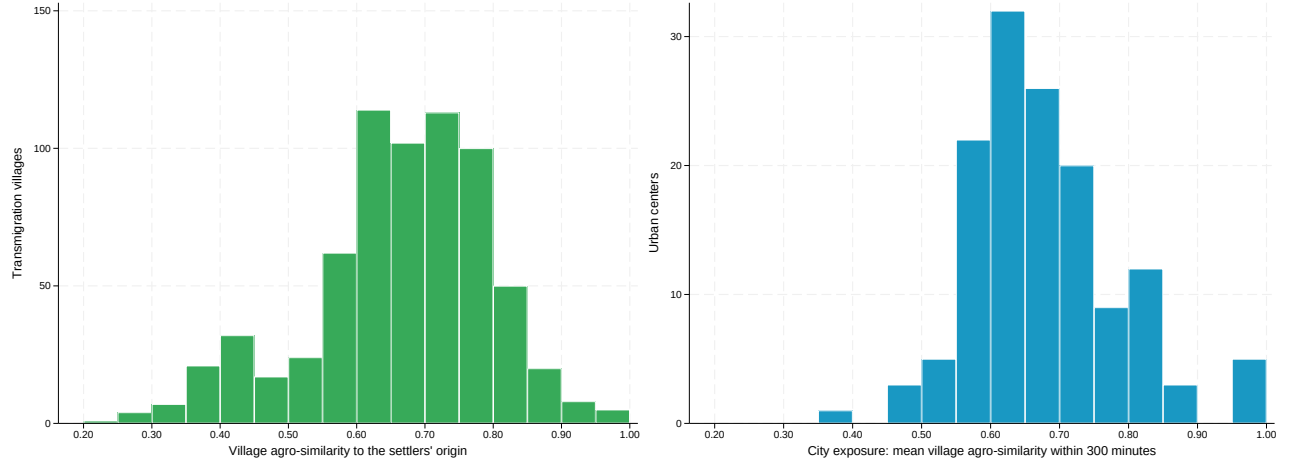
$$AgSim_c = \frac{1}{|V_c|} \sum_{v \in V_c} AS_v, \quad V_c = \{v : d_{cv} \leq 300 \text{ minutes}\}, \quad (2)$$

where  $d_{cv}$  is the travel time used to construct exposure and  $|V_c|$  is the number of villages in the catchment.<sup>8</sup> The city mean remains between 0 and 1. For estimation, I subtract its mean and divide by its standard deviation across all urban centers with measured exposure, before restricting the sample to cities with complete baseline controls. A one-standard-deviation increase therefore refers to city agro-similarity; it does not measure a proportional increase in agricultural productivity. The two sample regions in [Figure 2](#) illustrate the comparison, and [Figure 6](#) shows the village and city distributions.

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<sup>8</sup>For exposure construction, travel time is capped at 1 minute for villages with more than 0.1% of their polygon area inside a city polygon. Times implying speeds below 2 kilometers per hour are capped at the time implied by great-circle distance from the city centroid to the village at that speed. Sample eligibility uses travel times before these adjustments.

Figure 6: Transmigration-Village Agro-Similarity and Aggregated Urban Exposure



*Notes:* Left panel: agro-similarity to the region of origin of the settlers, an index from 0 to 1, for the 680 of the 814 transmigration villages that lie within 300 travel minutes of an outer-island urban center; the other 134 lie beyond 300 minutes of every urban center. Right panel: the exposure of each of the 138 urban centers with a village in that window, the mean agro-similarity of those villages. Averaging narrows the city distribution, which starts at 0.38 where the village index starts at 0.22. Bins are 0.05 wide on the same horizontal scale in both panels.

**Settler counts and weighting.** I also measure the scale of settlement around each city using the number of individuals initially placed in nearby villages. Let  $N_v$  denote that initial count in thousands. The control sums these counts with inverse-travel-time weights over the same catchment:

$$TransMigrants_c^{IDW} = \sum_{v \in V_c} \frac{N_v}{\max\{1, d_{cv}\}}, \quad (3)$$

where travel time is floored at 1 minute. Its unit is thousands of settlers per travel minute. It increases with the number and size of nearby settlements and gives greater weight to closer villages.

Equal village weights measure the average match quality of nearby settlements. Conditional on the settler-count and destination controls, the comparison is between cities whose catchments offer different average conditions for transferring farming skills. The contribution of each village to urban food supply can also depend on its size, marketed output, and transport costs. I therefore compare equal weights with weights based on initial settlers, inverse travel time, and their combination. Holding the main controls fixed, alternative weights within 300 minutes preserve the negative agricultural-share estimate, while travel-time weights reduce the positive modern-employment estimate and its statistical significance. Wider catchments preserve these signs but weaken the agricultural-share evidence; the 180-minute comparisons use fewer cities and do not reproduce the main employment findings (Appendix D.2).

**Measured exposure and the main city sample.** Applying the 300-minute cutoff to unadjusted nearest-village travel time includes 138 of the 187 urban centers, of which 134 have a 1980 baseline recorded directly in the census. The main city sample consists of the 134 cities with a nearest-village

travel time within the window, measured exposure, the required baseline controls, and measured 2010 outcomes. When  $V_c$  is empty, the catchment mean is undefined and exposure is left unmeasured. The regressions compare degrees of exposure among cities with measured catchments. The distribution of nearest-village travel times and the full sample selection are reported in [Figure A1b](#) and [Appendix Table A1](#).

### 3.4 Baseline Controls

The main specification includes the settler-count control and 16 destination agro-climatic characteristics averaged over the same villages as exposure. For each village characteristic  $X_v$ , I construct  $X_c = |V_c|^{-1} \sum_{v \in V_c} X_v$ . They comprise three slope measures, ruggedness, log elevation, soil organic carbon, exchangeable sodium percentage, soil pH, two soil-texture and two drainage measures, distances to coasts and rivers, and mean rainfall and temperature over 1948-1978. They condition on destination characteristics that also enter the origin-destination match. The tables refer to them as agro-climatic transmigration-village controls.

The city controls are four 1980 baselines: log population, log employment, the urban population share, and log agricultural employment. Some villages were settled in 1979 and 1980, before the census that records these baselines; [Section 4.3](#) discusses the timing. A sensitivity check adds the 1980 birthplace shares of the workforce ([Appendix Table C3](#)).

The urban population share distinguishes the census administrative classification from city membership. For city  $c$  it is

$$UrbanShare_c = \frac{\sum_{k \in c} Pop_k^{\text{urban}}}{\sum_{k \in c} Pop_k},$$

where  $k$  indexes its constituent subdistricts and  $Pop_k^{\text{urban}}$  is their 1980 population in villages classified as urban by BPS. The satellite boundary determines membership in the city; the census village classification determines the urban share within that geography.

### 3.5 Trade and Openness Measures

Access to imported rice can reduce a city's dependence on its hinterland. I measure this access using international rice routes and realized rice imports, and describe general port access using domestic travel times. [Appendix B.2](#) reports the construction and comparison of these measures.

**Route openness.** The preferred measure adds least-cost travel time from the city to its nearest customs port of entry to the onward sea leg to the nearer of Bangkok and Ho Chi Minh City, following the rice-route approach in [Bazzi \(2017\)](#). Both legs are expressed in hours. The domestic leg is bounded below at half an hour, and the international leg approximates sea travel by great-circle distance divided by a shipping speed of 40 kilometers per hour. Shorter total travel time means greater openness. The high-openness group lies below the median travel time among cities in the exposure window; continuous route proximity reverses the sign and is standardized so that higher values mean greater openness.

The international leg distinguishes cities near different ports by their distance from the rice exporters. The port set includes the full customs list with coordinates, without selecting on realized rice imports. Its opening dates are unrecorded. Historical maps identify ports with cargo activity in 1973 or a harbor inventory in 1986 (Table B1). The domestic leg uses the land and shipping inputs described in Section 3.3, so the measure combines international geography with domestic links observed after settlement.

**Realized-import access and general port access.** Gravity market access sums port rice imports recorded in 2000-2010, dividing each port’s volume in kilograms by travel time from the city in hours, bounded below at half an hour. I take  $\ln(1+x)$  of this sum and standardize it across the full set of cities in the source data, with higher values indicating greater access. It supplies the alternative openness split in Table 5, with high openness defined above the median within the exposure window. Realized imports respond to demand and supply, so this measure is partly an equilibrium outcome. I use it to show which way that bias runs, and the openness result relies on the route measure (Appendix B.2). The balance comparisons in Section 4.3 also use travel hours to the nearest customs port and the log of one plus summed inverse travel hours to all customs ports (with a half-hour lower bound in that sum). Both describe general port access.

Rice from other Indonesian regions provides another potential source of urban food. These import and port-access indices do not measure the availability of that supply. The appendix comparisons with domestic port access describe how much international-route geography adds to general port proximity. The instrumental-variable interpretation in Section 5.3 requires that, conditional on the main controls, route proximity affects urban outcomes and their response to exposure only through measured rice-import access. Access to other markets could violate that restriction.

## 4 Conceptual Framework and Empirical Strategy

Rural productivity can affect cities through food trade and migration. I first describe these mechanisms, then set out the estimating equation and the assumptions needed to interpret the city-level comparison causally.

### 4.1 Framework: Mechanisms Linking Rural Productivity to Urban Outcomes

The productivity shock directly affects food production in the villages that receive it. The region is the city and its rural hinterland. Food trade can then change the returns to urban agriculture and the demand for nonagricultural goods, while changes in rural earnings affect workers’ incentives to move. The relative strength of these responses is ambiguous.<sup>9</sup> Figure 7 summarizes the mechanisms within

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<sup>9</sup>The dual-economy and food-problem literatures emphasize the role of agricultural productivity in releasing labor and sustaining nonagricultural employment (Lewis 1954; Schultz 1953; Gollin et al. 2007). Trade opportunities and the factor bias of technical change can alter this relationship (Matsuyama 1992; Bustos et al. 2016). The Online Supplement discusses these accounts.

a city and its hinterland; the spatial equilibrium model in [Section 6](#) also allows trade and migration between urban regions.

Cheaper food can change urban employment even when workers remain in the city. If higher rural production lowers the price of food available to urban households, cities themselves do not need to produce as much food locally. Urban farmers then face lower returns, encouraging some to leave agriculture for manufacturing and services. Households can also reallocate spending to services and other nonagricultural goods when cheaper food reduces the cost of meeting their food needs. The resulting increase in demand can raise employment in those sectors. These supply and demand responses allow structural transformation among existing urban residents.

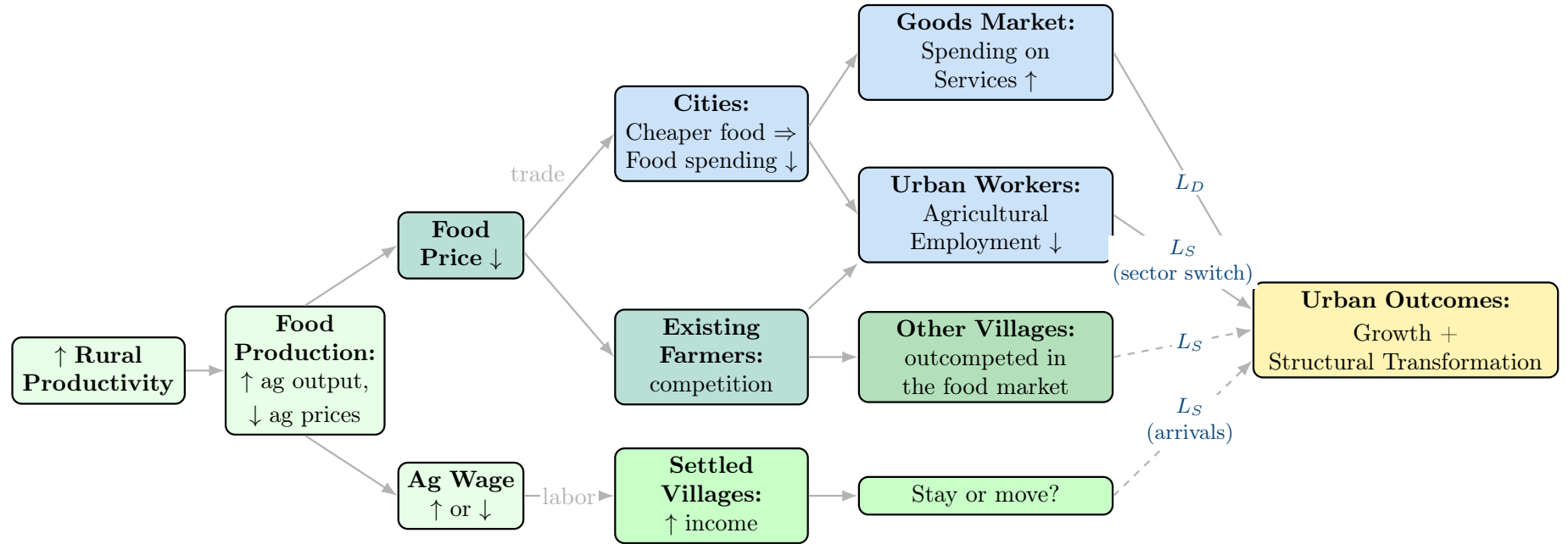
Rural wage and income effects also change workers' location choices. At given food prices and allocations of land and labor, higher agricultural productivity raises the marginal product of labor. The equilibrium response also reflects lower food prices and changes in land and labor allocations. When incomes rise, farmers have a stronger incentive to stay in the villages that received the shock, but may also be better able to finance migration ([Bryan et al. 2014](#)). Other villages face competition from the more productive settlements and may lose workers as farming becomes less profitable. Urban population growth therefore need not imply that workers came from the villages whose productivity increased.

The importance of local food production depends on access to outside markets. In a city-and-hinterland economy such as that of [Glaeser \(2014\)](#), cheaper food from nearby villages can have a larger effect when alternative sources are costly. A well-connected city can obtain food from elsewhere, limiting the effect of its own hinterland's productivity on urban food prices. Variation in outside market access therefore provides a way to assess the food-trade mechanism.

The joint response of food prices and employment helps distinguish these mechanisms. Lower urban food prices, a decline in agricultural employment, and an expansion of manufacturing and services are consistent with cheaper rural food reducing the cost of subsistence and raising demand for nonagricultural goods. With food-supply conditions held fixed, an inflow of workers increases food demand and does not imply this price decline. A stronger sectoral response in cities with limited outside food access would further support the local-food mechanism, provided that outside access is not also capturing differences in urban labor demand. Each employment response considered separately is consistent with several mechanisms.

Migration requires a separate comparison. A small net population response can conceal arrivals of modern-sector workers and departures of agricultural workers, producing sectoral reallocation without incumbent workers changing occupations. An explanation based entirely on arrivals must account for the increase in modern-sector employment using the number and employment composition of the migrants. Birthplace data help assess that requirement and whether arrivals came from the regions the hypothesis predicts, but repeated cross-sections do not identify individual moves or occupational transitions. I therefore combine this accounting evidence with the price and openness results, and use the model to compare responses under different trade and migration costs.

Figure 7: Mechanism Map



*Notes:* The figure summarizes adjustments within a city and its rural hinterland. Signs on the food-trade branch describe the case in which additional rural supply lowers urban food prices and the cost of meeting food needs. Agricultural wages depend on the food price and the marginal product of labor; wages and rural incomes can rise or fall as prices, land use, and employment adjust.  $L_S$  denotes labor available to urban nonagricultural sectors through sector switching or migration;  $L_D$  denotes labor demand through spending on services. Shading distinguishes village agriculture, urban markets, and competition faced by other farmers. Dashed arrows identify migration responses whose direction is ambiguous.

## 4.2 Estimating Equation

I compare urban outcomes across cities surrounded by villages with different agro-climatic match quality, conditional on settlement scale, destination characteristics, and the city’s 1980 baseline. The main population and employment outcomes are measured in the 2010 census. For urban center  $c$ , I estimate

$$y_c = \beta AgSim_c + \phi TransMigrants_c^{IDW} + C_{c,1980} \xi + X_c \zeta + \varepsilon_c \quad (4)$$

where  $y_c$  is an urban outcome aggregated from the city’s subdistricts and  $AgSim_c$  is standardized city agro-climatic similarity, constructed as in [Section 3.3](#). The coefficient  $\beta$  measures the outcome difference associated with a one-standard-deviation increase in that similarity measure, conditional on the other regressors. It is a reduced-form effect of match quality. An effect per unit of agricultural productivity would require a separate productivity measure and an additional estimation step; the appendix examines alternative crop and output measures ([Table E5](#)).

The settler-count control,  $TransMigrants_c^{IDW}$ , accounts for the scale and proximity of settlement, and  $X_c$  contains the catchment’s agro-climatic and geographic controls. Their construction and weighting are defined in [Section 3.3](#). The coefficient on the settler count,  $\phi$ , does not have a causal interpretation because the program chose settlement locations and sizes. Conditioning on these observed characteristics does not ensure that all variation in city-level match quality is exogenous, as discussed below.

The baseline vector  $C_{c,1980}$  contains log population, log total employment, the urban population share, and log agricultural employment. The specification compares endline outcomes conditional on these initial characteristics. For population and total employment, the corresponding baseline outcome enters directly with an estimated coefficient. Together, baseline agricultural and total employment also determine the agricultural employment share, even though that share does not enter separately. Prices, wages, and expenditure lack comparable 1980 observations, so their regressions cannot adjust for initial values of the same outcome. I interpret those estimates as suggestive evidence on the mechanisms.

The main specification is estimated without fixed effects. I use Conley standard errors allowing spatial correlation within 200 kilometers. Alternative specifications add island fixed effects and vary the treatment construction, controls, and inference ([Appendix Figure D2](#)).

## 4.3 Identification

The assignment argument concerns which settlers reached a given destination. Settlers had little influence over placement, and destinations depended on transport departures and which sites were ready to receive them, without systematic matching on agro-climate ([Bazzi et al. 2016](#)). This motivates comparisons among sites available in similar periods and regions. The empirical reassignment exercises approximate that set using provinces and settlement waves. They require assignments within each group to be unrelated to unobserved determinants of urban outcomes, conditional on the controls; the

historical account does not establish that every site in a province and wave was equally available. A causal interpretation of  $\beta$  also requires this assignment argument to remain valid when matches are aggregated across the sites surrounding a city.

Agro-climatic similarity predicts agricultural productivity. [Bazzi et al. \(2016\)](#) find that a one-standard-deviation increase in village similarity is associated with 20% higher rice productivity. The village-level validation for this paper also shows a positive relationship with the unitless crop-yield index (Appendix [Figure C2](#) and [Table C4](#)). The validation supports the productivity interpretation, but other consequences of origin-destination matching could also affect cities. I assess this interpretation using evidence on food prices, migration, and competing channels ([Section 5.2](#) and [Section 5.4](#)).

Averaging across villages can introduce a nonrandom component even when settler assignment is quasi-random. A city surrounded by sites with land more similar to the Java and Bali origin mix can have higher expected exposure regardless of how settlers are reassigned. The village locations, destination conditions, and the set of sites within each city’s catchment determine that expectation. Controlling for settlement scale and observed land characteristics addresses part of this concern, but a single settler-count measure does not hold the surrounding geography fixed.

I examine the expected component of exposure following [Borusyak & Hull \(2023\)](#). The first exercise reshuffles observed village similarity scores within provinces and, separately, within province-by-settlement-wave groups, keeping sites and catchments fixed. Averaging city exposure across draws gives the exposure expected under each reassignment rule. This exercise assumes that observed scores could have been exchanged across sites within those groups. Because a score reflects both origins and destination conditions, quasi-random assignment of settlers does not automatically imply exchangeability of the scores.

The origin-reassignment exercise instead exchanges settlers’ origin compositions across sites within the province-by-wave groups and recalculates each destination’s match using its own agro-climate. This follows the historical assignment argument more closely, while still assuming that those origin compositions could have been assigned to the alternative sites in the group. It assesses how the result changes when the destination-specific component of match quality is held fixed.

The regressions use these expected exposures in distinct ways. One specification adds the expectations from the province and province-by-wave score reshuffles as controls, allowing the outcome to vary separately with each expected exposure. The recentered specifications subtract the relevant expectation from realized exposure and standardize the remainder. The design-based interpretation requires the stated assignment assumptions. These regressions do not test those assumptions. The control specification estimates the coefficients on realized and expected exposure separately. Subtraction restricts them to be equal in magnitude and opposite in sign before standardization. Controlling for an expectation also removes its linear association with realized exposure. Their coefficient magnitudes therefore cannot be compared as effects of the same one-standard-deviation change. The estimates follow the main results in [Section 5.1](#), with construction details in [Appendix C.1](#).

Earlier roads limit the scope for subsequent investment responding to settlement or productivity gains to determine measured exposure. This advantage requires the earlier road network to be represented in the inherited land surface. The exact source version remains uncertain, and the added shipping observations follow settlement ([Appendix B.1](#)). These limitations prevent treating the combined transport surface as predetermined. If transport responses affected catchment membership or nearest-village eligibility, holding catchments fixed in the expected-exposure exercises would not address the resulting selection.<sup>10</sup>

The 1980 census allows me to examine whether cities with different eventual exposure already differed near the start of the program. Some villages had been settled before that census, so the baseline is not strictly pre-program. I therefore repeat the balance tests using exposure constructed only from villages settled after the census ([Appendix Table C2](#)). This addresses the timing concern for the balance exercise without treating 1980 as an untreated year for every city.

For 13 of the 14 city characteristics in [Figure 8](#), the confidence interval for the exposure coefficient includes zero. The exception is the 1980 urban population share, which is higher in more exposed cities, with a standardized coefficient of +0.25 and a standard error of 0.11, and which enters the main specification as a control. Some of the other 13 intervals also allow substantial baseline differences. For example, the agricultural employment share has a coefficient of  $-0.15$  standard deviations and a standard error of 0.15. Nonsignificance therefore does not establish that initial sectoral composition was the same. The post-census exposure and alternative weighting schemes give similar balance patterns ([Appendix Table C2](#)), and adding baseline birthplace shares leaves the compositional results similar ([Appendix Table C3](#)).<sup>11</sup> These checks address observed initial conditions; they cannot establish balance in unobserved determinants of urban growth.

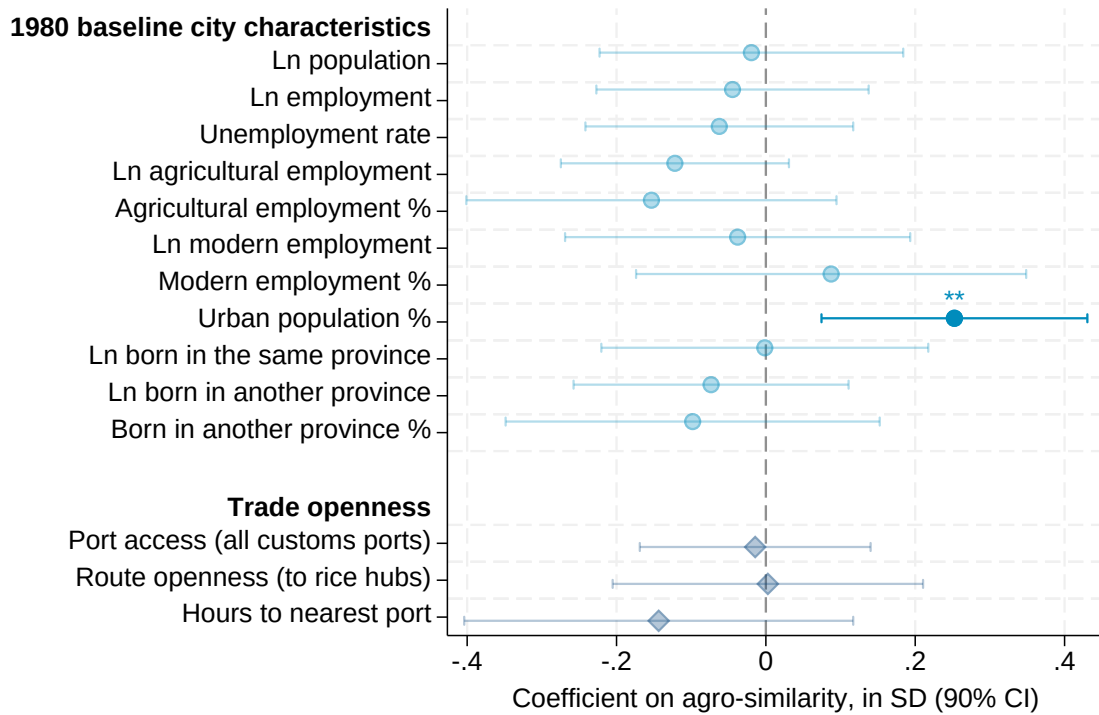
The urban geography poses a separate limitation. The sample is defined using 2020 GHS footprints, and I measure earlier outcomes within those fixed boundaries. This provides consistent geographic units across census years, but it cannot rule out selection into the sample if productivity affected which places eventually qualified as urban. The baseline comparisons and the alternative urban definitions examined in [Section 5.1](#) address observable implications of this concern. The estimates describe outcomes in the sampled urban centers and do not identify the effect on the formation of new cities.

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<sup>10</sup>District price differences provide descriptive evidence of food-market frictions. Using 2011 Susenas unit values for pairs of outer-island districts, the coefficient on log travel time in a 95th-quantile regression of the absolute log rice-price gap is 0.087; the corresponding coefficient for the food basket is 0.065. Travel time explains at most 12% of the variation in pairwise price gaps. These associations do not identify trade costs or establish that transport links are predetermined, and the quantitative exercise in [Section 6](#) uses the trade-cost range from the literature.

<sup>11</sup>The 1980 census records province of birth, which limits the baseline comparison to a broader geographic classification than the district-level decomposition available in 2010.

Figure 8: Baseline Balance: 1980 City Characteristics and Exposure



*Notes:* Each row is one regression of a standardized city characteristic on standardized agro-similarity, with the settler count and 16 agro-climatic covariates within 300 minutes as controls, as in the levels test of Appendix Table C2. Bars are 90% confidence intervals from Conley SE within 200km, a solid marker is a row whose interval excludes zero, and stars above a marker give \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The upper block (circles) holds the 10 1980 baseline outcomes of the placebo table, in its order (the birthplace rows split the 1980 population by province of birth, so the share born in the same province is one minus the plotted share born in another province), and the 1980 urban population share, a main specification control. The lower block (diamonds) reports trade-access measures: port access (the log of summed inverse travel hours to every customs port of entry), route openness to the Thai and Vietnamese rice hubs, and travel hours to the nearest customs port of entry. Every row uses the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline).

## 5 Reduced-Form Results

This section establishes the reduced-form case in five steps. Cities surrounded by more productive villages expand modern-sector employment substantially and move workers out of agriculture, without statistically significant growth in population or total employment (Section 5.1). Under the fixed-incumbent accounting assumptions, the net addition of workers could fill only a small part of the new modern-sector jobs; it runs through the goods market, where the food expenditure share falls and households spend more on non-agricultural items, which raises the demand for the modern sector (Section 5.2). The effects are stronger in cities less connected to the global trade network, which rely more on local food supply (Section 5.3). The other channels through which a rural productivity gain could reach a city, capital, public investment, new district capitals and rural demand, are tested directly, and in the city exposure predicts neither the roads and facilities the state builds nor a new district seat (Section 5.4). The villages that received the gain, if anything, keep more of their workers in farming, and the rural

subdistricts around them have higher real wages (Section 5.5). These facts identify the shock that the structural model in Section 6 then decomposes into its food-trade and migration components.

## 5.1 Urban Structural Transformation

Across the cities in the main sample, exposure to more productive villages shifts employment substantially from urban agriculture to manufacturing and services, without statistically significant growth in population or employment. Every estimate controls for the 1980 baselines, so the 2010 outcomes can be read as changes over the three decades since the program began, and each column compares cities whose exposure differs by one standard deviation. Agricultural employment is declining and modern-sector employment is growing, in both levels and shares. The agricultural employment share falls by 5.1 percentage points, against a 2010 mean of 37.4%, and agricultural employment falls by about 15%, a marginally significant estimate; modern-sector employment (combined manufacturing and services) rises by about 17%, and its share of employment by 6.6 percentage points, against a mean of 51.6% (Table 3).<sup>12</sup>

Population and total employment rise by about 5% and 2% per standard deviation of exposure, and neither effect is statistically significant. There is no effect on the unemployment rate, so cities are not growing at the expense of people being jobless. The modern share divides modern-sector jobs by total employment, so its rise of 6.6 percentage points means that modern-sector jobs grew faster than total employment.

In 1980, as the villages were being settled, city population, employment, sectoral employment and migrant composition are uncorrelated with exposure. The cities differ only in their 1980 urban population share (Figure 8), and adding the 1980 birthplace shares leaves every headline estimate unchanged (Appendix Table C2; Appendix Table C3).

One standard deviation of exposure lowers the agricultural share by about a quarter of the average city's fall between 1980 and 2010. Across the main city sample the average city's agricultural employment share fell 18.6 percentage points over those three decades, from 56.0% to 37.4%, and cities at the 90th percentile of exposure sit 2.3 standard deviations above those at the 10th, a gap that implies a share 11.6 points lower, about 63% of the average city's fall over the period. Here I compare more exposed cities against less exposed ones, and zero exposure lies far outside the range the estimate is identified on. Manufacturing is a small sector in these cities: services account for most of the rise in the modern share and manufacturing for about one percentage point, so the shift is primarily into services (Appendix Table E4).

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<sup>12</sup>Appendix Table D6 re-runs the same columns at 150, 450, and 500 kilometers and with clustering on the catchment components; the three compositional columns reject at the 5% level under every Conley radius that returns a standard error and under the component clustering, and the agricultural level at the 10% level or better under each of them; under the wild cluster bootstrap over the catchment components, which has few clusters and little power, only the two modern-sector columns stay significant at the 10% level.

Table 3: Urban Growth and Structural Transformation: From Agriculture to the Modern Sector

|                        | Population + Employment |                  |                   | Agric Empl         |                      | Modern Empl        |                     |
|------------------------|-------------------------|------------------|-------------------|--------------------|----------------------|--------------------|---------------------|
|                        | (1)<br>ln pop           | (2)<br>ln emp    | (3)<br>unemp rate | (4)<br>ln L agric  | (5)<br>agric %       | (6)<br>ln L modern | (7)<br>modern %     |
| Agro-similarity        | 0.051<br>(0.059)        | 0.024<br>(0.057) | 0.008<br>(0.007)  | -0.166*<br>(0.093) | -0.051***<br>(0.018) | 0.158**<br>(0.062) | 0.066***<br>(0.015) |
| Settler count          | ✓                       | ✓                | ✓                 | ✓                  | ✓                    | ✓                  | ✓                   |
| Baseline Pop 1980      | ✓                       | ✓                | ✓                 | ✓                  | ✓                    | ✓                  | ✓                   |
| Urban share 1980       | ✓                       | ✓                | ✓                 | ✓                  | ✓                    | ✓                  | ✓                   |
| Total Empl 1980        | ✓                       | ✓                | ✓                 | ✓                  | ✓                    | ✓                  | ✓                   |
| Agri Empl 1980         | ✓                       | ✓                | ✓                 | ✓                  | ✓                    | ✓                  | ✓                   |
| Agro-climatic controls | ✓                       | ✓                | ✓                 | ✓                  | ✓                    | ✓                  | ✓                   |
| Outcome mean           | 11.842                  | 10.933           | 0.122             | 9.822              | 0.374                | 10.232             | 0.516               |
| Outcome SD             | 0.854                   | 0.840            | 0.040             | 0.896              | 0.183                | 0.975              | 0.156               |
| Observations           | 134                     | 134              | 134               | 134                | 134                  | 134                | 134                 |

Notes: Each column is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample, with the main specification controls marked by the check marks: the settler count and 16 agro-climatic covariates within 300 minutes, and four 1980 baselines. Outcomes are log population, log total employment and the unemployment rate (columns 1 to 3), log agricultural employment and the agricultural employment share (columns 4 and 5), and log modern-sector (manufacturing and services) employment and the modern-sector employment share (columns 6 and 7); shares and the unemployment rate are fractions, and the four sector columns are shaded. In the labels: ln (the natural log), L (a count of workers), emp and Empl (employment), unemp rate (the unemployment rate), and agric % and modern % (the two sector shares); Baseline Pop 1980, Total Empl 1980 and Agri Empl 1980 are the logs of 1980 population, total employment and agricultural employment, and Urban share 1980 is the urban share of the 1980 population; SD is the standard deviation. Conley SE within 200km in parentheses, with stars from the same standard errors: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . [Appendix C.1](#) reports randomization inference for the design-based estimates, the Online Supplement every randomization-inference  $p$ -value, and [Appendix Table D6](#) other spatial-inference radii; variants with polynomial terms in the settler count are in the appendix.

Most of the shift remains after I control for where the government placed the settlements. The government matched settlers to villages at random, but it placed the villages where there was greenfield land to develop, so some cities sit near many settlement sites and their exposure would be high under almost any assignment of settlers. I therefore control for each city's expected exposure, the exposure it would have if settlers were reassigned at random among the sites opened in the same province and period. The agricultural employment share then falls by 0.040 and the modern employment share rises by 0.052, against  $-0.051$  and  $+0.066$  in the main specification, about four fifths of each, and both remain statistically significant ([Appendix Table C1](#)). A test that reshuffles settlers among the same sites rejects the hypothesis of no effect on the modern employment share and does not reject it for the agricultural share.<sup>13</sup> Subtracting the expected exposure from exposure instead leaves little variation, because the two move closely together, so the estimates shrink toward zero ([Appendix Figure C1](#)); the precision of the main estimates comes from the Conley standard errors in [Table 3](#).

The shift out of agriculture and into the modern sector holds under alternative control sets, alternative weights on the surrounding villages, wider catchments, and standard errors clustered by province. The agricultural employment share falls and modern-sector employment rises in all 12 alternative specifications of [Table 4](#), statistically or marginally significant in 10 and 10 of them, and the population estimate is positive and statistically significant in 3. Different distance weights do not affect the agricultural share much, and modern-sector employment loses significance only when the nearest

<sup>13</sup>The randomization  $p$ -values, computed on the  $t$ -statistic of the controlled coefficient, are 0.050 for the modern employment share and 0.196 for the agricultural share (Online Supplement).

villages take almost all the weight, at 0.075 log points under inverse squared distance. At catchments of 400 and 500 minutes modern-sector employment rises more, by 0.188 and 0.268 log points, and population rises by 0.118 and 0.137 log points, statistically significant at both, while the agricultural share moves by -0.023 at 400 minutes, a statistically insignificant estimate, and by -0.044 at 500, marginally significant. Clustering the standard errors by province keeps the agricultural share statistically significant and modern-sector employment marginally significant. The estimates are also insensitive to whether the settler-count control enters linearly, in logs, or with a quadratic term (Online Supplement), and to controlling in addition for the settlers' size relative to the pre-existing rural population of the city's hinterland; only log agricultural employment becomes marginally insignificant under that control (Appendix Table D1). The full set of specifications is defined variant by variant in Appendix Table D2 and extended to every outcome in Appendix Figure D2, and the extended robustness material, including the price-measurement checks behind Section 5.2, is collected in Appendix D.

Table 4: Estimates' Robustness Under Main Specification Variations

|                             | (1)<br>Log<br>population | (2)<br>Agricultural<br>employment share | (3)<br>Log modern-sector<br>employment | Observations |
|-----------------------------|--------------------------|-----------------------------------------|----------------------------------------|--------------|
| Main specification          | 0.051<br>(0.059)         | -0.051***<br>(0.018)                    | 0.158**<br>(0.062)                     | 134          |
| <i>Controls</i>             |                          |                                         |                                        |              |
| Unweighted settler count    | 0.047<br>(0.071)         | -0.053***<br>(0.018)                    | 0.150**<br>(0.074)                     | 134          |
| Weighted covariates         | 0.101*<br>(0.060)        | -0.045***<br>(0.014)                    | 0.187***<br>(0.060)                    | 134          |
| Own-outcome baseline        | 0.051<br>(0.059)         | -0.051***<br>(0.011)                    | 0.177***<br>(0.063)                    | 134          |
| <i>Exposure weights</i>     |                          |                                         |                                        |              |
| Population-weighted         | 0.057<br>(0.057)         | -0.044***<br>(0.015)                    | 0.155**<br>(0.061)                     | 134          |
| 1/d                         | 0.002<br>(0.045)         | -0.052***<br>(0.018)                    | 0.110*<br>(0.062)                      | 134          |
| 1/d <sup>2</sup>            | -0.023<br>(0.036)        | -0.047***<br>(0.017)                    | 0.075<br>(0.057)                       | 134          |
| <i>Catchment radius</i>     |                          |                                         |                                        |              |
| Within 400 min              | 0.118**<br>(0.059)       | -0.023<br>(0.032)                       | 0.188**<br>(0.094)                     | 134          |
| Within 500 min              | 0.137***<br>(0.048)      | -0.044*<br>(0.026)                      | 0.268***<br>(0.052)                    | 134          |
| <i>FE and inference</i>     |                          |                                         |                                        |              |
| Province FE                 | 0.060<br>(0.074)         | -0.017<br>(0.017)                       | 0.084<br>(0.056)                       | 134          |
| Island FE                   | 0.043<br>(0.059)         | -0.049**<br>(0.020)                     | 0.151**<br>(0.060)                     | 134          |
| Cluster: province           | 0.051<br>(0.077)         | -0.051**<br>(0.022)                     | 0.158*<br>(0.084)                      | 134          |
| Island FE, cluster province | 0.043<br>(0.078)         | -0.049*<br>(0.025)                      | 0.151*<br>(0.084)                      | 134          |
| Outcome mean                | 11.842                   | 0.374                                   | 10.232                                 |              |
| Outcome SD                  | 0.854                    | 0.183                                   | 0.975                                  |              |

Notes: Each cell is the coefficient on standardized agro-similarity from one city-level regression of the 2010 census outcome in the column heading, where the modern sector is manufacturing plus services, with its standard error in parentheses. The first row is the main specification of Table 3, on the main city sample, with the full set of controls and Conley SE within 200km. Each row below it changes the design choice its label names, as Appendix Table D2 defines; the exposure-weight rows replace the equal weights of the villages within 300 minutes, with  $d$  the travel time from a city to a transmigration village, and the fixed-effect rows add an indicator for each province or for each island. Standard errors are clustered by province (17 provinces) in the two rows labeled as clustered, whose stars use the  $t$  distribution. Shaded cells clear the 10% level, and the shade deepens at the 5% and 1% levels. The Observations column gives the number of cities in each row. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Province fixed effects make both sectoral results statistically insignificant, and every other check leaves at least one of them statistically significant. Within provinces the agricultural share moves by -0.017, about a third of the main estimate, and modern-sector employment by 0.084 log points, about half of it, while the standard errors stay about the same size, so the coefficients shrink. Island fixed effects keep both results statistically significant.

The sectoral results hold when the proximity window is moved and when the cities are redrawn without the areas reclassified as urban. I re-estimate the effects at every proximity window from 250 to 500 minutes, with the exposure itself reconstructed at each radius, and the estimates keep their sign throughout, the sectoral responses hold at most windows of 300 minutes and wider, and the tighter window is imprecise (Appendix Figure D4). Without the reclassified urban areas, on 114 cities, the agricultural share moves by -0.059 and modern-sector employment by +0.179 log points, and under stricter urban criteria, on 81 cities, by -0.067 and +0.092, the last marginally significant (Appendix Table A2). The modern-employment result does not differ between the cities that entered 1980 more and less agricultural, while splitting at the median 1980 population separates the cities: modern-sector employment rises by 0.152 log points more in the larger half and population by 0.150 more, and the agricultural share moves alike in both halves (Appendix Table D5). Under every window and both alternative boundaries the share falls and modern-sector employment rises, and the population estimate stays small and mostly statistically insignificant.

## 5.2 The Goods-Market Mechanism: Food Trade, Not Migration

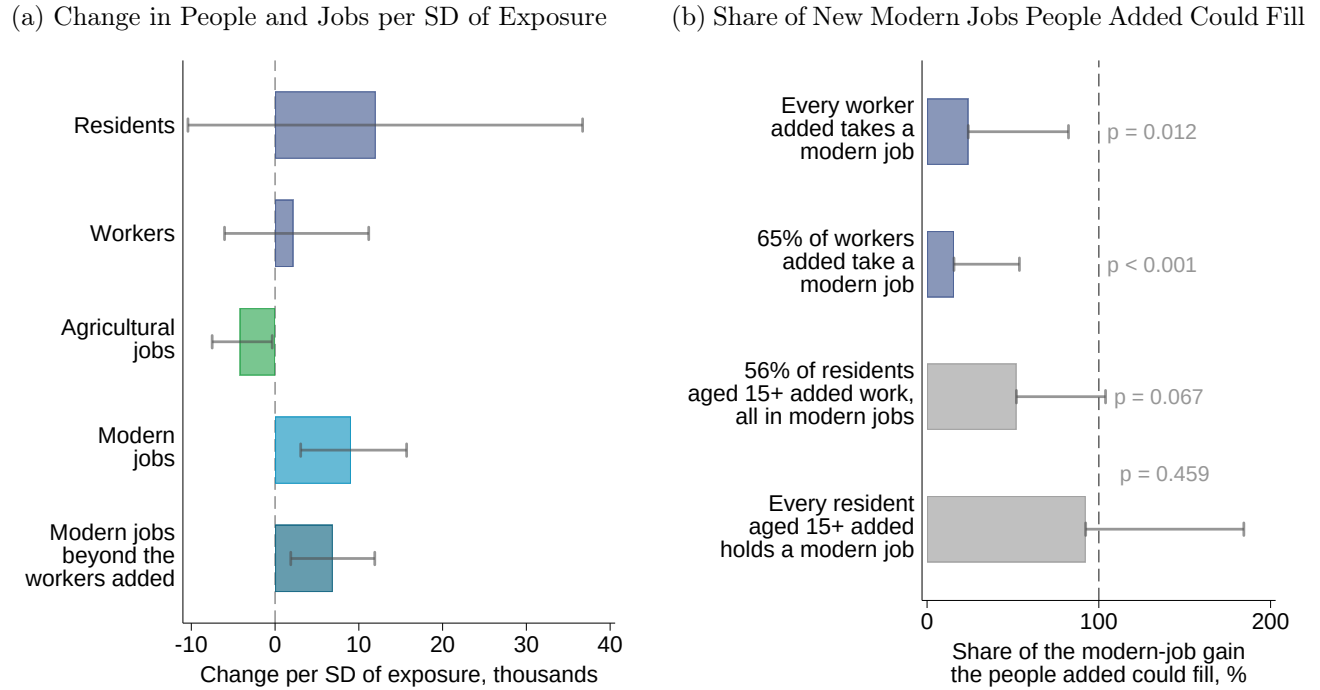
Per standard deviation of exposure, modern-sector employment grows by about 17% and population by about 5%, and the cities add far fewer workers than modern-sector jobs. At the mean city, one standard deviation of exposure adds about 9,100 modern-sector jobs and removes about 4,200 agricultural jobs, while the workforce grows by about 2,200 workers, which leaves 6,900 modern-sector jobs (90% interval 1,900 to 11,900) that the growth of the workforce cannot account for. Migrants who move to cities most of the time do not move for agriculture. If net employment growth equals arrivals, with the incumbent workforce fixed and without offsetting departures or labor-force entry and exit, and all additional workers take modern-sector jobs, they would fill about a quarter of the new modern-sector jobs, 24% at the estimates and at most 82% at 95% confidence, and the data reject that they fill all of them; at the rate at which in-migrants work in modern-sector jobs they fill 16%, at most 54% (Figure 9). In-migrants farm too: workers born outside the city's district farm at a rate of 26.9% against 40.4% for the locally born (Appendix E.2).

The restricted arrivals hypothesis is rejected even after allowing for how imprecisely the employment response is estimated.<sup>14</sup> The responses of total employment and of modern-sector employment move together across cities, so the test uses their joint precision, and it rejects that hypothesis whether every in-migrant works in the modern sector or in-migrants work in modern-sector jobs at their measured rate. Under the design-based specifications of Section 4.3, which hold expected exposure fixed,

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<sup>14</sup>At the 5% level if every in-migrant works in the modern sector (one-sided  $p = 0.012$ ) and at the 1% level at the rate at which in-migrants work in modern-sector jobs ( $p < 0.001$ ), the two  $p$ -values printed in panel (b) of Figure 9.

Figure 9: Workers Added and Modern Jobs Gained at the Mean City



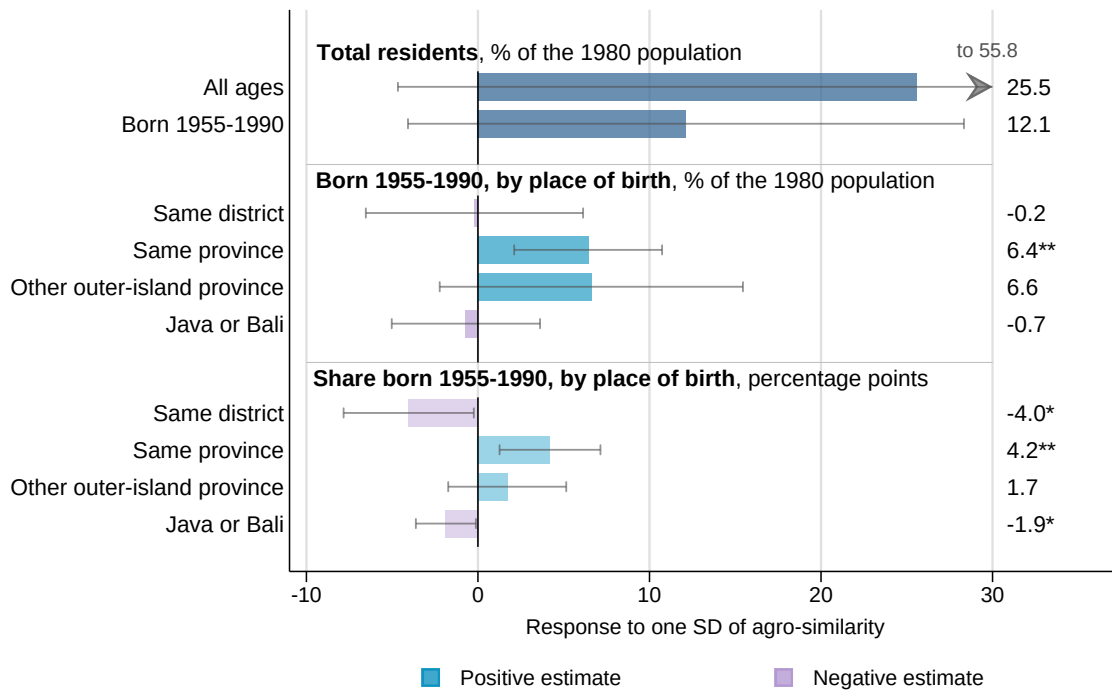
*Notes:* The main city sample, in the main specification with the full set of controls and Conley SE within 200km. Panel (a) turns columns 1, 2, 4 and 6 of Table 3 into the change in each outcome's level at the mean city per SD of exposure (the 2010 mean times  $e^b - 1$ ), with 90% intervals; the last bar is the modern jobs gained less the workers added, with its interval from the joint covariance of the two coefficients. Panel (b) gives the share of the modern-job gain the people added could fill if nobody already working in the city changed sector, its one-sided 95% upper bound (whisker), and the one-sided  $p$ -value of a share of 100% or more (dashed line); the first two bars count workers and the last two residents aged 15 and over. Appendix E.2 gives the construction. Source: 1980 and 2010 population censuses.

the share the workers added could fill is no longer bounded below the whole gain, because the two responses lose precision together, so only the main specification rejects when every additional worker takes a modern-sector job (Appendix E.2). Counted in residents aged 15 and over, the data do not reject that the people added fill the whole gain: residents of those ages grow by more than workers, about 8,400 at the mean city, imprecisely, so if every one of them held a modern-sector job they could fill most of it (92%, at most 184%). Yet a new resident is not a new worker: at the city's 0.563 workers per resident aged 15 and over, the residents added fill about half of the gain (52%, at most 104%), a marginally significant rejection of the whole gain.

Employment among workers born in the city's own district and the numbers of children and youths per native-born adult do not rise detectably with exposure. These proxies do not rule out fertility responses or labor-force entry and exit (Appendix Table E11).

I split the 2010 residents born from 1955 to 1990 by birthplace into four groups: born in the same district as the city, born elsewhere in the same province, born in another outer-island province, and born on Java or Bali; the row labels of Figure 10 abbreviate these four groups.

Figure 10: 2010 Urban Residents by Birthplace



*Notes:* Each bar is one regression of the row's outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Whiskers are 90% confidence intervals from the same standard errors; the whole-population interval runs past the axis and ends in an arrowhead, with its upper bound printed. The first block is the city's whole population and its residents born from 1955 to 1990 in the 2010 census, in % of its 1980 census population; the whole-population response is 10% of that outcome's mean of 244%, against a log population response of 5% in Table 3, since the two outcomes weight the cities differently and a regression in levels gives more weight to the cities that grew most. The second block splits that cohort, in the same units, into four birthplace groups that add up to it: born in the same district as the city, elsewhere in the same province, in another outer-island province, and on Java or Bali; the third block gives each group's share of the cohort in percentage points, with standard errors in Appendix Table C3. The hue of a bar marks the sign of its estimate, and its shade marks the block. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The policy moved two million people, so a mechanical effect is possible. Settlers could live in these villages for a few years and then move to the cities. The data show otherwise. Cities surrounded by more productive villages have fewer residents born in Java or Bali at the end of the period. That share is the best proxy for transmigrants I have in the city, so if anything, the cities surrounded by more productive villages do not have more of those transmigrants. Other rural migrants are moving in from elsewhere in the same province. They are likely primarily residents of other villages, some of whom could be transmigrants, and children already born on the Outer Islands (Figure 10).

Most of the fall in the agricultural share happens among workers of the same birthplace, and a smaller part comes from the rising share of in-migrants in the workforce. Crossing each city's 2010 workforce by birthplace and by industry splits the fall into a composition component, from the rising in-migrant share of the workforce, and a within component, from differences in agricultural employment shares within birthplace groups: 84% of the fall is within birthplace groups and 16% is composition,

with an interaction of 1%, and the within share is above the 80% that [Eckert & Peters \(2025\)](#) attribute to within-region change in the 1880-1920 United States (Appendix [Figure E2](#)). The census is a repeated cross-section, so the within component mixes two mechanisms, workers changing sector and one generation replacing another with nobody changing jobs, which [Budi-Ors \(2026\)](#) puts at 68% of Indonesia’s aggregate agricultural-share fall.<sup>15</sup>

The accounting rejects an arrivals explanation under the fixed-incumbent assumptions above. The census birthplace data cannot trace the in-migrants to particular rural areas. The model’s counterfactuals confirm that the sectoral reallocation can occur without migration and cannot occur without trade, and the model in [Section 6](#) captures the small role of migration with a low location-choice elasticity, consistent with the high internal mobility costs that [Bryan & Morten \(2019\)](#) document for Indonesia; the analysis of urban emergence in the Online Supplement finds that exposure transforms existing cities near the productive villages.

In more exposed cities the food expenditure share falls, and households do not consume less food. In the 1996 household survey, before the 2004 rice import restrictions, the food share of spending moves by  $-0.462$  of a standard deviation per standard deviation of exposure, calories per person by  $0.719$  and staple kilograms per person by  $0.407$ , all three statistically significant, and total household spending is higher, a marginally significant estimate; in 2011, staples per person do not differ detectably with exposure (Appendix [Table E6](#)). The rural food production has more comparative advantage in these higher-productivity villages, so urban dwellers now buy the food. Their consumption does not fall, while their rice imports, consumption minus the city’s own production in milled-rice weight, are higher, imprecisely. Importantly, the food expenditure share is going down while households consume more calories, which is consistent with the idea of subsistence in food consumption, that once basic food consumption needs are satisfied, it will not grow as much. The residents, through this potentially cheaper food, satisfy those basic needs and spend more money on non-agricultural items, raising demand for services and manufacturing products, which then drives up the demand for labor in that market. The model in [Section 6](#) formalizes this demand structure, non-homothetic preferences with a subsistence term in food ([Figure 11](#)).

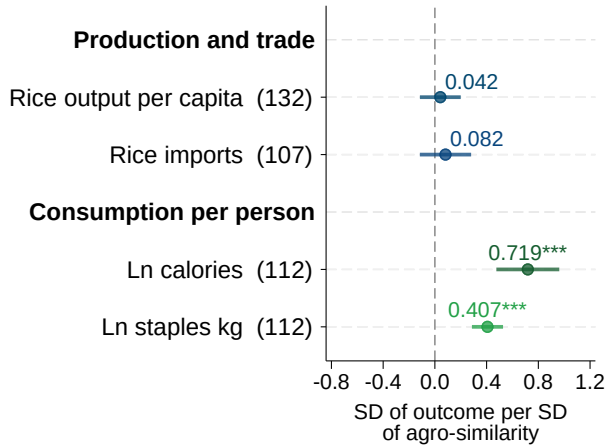
In the long run, the changes in wages and housing rents in the more exposed cities are statistically insignificant. The non-food expenditure share in 2011 rises by 1.6 percentage points, a statistically significant estimate, while housing rents and all three wage measures, the non-agricultural wage, the overall wage, and the real wage, are statistically insignificant in that wave, and rents do not move detectably in the 1996 wave either (Appendix [Table E2](#); Appendix [Table E7](#)). Flat wages alongside expanding modern-sector employment are the pattern the model must produce through its offsetting price and land terms, since the headline calibration sets the agglomeration elasticity to zero ([Section F.6](#)), and they are why the reallocation shows up in employment composition.

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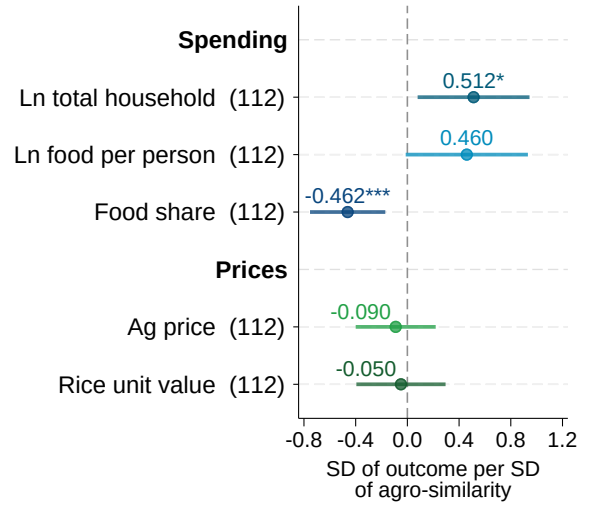
<sup>15</sup>[Hamory et al. \(2021\)](#) use the Indonesian Family Life Survey (IFLS) panel to study sectoral switching and find that individual fixed effects remove over 80% of the estimated agricultural productivity gap, so the gap largely reflects selection into sectors. The IFLS sample with sectoral histories in the outer-island cities that I study is too small, covering too few people and too few unique cities, the level at which I define treatment.

Figure 11: The Goods-Market Mechanism: Responses to Exposure, in SD

(a) Production, Trade, and Consumption



(b) Spending and Prices



*Notes:* Each row is one regression of the standardized outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Bars are 90% confidence intervals from the same standard errors; the number above each marker is the coefficient, and parentheses give the number of urban centers. Rice output per capita is 2003 village-census (PODES) rice production in tons of unhusked paddy over the population the same census records for the villages with a rice output figure, and rice imports are 2002 household-survey (Susenas) rice consumption minus that production in milled-rice weight, at the official rate of 0.6402 tons of milled rice per ton of paddy, both summed over the subdistricts with surveyed households and a rice output figure. Every other row is from the 1996 Susenas wave, before the 2004 rice import ban: calories over the whole diet and staple kilograms per person, total household spending, food spending per person, the food share of spending, the city's raw-food price index, and the rice unit value (purchase value over purchase kilograms). Logs are  $\ln(x+1)$ . \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

There is no statistically significant effect on food prices, which are measured imprecisely, as unit values from the household consumption module: in the 1996 wave the city's raw-food price index moves by  $-0.090$  of a standard deviation and the rice unit value by  $-0.050$ , both negative.<sup>16</sup> The 1996 household survey records rice consumption by grade, and the grade shares are consistent with more households buying higher-quality rice (Table S24 of the Online Supplement). The size of the price decline is model-implied.

I do not have data on food trade between a city and its hinterland, and the closest measures do not show a detectable change in either direction. The city's rice output per person moves by 0.042 of a standard deviation and its rice imports, consumption minus own production in milled-rice weight, by 0.082, both statistically insignificant, while the city's own rice production, in logs, is higher by +0.735 (Appendix Table E3; Figure 11). The food-trade mechanism predicts the largest response in the cities that depend most on food from their own hinterland, and Section 5.3 tests that prediction.

<sup>16</sup>The appendix re-estimates the price response on five 1996 price measures, four unit-value indices and the rice unit value, and the estimates are small, negative and statistically insignificant throughout (Figure D3).

### 5.3 Heterogeneity by Trade Openness

A city that can already import food cheaply gains little from a more productive hinterland. Its food price barely depends on the local harvest, so its labor allocation does not respond either. A city weakly connected to outside food markets depends on its own hinterland, so the same shock has its largest effect there. This is the price channel’s own prediction, and it matches the closed economy of Glaeser (2014), in which urbanization rises with the productivity of the farms around the city. The structural transformation is expected to concentrate in trade-closed cities and to be muted in trade-open ones. Table 5 runs the test on route proximity, each city’s route-aware cost-distance to the Thai and Vietnamese hubs that ship the bulk of Indonesia’s rice imports (Dawe 2008), following Bazzi (2017). A second measure, gravity market access built from realized rice-import volumes, is in part an equilibrium outcome, and Panel B of Table 5 reports it to show which way that bias runs. The specification interacts exposure with an indicator for high openness:

$$y_c = \beta_1 AgSim_c + \beta_2 AgSim_c \times HighOpen_c + \delta HighOpen_c + \phi TransMigrants_c^{IDW} + C_{c,1980} \xi + X_c \zeta + \varepsilon_c \quad (5)$$

where  $HighOpen_c$  equals one for a city on the more open side of the median of the panel’s openness measure and the other terms are those of (4), so  $\beta_1$  is the exposure effect in trade-closed cities,  $\beta_2$  the high-openness offset, and  $\beta_1 + \beta_2$  the net effect in trade-open ones. Panels A and B of Table 5 report  $\beta_1$ ,  $\beta_1 + \beta_2$  and  $\delta$ .

The market-access measure of Panel B might be endogenous, so in Panel C I leverage the more exogenous component of its variation, instrumenting market access by route proximity to the Thailand and Vietnam trading centers. The second stage replaces the indicator with the continuous measure:

$$y_c = \beta_1 AgSim_c + \beta_2 AgSim_c \times MA_c + \delta MA_c + \phi TransMigrants_c^{IDW} + C_{c,1980} \xi + X_c \zeta + \varepsilon_c \quad (6)$$

where  $MA_c$  is gravity market access in standard deviations and the other terms are those of (4), so  $\beta_1$  is the exposure effect at mean market access and  $\beta_2$  its change per standard deviation of market access. Market access and its interaction with exposure are instrumented by route proximity and its interaction with exposure, with first stages

$$\begin{aligned} MA_c &= \pi_1 Route_c + \pi_2 AgSim_c \times Route_c + \pi_3 AgSim_c + W_c \kappa + u_c \\ AgSim_c \times MA_c &= \rho_1 Route_c + \rho_2 AgSim_c \times Route_c + \rho_3 AgSim_c + W_c \lambda + v_c \end{aligned} \quad (7)$$

where  $Route_c$  is route proximity to the rice hubs in standard deviations, a higher value of either measure means a more open city, and  $W_c$  collects the settler count, the 1980 baselines and the agro-climatic controls of (4). The exclusion restriction is that, conditional on these controls, route proximity

shapes urban outcomes and their response to exposure only through market access. Route proximity predicts market access, and the first-stage Kleibergen-Paap  $F$  statistic is 8.13. Panel C reports the exposure effect at the mean market access of the more closed and of the more open cities,  $\beta_1 + \beta_2 MA_c$  evaluated at  $-0.84$  and  $0.86$  standard deviations, and  $\delta$ . With the two median splits in place of the continuous measures, the rows match those of Panels A and B one for one, with a weaker first stage (Appendix Table B2).

The effects concentrate in cities less connected to global markets, which rely more on local food supply. In the trade-closed cities a one standard deviation increase in exposure lowers the agricultural employment share by 6.5 percentage points and raises modern-sector employment by 0.269 log points, about 2.7 times the open-city estimate, and both are statistically significant. In the more open cities the agricultural share still falls, by 4.4 points, and the modern share still rises, while the estimate for modern-sector employment, 0.099 log points, is statistically insignificant. The offset on modern-sector employment,  $\beta_2$ , is  $-0.169$  log points (s.e. 0.095) and marginally significant, the offset on the modern share is marginally significant as well, and the offset on the agricultural share, 0.021 (s.e. 0.025), is statistically insignificant. On the gravity split of Panel B the agricultural-share offset is larger and statistically significant, at 0.053 (s.e. 0.021). That is the direction in which a realized-trade measure is biased: an exposed city that buys more of its food nearby imports less rice and is classified as more closed. On the modern-sector outcomes the two splits agree (Appendix B.2). Under the realized market-access split of Panel B, the agricultural share falls in the more closed cities and does not fall detectably in the more open ones. The agricultural level falls marginally significantly in the open cities and imprecisely in the closed ones, and city size does not respond detectably in either half. In Panel C the instrumented offset,  $\beta_2$ , is  $-0.153$  log points on modern-sector employment per standard deviation of market access (s.e. 0.080), marginally significant, and the offset on the modern share is marginally significant as well. On the agricultural share and on modern-sector employment, the offset is attenuation, not reversal. The open-city effects keep the closed-city sign, as the price channel implies, where the open economy of Glaeser (2014), his Propositions 2 and 3, would flip it. In the more closed cities the population and employment estimates are positive and larger than in the more open ones, and neither difference is precisely estimated; the pattern is consistent with urbanization rising with farm productivity in Glaeser’s closed economy. In his model the urban sector produces traded manufacturing goods; here the added modern-sector workers are mostly in services, so cities can also grow in services (Gollin et al. 2016; Jedwab & Vollrath 2015) once their residents are well fed (Appendix Table E4). The split by openness is largely a split between the western and the eastern cities of the sample: with island indicators interacted with exposure, the modern-employment difference between the two halves keeps its size, at  $-0.203$  log points (s.e. 0.126), and loses precision. Agricultural productivity of the surrounding hinterland does not play as much of a role in shaping urban outcomes in cities with easier access to open markets, which may be because food supply is much less of a binding constraint in these cities at baseline. There are no baseline data on prices and food availability, so I cannot test that directly. The split is difficult to rationalize outside the goods market. A migration or amenity mechanism, generic income growth, or land conversion at the urban fringe does not give a city’s rice-import connections any role in whether its labor market responds to the hinterland, while a food-trade mechanism requires

exactly that switch, so each would predict effects in open and closed cities alike; the direct tests of the other channels are in [Section 5.4](#). Among the reduced-form results, the split by trade openness ties the response most directly to the cost of food in the city. Construction and validation of both openness measures, including a validation of the route measure against the price response to Indonesia’s 2004 rice-import ban, are in [Appendix B.2](#).

Table 5: Agro-Similarity by Trade Openness: Exogenous Route Geography and Realized Market Access

|                                                                                                  | (1)          | (2)          | (3)            | (4)             | (5)              | (6)             | (7)             |
|--------------------------------------------------------------------------------------------------|--------------|--------------|----------------|-----------------|------------------|-----------------|-----------------|
|                                                                                                  | ln pop       | ln emp       | unemp rate     | ln L agric      | agric %          | ln L modern     | modern %        |
| <i>Panel A: High openness = below-median route distance to the Thai and Vietnamese rice hubs</i> |              |              |                |                 |                  |                 |                 |
| <b>Agro-similarity, more closed cities: <math>\beta_1</math></b>                                 | <b>0.118</b> | <b>0.101</b> | <b>0.004</b>   | <b>-0.164</b>   | <b>-0.065***</b> | <b>0.269***</b> | <b>0.093***</b> |
|                                                                                                  | (0.086)      | (0.082)      | (0.009)        | (0.132)         | (0.024)          | (0.089)         | (0.022)         |
| Agro-similarity, more open cities: $\beta_1 + \beta_2$                                           | 0.014        | -0.018       | 0.011          | -0.170*         | -0.044**         | 0.099           | 0.052***        |
|                                                                                                  | (0.059)      | (0.060)      | (0.007)        | (0.092)         | (0.021)          | (0.070)         | (0.016)         |
| High openness (=1)                                                                               | 0.028        | 0.009        | -0.004         | 0.053           | 0.004            | -0.007          | -0.013          |
|                                                                                                  | (0.067)      | (0.083)      | (0.010)        | (0.151)         | (0.028)          | (0.083)         | (0.019)         |
| <i>Panel B: High openness = above-median market access over realized rice-import volumes</i>     |              |              |                |                 |                  |                 |                 |
| <b>Agro-similarity, more closed cities: <math>\beta_1</math></b>                                 | <b>0.109</b> | <b>0.073</b> | <b>0.014**</b> | <b>-0.223**</b> | <b>-0.078***</b> | <b>0.231**</b>  | <b>0.083***</b> |
|                                                                                                  | (0.085)      | (0.085)      | (0.007)        | (0.105)         | (0.018)          | (0.092)         | (0.016)         |
| Agro-similarity, more open cities: $\beta_1 + \beta_2$                                           | -0.002       | -0.022       | 0.002          | -0.106          | -0.025           | 0.085           | 0.049***        |
|                                                                                                  | (0.068)      | (0.057)      | (0.007)        | (0.084)         | (0.020)          | (0.077)         | (0.018)         |
| High openness (=1)                                                                               | 0.025        | 0.010        | -0.005         | 0.084           | 0.008            | -0.036          | -0.008          |
|                                                                                                  | (0.074)      | (0.067)      | (0.011)        | (0.082)         | (0.026)          | (0.096)         | (0.023)         |
| <i>Panel C: Two-stage least squares on market access (z), at the mean closed and open city</i>   |              |              |                |                 |                  |                 |                 |
| <b>Agro-similarity, mean closed city</b>                                                         | <b>0.141</b> | <b>0.125</b> | <b>0.004</b>   | <b>-0.138</b>   | <b>-0.061***</b> | <b>0.307***</b> | <b>0.103***</b> |
|                                                                                                  | (0.099)      | (0.101)      | (0.010)        | (0.161)         | (0.021)          | (0.096)         | (0.021)         |
| Agro-similarity, mean open city                                                                  | -0.032       | -0.071       | 0.014          | -0.233**        | -0.050**         | 0.046           | 0.047**         |
|                                                                                                  | (0.067)      | (0.075)      | (0.009)        | (0.118)         | (0.023)          | (0.075)         | (0.021)         |
| Market access (z)                                                                                | 0.122        | 0.147        | -0.015         | 0.322*          | 0.045            | 0.019           | -0.057**        |
|                                                                                                  | (0.120)      | (0.125)      | (0.015)        | (0.171)         | (0.028)          | (0.133)         | (0.025)         |
| First-stage $F$ (Kleibergen-Paap)                                                                | 8.13         | 8.13         | 8.13           | 8.13            | 8.13             | 8.13            | 8.13            |
| Outcome mean                                                                                     | 11.842       | 10.933       | 0.122          | 9.822           | 0.374            | 10.232          | 0.516           |
| Outcome SD                                                                                       | 0.854        | 0.840        | 0.040          | 0.896           | 0.183            | 0.975           | 0.156           |
| Observations                                                                                     | 134          | 134          | 134            | 134             | 134              | 134             | 134             |

*Notes:* Each column of each panel is one regression of a 2010 outcome of [Table 3](#) on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Panels A and B interact agro-similarity with an indicator for high openness, as in (5), split at the median over all 138 urban centers within the 300-minute window: route distance ([Section 3.5](#)) is the travel time from a city through its nearest customs port to the nearer of Bangkok and Ho Chi Minh City, and market access ([Appendix B.2](#)) sums the rice imports of every customs port, discounted by the travel time from the city. Realized imports respond to the substitution the paper studies, so Panel B is reported to show the sign of the bias that realized trade introduces, and Panel A, the route split, is the paper’s openness result. Panel C is two-stage least squares, with standardized market access and its interaction with agro-similarity instrumented by standardized route proximity and its interaction; its two slopes are the effect of agro-similarity at the mean market access of the cities below and above the median of market access ( $-0.84$  and  $0.86$  SD), and the Kleibergen-Paap first-stage  $F$  under the Conley variance, 8.13, is above the Stock-Yogo 10% maximal-size critical value of 7.03. SE in parentheses; the four sector columns are shaded. (z) means standardized to mean zero and SD one, and the beta terms are the coefficients of (5). \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 5.4 Alternative Channels

A rural productivity gain can reach a city through more than the food market, and the theory itself names the rival explanations: workers released off the farm, capital moving out through the banking system ([Bustos et al. 2020](#)), and a factor-biased gain that pulls labor into farming at a fixed outside

price (Bustos et al. 2016); outside the theory sits a fourth possibility, that the state, having settled the villages, built the roads and facilities that transformed the cities. Two further possibilities are a new district capital, which brings government offices to the town chosen as its seat, and the spending of rural households whose incomes rose. Each is tested. The labor release is a prediction about the villages that received the gain, and the village agricultural employment share and measured settler stay-share do not fall detectably with agro-similarity (Section 5.5). A factor-biased mechanism does not give a city’s food-import connections any role in whether its labor market responds to the hinterland, so it fails to predict the concentration of the response in trade-closed cities that Section 5.3 establishes. If capital moved out through the banking system, banks would expand where the gain arrived. At a given 1980 level, banks per person and bank presence do not move with exposure in the city or in the countryside with settlements, and only the countryside without settlements gains banks per person; the village censuses count banks and do not record deposits or lending. The finance and insurance share of city employment does rise with exposure from 1980 to 2010, the one result consistent with a capital channel in the city (Table 6). The state-provision explanation is tested directly, with each item’s own 1980 level held fixed. In the city, exposure does not predict the population shares on an asphalt main road and on a road passable by car, health centers per person, road density around the city at its 1950 density, road quality on the surveyed corridors over 1990 to 2007, or the placement of the INPRES (Instruksi Presiden, presidential instruction) school-building program of the 1970s (Table 6, Appendix E.4). Exposure does not predict which cities became the seat of a district created after 1999, and modern-sector employment without education, health, social and government services rises with exposure as the headline does (Table 6); the headline estimates also hold without the 33 cities that host a new seat (Appendix E.5).

The countryside is where public provision did respond. At a given 1980 level, the countryside with settlements around more exposed cities gained asphalt roads by 1990 and grid connections and senior secondary schools by 2011, and the countryside without settlements gained roads, grid connections and banks per person; with each rural subdistrict as a unit, roads and electricity rose with exposure in subdistricts with and without a settlement alike (Table 6, Appendix E.5). From 1990 on, the share of villages on the state grid rose faster in the more exposed rural subdistricts with a settlement, partly as the grid replaced other suppliers, while households in villages already on the grid did not connect at a higher rate; with the grid measured at both ends, exposed places were not less electrified in 1980. A grid line can follow demand that is already there, so these tests cannot tell the state’s supply apart from household demand, and they leave public provision open as an explanation of the changes in the countryside and, through food production and trade, the cities. Rural households around more exposed cities do not spend more per person in 2011, and the non-food share of their spending does not move, so these data do not show the rise in rural demand for city goods that a hinterland-demand channel requires (Table 6). A gain that arrived as export crops would reach a city as income in its hinterland. The gain the settlement villages receive shows up in rice, the staple the cities consume. A village more similar to its settlers’ origins is more likely to grow rice at all, under the village controls of this paper

Table 6: Alternative Channels in the City and Its Countryside

|                                                                                                 | The city                           |        | The countryside                   |     |                                   |     |
|-------------------------------------------------------------------------------------------------|------------------------------------|--------|-----------------------------------|-----|-----------------------------------|-----|
|                                                                                                 | Estimate                           | Cities | with settlements                  |     | without settlements               |     |
| Asphalt main road<br>population share, 1980 to 1990                                             | 0.0271<br>(0.0172)<br>[0.0280]     | 131    | 0.0590***<br>(0.0207)<br>[0.0218] | 129 | 0.0679***<br>(0.0161)<br>[0.0175] | 123 |
| Main road passable by car all year<br>population share, 1980 to 2011                            | 0.0193<br>(0.0127)<br>[0.0136]     | 122    | 0.0201*<br>(0.0106)<br>[0.0091]   | 121 | 0.0286***<br>(0.0084)<br>[0.0083] | 116 |
| Road density on the current road map<br>1950 density held fixed                                 | -0.0764<br>(0.0894)<br>[0.0991]    | 131    |                                   |     |                                   |     |
| Households on the state grid<br>household share, 1980 to 2011                                   |                                    |        | 0.0553*†<br>(0.0284)<br>[0.0262]  | 131 | 0.0792***<br>(0.0194)<br>[0.0198] | 125 |
| Population in a village with grid power<br>population share, 1980 to 2011                       |                                    |        | 0.0719***<br>(0.0216)<br>[0.0215] | 123 | 0.0552***<br>(0.0168)<br>[0.0168] | 119 |
| Senior secondary schools<br>per 1,000 people, 1980 to 2011                                      |                                    |        | 0.0424***<br>(0.0098)<br>[0.0120] | 131 | 0.0030<br>(0.0037)<br>[0.0043]    | 125 |
| Health centers<br>per 1,000 people, 1980 to 2011                                                | 0.0095<br>(0.0097)<br>[0.0106]     | 128    | 0.0072<br>(0.0186)<br>[0.0174]    | 131 | -0.0066<br>(0.0155)<br>[0.0146]   | 125 |
| Banks<br>per 1,000 people, 1980 to 2011                                                         | 0.0110<br>(0.0092)<br>[0.0093]     | 133    | 0.0119*†<br>(0.0072)<br>[0.0060]  | 131 | 0.0077***<br>(0.0026)<br>[0.0034] | 124 |
| Bank presence<br>population share, 1980 to 2011                                                 | 0.0251<br>(0.0248)<br>[0.0222]     | 122    | -0.0138<br>(0.0137)<br>[0.0130]   | 123 | 0.0185*†<br>(0.0112)<br>[0.0118]  | 117 |
| Finance and insurance employment<br>share of employment, 1980 to 2010                           | 0.0009**<br>(0.0004)<br>[0.0005]   | 134    |                                   |     |                                   |     |
| Seat of a district created 1999 to 2010<br>seat in 1998 held fixed                              | -0.0064<br>(0.0518)<br>[0.0558]    | 134    |                                   |     |                                   |     |
| Log modern-sector employment, 2010<br>less education, health, social<br>and government services | 0.1652***†<br>(0.0711)<br>[0.0660] | 134    |                                   |     |                                   |     |
| Log spending per person, 2011<br>rural households<br>rightmost estimate: whole hinterland       |                                    |        | -0.0225<br>(0.0393)<br>[0.0353]   | 131 | -0.0306<br>(0.0273)<br>[0.0264]   | 128 |
| Non-food share of spending, 2011<br>rural households<br>rightmost estimate: whole hinterland    |                                    |        | -0.0022<br>(0.0078)<br>[0.0080]   | 131 | -0.0060<br>(0.0054)<br>[0.0047]   | 128 |

*Notes:* Each cell regresses the row's outcome, measured over the column's area, on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km plus the outcome's own 1980 value or the earlier value its label names; the modern-sector row, which counts workers with a recorded industry, holds the log of its 1980 count fixed, and the two spending rows do not add a 1980 value, since the household survey (Susenas) waves held for the paper start in 1994. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets; Cities counts the cities in each cell. The countryside with settlements is the rural subdistricts with a settlement village within 300 travel minutes of the city, its own subdistricts left out; the countryside without settlements is the rural subdistricts within 300 travel minutes of the city with no settlement village, and the whole hinterland is every rural subdistrict within those 300 minutes. Road density is measured over the subdistricts that hold the city's catchment villages, every subdistrict that belongs to a city left out; electricity and schools are tested in the countryside only. A presence share is the population share in villages with the item (for roads, of villages reached by land); health centers are puskesmas and auxiliary centers (pustu). The presence shares to 2011 leave out the 8 cities of West Kalimantan, whose villages were merged between the 1990 and 1993 village censuses; the car-access and bank-presence shares also leave out the West Sumatra subdistricts recorded as nagari in 2011, a larger unit than the 1980 village, which the grid-village share keeps, measured on their hamlets. Sources: village censuses (PODES) 1980 to 2014, population censuses 1980 and 2010, Susenas 2011, the current national and provincial road map and the 1950 atlas, and the World Bank district proliferation crosswalk. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test the table does not print), or with both.

and under those of [Bazzi et al. \(2016\)](#), whose Table 7 finds the less similar villages tilting their crop revenue toward cash crops.<sup>17</sup>

Three further possibilities concern the cities themselves. The communal and separatist conflicts of around 2000 in Aceh and in West and Central Kalimantan ([Barter & Côté 2015](#)), and the Poso conflict in Central Sulawesi ([van Klinken 2007](#)), could have pushed displaced people into the towns of those provinces, which hold 26 of the 134 cities. Interacting exposure with an indicator of the four provinces, the effect does not differ measurably inside them, and the slope outside stays near the headline; leaving the four provinces out instead halves the structural-change estimates and leaves them imprecise, because the control coefficients are then estimated on 108 cities and the exposure variation left after the controls falls by a third ([Appendix E.5](#)). An exposed city could also grow by drawing people and firms from less exposed cities nearby. With the exposure of neighboring cities, weighted by inverse distance, added beside a city's own, the sector shares hold under every variant of the measure: the farm share and the modern share do not move with the exposure of neighbors, and the own-exposure estimates on both stay at the headline. The size results are mixed. Log population and employment fall with the exposure of neighbors under the weighted measure. Both estimates remain negative but smaller without three pairs of twin cities or with province effects, and are statistically insignificant except for log employment without the twin cities, which stays marginally significant. The plain mean within 100 km of the city is statistically insignificant on every outcome ([Appendix E.5](#)). Subsetting each city more strictly, to exclude its surrounding less urban and more agricultural subdistricts, does not weaken the sectoral reallocation of employment out of agriculture, which is therefore not a product of the farmland that whole subdistricts add to a city ([Appendix Table D4](#)). Rebuilt on its member subdistricts that were at least one quarter non-agricultural in 1980, a city's agricultural employment share falls by 6.2 percentage points per standard deviation of agro-similarity, against 5.1 on all of its subdistricts, its farm employment falls by more, and its modern-sector employment rises. At one half, a cut that also removes old town centers still mostly agricultural in 1980, the share falls by 7.1 points while the modern-sector estimate loses precision, so the shift out of agriculture is strongest where the city is most urban, which speaks against land conversion at the urban fringe as well. Three channels stay untested here: processing of the farm output, commodity booms, and other government programs. The tables of every test are in [Appendix E.5](#). The Online Supplement sets out the rival explanations named by the theory in full.

## 5.5 Winners and Losers of Rural Productivity Growth

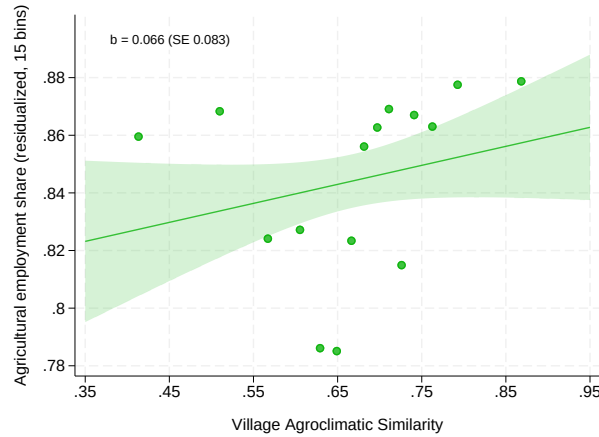
The newly settled transmigration villages show the flip side of the city result. If anything, more of their workers farm where agro-similarity is higher, while in the nearby cities the agricultural share falls ([Figure 12](#)).

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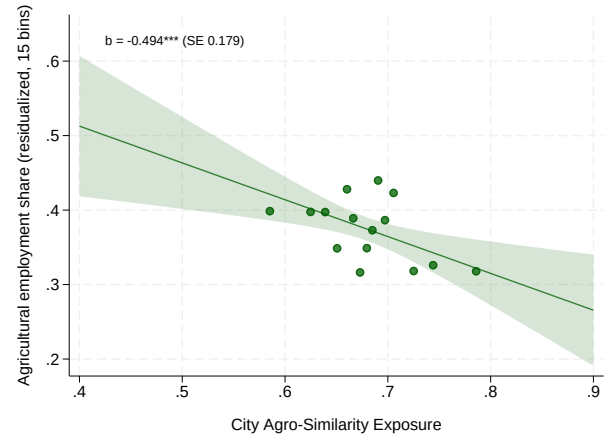
<sup>17</sup>Per standard deviation of agro-similarity, the probability that a settlement village grows any rice rises by 0.079 (standard error 0.017) with the village geography controls of the paper's village results, by 0.058 (0.026) with province-by-placement-year cells, and by 0.127 (0.022) under the Table 7 controls of [Bazzi et al. \(2016\)](#), on the 814 settlement villages with Conley standard errors at 200 km; under the village controls the estate-crop outcomes do not move.

Figure 12: The Agricultural Employment Share Against Agro-Similarity in the Transmigration Villages and in the Nearby Cities

(a) Transmigration Villages, 2000



(b) Nearby Cities, 2010



*Notes:* Binned scatter plots: dots are 15 quantile-bin means of the outcome and of agro-similarity on its 0 to 1 index scale, both residualized on the panel's controls with means added back. The line and its 90% band come from ordinary least squares on every residualized unit of the sample, and the in-plot coefficient is its slope with the panel's standard error and stars under that standard error (\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ ). **(a)** The share of a village's workers in agriculture in the 2000 Population Census against the village's own agro-similarity, on the 808 transmigration villages with a measured agricultural share and a measured settler stay-share below 2, with village terrain, geographic and climate controls; the standard error is clustered by island (wild cluster bootstrap over the four island clusters, Webb weights,  $p = 0.83$ ; Conley SE within 200km,  $p = 0.32$ ). **(b)** The agricultural employment share in the 2010 population census against city exposure, the mean agro-similarity of the transmigration villages within 300 travel minutes, on the main city sample, in the main specification with the full set of controls and Conley SE within 200km; column (5) of Table 3 gives the same relationship in standardized units.

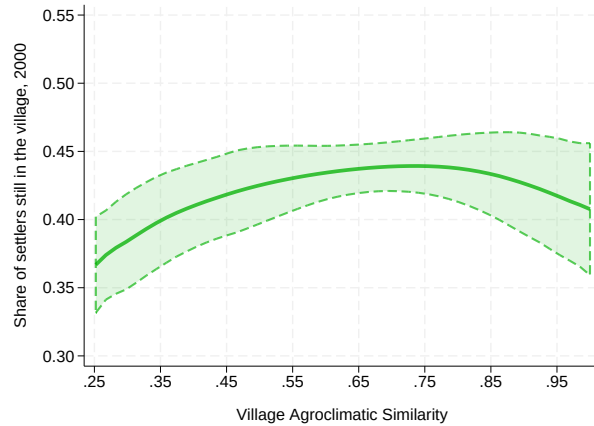
In the transmigration villages, panel (a), there is no sign of the negative gradient a release of labor requires. Well-matched villages keep farming. The rise of the agricultural share with agro-similarity is statistically insignificant, and the fitted curve is flat over the lower half of the similarity range and rises over the upper half, most steeply among the best-matched villages (panel (b) of Figure 13). In the nearby cities, panel (b), the same variation lowers the agricultural employment share (Table 3, column 5). The result within cities is almost the exact mirror of that in the villages themselves, where the more productive villages are retaining more workers in agriculture. In response to the positive productivity shock, the villages focus on agriculture and trade this food with the cities, and the cities then reduce agricultural employment, freeing up workers who no longer need to produce food within the city given the higher food supply from the villages.<sup>18</sup> In Matsuyama's world, the villages are the open economy that receives the shock: they trade their food with the cities and, if anything, specialize further in agriculture, their comparative advantage. The cities they feed respond as a city does in the closed economy of Glaeser (2014), moving employment out of agriculture and into the modern sector, and a test confined to the shocked villages misses this pairing.

<sup>18</sup>The village side is suggestive. The conditional gradient of the village agricultural share on agro-similarity is positive under every specification (0.066 on the common sample of  $N = 808$ ), and Conley standard errors and a wild cluster bootstrap both leave it indistinguishable from zero, with or without a quadratic; the city side, panel (b), is the headline estimate and is precise.

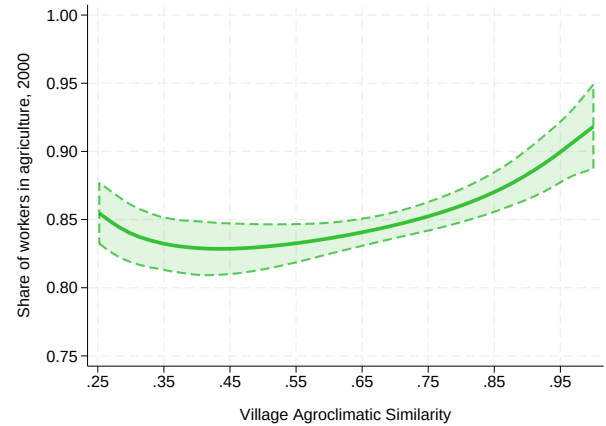
The measured settler stay-share does not vary detectably with agro-similarity. The slope is statistically insignificant, and if anything settlers are slightly more likely to stay in the more productive villages (Figure 13). The numerator does not identify original settlers exactly and excludes their locally born children, so the measure cannot establish where urban in-migrants came from.

Figure 13: Settler Stay-Share and the Agricultural Employment Share Against Agro-Climatic Similarity

(a) Settler Stay-Share



(b) Agricultural Employment Share, 2000



*Notes:* Semiparametric local linear fit (Epanechnikov kernel, bandwidth 0.25) of the village settler stay-share in panel (a), and of the share of a village's workers in agriculture in the 2000 Population Census in panel (b), on the village's agro-similarity (0 to 1 scale), with their pointwise 90% confidence bands, partialing out the village's ruggedness, slope classes, log altitude, distance to the coast and 1948 to 1978 mean temperature, estimated on the 808 transmigration villages with a measured 2000-census agricultural employment share and a measured stay-share below 2 and plotted over the 0.25 to 1 similarity range. The bands treat the villages as independent; panel (a) of Figure 12 reports the linear gradient of the agricultural share with its island-clustered, wild-bootstrap and Conley inference. Appendix Figure C3 draws both panels at a bandwidth of 0.30 and with each village weighted by its 2000 population. The plotted stay-share is a village's 2000-census residents born in Java or Bali over the settlers the program placed there; the stay-share named in the text also removes residents who lived in Java or Bali five years before the census and residents born after the village was settled. The linear gradient under each definition, with island-clustered, wild-bootstrap, and Conley inference, is in the Online Supplement.

The opposite responses in the villages and in the cities are consistent with these two cases. The village gradient is imprecisely estimated, so the evidence from the villages is suggestive. The city, with less of a comparative advantage in agriculture and enough food available, is where the economy transforms, while the directly shocked villages continue to focus on agriculture. The measured stay-share has a statistically insignificant gradient.

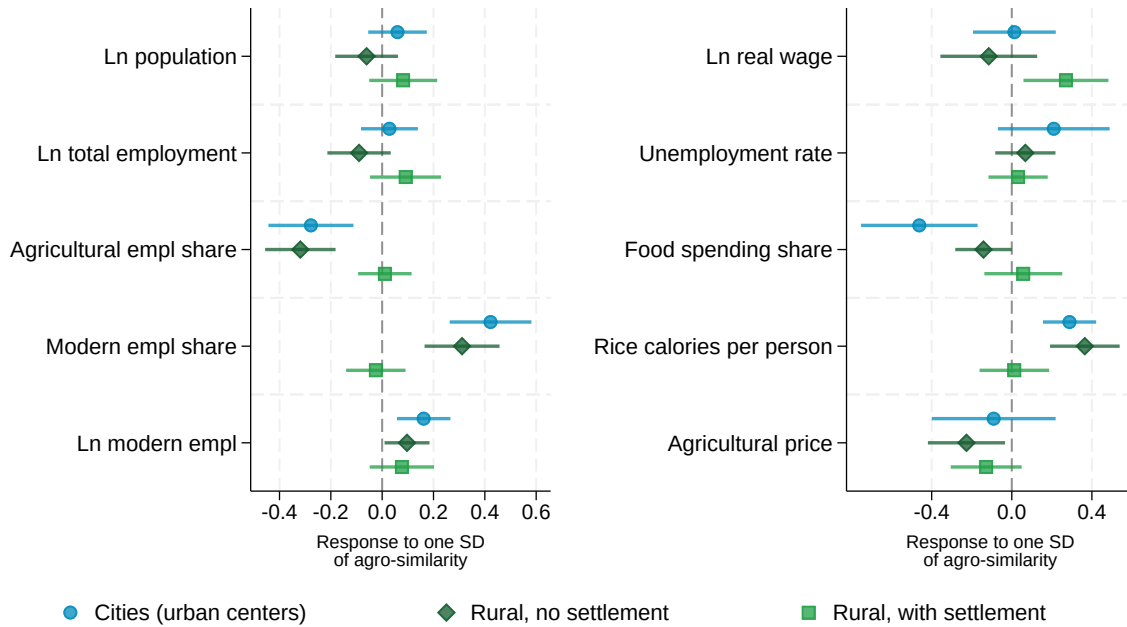
A national average of agricultural contraction and modern-sector expansion conceals within-country variation in how the gains to agricultural productivity accrue, differing between the regions that experience the growth and the regions that benefit from it indirectly through trade. Figure 14 separates the two. It compares the urban centers with rural subdistricts with and without a settlement. Both rural groups lie outside every city and within 300 travel minutes of a settlement village. Outcomes and agro-similarity are standardized separately among cities and among the pooled rural subdistricts. The regressions use the same controls, with the settler count measured within 300 travel minutes for cities and within 1,000 travel minutes for rural subdistricts.

The cities show the strongest structural transformation, and modern-sector employment growth outpaces population growth. The modern employment share rises by 0.42 of a standard deviation and log modern employment by 0.16. The population estimate is positive, at 0.06 of a standard deviation, and statistically insignificant.

Figure 14: The Response by Location Type: Cities and Rural Subdistricts With and Without a Settlement

(a) Structural Transformation

(b) Wages, Unemployment, Food, and Prices



*Notes:* Each marker is the response to a one-SD increase in agro-similarity, the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, and its bar is the 90% confidence interval from Conley SE within 200km. Cities are the main city sample; the rural subdistricts with and without a settlement lie outside every city and within 300 travel minutes of a transmigration village. Outcomes and agro-similarity are standardized once among the cities and once among the pooled rural subdistricts, at exposure SDs of 0.103 (138 cities with a measured exposure) and 0.100 (787 subdistricts), so the three areas are read on one scale. The city regressions are in the main specification with the full set of controls; the subdistrict regressions have the same controls with the settler count taken over villages within 1,000 travel minutes. **(a)** Every row is a 2010 census outcome on the main city sample, 191 rural subdistricts with a settlement and 596 without. **(b)** The unemployment rate is from the 2010 census; the real wage is the median earner's wage over the local price index, from the 2011 household-survey (Susenas) wave, the only wave with wages for the subdistricts; the food share of spending, rice calories per person (rice kilograms at a fixed calorie content) and the agricultural price (the raw-food index) are from the 1996 wave, before the 2004 rice import ban. Each row uses the units its wave covers: 112 to 134 cities, 82 to 191 subdistricts with a settlement and 236 to 596 without. Every outcome is in Appendix Table E1 with its SE.

The rural subdistricts without a settlement match the cities to an extent, in a similar sectoral reallocation of employment. The agricultural employment share falls by 0.32 of a standard deviation there and the modern share rises by 0.31, both precisely estimated. Yet the modern-sector growth is more contained there, at 0.10 in logs against 0.16 in the cities, and the estimate on overall employment is negative, at  $-0.09$ , so this does not look like overall growth in those areas.

The areas that experienced the positive productivity shock are effectively not changing. The estimates on population and on overall employment are positive, at 0.08 and 0.09, which is consistent with the historical accounts, and on 191 subdistricts both are indistinguishable from zero. There is no reallocation out of agriculture into the modern sector. The standard errors on the three composition

outcomes are smaller here than in the larger group with no settlement, and the intervals exclude the reallocation the other two areas show. The pattern is consistent with the same hypothesis, in which village economies are open economies. They trade with the cities, and facing more demand from the cities they continue in agriculture .

The comparison extends to prices and wages, and on the real wage the gain accrues to the rural subdistricts with a settlement. Where the transmigration villages are more productive, the subdistricts around them have higher real wages, by 0.27 of a standard deviation in the 2011 wave. The nominal wage there rises by 0.15 and is not distinguishable from zero, so the real gain is a price effect. The raw-food price index of the 1996 wave is lower too, and imprecisely estimated: it falls by 0.13 of a standard deviation in the subdistricts with a settlement, an estimate not distinguishable from zero, and by 0.23 in the subdistricts without one, a marginally significant estimate (Appendix [Table E1](#)).

There is no real wage growth in the cities. The sectoral reallocation there therefore comes without as clear a gain in what a worker earns, and the effect is confined to the movement of workers out of agriculture and into the modern sector. In the rural subdistricts with no settlement the real wage effect is, if anything, negative, at  $-0.11$  of a standard deviation, which the data do not separate from zero.

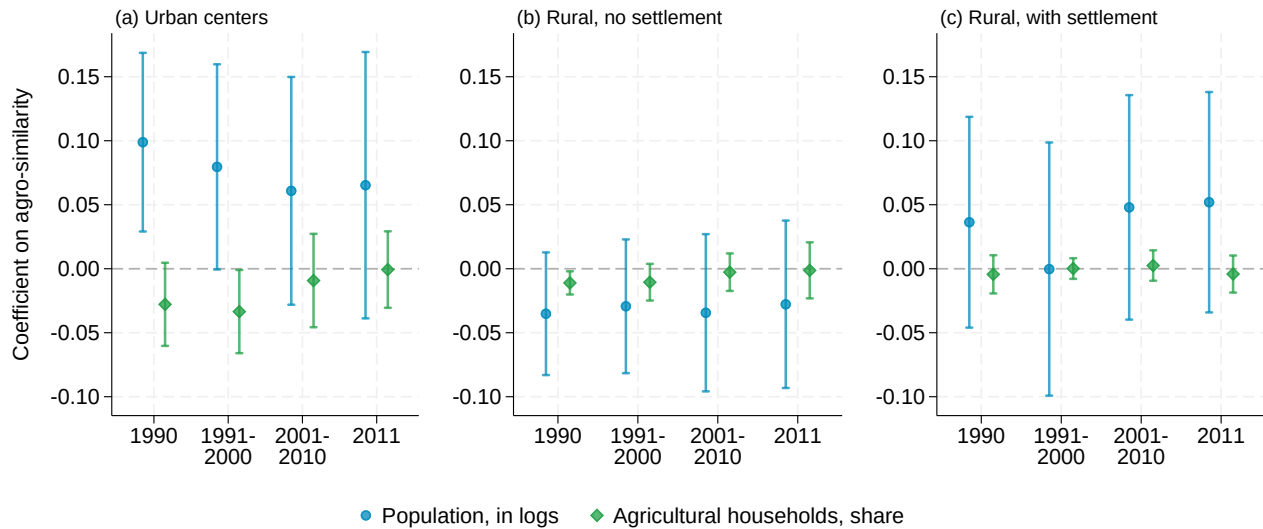
The food share of spending falls in the cities and in the rural subdistricts without a settlement, the two areas where employment moves out of agriculture. In the 1996 wave it falls by 0.46 of a standard deviation in the cities and by 0.14 in the subdistricts without a settlement, a marginally significant estimate, and the change in the subdistricts with a settlement,  $+0.06$ , is not distinguishable from zero. Calories per person rise by 0.72 of a standard deviation in the cities and do not move detectably in either rural group (Appendix [Table E1](#)). There is no unemployment response in any of the three areas.

The productivity gain and its consequences fall in different subdistricts. Where the settlements are, the gain appears in the real wage, and employment does not reallocate. The cities and the rural subdistricts without a settlement are the regions surrounded by more productive hinterlands, which benefit from the higher availability of food and expand their modern sector once the subsistence constraint is satisfied.

The village census (PODES) allows me to trace the key effects over time. In urban centers, both the population response and the decline in the share of agricultural households occur early and dissipate over time. In the first two periods the effect on population is positive and the effect on the agricultural share is negative, and each is statistically significant in one of the two periods. In the later periods both effects are closer to zero and the population estimates are noisier. Population growth does not occur in the other rural areas, as expected from the comparison by location type, yet the rural areas with no settlement lose agricultural households in the early periods. This pattern is consistent with production in the transmigration villages replacing some of the farming by pre-existing households in these areas. The rural areas that contain transmigration villages do not show statistically significant effects. The 1980 baseline measures are reported at the subdistrict level, so I aggregate these outcomes

to subdistricts without separating the settlements from their surrounding areas, and the absence of an effect may conceal some co-movement within a subdistrict. The village census counts households as reported by village heads (the population census counts individual workers), so its estimates are subject to more measurement error than the census outcomes and differ from them in composition (Figure 15).

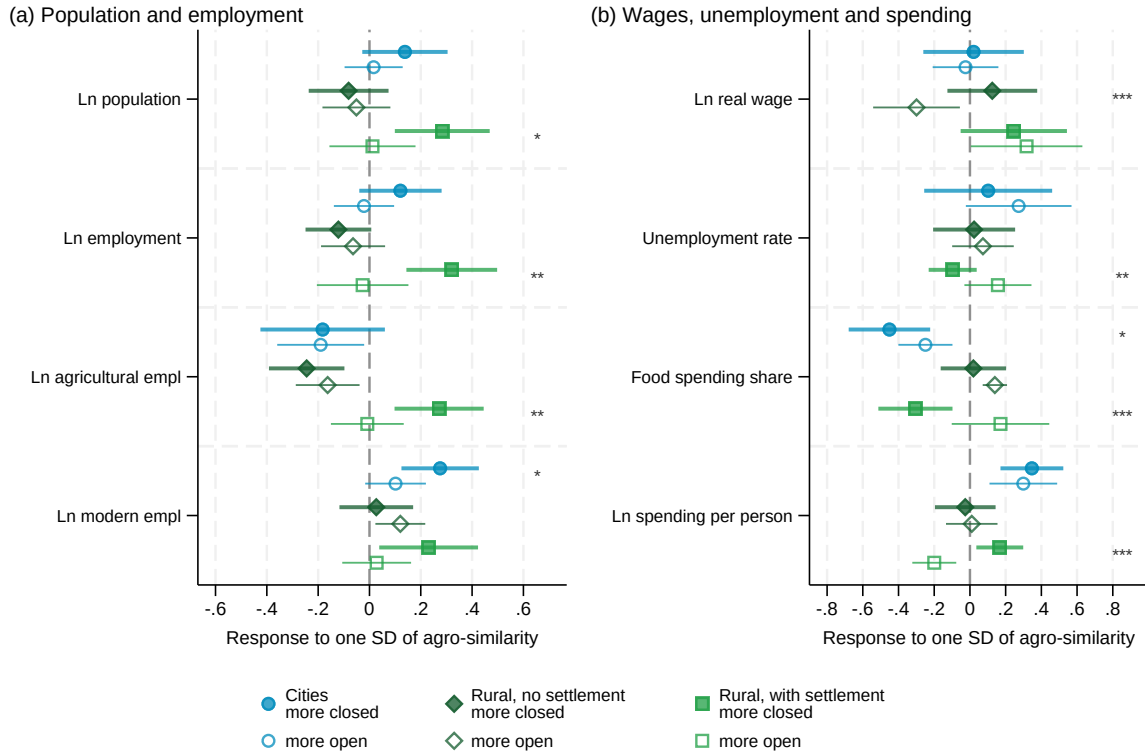
Figure 15: Population and the Agricultural Household Share, 1990 to 2011, by Location Type



*Notes:* Each point is the coefficient on standardized agro-similarity from one regression, in the main specification with the full set of controls, on one cell of village-census (PODES) waves, with 90% confidence intervals from Conley SE within 200km. The cells are the 1990 wave, the 1993 and 1996 waves, the 2003, 2005 and 2008 waves, and the 2011 wave. Each panel keeps the units that report both outcomes in every cell: in panel (a) 125 urban centers of the main city sample, and in panels (b) and (c) 588 rural subdistricts without a settlement and 191 with one, lying outside every city and within 300 minutes of a transmigration village. The series start in 1990, because earlier waves cover the agricultural share for less than two-thirds of the sample, and end in 2011, the last wave that releases population and household counts.

Across the cities and the rural subdistricts around them, structural transformation and sectoral reallocation are happening differently in different areas, and the cities adjust their sectoral employment in response to rural productivity.

Figure 16: Trade Openness and the Response by Location Type: Population, Employment, Wages, and Spending



Notes: Each marker is the response of its row's outcome to a one-SD increase in agro-similarity, in the specification and samples of Figure 14: the main city sample and the rural subdistricts with and without a settlement, which lie outside every city, with each outcome standardized over the cities or over the pooled rural subdistricts. Each location type is split at the median route distance to the Thai and Vietnamese rice hubs among its units within 300 travel minutes of a transmigration village, and one regression per outcome and location type interacts exposure with the two halves, so each coefficient is the response of its own half. Filled markers on thick intervals are the more closed half and hollow markers on thin intervals the more open half; intervals are 90%, from Conley SE within 200km. The star beside each pair is the significance of the difference between the two halves, from one test inside that regression: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Panel (a): the 2010 census outcomes. Panel (b): the unemployment rate (2010 census) and three household-survey outcomes; the real wage and the two spending rows are from the 2011 household-survey (Susenas) wave, which covers fewer subdistricts than the census. The Online Supplement reports each row's coverage and the test of each difference under two further variance estimators. Regressions are in the main specification with the full set of controls, except that the subdistrict settler count is taken over every village within 1,000 travel minutes.

## 6 A Spatial Equilibrium Model with Trade and Migration

The empirical results show a substantial shift from agriculture to modern employment in cities exposed to more productive rural hinterlands, alongside a modest and imprecisely estimated population response. I use a spatial model to examine how goods-market adjustment can generate this urban reallocation without migration. The model allows workers to change sector within a location, links locations through food and modern-output trade, and incorporates a subsistence food requirement. The central counterfactual holds each location's population fixed while wages, prices, employment composition, and land use adjust.

The model combines spatial trade and location choice (Allen & Arkolakis 2014; Redding 2016) with non-homothetic food demand (Matsuyama 1992; Kongsamut et al. 2001). Different elasticities

govern sector and location choices (Khanna et al. 2025). Cheaper food reduces the cost of meeting subsistence needs and changes relative sectoral returns; trade in modern output provides a source of earnings for cities purchasing food from their hinterlands.

Eckert & Peters (2025) combine non-homothetic demand, trade, and sector and location choices in a dynamic model of structural change. I take rural productivity gains as given and study their effects on cities while holding urban productivity fundamentals fixed. The transmission through food markets complements the capital-flow mechanism studied by Bustos et al. (2020). The counterfactuals separate adjustment in urban employment composition from adjustment in city size.

## 6.1 Setup and Equilibrium

The economy consists of  $J$  locations and two employment sectors: agriculture,  $A$ , and a modern sector,  $S$ , combining manufacturing and services. Modern employment is divided between a traded activity and a local activity. Food and traded-modern output connect urban and rural locations, while local modern output is consumed where it is produced. This is the economy used for the channel comparisons. The specification with only food traded is a restricted comparison.

A fixed mass  $\bar{L}$  of workers chooses where to live and in which sector to work. Total employment at location  $j$  is  $N_j = L_{jA} + L_{jS}$ , with  $\sum_j N_j = \bar{L}$ . The model abstracts from unemployment and natural population growth, so changes in  $N_j$  represent migration. Within modern employment,  $L_{jM}$  produces traded output and  $L_{jS} - L_{jM}$  produces local output; both activities pay the common wage  $w_{jS}$ . Each location has a fixed land endowment  $\bar{T}_j$ , divided between agriculture,  $T_{jA}$ , and local modern uses,  $T_{jH}$ . Production is

$$\begin{aligned} Y_{jA} &= (A_j^A T_{jA})^{1-\beta} L_{jA}^\beta, \\ Y_{jS} &= \bar{A}_j^S N_j^\eta (T_{jH})^{1-\gamma_S} (L_{jS} - L_{jM})^{\gamma_S}, \quad Y_{jM} = A_{jM} L_{jM}, \end{aligned} \tag{8}$$

where  $Y_{jS}$  denotes local-modern output and  $0 < \beta, \gamma_S < 1$ . Agricultural productivity is  $A_j^A$ ;  $\bar{A}_j^S$  and  $A_{jM}$  govern local and traded-modern productivity. The agricultural and local-modern activities use land, while traded-modern output is linear in labor. I set  $\eta = 0$  in the channel comparisons, so population changes do not raise modern productivity through agglomeration. The experiment raises agricultural productivity in rural locations hosting transmigration settlements and holds urban productivity fixed.

A common land market allocates  $T_{jA} + T_{jH} = \bar{T}_j$  across active uses. Land rental revenue accrues to local residents and is distributed equally as a per-person rebate  $R_j$ . Land can therefore adjust between sectors even when population is held fixed. Both consumption-basket variants in the decomposition use this local rebate. Appendix F.1 describes technology and endowments, and Appendix Section F.7 gives the land-market conditions.

**Preferences and expenditure.** Households consume food,  $c_A$ , traded-modern output,  $c_M$ , and local modern output,  $c_S$ . Stone-Geary preferences introduce a subsistence food requirement (Matsuyama

1992; Kongsamut et al. 2001):

$$U(c_A, c_M, c_S) = (c_A - \bar{a})^{\psi_A} c_M^{\psi_M} c_S^{\psi_S}, \quad \psi_A + \psi_M + \psi_S = 1, \quad \bar{a} > 0 \quad (9)$$

The parameter  $\bar{a} > 0$  is the minimum food requirement. Let  $y$  denote the resources available for expenditure.<sup>19</sup> For  $y > p_j^A \bar{a}$ , expenditure above the subsistence bill is divided among the three goods in proportions  $\psi_A$ ,  $\psi_M$ , and  $\psi_S$ . At given prices, the food expenditure share falls with income:

$$\frac{p_j^A c_A}{y} = \psi_A + (1 - \psi_A) \frac{p_j^A \bar{a}}{y} \quad (10)$$

At a given income, cheaper food lowers the subsistence bill and raises expenditure on modern output. Indirect utility is proportional to  $(y - p_j^A \bar{a})/P_j$ , where  $P_j = (p_j^A)^{\psi_A} (p_j^M)^{\psi_M} (p_j^S)^{\psi_S}$ . Setting  $\bar{a} = 0$  gives Cobb-Douglas preferences with expenditure shares independent of income. Appendix Section F.2 gives the household demand derivation, and Appendix Appendix F.12 describes the traded-modern specification.

**Sector and location choice.** Worker heterogeneity coming from Fréchet shocks allows workers facing the same wages and prices to make different choices, while resulting in the closed-form choice probabilities that determine the aggregate allocation of workers across locations and sectors. The probability  $\pi_{A|j}$  gives the agricultural employment share conditional on living in location  $j$ , with  $1 - \pi_{A|j}$  giving the corresponding modern-sector share. The sectoral allocation rule is

$$\pi_{A|j} = \frac{w_{jA}^{\theta_S}}{w_{jA}^{\theta_S} + w_{jS}^{\theta_S}}, \quad L_{jA} = N_j \pi_{A|j}, \quad L_{jS} = N_j (1 - \pi_{A|j}), \quad (11)$$

where  $w_{js}$  is the sectoral wage. The first parameter  $\theta_S$  governs the response of employment composition to relative sectoral wages, the sectoral choice elasticity. A larger  $\theta_S$  implies a larger response of the agricultural employment share to a given change in  $w_{jA}/w_{jS}$ .

The probability  $\pi_j$  gives the share of all workers in the economy choosing location  $j$ , so that  $N_j = \bar{L} \pi_j$ . Let  $v_j$  denote the deterministic location value, incorporating sectoral earning opportunities, the subsistence food bill, local consumption prices, land income, and amenities  $\Phi_j^L$ . The location allocation rule is

$$\pi_j = \frac{v_j^{\theta_L}}{\sum_k v_k^{\theta_L}}, \quad N_j = \bar{L} \pi_j \quad (12)$$

The parameter  $\theta_L$  governs the population response to differences in location values. A lower value implies less population adjustment for a given change in relative location values. Appendix Appendix F.13

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<sup>19</sup>The calculations use  $W_j + R_j$  for spending, where  $W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}$  summarizes sectoral earnings opportunities. This differs from average wage payments. Food supply prices use inverse marginal agricultural productivity. These conventions do not impose a complete balance between factor income and goods expenditure; Appendix Section F.9 gives the accounting details.

considers explicit origin-destination moving costs informed by IFLS migration flows; Appendix [Section F.8](#) gives the choice rules.

**Food trade and prices.** Food varieties are differentiated by origin and combined with elasticity of substitution  $\sigma > 1$ . Shipping from  $k$  to  $j$  incurs an iceberg cost  $\tau_{kj} \geq 1$ . I parameterize these costs using the travel-time measure introduced in [Section 3.3](#):  $\tau_{kj} = \max\{d_{kj}, 1\}^\delta$  for  $k \neq j$ , with  $\tau_{jj} = 1$ . Let  $q_k^A$  denote the origin food-supply index, constructed from inverse agricultural marginal productivity. The food price index at location  $j$  is

$$p_j^A = \left[ \sum_k (\tau_{kj} q_k^A)^{1-\sigma} \right]^{1/(1-\sigma)}, \quad \sigma = 4 \quad (13)$$

The sensitivity of  $p_j^A$  to an origin's supply price equals that origin's share of food expenditure at  $j$ . Productivity gains therefore have greater scope to affect a city when the affected villages account for a larger share of its food purchases. This sourcing relationship provides the link between the geography of the shock and the geography of urban exposure. Appendix [Section F.4](#) describes the trade-cost specification and the origin supply index. The model's endogenous food-expenditure weights differ from the geographic exposure weights used in the empirical regression.

**Equilibrium and transmission.** The model jointly determines wages, prices, land allocation, rents, and employment across sectors and locations. The local modern good clears within each location, while food-price indices and traded-modern sales connect locations. The modern wage in the median-population urban location is the numeraire.

A fall in the urban food price lowers the subsistence bill and changes demand for modern output. Competition from rural origins also changes the return to urban agriculture. Workers respond to relative sectoral wages, while differences in location values induce migration. Household income, land rents, and modern-good prices adjust together, so the direction and size of employment changes require the full calculation.

**Comparative statics.** A rural productivity gain affects cities directly through their food purchases. Consider an increase in agricultural productivity  $A_v^A$  in village  $v$  and its effect on a city  $j \neq v$ , holding the city's productivity fundamentals fixed. At given agricultural land and employment in the village, higher  $A_v^A$  lowers its food-supply index  $q_v^A$ . Holding other origins' indices fixed, the CES food-price index implies

$$\frac{\partial \ln p_j^A}{\partial \ln A_v^A} = s_{vj} \frac{\partial \ln q_v^A}{\partial \ln A_v^A} < 0, \quad s_{vj} = \frac{(\tau_{vj} q_v^A)^{1-\sigma}}{\sum_k (\tau_{kj} q_k^A)^{1-\sigma}}, \quad (14)$$

where  $s_{vj}$  is the share of city  $j$ 's food expenditure allocated to village  $v$ , and  $\sigma > 1$  is the elasticity of substitution across food origins. A given reduction in the village's supply index therefore lowers food prices more in cities that buy a larger share of their food from that village. Access to alternative

suppliers can weaken this transmission by reducing the city’s reliance on the affected villages. This sourcing relationship connects the model to the openness heterogeneity in [Section 5.3](#).

At a given nominal household income, cheaper food reduces the subsistence bill and increases expenditure on modern output in the city. Urban employment responds to the change in relative wages induced by the rural productivity gain. The sector-choice equation gives:

$$\frac{d\pi_{A|j}}{d\ln A_v^A} = \theta_S \pi_{A|j} (1 - \pi_{A|j}) \frac{d\ln(w_{jA}/w_{jS})}{d\ln A_v^A}, \quad (15)$$

where  $\pi_{A|j}$  is the city’s agricultural employment share,  $w_{jA}/w_{jS}$  is its relative agricultural wage, and  $\theta_S$  is the sector-choice elasticity. The village productivity gain shifts urban employment toward the modern sector if it lowers the relative agricultural wage in the city. This sectoral adjustment can occur with the city’s population held fixed.

These expressions separate the direct food-price effect from the subsequent adjustment of household income, relative wages, land allocation, and population. The counterfactuals in [Section 6.3](#) compare those adjustments with and without population movement.

## 6.2 Taking the Model to the Data

I represent urban centers and rural district remainders using population and employment from the 2010 census, wages and consumption from Susenas 2011, planted crop area from PODES 2003, and geographic information. The rural aggregation allows the model to include the settlements supplying cities with food. The channel exercise compares this end-period economy before and after a rural productivity gain.<sup>20</sup>

**Preferences, technology, and local fundamentals.** Susenas expenditure shares discipline the food and traded-modern consumption weights. The subsistence requirement uses the BPS food-poverty threshold for 2000 normalized by the reference wage. Labor shares govern production; the elasticity of substitution across food origins is  $\sigma = 4$ . Household service flows from IFLS inform the traded-modern distance decay. In the absence of bilateral food shipments at this spatial resolution, food trade costs are assumed and varied across specifications.

Location characteristics determine productivity and amenities. Agricultural productivity uses agricultural employment per unit of crop land, local-modern productivity uses the modern employment share and land scale, and amenities combine satellite measures around 1980 with village infrastructure in 1983. Traded-modern productivity is calibrated against census traded-modern employment valued at the reference allocation’s model wages, with total sales scaled to expenditure on that good. This construction disciplines cross-location differences in production and consumption opportunities.

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<sup>20</sup>The channel calculations contain 287 locations, including 118 urban centers, and use empirical targets from 123 cities. These samples differ from the main empirical sample. [Appendix F.11](#) provides data definitions and additional calibration exercises.

**Estimating the choice elasticities.** I choose the sector and location elasticities jointly using urban exposure gradients. Population is informative about adjustment across locations, while the agricultural employment share is informative about adjustment within locations. Both parameters affect both outcomes through equilibrium wages and prices.

For each candidate pair  $\theta = (\theta_L, \theta_S)$ , the model is solved before and after a rural agricultural productivity change. The resulting urban outcomes are related to exposure, and the parameters are chosen to minimize the distance between the model and empirical coefficients. This is indirect inference: the coefficients of an auxiliary regression are the estimation targets (Hansen 1982; Gouriéroux et al. 1993; Smith 1993). Let  $\hat{b}$  collect the targeted empirical coefficients and  $b(\theta)$  the corresponding model coefficients. The estimates minimize

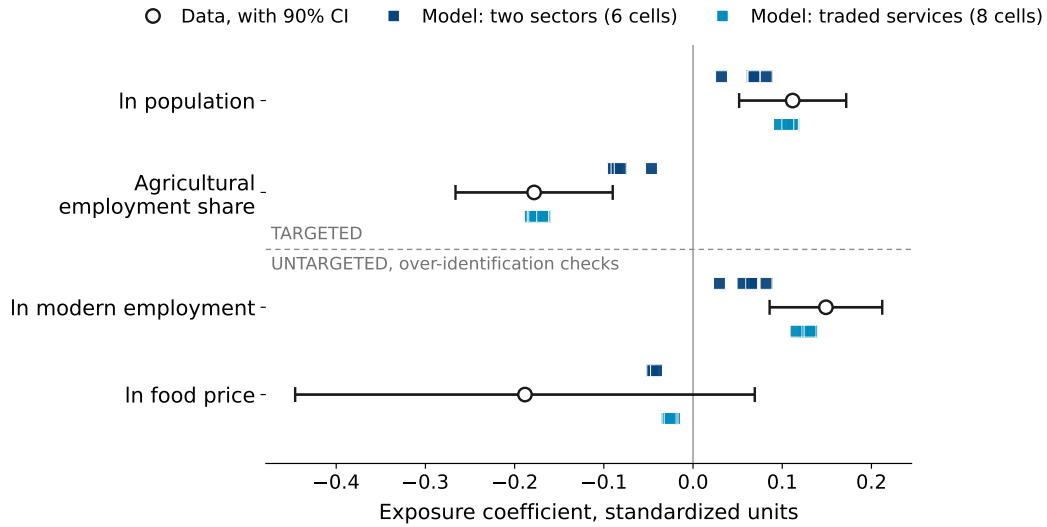
$$\hat{\theta} = \arg \min_{\theta} [b(\theta) - \hat{b}]' \Omega [b(\theta) - \hat{b}], \quad (16)$$

where  $\Omega$  weights the coefficients by their empirical precision. Figure 17 compares population, employment, and food-price predictions across the food-only and traded-modern specifications. The channel comparisons use the traded-modern economy with subsistence demand, a common wage across modern activities, local land-rent rebates, and zero agglomeration. Within each consumption basket, the full and restricted calculations hold the productivity shock and parameter values fixed.<sup>21</sup>

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<sup>21</sup>The fit figure uses population and the agricultural employment share as targets. The channel parameter selection also used modern employment and simple exposure slopes without the full empirical control set. The channel calculations use  $\theta_L = 0.25$ , with  $\theta_S = 0.82$  in the services basket and  $\theta_S = 0.84$  when prepared foods enter the traded basket; the food and traded-modern distance elasticities are 0.5 and 1.0. The comparisons are conditional on these parameterizations. Separate outcome standardization limits information about elasticity levels, and parameter uncertainty is not estimated.

Figure 17: Model Comparisons for Employment, Population, and Food Prices



*Notes:* Circles report empirical coefficients with 90% confidence intervals; squares report model comparisons across the single-tradable and traded-modern specifications and food trade costs. The population and agricultural-share coefficients are targets; modern employment is untargeted for this two-target parameter selection. These model quantity rows relate separately standardized outcome changes to standardized exposure. The food-price row instead relates post-shock log food prices to exposure with controls on the empirical price sample. It is a distinct comparison. Appendix [Table B3](#) lists empirical moments.

**Additional outcome comparisons.** The fit comparison also reports modern employment and food prices. Modern employment is untargeted in that comparison, although it is partly implied by the population and agricultural-share targets. Food prices test a distinct prediction of the mechanism, with imprecise empirical estimates limiting their ability to distinguish specifications.

I use the calibrated economy to compare sectoral adjustment with and without population movement. This comparison isolates an implication of the trade mechanism that the reduced form cannot observe directly.

### 6.3 Counterfactuals

With population held fixed, the model reproduces nearly the full standardized exposure gradient in urban agricultural employment shares. This finding is stable across both displayed specifications. I obtain it from a common increase of 0.05 log points in the agricultural-productivity parameter in rural locations hosting transmigration settlements, holding urban productivity and the remaining exogenous inputs fixed. Agro-similarity measures cities' exposure to the affected locations; the common host gain defines the experiment.

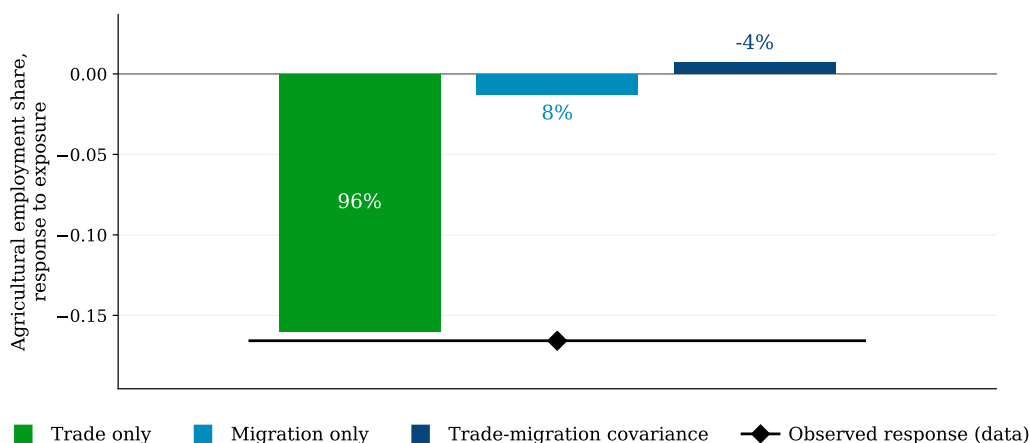
In the full calculation, population, wages, prices, land use, and sectoral employment adjust. The trade-only calculation holds population at each location's unshocked allocation and allows prices, wages, land use, and employment shares to respond. The migration-only calculation instead fixes food and

traded-modern prices and traded-modern employment, while population and the local economy adjust. This restriction suppresses the specified goods-market adjustment without imposing autarky.

The figures compare the association between exposure and the urban response. Each outcome change is standardized within its own calculation, so the coefficients describe the strength of that association. The interaction is the full coefficient less the two restricted coefficients. These terms summarize standardized exposure gradients; they do not allocate a common-unit employment change between channels.

The main specification uses the services basket. The companion comparison also assigns prepared foods to traded-modern consumption; both distribute land income locally.

Figure 18: Channel Decomposition of the Agricultural-Share Response



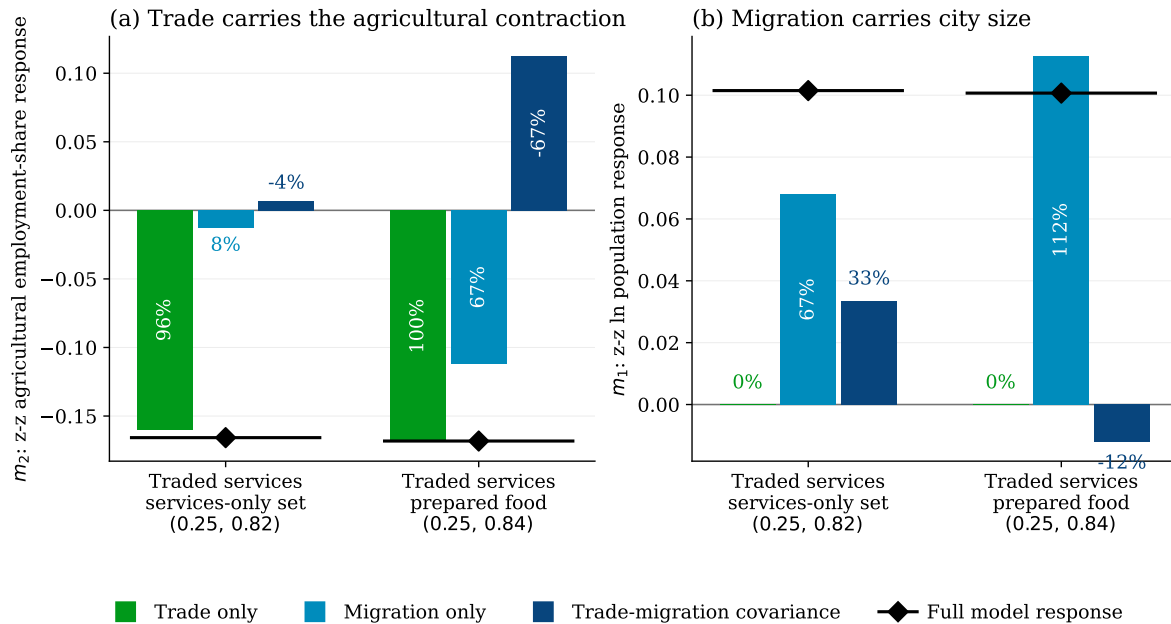
*Notes:* The bars are standardized exposure slopes for changes in the agricultural employment share. Population is fixed in the trade-only calculation; food and traded-modern prices and traded-modern employment are fixed in the migration-only calculation. Each outcome change is standardized separately. The third bar, labeled covariance in the image, is the full coefficient less the other two coefficients. The diamond is the model's full response despite its data label in the image. Percentages are ratios to that model coefficient; they do not measure shares of a common-unit employment decline. The experiment raises agricultural productivity in settlement-hosting rural locations and uses a local land-rent rebate. These model calculations do not report sampling uncertainty.

**Trade with population fixed.** Figure 18 shows that the exposure gradient remains nearly intact when adjustment is confined to goods markets and employment within locations. The comparison in Figure 19 gives the same conclusion under the alternative specification with prepared foods in the traded basket and its associated demand and sector-choice parameters. The migration-only gradient is larger in that specification. The stable finding is that trade can reproduce the standardized exposure gradient in urban sectoral composition without population movement. Separately standardized coefficients do not establish how much of the employment decline trade accounts for, or that migration is unimportant.

This comparison connects the model to the empirical combination of urban sectoral change and limited net population adjustment. Lower food costs change urban demand and relative sectoral

returns, allowing employment to move within existing cities. A separate comparison with the subsistence requirement removed would determine how much of this response requires non-homothetic demand.

Figure 19: Both Outcomes, Both Traded-Services Specifications



Notes: Panels compare a services-only traded-consumption basket with one that also includes prepared foods. Both use a local land-rent rebate. Bars are simple standardized exposure slopes under the restrictions described in the text; each outcome change has its own standard deviation. The diamond gives the full model coefficient, and the interaction is that coefficient less the other two. The trade-only population response is imposed to be zero by fixing populations. It is not an independent test of the model. These calculations do not report sampling uncertainty.

**Historical magnitudes.** Between 1980 and 2010, the aggregate agricultural employment share fell from 48.27% to 29.92% in the 134 main-sample cities, a change of -18.35 percentage points. Agricultural employment rose by 1,350,646 workers over the same period.<sup>22</sup>

Regressing the change in the agricultural share on exposure with the main specification's controls gives -5.13 percentage points per standard deviation of exposure, with a 95% interval of [-7.23, -3.04]. A lower-match comparison illustrates its scale. Suppose each city's exposure above the full main sample's tenth percentile is reduced to that reference, with lower exposures held fixed. Applying the estimated slope and holding observed 2010 employment weights fixed gives a difference of -5.54 percentage points between the observed and lower-match profiles, equal to 30.20% of the observed aggregate decline. The coefficient-based 95% interval is [-7.81, -3.28] percentage points. This reference is an observed exposure level; it has not been established as an absence of match-related productivity gains. Shared

<sup>22</sup>The denominator includes all workers with a reported industry, and the aggregate share divides total agricultural employment by total employment. Counts follow the main analysis's fixed city membership, historical crosswalk, and allocation weights. Some later-created member codes lack a separate historical observation because their predecessor population is assigned to another member. A sensitivity requiring every member code to be observed retains 121 cities and gives a change of -18.68 percentage points; label completeness does not establish more accurate historical boundaries.

village catchments also mean this city-level comparison need not correspond to a feasible reassignment of settlers.

Historical attribution requires the match-related productivity component at both endpoints and its effect after prices, wages, and employment weights adjust. A persistent matching advantage can have different urban effects as other economic conditions change. Cross-city exposure differences do not identify an effect common to all cities. The magnitude comparison therefore does not identify the share of the historical decline caused by agricultural similarity. A model calculation of that share requires productivity mappings and the corresponding counterfactual at both endpoints, with settlement placements and initial settler counts held fixed.

## 7 Conclusion

Governments and development organizations spend large amounts of money on programs that raise agricultural productivity and reduce rural poverty. Most existing research tells us how agricultural productivity within a region affects outcomes in that same region; we know much less about how it affects cities outside those areas. Investing in rural areas could create incentives for people to stay there, or it could help cities by producing enough food to supply a growing urban population. I use a population resettlement program in Indonesia that generated as-good-as-random variation in productivity across villages. Cities surrounded by more productive villages barely grow in population, but they change the types of jobs people have, shifting from agricultural to non-agricultural jobs in manufacturing and services. The policy only targeted villages; the cities around them restructured their employment anyway.

Rural productivity transforms cities through trade, not migration. Urban residents in the exposed cities spend relatively less on food, without lower consumption, and more on non-agricultural items. I do not observe trade flows directly, so I compare cities more and less likely to rely on local trade, and the modern-sector response concentrates in cities weakly connected to outside food markets and is much smaller in cities better connected to the Thai and Vietnamese rice hubs. Migration exists, but it is small: there is no distinct migration response among the settlers by village productivity, and the new arrivals cannot account for the shift. The classical theories of structural transformation view rising agricultural productivity in villages as releasing workers who move to cities; here it reaches the city through goods.

To interpret and quantify these patterns, I develop a spatial general equilibrium model of multiple locations, villages and cities, with two sectors: agriculture, which produces food, and the modern sector, combining manufacturing and services. Households make two choices, the location and the sector, each governed by its own elasticity, and the main modeling choice is the Stone-Geary utility with a subsistence term in the demand for food, which lets a positive productivity shock in the villages generate agricultural contraction in the cities. Food prices are determined endogenously through costly inter-regional trade, and the modern-sector agglomeration externality is set to zero in the headline

calibration. I choose two data moments to identify the two parameters I have, the population growth and the agricultural employment decline, and the model’s discrete-choice structure gives estimating equations that map directly to the reduced-form regressions. I then shut down trade, meaning the trade costs rise substantially, and I shut down migration separately; in the real world both forces are in play and I observe the general-equilibrium effect. Trade is sufficient to reproduce nearly the entire agricultural contraction, and the trade-only response on city size is zero by construction. Migration enters as an estimated friction, disciplined by a gravity-based moving-cost estimate, and it moves the city’s size, not its composition.

When governments invest in rural areas, they rarely consider the consequences for the cities around them; I show these are interconnected. As governments think about reducing barriers to growth and structural transformation, trade barriers and migration barriers may affect urban growth very differently. When cities lack sufficient access to food, investments in rural areas have more direct impacts on the structure of urban economies.

## References

- Allcott, H. & Keniston, D. (2018). Dutch Disease or Agglomeration? The Local Economic Effects of Natural Resource Booms in Modern America. *Review of Economic Studies*, 85(2), 695–731.
- Allen, T. & Arkolakis, C. (2014). Trade and the Topography of the Spatial Economy. *Quarterly Journal of Economics*, 129(3), 1085–1140.
- Asher, S., Chauvin, J. P., Grigoryeva, I., & Novosad, P. (2026). Rural Spillovers of Urban Growth. Working paper.
- Badan Pusat Statistik (2011). Survei Sosial Ekonomi Nasional (Susenas): National Socioeconomic Survey. *Badan Pusat Statistik (Statistics Indonesia), household survey microdata*. Consumption, price, wage, and housing modules. This paper uses the 2002 and 2011 waves; the 2011 wave covers quarters Q1–Q3, the only quarters released.
- Barter, S. J. & Côté, I. (2015). Strife of the soil? Unsettling transmigrant conflicts in Indonesia. *Journal of Southeast Asian Studies*, 46(1), 60–85.
- Bazzi, S. (2017). Wealth Heterogeneity and the Income Elasticity of Migration. *American Economic Journal: Applied Economics*, 9(2), 219–255.
- Bazzi, S., Gaduh, A., Rothenberg, A. D., & Wong, M. (2016). Skill Transferability, Migration, and Development: Evidence from Population Resettlement in Indonesia. *American Economic Review*, 106(9), 2658–2698.
- Bazzi, S., Gaduh, A., Rothenberg, A. D., & Wong, M. (2019). Unity in Diversity? How Intergroup Contact Can Foster Nation Building. *American Economic Review*, 109(11), 3978–4025.
- Borusyak, K. & Hull, P. (2023). Nonrandom Exposure to Exogenous Shocks. *Econometrica*, 91(6), 2155–2185. Previously NBER WP 27845, “Non-Random Exposure to Exogenous Shocks: Theory and Applications”.
- Bryan, G., Chowdhury, S., & Mobarak, A. M. (2014). Underinvestment in a Profitable Technology: The Case of Seasonal Migration in Bangladesh. *Econometrica*, 82(5), 1671–1748.

- Bryan, G. & Morten, M. (2019). The Aggregate Productivity Effects of Internal Migration: Evidence from Indonesia. *Journal of Political Economy*, 127(5), 2229–2268.
- Budi-Ors, T. (2026). *Rural-Urban Migration and Structural Change: A Reinterpretation*. Working Paper, Bank of Spain. March 2026; formerly CEMFI job market paper.
- Bustos, P., Caprettini, B., & Ponticelli, J. (2016). Agricultural Productivity and Structural Transformation: Evidence from Brazil. *American Economic Review*, 106(6), 1320–1365.
- Bustos, P., Garber, G., & Ponticelli, J. (2020). Capital Accumulation and Structural Transformation. *Quarterly Journal of Economics*, 135(2), 1037–1094.
- Dawe, D. (2008). Can Indonesia Trust the World Rice Market? *Bulletin of Indonesian Economic Studies*, 44(1), 115–132.
- Dijkstra, L., Florczyk, A. J., Freire, S., Kemper, T., Melchiorri, M., Pesaresi, M., & Schiavina, M. (2021). Applying the Degree of Urbanisation to the Globe: A New Harmonised Definition Reveals a Different Picture of Global Urbanisation. *Journal of Urban Economics*, 125, 103312.
- Dufo, E. (2001). Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment. *American Economic Review*, 91(4), 795–813.
- Eckert, F. & Peters, M. (2025). *Spatial Structural Change*. NBER Working Paper 30489, National Bureau of Economic Research. September 2022, revised July 2025. Under revision at the Review of Economic Studies.
- Edwards, R. B. (2024). *Spillovers from Agricultural Processing*. Departmental Working Paper 2024-6, Arndt-Corden Department of Economics, Australian National University. July 2024; first circulated 2019.
- Egger, D., Haushofer, J., Miguel, E., Niehaus, P., & Walker, M. (2022). General Equilibrium Effects of Cash Transfers: Experimental Evidence from Kenya. *Econometrica*, 90(6), 2603–2643.
- Emerick, K. (2018). Agricultural Productivity and the Sectoral Reallocation of Labor in Rural India. *Journal of Development Economics*, 135, 488–503.
- Feyrer, J., Mansur, E. T., & Sacerdote, B. (2017). Geographic Dispersion of Economic Shocks: Evidence from the Fracking Revolution. *American Economic Review*, 107(4), 1313–1334.
- Foeken, D. & Mwangi, A. M. (1998). *Farming in the City of Nairobi*. ASC Working Paper 30, African Studies Centre, Leiden.
- Foster, A. D. & Rosenzweig, M. R. (2004). Agricultural Productivity Growth, Rural Economic Diversity, and Economic Reforms: India, 1970–2000. *Economic Development and Cultural Change*, 52(3), 509–542.
- Gallaher, C. M., WinklerPrins, A. M. G. A., Njenga, M., & Karanja, N. K. (2015). Creating Space: Sack Gardening as a Livelihood Strategy in the Kibera Slums of Nairobi, Kenya. *Journal of Agriculture, Food Systems, and Community Development*, 5(2), 155–173.
- Gertler, P. J., Gonzalez-Navarro, M., Gračner, T., & Rothenberg, A. D. (2024). Road Maintenance and Local Economic Development: Evidence from Indonesia’s Highways. *Journal of Urban Economics*, 143, 103687.
- Glaeser, E. L. (2014). A World of Cities: The Causes and Consequences of Urbanization in Poorer Countries. *Journal of the European Economic Association*, 12(5), 1154–1199.

- Gollin, D. (2023). Agricultural Productivity and Structural Transformation: Evidence and Questions for African Development. *Oxford Development Studies*, 51(4), 375–396. First circulated as STEG Pathfinding Paper PP04, April 2021.
- Gollin, D., Hansen, C. W., & Wingender, A. M. (2021). Two Blades of Grass: The Impact of the Green Revolution. *Journal of Political Economy*, 129(8), 2344–2384.
- Gollin, D., Jedwab, R., & Vollrath, D. (2016). Urbanization with and without Industrialization. *Journal of Economic Growth*, 21(1), 35–70.
- Gollin, D., Lagakos, D., & Waugh, M. E. (2014). The Agricultural Productivity Gap. *Quarterly Journal of Economics*, 129(2), 939–993.
- Gollin, D., Parente, S. L., & Rogerson, R. (2007). The Food Problem and the Evolution of International Income Levels. *Journal of Monetary Economics*, 54(4), 1230–1255.
- Gourieroux, C., Monfort, A., & Renault, E. (1993). Indirect Inference. *Journal of Applied Econometrics*, 8(S1), S85–S118.
- Halpern, B. S., Walbridge, S., Selkoe, K. A., Kappel, C. V., Micheli, F., D’Agrosa, C., Bruno, J. F., Casey, K. S., Ebert, C., Fox, H. E., Fujita, R., Heinemann, D., Lenihan, H. S., Madin, E. M. P., Perry, M. T., Selig, E. R., Spalding, M., Steneck, R., & Watson, R. (2008). A Global Map of Human Impact on Marine Ecosystems. *Science*, 319(5865), 948–952.
- Hamory, J., Kleemans, M., Li, N. Y., & Miguel, E. (2021). Reevaluating Agricultural Productivity Gaps with Longitudinal Microdata. *Journal of the European Economic Association*, 19(3), 1522–1555.
- Hansen, L. P. (1982). Large Sample Properties of Generalized Method of Moments Estimators. *Econometrica*, 50(4), 1029–1054.
- Henderson, J. V. & Turner, M. A. (2020). Urbanization in the Developing World: Too Early or Too Slow? *Journal of Economic Perspectives*, 34(3), 150–173.
- Herrendorf, B., Rogerson, R., & Valentinyi, Á. (2014). Growth and Structural Transformation. In P. Aghion & S. N. Durlauf (Eds.), *Handbook of Economic Growth*, volume 2 (pp. 855–941). Amsterdam: Elsevier.
- Hornbeck, R. & Keskin, P. (2015). Does Agriculture Generate Local Economic Spillovers? Short-Run and Long-Run Evidence from the Ogallala Aquifer. *American Economic Journal: Economic Policy*, 7(2), 192–213.
- Jedwab, R., Ianchovichina, E., & Haslop, F. (2025). The Employment Profile of Cities around the World: Consumption vs. Production Cities and Economic Development. *World Development*, 188, 106883.
- Jedwab, R. & Vollrath, D. (2015). Urbanization without Growth in Historical Perspective. *Explorations in Economic History*, 58, 1–21.
- Khanna, G., Medina, C., Nyshadham, A., Ramos-Menchelli, D., Tamayo, J., & Tiew, A. (2025). *Spatial Mobility, Economic Opportunity, and Crime*. Borradores de Economía 1334, Banco de la República.
- Kodde, D. A. & Palm, F. C. (1986). Wald Criteria for Jointly Testing Equality and Inequality Restrictions. *Econometrica*, 54(5), 1243–1248.
- Kongsamut, P., Rebelo, S., & Xie, D. (2001). Beyond Balanced Growth. *Review of Economic Studies*, 68(4), 869–882.

- Kraus, S., Heilmayr, R., & Koch, N. (2024). Spillovers to Manufacturing Plants from Multimillion Dollar Plantations: Evidence from the Indonesian Palm Oil Boom. *Journal of the Association of Environmental and Resource Economists*, 11(3), 613–656.
- Lagakos, D., Mobarak, A. M., & Waugh, M. E. (2023). The Welfare Effects of Encouraging Rural–Urban Migration. *Econometrica*, 91(3), 803–837.
- Lewis, W. A. (1954). Economic Development with Unlimited Supplies of Labour. *Manchester School*, 22(2), 139–191.
- Liu, M., Shamdasani, Y., & Taraz, V. (2023). Climate Change and Labor Reallocation: Evidence from Six Decades of the Indian Census. *American Economic Journal: Economic Policy*, 15(2), 395–423.
- Marden, S. (2016). *The Agricultural Roots of Industrial Development: ‘Forward Linkages’ in Reform Era China*. Working Paper 09/2016, Department of Economics, University of Sussex.
- Matsuyama, K. (1992). Agricultural Productivity, Comparative Advantage, and Economic Growth. *Journal of Economic Theory*, 58(2), 317–334.
- McArthur, J. W. & McCord, G. C. (2017). Fertilizing Growth: Agricultural Inputs and Their Effects in Economic Development. *Journal of Development Economics*, 127, 133–152.
- McGee, T. G. (1991). The Emergence of Desakota Regions in Asia: Expanding a Hypothesis. In N. Ginsburg, B. Koppel, & T. G. McGee (Eds.), *The Extended Metropolis: Settlement Transition in Asia*. Honolulu: University of Hawaii Press.
- Mitchell, H. (2026). The Curse of Connectivity: Evidence from Village Resettlement in Indonesia. Job market paper in progress, University of California San Diego.
- Moreno-Monroy, A. I., Schiavina, M., & Veneri, P. (2021). Metropolitan Areas in the World. Delineation and Population Trends. *Journal of Urban Economics*, 125, 103242.
- Moretti, E. (2010). Local Multipliers. *American Economic Review*, 100(2), 373–377.
- Morten, M. & Oliveira, J. (2024). The Effects of Roads on Trade and Migration: Evidence from a Planned Capital City. *American Economic Journal: Applied Economics*, 16(2), 389–421. Previously NBER WP 22158, “Paving the Way to Development: Costly Migration and Labor Market Integration”.
- Moscona, J. (2019). *Agricultural Development and Structural Change, Within and Across Countries*. Working Paper, Massachusetts Institute of Technology.
- Nath, I. (2025). Climate Change, the Food Problem, and the Challenge of Adaptation through Sectoral Reallocation. *Journal of Political Economy*, 133(6), 1705–1756.
- Nelson, A. (2008). *Travel Time to Major Cities: A Global Map of Accessibility*. Technical report, Global Environment Monitoring Unit, Joint Research Centre of the European Commission, Ispra, Italy.
- Nunn, N. & Qian, N. (2011). The Potato’s Contribution to Population and Urbanization: Evidence From A Historical Experiment. *Quarterly Journal of Economics*, 126(2), 593–650.
- Olken, B. A. (2009). Matched Set of Indonesian BPS Subdistrict (Kecamatan) Codes. National Bureau of Economic Research data set, <https://www.nber.org/research/data/matched-set-indonesian-bps-subdistrict-kecamatan-codes>.

- Redding, S. J. (2016). Goods Trade, Factor Mobility and Welfare. *Journal of International Economics*, 101, 148–167.
- Restuccia, D., Yang, D. T., & Zhu, X. (2008). Agriculture and Aggregate Productivity: A Quantitative Cross-Country Analysis. *Journal of Monetary Economics*, 55(2), 234–250.
- Rutz, W. (1987). Indonesia’s Sea Transport System - A Series of Maps. *GeoJournal*, 14(4), 491–502.
- Santangelo, G. (2019). *Firms and Farms: The Local Effects of Farm Income on Firms’ Demand*. Cambridge Working Papers in Economics 1924, University of Cambridge.
- Schultz, T. W. (1953). *The Economic Organization of Agriculture*. New York: McGraw-Hill.
- Smith, A. A. (1993). Estimating Nonlinear Time-Series Models Using Simulated Vector Autoregressions. *Journal of Applied Econometrics*, 8(S1), S63–S84.
- Tombe, T. (2015). The Missing Food Problem: Trade, Agriculture, and International Productivity Differences. *American Economic Journal: Macroeconomics*, 7(3), 226–258.
- Tombe, T. & Zhu, X. (2019). Trade, Migration, and Productivity: A Quantitative Analysis of China. *American Economic Review*, 109(5), 1843–1872.
- van Klinken, G. (2007). *Communal Violence and Democratization in Indonesia: Small Town Wars*. London: Routledge.
- Weiss, D. J., Nelson, A., Gibson, H. S., Temperley, W., Peedell, S., Lieber, A., Hancher, M., Poyart, E., Belchior, S., Fullman, N., Mappin, B., Dalrymple, U., Rozier, J., Lucas, T. C., Howes, R. E., Tusting, L. S., Kang, S. Y., Cameron, E., Bisanzio, D., Battle, K. E., Bhatt, S., & Gething, P. W. (2018). A Global Map of Travel Time to Cities to Assess Inequalities in Accessibility in 2015. *Nature*, 553(7688), 333–336.
- Wolak, F. A. (1989). Testing Inequality Constraints in Linear Econometric Models. *Journal of Econometrics*, 41(2), 205–235.
- World Bank (2007). *World Development Report 2008: Agriculture for Development*. Washington, DC: The World Bank.

# Appendix

## Contents

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## A Data and Sample

This part documents the sources behind every number in the paper and the rule that defines the estimation sample.

### A.1 Urban Sample Construction

*Cited in the main text. This section:* [Section 2.2, p. 7](#); [Section 3.2, p. 12](#). Figure A1a: [Section 2.1, p. 6](#). Figure A1b: [Section 3.3, p. 16](#). Figure A2a: [Section 2.2, p. 7](#). Table A1: [Section 3.3, p. 16](#). Table A2: [Section 5.1, p. 27](#).

**Single inclusion rule.** The urban sample is defined only by Urban Center overlap. A subdistrict enters the urban sample if it overlaps a GHS Urban Center polygon; overlapping subdistricts are aggregated into one analysis unit per urban center; and the objective reclassification described in [Section 2.2.1](#) re-attaches dense administratively urban edge subdistricts and adds the new nodes, giving the 187-center sample, every center of which has a travel time to a program village. The Urban Center layer itself follows one global rule: on a one-kilometer population and built-up grid, contiguous high-density cells (at least 1,500 inhabitants per square kilometer) forming clusters of at least 50,000 people are classified as urban centers, and the polygon layer collects these clusters ([Dijkstra et al. 2021](#)), so a single satellite-based rule identifies cities consistently across the world.

**Aggregation to the urban center.** To identify what an urban center is, I combine data at three levels. The outcome data are available at the village and urban-neighborhood level (administrative level 4), the finest spatial units available; these comprise the subdistricts (administrative level 3) that the 1980 baseline data cover. Since an important part of the identification is to control for baseline levels, I need to combine the baseline and endline units in a comparable way, so the finest level the data afford is the level-3 subdistrict. The household surveys are at most available at the subdistrict level as well. Multiple contiguous subdistricts overlap with the same urban area, and since the analysis looks at the market overall, with exposure based on distance around the city, there is little meaningful variation across subdistricts within a city. I therefore dissolve contiguous subdistricts into one urban-center area, the unit that corresponds to the meaningful urban labor market.

**The 2010 totals each city is built from.** A city’s 2010 population and employment are the sums of its member year-2000 subdistricts, and those subdistrict totals rely on the village-level link described in the Online Supplement. Under the boundary overlay, 203 outer-island subdistricts recorded a 2010 population under half or over two and a half times their year-2000 population, against a median move of 1.15; under the administrative link that falls to 54, of which 38 are in Aceh, in northern Sumatra, where the year-2000 census itself is short.<sup>23</sup>

Table A1: The Urban Sample, From the Satellite Layer to the Main City Sample

|                                                                         | Urban centers |
|-------------------------------------------------------------------------|---------------|
| Urban Center polygons on the Outer Islands (GHS layer)                  | 182           |
| Less a census twin, two polygons on one subdistrict                     | −1            |
| Satellite-defined urban centers                                         | 181           |
| Plus centers the reclassification adds as standalone units              | +6            |
| Analysis universe                                                       | 187           |
| Less centers more than 300 travel minutes from a transmigration village | −49           |
| Within the 300-minute window                                            | 138           |
| Less centers without a 1980 baseline the census recorded directly       | −4            |
| Main city sample                                                        | 134           |

*Notes:* Counts of outer-island urban centers at each step of the sample construction. The Urban Center polygons are the GHS Urban Center Database layer; the census twin is two polygons that fall on a single subdistrict in the census geography and are treated as one unit. The reclassification adds two kinds of standalone center: dense administrative centers that fall below the satellite population threshold, and rural remainders of administrative cities (kota) whose 1980 urban population share exceeds one half. It also re-attaches 42 dense edge subdistricts to 22 existing centers, which leaves the count unchanged. The window keeps the centers at most 300 travel minutes from a transmigration village. The main city sample keeps the in-window centers whose three 1980 baselines, log population, log employment and the urban-population share, the census records directly; the text of this subsection gives the reason each of the other four is left out.

**From the satellite layer to the analysis sample.** The GHS layer has 182 Urban Center polygons on the Outer Islands, and one of them leaves before the analysis, a census twin: two polygons fall on a single subdistrict in the census geography, so they are treated as one unit throughout. That leaves 181 unique satellite-defined centers. A year-2000-population size floor had separately removed one Aceh city on Sumatra, Bireuen: its year-2000 census count of 357 is a known SP2000 enumeration shortfall, against a measured 78,876 people in 1980 and 78,997 in 2010. No size, inclusion, or outlier cutoff reads a year-2000 quantity, so Bireuen is in the satellite-defined count above. The reclassification rule (1980 population density above 500 persons per square kilometer with a 1980 agricultural employment share below one half) adds six dense administrative centers as standalone units, for 187 urban centers, and merges 42 dense urban edge subdistricts, drawn from 25 districts, into 22 parent rows. [Figure A2a](#) shows the rule in the data. One center, in the Sangihe island group off North Sulawesi, has no transmigration village within reach of its island group. Its travel time to the nearest program village, 1,120 minutes, is recovered from the all-subdistrict travel-time matrix rather than the urban-center matrix, and places it far outside any window used here. The analysis universe is therefore all 187 centers. Two restrictions then define the estimation sample, and they are separate. The 300-minute travel-time window leaves 138 centers, of which 134 have a 1980 baseline the census recorded directly, which is the estimation sample ([Table A1](#)). A center enters only if all three of the 1980 baselines that define the rule are observed, log population, log employment, and

<sup>23</sup>The 2000 census enumerated only 40.3% of Aceh’s villages, so a 2000-to-2010 comparison for Aceh cannot be repaired by any subdistrict link. The paper uses the year-2000 census as [Bazzi et al. \(2016\)](#) do, for the settled villages: the village agricultural employment share of [Figure 12](#) and the settler stay-share of [Figure 13](#). Its subdistrict map and its village list are the geography every city total is built on, and its counts serve here only as a check on that link. Every city-level quantity taken from the population census comes from its 1980 and 2010 rounds.

the urban-population share, and a partial baseline does not qualify it. The main specification conditions on a fourth 1980 baseline, log agricultural employment, which is a control and not part of the inclusion rule; it is missing in the stored file for two in-sample centers whose 1980 agricultural employment level is measured, and I rebuild it from that level. This is an inclusion rule rather than an incidental data gap: the specification conditions each 2010 outcome on the pre-program level of the same quantity, which a center without a 1980 measurement cannot support, so the requirement is not relaxed on any other ground. Four in-window centers are excluded, and for two distinct reasons. Three, in Bengkulu on Sumatra, West Kalimantan and Central Kalimantan, have no 1980 census record at all, and their baselines cannot be recovered from the earlier village censuses, which count households by activity and do not count workers. The fourth, a reclassification node in West Sumatra, has a 1980 baseline supplied by the apportioned-parent fill, and a filled baseline does not qualify a center either. No Aceh center is excluded on this ground: the extension to the 1980 crosswalk described in [Section 3.1](#) recovered them. Of the six added units, three are dense administrative centers that fall below the satellite population threshold, and a second, equally pre-treatment screen applies the same logic to the rural remainders of administrative cities (*kota*) whose 1980 urban population share exceeds one half: three enter as their own units (Palangka Raya on Kalimantan, Sawahlunto on Sumatra, and Batam off Sumatra), two re-attach to their city’s existing urban center (Palu on Sulawesi and Samarinda on Kalimantan), and one is dropped because its *kota* is fully urban, so the node would duplicate the city itself (Pangkalpinang).

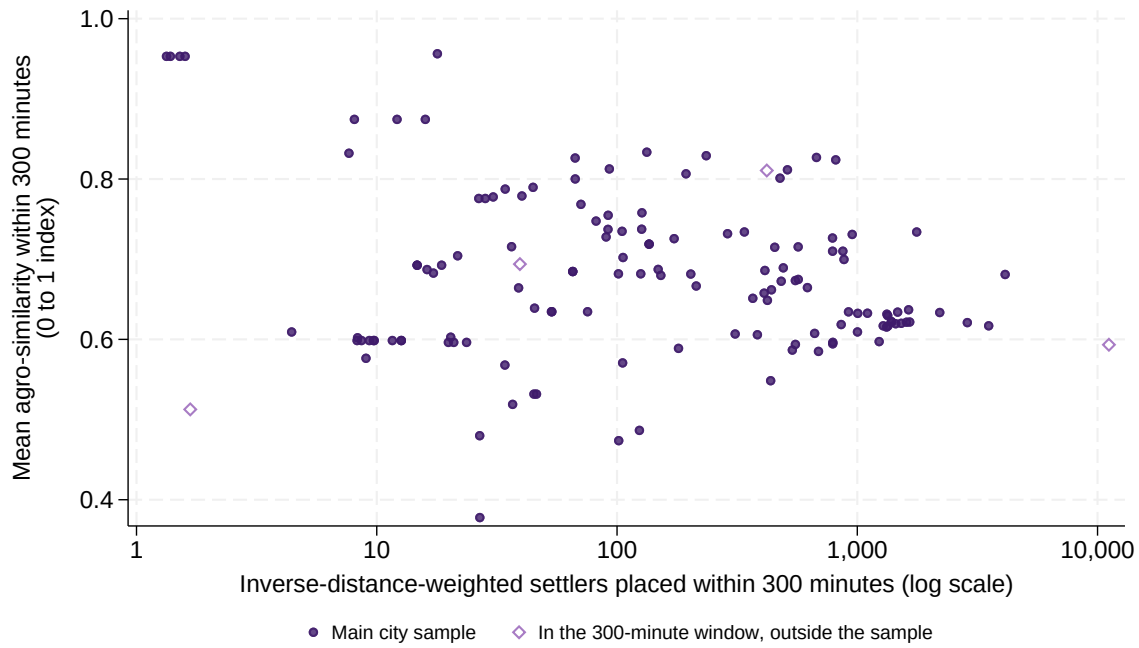
**Distance to the nearest transmigration village, and the 300-minute cutoff.** The 300-minute cutoff falls at the 73.8th percentile of travel time to the nearest transmigration village across the 187 centers, keeping the closer three-quarters of the sample. The cutoff is not a natural break in this distribution; it is chosen as a meaningful, achievable travel distance. The median center is 145 minutes from its nearest program village ([Figure A1b](#)).

**The results do not turn on the sample construction.** A reader may reasonably ask whether the reclassification accounts for the result, or whether the estimates rest on centers that were barely urban to begin with. [Figure A2b](#) re-estimates the main effects on subsamples that answer both questions: first dropping every center the reclassification touched, then imposing a 1980 population-density floor, a 1980 agricultural-share ceiling, and finally both. The two composition responses, the agricultural and modern employment shares, hold their sign, magnitude and significance in every one of them, and each estimate falls inside the interval of the headline it is being compared against. The agricultural-share screen is the demanding one, since it conditions on the 1980 level of the agricultural share itself and so works against finding a response in that share, and the response survives it.

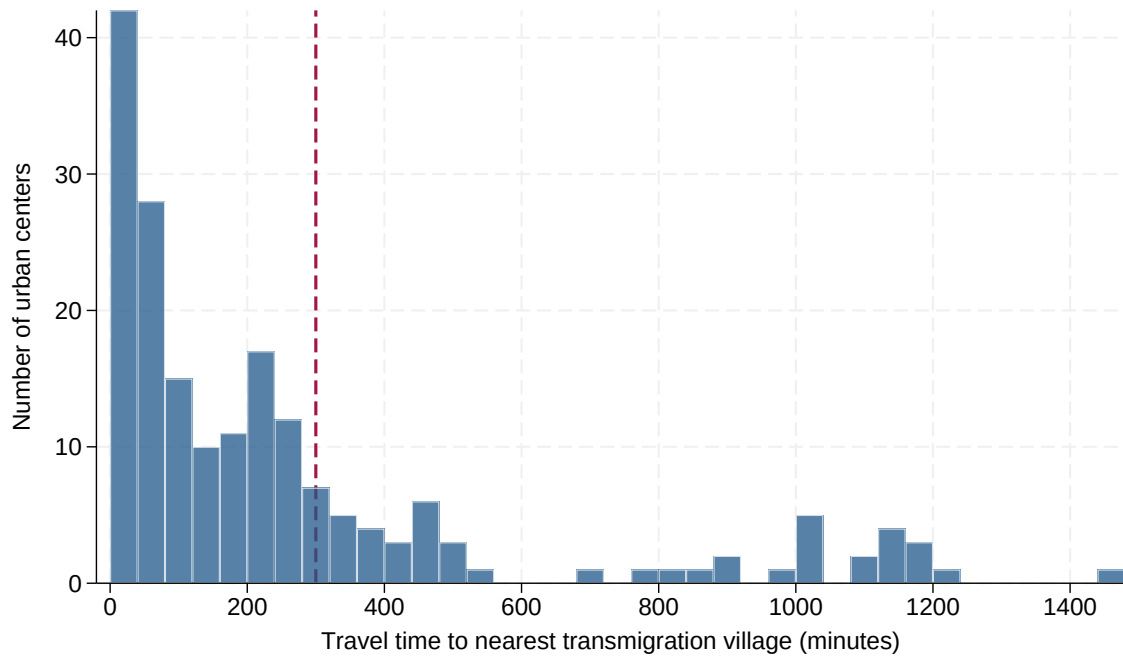
The same comparison, with the coefficient and its standard error for every outcome, is also given as a table for each of the three samples: the headline, the sample with every reclassified center removed, and the denser, less agricultural centers within that sample ([Table A2](#)).

Figure A1: Settler Placement and the Travel Time From Each City to the Nearest Program Village

(a) Agro-Similarity Against the Settler Count, Cities Within 300 Minutes



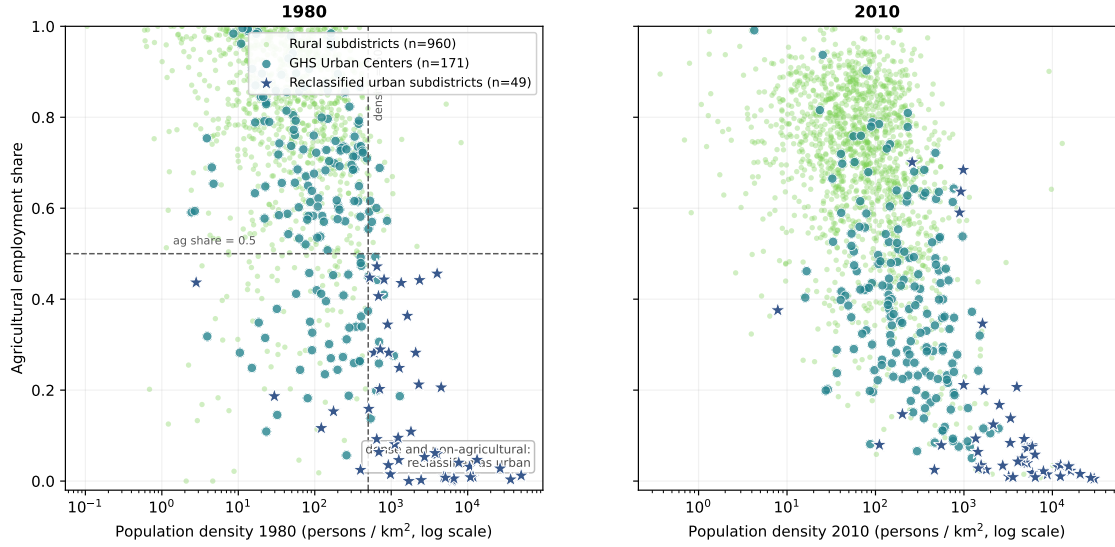
(b) Urban Centers by Travel Time to the Nearest Transmigration Village



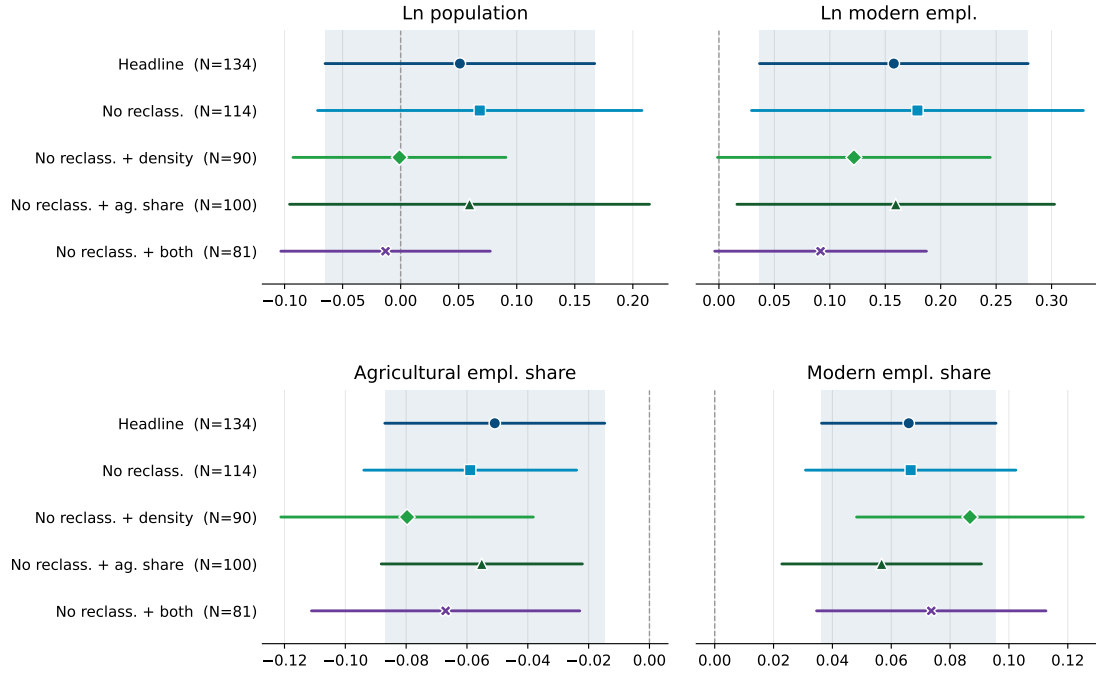
*Notes:* Panel (a): the 138 urban centers within 300 travel minutes of a transmigration village, with the main city sample (those with a 1980 census baseline) drawn as solid circles and the four in-window centers it excludes as hollow diamonds. The vertical axis is agro-similarity, the mean agro-climatic similarity of the transmigration villages within 300 minutes on its 0 to 1 index scale, and the horizontal axis is the number of settlers placed in those villages, weighted by inverse travel time, on a log scale. Panel (b): the 187-center analysis universe by travel time in minutes to the nearest transmigration village, in 40-minute bins. The dashed line marks the 300-minute cutoff, which keeps 138 of the 187 centers, the 73.8th percentile of this distribution; the median center is 145 minutes away.

Figure A2: The Reclassification Rule in the Data and the Results Under Tighter Urban Samples

(a) Reclassification Diagnostics: Density and Agricultural Employment Share



(b) Main Effects Under Tighter Urban-Sample Restrictions



Notes: Panel (a): each point is a place, positioned by population density (horizontal axis, log scale) and agricultural employment share (vertical axis) in 1980 (left) and 2010 (right), the two observables the reclassification rule uses; small circles are rural subdistricts, larger circles the satellite-defined urban centers, and stars the reclassified urban subdistricts. The dashed lines on the left mark the rule, density above 500 persons per square kilometer and agricultural employment share below 0.5 in 1980, so every star falls in the dense, non-agricultural quadrant. Panel (b): each marker is the coefficient on agro-similarity, standardized once so that every row shares one scale, from one regression of the 2010 outcome named above it, in the main specification with the full set of controls and Conley SE within 200km; bars are 95% confidence intervals from the same standard errors. The top row, repeated as the shaded band, is the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline); the four restricted samples drop the centers the reclassification created or altered, and three of them add a 1980 population-density floor of 25 residents per square kilometer, a 1980 agricultural employment-share ceiling of 80%, or both.

Table A2: Main Estimates Under Tighter Urban-Sample Definitions

|                                                                                                                                              | Population + Employment |                   |                  | Agric Empl           |                      | Modern Empl        |                     |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------|------------------|----------------------|----------------------|--------------------|---------------------|
|                                                                                                                                              | (1)                     | (2)               | (3)              | (4)                  | (5)                  | (6)                | (7)                 |
|                                                                                                                                              | ln pop                  | ln emp            | unemp rate       | ln L agric           | agric %              | ln L modern        | modern %            |
| <i>Panel A: the main city sample</i>                                                                                                         |                         |                   |                  |                      |                      |                    |                     |
| Agro-similarity                                                                                                                              | 0.051<br>(0.059)        | 0.024<br>(0.057)  | 0.008<br>(0.007) | -0.166*<br>(0.093)   | -0.051***<br>(0.018) | 0.158**<br>(0.062) | 0.066***<br>(0.015) |
| Observations                                                                                                                                 | 134                     | 134               | 134              | 134                  | 134                  | 134                | 134                 |
| <i>Panel B: excluding every area the reclassification created or enlarged</i>                                                                |                         |                   |                  |                      |                      |                    |                     |
| Agro-similarity                                                                                                                              | 0.068<br>(0.071)        | 0.036<br>(0.067)  | 0.012<br>(0.008) | -0.152<br>(0.093)    | -0.059***<br>(0.018) | 0.179**<br>(0.076) | 0.067***<br>(0.018) |
| Observations                                                                                                                                 | 114                     | 114               | 114              | 114                  | 114                  | 114                | 114                 |
| <i>Panel C: Panel B further restricted to at least 25 residents per km<sup>2</sup> and at most 80% agricultural employment, both in 1980</i> |                         |                   |                  |                      |                      |                    |                     |
| Agro-similarity                                                                                                                              | -0.013<br>(0.046)       | -0.052<br>(0.048) | 0.012<br>(0.008) | -0.304***<br>(0.099) | -0.067***<br>(0.023) | 0.092*<br>(0.049)  | 0.074***<br>(0.020) |
| Observations                                                                                                                                 | 81                      | 81                | 81               | 81                   | 81                   | 81                 | 81                  |
| Settler count                                                                                                                                | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                  | ✓                   |
| Baseline Pop 1980                                                                                                                            | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                  | ✓                   |
| Urban share 1980                                                                                                                             | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                  | ✓                   |
| Total Empl 1980                                                                                                                              | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                  | ✓                   |
| Agri Empl 1980                                                                                                                               | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                  | ✓                   |
| Agro-climatic controls                                                                                                                       | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                  | ✓                   |

*Notes:* Each cell is the coefficient on standardized agro-similarity from one regression of the 2010 census outcome in its column, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Agro-similarity is standardized once over every urban center with a measured exposure, so the panels share one scale. Panel A is the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline); Panel B drops every center the reclassification created or enlarged; Panel C keeps the Panel B centers with at least 25 residents per square kilometer and at most 80% of employment in agriculture in 1980. Columns 1 to 3 are population, employment and unemployment, columns 4 and 5 agriculture, and columns 6 and 7 the modern sector (manufacturing plus services); the four sector columns are shaded. In the labels: ln (the natural log), L (a count of workers), emp and Empl (employment), unemp rate (the unemployment rate), and agric % and modern % (the two sector shares); Baseline Pop 1980, Total Empl 1980 and Agri Empl 1980 are the logs of 1980 population, total employment and agricultural employment, and Urban share 1980 is the urban share of the 1980 population. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## A.2 Transmigration-Village and Sample-Group Descriptives

*Cited in the main text.* Table A3: [Section 2.2, p. 8](#). Table A4: [Section 2.2, p. 8](#). Table A5: [Section 2.2, p. 8](#). Table A6: [Section 3.2, p. 13](#).

The transmigration villages whose agro-climatic similarity to the settlers' origin regions generates the city-level agro-similarity exposure are small agricultural settlements (mean 1,885 settlers placed at founding, 69% of employment in agriculture) established between 1979 and 1988. The program-wide total of more than two million resettled people cited in [Section 2](#) is the official aggregate, which includes placements outside this village census. The village sample underlying the exposure measure records approximately 1.5 million placed individuals, matching the annual placements plotted in the Online Supplement. The summary statistics cover all 814 program villages, with the village-population rows available for the 716 villages with digitized polygon footprints ([Table A3](#)).

Table A3: Summary Statistics: The Transmigration Villages

| Panel A: Levels (settlers placed, village population) |       |       |        |     |         |     |
|-------------------------------------------------------|-------|-------|--------|-----|---------|-----|
|                                                       | Mean  | SD    | Median | Min | Max     | N   |
| Settlers placed (count)                               | 1,885 | 1,011 | 1,792  | 136 | 8,852   | 814 |
| Village population 1980                               | 1,279 | 6,057 | 282    | 0   | 130,914 | 716 |
| Village population 2010                               | 2,235 | 5,228 | 817    | 0   | 85,296  | 716 |

| Panel B: Economic structure, similarity, and agro-climatic covariates |         |        |         |        |         |     |
|-----------------------------------------------------------------------|---------|--------|---------|--------|---------|-----|
|                                                                       | Mean    | SD     | Median  | Min    | Max     | N   |
| Agro-climatic similarity to migrant origins                           | 0.667   | 0.136  | 0.684   | 0.000  | 1.000   | 814 |
| Agricultural employment share                                         | 0.689   | 0.241  | 0.758   | 0.009  | 0.993   | 814 |
| Urban village (share)                                                 | 0.029   | 0.169  | 0.000   | 0.000  | 1.000   | 814 |
| Years of schooling                                                    | 3.998   | 0.899  | 3.978   | 0.713  | 7.653   | 814 |
| Agricultural productivity (normalized)                                | 0.996   | 2.647  | 0.435   | 0.001  | 27.316  | 770 |
| Ethnolinguistic fractionalization                                     | 0.428   | 0.202  | 0.440   | 0.021  | 0.889   | 814 |
| Ruggedness                                                            | 0.311   | 0.129  | 0.310   | 0.006  | 0.691   | 814 |
| Distance to coast (index)                                             | 10.553  | 1.138  | 10.844  | 5.795  | 11.925  | 814 |
| Soil organic carbon                                                   | 4.765   | 6.214  | 1.685   | 0.584  | 25.065  | 814 |
| Soil pH                                                               | 5.069   | 0.435  | 4.950   | 4.256  | 7.790   | 814 |
| Rainfall, 1948–78                                                     | 224.998 | 35.188 | 227.616 | 90.999 | 331.816 | 814 |
| Temperature, 1948–78                                                  | 26.253  | 1.709  | 26.748  | 17.225 | 28.174  | 814 |

*Notes:* Unit of observation is the transmigration village: the 814 villages from which the agro-similarity exposure of the urban centers is built, settled between 1979 and 1988 (mean 1983). Columns are the mean, standard deviation, median, minimum, maximum and number of villages with a value. Village population in 1980 and 2010 is estimated from the Global Human Settlement population grid for the 716 villages with digitized settlement footprints, and normalized agricultural productivity is available for 770 villages. Agricultural productivity is the unitless yield index of [Bazzi et al. \(2016\)](#), equal to one for a village that farms every crop at the benchmark yield of that crop. Similarity to the origin regions of the settlers is an index from 0 to 1, soil organic carbon is a percentage and temperature is in degrees Celsius. Settler counts, placement years, agro-climatic similarity and the covariates come from the transmigration program records and the construction of [Bazzi et al. \(2016\)](#). N is the number of villages with a value; the rainfall and temperature rows are averages over 1948 to 1978, and the urban village row is the share of villages classified as urban.

The three sample groups entering the structural-transformation regressions differ on observable characteristics: cities are larger and far less agricultural than rural subdistricts, while the agro-similarity exposure is balanced across all three groups, consistent with the quasi-random settler allocation ([Table A4](#)).

**The main city sample and the other outer-island urban centers.** [Table A5](#) sets the main city sample beside the urban centers outside it, on the 1980 baselines and the 2010 outcomes of [Table 1](#).

Table A4: Descriptive Statistics by Location Type: Cities and Rural Subdistricts With and Without a Settlement

|                                  | Cities  |           | Rural, no TMV |          | Rural, with TMV |          | All    |           |
|----------------------------------|---------|-----------|---------------|----------|-----------------|----------|--------|-----------|
|                                  | mean    | sd        | mean          | sd       | mean            | sd       | mean   | sd        |
| Population 1980                  | 118,652 | (220,180) | 25,601        | (18,604) | 17,949          | (13,326) | 37,552 | (91,647)  |
| Population 2010                  | 229,366 | (404,467) | 36,812        | (24,079) | 52,180          | (35,899) | 68,015 | (169,605) |
| Employed 2010                    | 90,259  | (152,082) | 16,101        | (10,894) | 22,864          | (15,625) | 28,293 | (64,284)  |
| Agric employment share 2010      | 0.374   | (0.183)   | 0.656         | (0.168)  | 0.694           | (0.130)  | 0.623  | (0.193)   |
| Manuf employment share 2010      | 0.047   | (0.042)   | 0.032         | (0.038)  | 0.025           | (0.025)  | 0.032  | (0.037)   |
| Services employment share 2010   | 0.470   | (0.145)   | 0.253         | (0.120)  | 0.218           | (0.085)  | 0.277  | (0.143)   |
| Urban population share 1980      | 0.260   | (0.228)   | 0.027         | (0.102)  | 0.011           | (0.052)  | 0.058  | (0.148)   |
| Agro-similarity (raw)            | 0.680   | (0.103)   | 0.677         | (0.098)  | 0.675           | (0.107)  | 0.677  | (0.101)   |
| Travel time to nearest TMV (min) | 109     | (86)      | 162           | (79)     | 1               | (0)      | 121    | (96)      |
| Observations                     | 134     |           | 596           |          | 191             |          | 921    |           |

*Notes:* Columns are the samples of the structural-transformation regressions by location type: cities (the main city sample: outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), rural subdistricts without a settlement, rural subdistricts with a settlement (one of the 814 transmigration villages), and all three pooled. The rural subdistricts are in 2000 boundaries, lie outside every city and within 300 travel minutes of a transmigration village, and have a measured 1980 baseline and 2010 outcomes. Cells give the group mean with the standard deviation in parentheses; the final row gives the number of observations, and every row is measured on all of them. Employment counts those who worked or ran a business in the previous week, and employment shares divide each sector count by everyone with an industry recorded. Agro-similarity is the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, before standardization. Travel time is in minutes to the nearest transmigration village, set to one in a subdistrict with a settlement. The population and employment rows come from the 1980 and 2010 population censuses. In the labels, TMV is a transmigration village.

**Coverage of the household survey by wave.** The 2011 wave of the household survey (Susenas) includes the whole main city sample but follows Indonesia’s 2004 rice import ban, which moved food prices nationwide for reasons unrelated to local agro-climatic conditions. The agricultural price in the main results therefore comes from a wave before the ban, and the two earlier waves also match the timing of the 2003 village-census (PODES) production data. Of the two, the 1996 wave covers more cities and more rural subdistricts, with and without a settlement, than the 2002 wave (Table A6).

**Price measures from household unit values.** The price measures of Figure D3 are built from the 1996 household survey (Susenas). A household’s unit value for a food item is the value of its consumption of the item, own production included, divided by the quantity; within each item the unit values are winsorized at the 5th and 99th percentiles within province and urban status, and the survey-weighted median over the city’s households gives the city’s price of the item. An index divides each item’s city price by the item’s median over the households of all the urban centers the survey covers, the national price, and aggregates the price relatives over the raw agricultural food items under those urban centers’ pooled expenditure weights (Laspeyres), under the city’s own expenditure weights (Paasche) and under equal weights (a geometric mean); the crops index takes the geometric mean over the rice, corn and cassava items and their processed forms. The rice unit value is the value of the rice consumed, the local and the higher-quality varieties with own production valued by BPS, over the kilograms consumed, each summed over the city’s households with survey weights. As unit values, every measure reflects price and quality together.

Table A5: The Main City Sample Against the Other Outer-Island Urban Centers

|                                      | Main city sample | Other centers | Difference | SE      | N sample | N other |
|--------------------------------------|------------------|---------------|------------|---------|----------|---------|
| <i>1980 baselines</i>                |                  |               |            |         |          |         |
| Log population, 1980                 | 11.093           | 11.139        | -0.046     | (0.205) | 134      | 50      |
| Log employment, 1980                 | 9.846            | 9.861         | -0.015     | (0.200) | 134      | 50      |
| Log agricultural employment, 1980    | 9.249            | 9.280         | -0.031     | (0.204) | 134      | 50      |
| Urban population share, 1980         | 0.260            | 0.237         | 0.023      | (0.032) | 134      | 50      |
| Agricultural employment share, 1980  | 0.560            | 0.548         | 0.012      | (0.036) | 134      | 50      |
| <i>2010 outcomes</i>                 |                  |               |            |         |          |         |
| Log population, 2010                 | 11.842           | 11.717        | 0.125      | (0.168) | 134      | 52      |
| Log employment, 2010                 | 10.933           | 10.779        | 0.154      | (0.177) | 134      | 52      |
| Log agricultural employment, 2010    | 9.822            | 9.753         | 0.069      | (0.223) | 134      | 52      |
| Log modern-sector employment, 2010   | 10.232           | 10.069        | 0.163      | (0.171) | 134      | 52      |
| Agricultural employment share, 2010  | 0.374            | 0.406         | -0.032     | (0.035) | 134      | 52      |
| Modern-sector employment share, 2010 | 0.516            | 0.510         | 0.007      | (0.030) | 134      | 52      |
| Unemployment rate, 2010              | 0.122            | 0.133         | -0.012     | (0.013) | 134      | 52      |

*Notes:* Each row compares the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline) with the 53 other outer-island urban centers: 49 lie farther than 300 travel minutes from every transmigration village, and 4 lie inside that window and lack either a 1980 census baseline or a 2010 endline. The first two columns are group means; the third is the coefficient from a regression of the row variable on an indicator for the main city sample, with its Conley SE within 200km (uniform kernel) in parentheses; the last two count the centers in each group with a value for the row. Population and employment counts are in natural logs, and shares and the unemployment rate are fractions; the modern sector is manufacturing plus services. A Wald test that the five 1980 rows jointly predict membership in the main city sample has  $p = 0.735$  under the uniform kernel, which weights every pair of centers inside the cutoff equally, and  $p = 0.727$  under the Bartlett kernel, which weights a pair by one minus its distance over the cutoff. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A6: Units Each Household Survey Wave Reaches, by Sample

|                      | Main city sample | Rural subdistricts   |                   |
|----------------------|------------------|----------------------|-------------------|
|                      |                  | Without a settlement | With a settlement |
| Universe             | 134              | 596                  | 191               |
| Sample, Susenas 1996 | 112              | 285                  | 117               |
| Sample, Susenas 2002 | 109              | 204                  | 104               |
| Sample, Susenas 2011 | 134              | 576                  | 188               |

*Notes:* Counts of units. Universe: the units of each sample that hold agro-similarity and every control of the main specification. The main city sample is the outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline; the rural subdistricts with and without a settlement lie outside every city and within 300 travel minutes of a transmigration village. Sample: the units of the universe that a wave's surveyed households live in, pooled to the city from its own households and to the year-2000 subdistrict, on the wave's fullest outcome; a few outcomes cover fewer. The 2011 wave follows the 2004 ban on rice imports.

### A.3 Cross-Country Food Consumption and Urbanization

*Cited in the main text. This section:* [Section 2.2, p. 11.](#)

[Figure 5](#) plots national food expenditure shares against urban population shares over the years available for each country, between 1960 and 2025. The main food-share source is the United Nations National Accounts household final consumption expenditure by purpose (Table 3.2). I divide expenditure on food and non-alcoholic beverages by total

household consumption expenditure. The World Bank’s World Development Indicators supply population and the urban population share, using each country’s administrative definition of urban. Marker size represents population and darker shades indicate later years. Differences in national urban definitions and food-expenditure measurement limit comparisons of levels across countries.

The country set starts with the 20 most populous countries in the United Nations panel, ranked by the latest available population observation. Poland and Iraq are omitted because their urban shares change little over the period. China, the United States, Pakistan, Bangladesh, and South Africa are added from national statistical sources, giving 23 countries. Russia is omitted because of price-measurement problems in its household data. Egypt, the Democratic Republic of the Congo, Viet Nam, Türkiye, and Thailand are absent because the United Nations source lacks their household-consumption breakdown.

Published household-survey rounds supply the United States (1984 to 2024), Pakistan (2001 to 2024), South Africa (2006 to 2023), Nigeria (1986 to 2019), and Bangladesh (1981 to 2022). The United States series uses the Bureau of Labor Statistics Consumer Expenditure Surveys: food at home is divided by the consumption subtotal, which excludes personal insurance, pensions, and cash contributions. Pakistan’s Household Integrated Economic Survey series removes tobacco and ready-made food from earlier rounds using the expenditure components in the published reports; later rounds report the narrower food category directly. South Africa’s published Income and Expenditure Survey and Living Conditions Survey rounds report food and non-alcoholic beverages relative to household consumption expenditure. Nigeria and Bangladesh retain their broader published food shares, which include meals away from home and tobacco. Nigeria’s change from diary collection to a shorter recall period further limits comparisons over time.

China’s series is the Engel coefficient published by its National Bureau of Statistics, covering 1978 to 2025. Before the national series begins in 2013, I combine the published urban and rural coefficients using urban and rural population weights. The urban and rural expenditure definitions differ, and the 2013 survey change introduces a break. China’s food category includes tobacco, liquor, and meals away from home. Dashed paths and asterisks identify China, Nigeria, and Bangladesh because these broader definitions prevent direct comparison of their levels with the narrower food shares.

Indonesia (IDNS) uses the national household-survey series from Susenas for 1994 to 2017. I remove catering, alcohol, and tobacco using item-level consumption data available in 7 of the 20 observed years. The adjustment is the share of reported food spending retained after these exclusions: it is measured in 1996, 2002, 2011, and 2013 to 2016, interpolated linearly between those years, and held at the nearest measured value outside that interval. I multiply each year’s broad food share by this retained share. I then add the mean difference between the United Nations and adjusted Susenas shares over their overlapping years, +6.3 percentage points, to the entire Susenas series. This constant adjustment aligns its level with the national-accounts source while preserving its changes in percentage points.

## B Measures

This part sets out how each measure the main text uses is constructed.

### B.1 Land-and-Sea Travel Times and Pairwise Distances

*Cited in the main text. This section:* [Section 2.1, p. 6](#); [Section 3.1, p. 12](#); [Section 3.3, p. 14](#).

Cost distance measures the minimum accumulated travel time between two locations, allowing the fastest route to differ from the shortest geographic path. I construct these measures for the paper’s mapped urban areas, subdistricts, and program villages before applying the estimation-sample restrictions.

**Inherited land surface.** I construct the pairwise travel measures used in this paper from the MAP 2015 land-travel friction surface, which assigns the time required to cross each location. The inherited surface combines information

on transport access, land cover, and terrain from global accessibility mapping (Weiss et al. 2018). The shipping speeds follow the construction described by Nelson (2008). My spatial construction starts with the land surface: I add the missing maritime connections, align the geographic layers, locate the paper’s origins and destinations, and compute their pairwise cost distances. The published distance to the nearest city is a separate output of the global accessibility studies; the distances used here are calculated for this paper’s locations.<sup>24</sup>

**Maritime connections and geographic alignment.** The land surface does not supply the shipping connections needed to travel between Indonesian islands. I add the commercial-shipping layer from Halpern et al. (2008), based on vessel observations during the 12 months beginning October 2004. I project the land and shipping layers into the same coordinate system for Southeast Asia, clip them to a common geographic extent, and align their grid cells. I express the land friction values in minutes per kilometer. For shipping, I set travel costs at 3 minutes per kilometer in cells with positive recorded intensity and 1.5 minutes per kilometer in major shipping corridors, corresponding to assumed speeds of 20 and 40 kilometers per hour. I add the land and water friction values where both are present, including coastal cells, and set costs at 30 minutes per kilometer in cells without an observed component. This last value is an assumed cost of travel outside the mapped connections. The shipping observations describe vessel activity after settlement, and their coverage is incomplete.

**Origins, destinations, and least-cost paths.** I link the census subdistrict polygons, GHS urban-center boundaries, and transmigration-village locations to their administrative identifiers and project their coordinates onto the combined surface. I calculate distances to program villages separately for subdistricts and urban locations. For the urban calculations, I intersect each constituent subdistrict with the union of mapped urban footprints and use the centroid of that intersection when it lies inside the footprint, otherwise an interior point. Where the intersection is empty, the calculation retains the subdistrict centroid. For the surrounding-subdistrict calculations, origins are the subdistrict centroids; destinations are the transmigration-village locations matched to the replication data of Bazzi et al. (2016).

For each origin, the least-cost calculation accumulates travel costs across adjacent cells and selects the path with the smallest sum to each destination. The original QGIS calculations use an approximately 1-kilometer grid and average the friction values of adjacent cells, with diagonal steps weighted by  $\sqrt{2}$ . They treat a horizontal or vertical step as one nominal kilometer.<sup>25</sup> I match the resulting pair records by subdistrict and village identifiers. For the revised urban endpoints, I calculate the change in least-cost travel time between the original and revised locations on the same surface and grid, calibrate it to the original QGIS scale, and add that change to the saved pairwise times. This preserves the original travel-cost convention while changing the location represented by the urban endpoint. Pairs newly included by the endpoint change use the calibrated revised calculation, and I round the resulting values to whole minutes with a minimum of one minute. Travel time from a city to a village is the minimum across the city’s constituent subdistrict locations, with membership updated to the city definition used in the analysis. For members without urban-origin pair records, I use the corresponding subdistrict-origin calculations.

**Stored pairs and exposure.** The pairwise calculations cover a broader set of locations than the estimation sample. The saved urban and subdistrict pair records retain journeys up to 2,000 travel minutes. This storage limit differs from the 300-minute catchment used to construct the main exposure measure. I separately calculate great-circle distances from geographic coordinates for the straight-line comparisons. I retain the original travel measure alongside the adjusted measure used in constructing exposure: the adjustments set travel time to one minute when more than

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<sup>24</sup>The saved land input agrees with the preserved MAP 2015 surface. The earlier VMap-based input has not been located. The earlier construction described by Nelson (2008) uses VMap0 roads, released in 1997 as an update to the Digital Chart of the World developed in 1992. The documentation accompanying the saved land input identifies the MAP 2015 surface associated with Weiss et al. (2018), which uses later road sources. The available records therefore do not establish that the road component of the saved travel times is restricted to VMap0.

<sup>25</sup>The projected cells are approximately 1.09 kilometers wide. The original QGIS calculation does not separately multiply its costs by that cell width. The reported travel-time measure retains this discretization convention.

0.1% of a village’s polygon area overlaps the urban footprint and bound excessive travel times using the city-centroid-to-village great-circle distance at 2 kilometers per hour. The city’s nearest-village eligibility test uses the original travel measure, while village catchment membership and inverse-time weights use the adjusted measure. The main exposure is the unweighted mean of village agro-similarity within the catchment; weighted variants are described in the main text.

**Other geographic pairs.** I also construct great-circle distances between urban-member subdistrict centroids and other subdistrict centroids in the same island group. For the spatial model, I construct pairwise land-and-sea travel costs among its urban and rural locations. The rural model locations aggregate subdistricts into the rural remainders of districts. These calculations have distinct geographic scopes from the urban-area-to-village and subdistrict-to-village travel measures described above.

**Road timing and identification.** Roads mapped closer to the settlement period limit the scope for subsequent investment responding to settlement or productivity gains to determine measured exposure. This advantage requires the earlier roads to be the ones represented in the inherited surface. It does not establish that those roads were unaffected by the program: roads mapped in the 1990s may already reflect investment during settlement. The later road sources and shipping observations prevent interpreting the combined surface as a transport network observed before the program. If transport responses affect catchment membership or nearest-village eligibility, holding those catchments fixed in the expected-exposure exercises does not remove the resulting selection.

**Travel time and straight-line distance.** Geographic connections matter for access around Tana Paser. Two transmigration villages inside the same 50-kilometer radius of the functional urban area have sharply different travel times (Figure B1). The northern village is labeled at about an hour, while the southwestern village is labeled at over five hours. This illustration uses travel time to the nearest functional urban area; the empirical distances between cities and villages use the urban-center definitions and pairwise construction described above.

## B.2 Trade-Openness Measure: Rice-Import Market Access

*Cited in the main text. This section:* Section 3.5, p. 16; Section 5.3, p. 33. Table B2: Section 5.3, p. 33. Table B1: Section 3.5, p. 17.

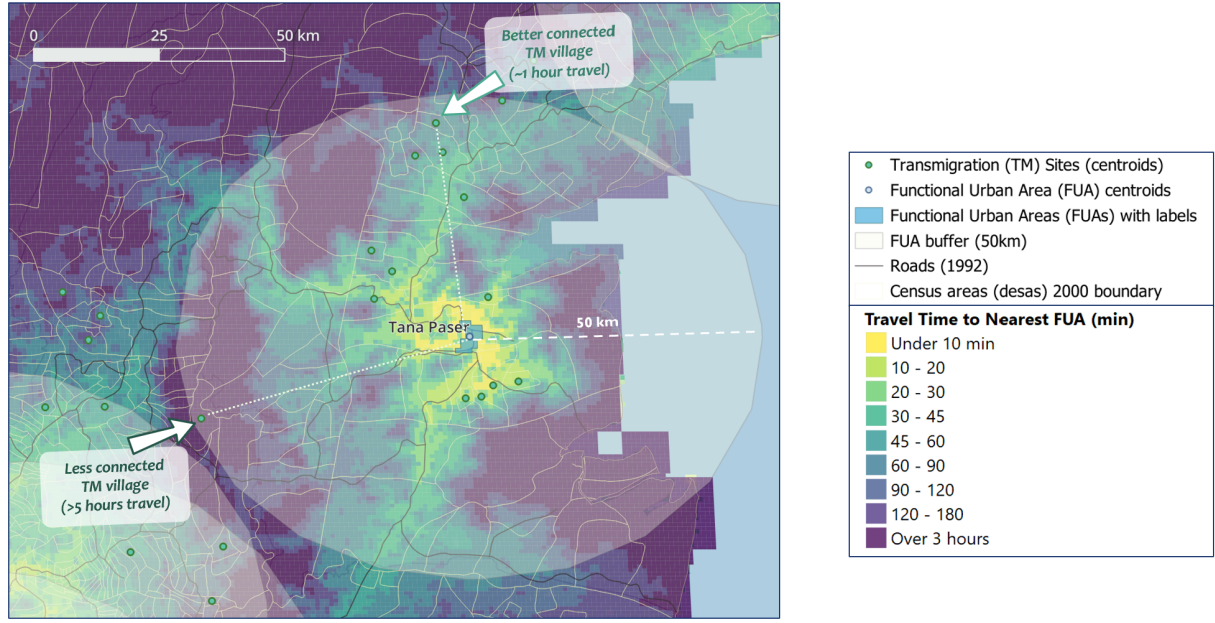
**Source data.** To measure how exposed each urban center is to imported food, I use administrative records of *rice imports by port of entry*: volumes (kg) and values disaggregated by importing port and origin for 2000-2010, with port coordinates (the “Ports & Prices” compilation). Because the traded commodity is rice, the resulting index is specifically a measure of *food-trade* openness, the form of openness most directly relevant to a local agricultural-productivity shock. Summing import throughput to the port level gives, for each port of entry  $p$ , a total volume  $\text{kg}_p$  and a location. The compilation stores port coordinates in degrees and minutes; taken as decimal degrees, each port falls 20 to 40 km from its berth, so every measure below uses the converted coordinates, which place the ports on the coast (median 2.8 km from the coastline against 13.0 km) and put the named terminals on their real berths.

**Construction.** Not every urban center is itself a port, so I separate *proximity* (being or abutting a port) from *trade intensity* (the volume of trade a city can reach), and compute three candidate metrics: (i) **proximity**, the great-circle distance  $\text{dist\_nport}_i$  from city  $i$  to its nearest customs port of entry (plus an indicator for a port within 50 km); (ii) **volume**, the import throughput of the reachable port(s); and (iii) a **gravity market-access** composite that combines the two,

$$\text{open}_i = \sum_p \frac{\text{kg}_p}{d_{ip}^\delta},$$

a distance-discounted sum of every port’s import volume (decay parameter  $\delta = 1$ ; the nearest-port distance is floored at 10 km so that port-cities do not diverge). The gravity variable is  $\ln \text{open}_i$ , standardized over the urban centers inside the headline cost-distance window ( $N = 138$ ). It uses information from *all* reachable ports weighted by their size, where

Figure B1: Straight-Line Distance and Travel-Time Connectivity Around Tana Paser



*Notes:* The background shows calculated travel time to the nearest functional urban area (FUA), Tana Paser in this region, in bins from under 10 minutes to over 3 hours. Filled dots identify transmigration-village locations, and the small ring identifies the urban-area centroid. The circle shows a 50-kilometer radius; dotted lines connect the two highlighted villages to the urban area. The 1992 roads are displayed for orientation and do not establish the road source underlying the travel-time surface. Village boundaries follow the 2000 census, and urban-area boundaries come from the Global Human Settlement Layer. Source: the author's geographic construction and travel-time calculations; transmigration-village locations from [Bazzi et al. \(2016\)](#). This is a descriptive map, with assumed travel costs and no statistical inference.

bare proximity collapses access to a single nearest-neighbor distance. Because it loads realized import volumes it is an equilibrium outcome in part, so it enters in Panel B of [Table 5](#) as a check on the sign of the bias that realized trade introduces; the preferred measure is route openness, constructed below.

**The gravity and port-distance measures, in brief.** The gravity measure loads realized trade. Administrative records of rice imports by port of entry over 2000-2010 give each port a total import volume, and the market-access index sums those volumes over every port, discounting by distance. Realized import volumes respond to the substitution this paper studies: a city whose hinterland becomes more productive buys more of its food there, its ports handle less imported rice, and the gravity measure classifies it as more closed. On the agricultural share the bias this builds into the split has a predicted sign. Closed cities are where the effect is large, so the gravity split is expected to overstate how much openness dampens the response, and it does so exactly where the route split finds no difference: on the agricultural employment share the openness offset is 0.053 (s.e. 0.021) on the gravity split against 0.021 (s.e. 0.025) on the route split. On the modern-sector outcomes the two splits agree in sign, and the gravity offsets are, if anything, slightly smaller:  $-0.034$  against  $-0.040$  on the modern employment share, both marginally significant, and  $-0.146$  against  $-0.169$  on log modern-sector employment, where only the route estimate is marginally significant. The gravity split therefore serves as a check on the sign of this bias, and the openness result relies on the route split. It is the alternative openness split in Panel B of [Table 5](#), and it correlates at 0.60 with route openness. [Rutz \(1987\)](#) records a 1973 cargo turnover at the ports through which 93.5% of the 2000 to 2010 rice tonnage moved, six years before the main placement wave of 1979 to 1984. The historical maps date part of the port list; the domestic travel-time input uses later roads and shipping observations. The simplest measure is the distance to the nearest customs port of entry. Bare distance is nearly uninformative about reachable trade, because ports of entry are common (the median city sits 45 km from one) and the measure ignores what moves through them. It enters the balance tests of [Section 4.3](#) as a trade-access covariate. The nearest port of entry is

a 1973 main port for 81 of the 134 cities, and a harbor the Directorate General of Sea Communications already listed in 1986 for 118 of them (Table B1).

Table B1: Vintage of the Ports of Entry Behind the Two Openness Measures

|                                                         | Route measure<br>(route hours)<br>cities | Balance covariate<br>(great circle)<br>cities | Rice imports<br>(2000 to 2010)<br>share of tons |
|---------------------------------------------------------|------------------------------------------|-----------------------------------------------|-------------------------------------------------|
| 1973 main port (Map 2)                                  | 88 (66%)                                 | 81 (60%)                                      | 93.5%                                           |
| Other 1986 harbor (Map 1)                               | 37 (28%)                                 | 37 (28%)                                      | 5.6%                                            |
| On neither map                                          | 9 (7%)                                   | 16 (12%)                                      | 0.9%                                            |
| In the 1986 inventory                                   | 125 (93%)                                | 118 (88%)                                     | 99.1%                                           |
| Cities                                                  | 134                                      | 134                                           |                                                 |
| <i>Route measure rebuilt on a restricted anchor set</i> |                                          |                                               |                                                 |
| 1973 main ports only                                    | same side of the median: 132 of 134      |                                               | correlation 0.989                               |
| 1986 inventory only                                     | same side of the median: 134 of 134      |                                               | correlation 0.998                               |

*Note:* The sample is the main city sample. Each city’s anchor is the nearest of the 183 customs ports of entry, by travel time over land and sea for the route measure and by great-circle distance for the port-distance balance covariate; the first two columns count cities by the category of that port, with the share of cities in parentheses, and the third gives the share of 2000 to 2010 rice-import tonnage through the ports of each category. A 1973 main port is the main port of one of 49 coastal regions at which Map 2 of Rutz (1987) draws a 1973 cargo circle; Map 1 prints the 237 harbors in their 1986 state. The main placement wave of the program ran 1979 to 1984. The lower panel rebuilds the route measure on the restricted anchor set and compares its below-median high-openness split with the full-list split. Each correlation is between the route measure rebuilt on the restricted ports and the full-list route measure across the cities.

**Proximity versus the gravity composite.** The gravity market-access index is negatively related to distance to the nearest customs port of entry, in the expected direction, but only weakly ( $\rho = -0.15$ ): at a given distance there is wide variation in gravity access, because the composite also loads the *size* of the reachable ports and the contribution of more distant ones, while bare proximity is nearly uninformative once every port of entry counts as a candidate port (the median city sits 45 km from one). Restricting the horizontal axis to the ports that recorded rice volume in 2000-2010 tightens it to  $\rho = -0.54$ , a tighter but conditioned comparison. Restricting to ports with realized rice volume conditions on the trade outcomes themselves, so the full-list figure is the one reported. Either way the conclusion is the same, and the full-list version shows it more clearly. The gravity measure is not a relabeling of raw proximity, and it contains independent trade-volume information. The alternative openness split in Panel B of Table 5 uses it for that reason.

**Route proximity to the Thailand/Vietnam rice supply.** The gravity composite loads realized import volumes, which are an equilibrium outcome. The preferred openness measure in Section 5.3 uses route proximity, following the “distance to Ho Chi Minh City/Bangkok (port)” construction in Bazzi (2017) (his original uses great-circle distance to the nearest port and averages the two hubs; I use least-cost travel time and the nearer hub): for each urban center, the route-aware least-cost travel time to its nearest customs port of entry (over the same land-and-sea friction surface used for every cost distance in the paper) plus the sea distance from that port onward to the nearer of Bangkok and Ho Chi Minh City, the hubs from which the bulk of Indonesian rice imports ship, converted to hours at the friction surface’s shipping-lane speed so the two legs share one unit. Smaller totals mean closer alignment with the rice-import routes, hence higher measured openness; the high-openness indicator in Table 5 is a below-median split inside the estimation window. The international geography is fixed. The domestic leg uses the land-and-sea surface described in Section B.1, whose road and shipping inputs postdate settlement. The customs list has no vintage field, and the Rutz maps date 88 of the 134 ports to a main port with a recorded 1973 cargo turnover and 125 of them to the 1986 harbor inventory or earlier, leaving 9 that may post-date the program (Table B1). The international leg dominates the measure. Three properties matter in practice. First, the international leg dominates the composite (median 42 hours against 1.2 domestic), which

adds the west-to-east proximity-to-the-routes gradient that nearest-port distance discards: route proximity correlates 0.60 with the gravity composite but only 0.18 with domestic-only nearest-port proximity, and the measure is insensitive to selecting the cheapest gateway rather than the nearest port (correlation 0.99). Second, the international leg is a great-circle approximation, which understates distances from western ports whose actual sea route rounds the Strait of Malacca. This compresses the open end of the gradient but preserves the ranking. Third, the customs-port set includes ports with and without recorded rice imports. Whether a given port appears in the 2000-2010 rice-import records is itself an equilibrium outcome, and one the mechanism of this paper could move, since a city that substitutes hinterland food for outside supply leaves its port less rice-import business. The port set is therefore the full customs port-of-entry list, every port with coordinates whether or not it ever recorded rice volume, which removes any dependence on realized import quantities. Restricting it to the 58 ports that did record rice volume changes little. The two sets place 134 of the 138 in-window cities on the same side of the openness median, and the four they disagree on split two in each direction. Restricting the port set instead to the 1973 main ports leaves 132 of the 134 on the same side of that median, at a correlation of 0.989 with the full-list measure. The high-openness offset in [Table 5](#) goes from +0.021 to  $-0.006$  on the agricultural share and from  $-0.040$  to  $-0.021$  on the modern share (standard errors 0.029 and 0.024 on the restricted set): the modern-share offset keeps its sign at about half its size and loses its significance at the 10% level, and the agricultural-share offset is near zero on both sets. The reason the two agree at all is the first property above, since the international leg accounts for 97% of the composite's cross-city variance and no port's rice-import status can move it. The historical-port comparison assesses sensitivity to port selection; it does not establish the timing of the domestic transport network. The measure is equally insensitive to the coordinate format. Rebuilding both panels with the port coordinates taken as decimal degrees, as they were before the degrees-and-minutes correction described above, leaves every coefficient of the gravity split (Panel B of [Table 5](#)) at its sign and significance, moves the market-access median split by ten of 138 cities, five in each direction, and leaves all three predetermined trade-openness rows of the balance test covering zero under either format.

**The gravity measure as the split.** The gravity market-access measure, in place of route proximity, gives a second openness split, above the median inside the estimation window. Built from realized rice-import volumes, the measure is in part an equilibrium outcome, so its estimates check the sign of the bias described above. They move in the same direction as the route split on both outcomes. On the agricultural share the offset is 0.053 ( $p = 0.013$ ) against 0.021 ( $p = 0.399$ ) on the route split, so the realized-trade measure finds a differential that the predetermined one does not, in the direction the substitution predicts. On modern employment the two agree, at  $-0.146$  ( $p = 0.229$ ) against  $-0.169$  ( $p = 0.074$ ) (Panel B of [Table 5](#), whose Panel A is the route split).

**The instrumented median splits.** The instrumented estimates of the main text use continuous market access and continuous route proximity. With the two median splits in their place, the gravity split of Panel B instrumented by the route split of Panel A, the rows are those of Panels A and B and can be compared one for one. The exposure effect on the modern employment share is 0.111 in the more closed cities and 0.021 in the more open ones ([Table B2](#)). The first stage of this specification is weak. Its Kleibergen-Paap  $F$  statistic is 3.37, below the critical value of 7.03, so the estimates lean toward the ordinary least squares estimates of Panel B and their standard errors understate the uncertainty. The Anderson-Rubin test stays valid with weak instruments, and it does not reject the hypothesis that both instrumented coefficients are zero at the 10% level in any of the seven columns.

**Validation: the 2004 rice-import ban.** The route measure has independent validation from the event it was designed around, Indonesia's 2004 rice-import ban. Post-ban (2002-2008) rice-price growth was faster in more route-open cities (+0.037 per standard deviation of proximity,  $p = 0.06$ ), the geography-driven price response documented by [Bazzi \(2017\)](#). The point estimate is the same on the rice-active port set, where the standard error is tighter and the response reaches  $p < 0.05$ ; the wider interval here comes from using the full customs list, and it is the interval I report. The same paper documents the pre-ban gradient with the sign reversed, an additional falsification the route measure inherits. The exposure treatment itself is orthogonal to that price shock: agro-similarity does not predict the ban-era price response (+0.006,  $p = 0.64$ ), and the route  $\times$  agro-similarity interaction is small and insignificant (its negative sign is consistent with productive hinterlands buffering the forced-autarky price jump, but the cross-sectional variation cannot establish

Table B2: Agro-Similarity by Trade Openness: The Gravity Split Instrumented by the Route Split

|                                                                      | (1)          | (2)          | (3)          | (4)           | (5)             | (6)            | (7)             |
|----------------------------------------------------------------------|--------------|--------------|--------------|---------------|-----------------|----------------|-----------------|
|                                                                      | ln pop       | ln emp       | unemp rate   | ln L agric    | agric %         | ln L modern    | modern %        |
| <b>Agro-similarity,<br/>more closed cities: <math>\beta_1</math></b> | <b>0.161</b> | <b>0.152</b> | <b>0.001</b> | <b>-0.163</b> | <b>-0.074**</b> | <b>0.342**</b> | <b>0.111***</b> |
|                                                                      | (0.122)      | (0.127)      | (0.012)      | (0.173)       | (0.032)         | (0.150)        | (0.032)         |
| Agro-similarity,<br>more open cities: $\beta_1 + \beta_2$            | -0.051       | -0.098       | 0.014        | -0.161        | -0.028          | -0.020         | 0.021           |
|                                                                      | (0.090)      | (0.108)      | (0.013)      | (0.139)       | (0.029)         | (0.112)        | (0.022)         |
| High openness (=1)                                                   | 0.045        | 0.008        | -0.008       | 0.100         | 0.010           | -0.028         | -0.029          |
|                                                                      | (0.116)      | (0.153)      | (0.018)      | (0.271)       | (0.052)         | (0.161)        | (0.039)         |
| First-stage $F$ (Kleibergen-Paap)                                    | 3.37         | 3.37         | 3.37         | 3.37          | 3.37            | 3.37           | 3.37            |
| Outcome mean                                                         | 11.842       | 10.933       | 0.122        | 9.822         | 0.374           | 10.232         | 0.516           |
| Outcome SD                                                           | 0.854        | 0.840        | 0.040        | 0.896         | 0.183           | 0.975          | 0.156           |
| Observations                                                         | 134          | 134          | 134          | 134           | 134             | 134            | 134             |

*Notes:* Two-stage least squares on the main city sample, with the main specification controls: the settler count and 16 agro-climatic covariates within 300 minutes, and four 1980 baselines. The high-openness indicator of the gravity measure (rice-import market access discounted by travel time, above its median among the centers with a transmigration village within 300 travel minutes) and its interaction with standardized agro-similarity are instrumented by the high-openness indicator of the route measure (travel time to the Thai and Vietnamese rice hubs, below its median among the same centers) and its interaction with agro-similarity. Rows give the agro-similarity effect in the more closed and in the more open cities and the high-openness indicator; the outcomes are measured in 2010, with the four sector columns shaded. Conley SE within 200km in parentheses. The Kleibergen-Paap  $F$  statistic of the first stage, under the same Conley variance, is below the Stock-Yogo 10% maximal-size critical value of 7.03 for two endogenous regressors and two instruments. Agro-similarity is standardized, and the beta terms are the coefficients of (5). \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

it). The route result is also stable across codings: with the continuous proximity in place of the high-openness dummy, the interaction is significant at  $p < 0.05$  or better on all four headline outcomes.

### B.3 Standardized ( $z$ -on- $z$ ) Reduced-Form Estimates: Model Calibration Targets

*Cited in the main text.* Table B3: [Section 6.2, p. 49](#).

The structural model is calibrated to match the reduced-form agro-similarity responses in standardized ( $z$ -on- $z$ ) units, with both the treatment and each outcome standardized, so that the model's predicted response slopes, which are unit-free, are directly comparable to the data. [Table B3](#) reports these targets: the reduced-form point estimates are at the headline estimation sample, with the Susenas- and PODES-based rows at the coverage each source reaches, reported per row, and the  $z$ - $z$  column reports the slope the structural model is calibrated to match. The two targeted moments ( $m_1$  and  $m_2$ ) are fit in the calibration, and with two parameters the estimator is exactly identified; the modern-employment slope ( $m_3$ ) and the remaining rows are over-identifying checks computed from the same solves.

The density-check regressions are run at the headline no-controls arm reported in the main text and at the model-matched control arm, so the gradient's robustness to conditioning is visible in one view (headline arm in [Figure F1](#); both arms in [Figure B2](#)).

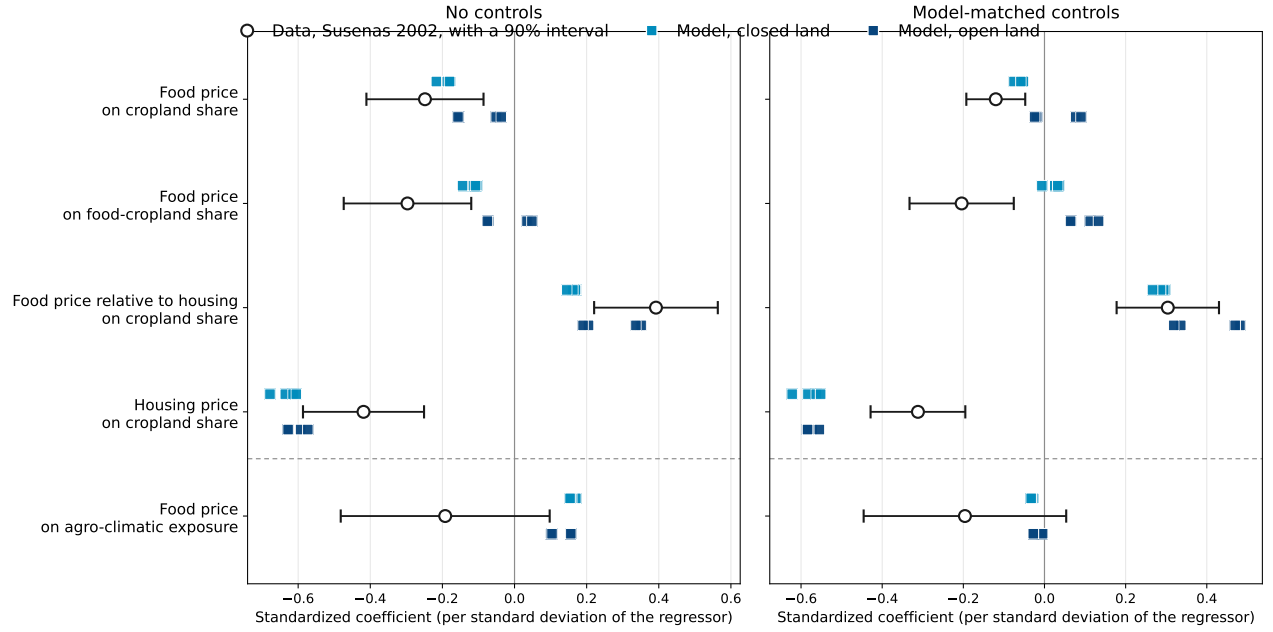
Table B3: Reduced-form AgSim estimates in standardized ( $z$ -on- $z$ ) units: the empirical moments matched in the structural model

|                                                             | Reduced form |         | $z$ - $z$ estimate |     |
|-------------------------------------------------------------|--------------|---------|--------------------|-----|
| Outcome                                                     | Coef.        | (SE)    | (unit-free)        | $N$ |
| <i>Targeted calibration moments (<math>m_1, m_2</math>)</i> |              |         |                    |     |
| $m_1$ ln population                                         | 0.060        | (0.059) | +0.070             | 133 |
| $m_2$ agric. employment share                               | −0.053***    | (0.018) | −0.287             | 133 |
| <i>Over-identifying moments (untargeted)</i>                |              |         |                    |     |
| $m_3$ ln modern empl. (serv+manuf), held out                | 0.169***     | (0.061) | +0.173             | 133 |
| $m_4$ ln agricultural empl.                                 | −0.167*      | (0.096) | −0.186             | 133 |
| modern employment share                                     | 0.068***     | (0.015) | +0.429             | 133 |
| $m_5$ housing rent                                          | −0.004       | (0.111) | −0.004             | 133 |
| $m_{pa}$ ag-goods price (level)                             | 0.018        | (0.022) | +0.148             | 133 |
| $m_6$ ln wage                                               | 0.039        | (0.035) | +0.133             | 128 |
| $m_{rw}$ ln real wage                                       | 0.002        | (0.031) | +0.008             | 132 |
| $m_7$ food expenditure share                                | −0.013**     | (0.006) | −0.263             | 132 |
| $m_8$ ln rice production                                    | 0.702***     | (0.252) | +0.350             | 127 |

*Notes:* Each “Coef.” is the coefficient on standardized AgSim ( $z$ -score over every urban center where AgSim is measured) from the headline Main Spec regression (controls: the settler count inverse-distance weighted inside the 300-minute window, 1980 ln population, 1980 ln employment and the 1980 urban-population share, and the matched agro-climatic covariates built on that same 300-minute rule; Conley spatial standard errors at 200 km), on outer-island urban centers inside the 300-minute cost-distance window of a transmigration village whose own 1980 baseline was measured and whose 2010 endline is present. The  $z$ - $z$  estimate column is the unit-free response slope (both treatment and outcome standardized), the form in which the structural model matches the data; the model’s AgSim-response slopes are computed in the identical  $z$ -on- $z$  form, so model and data slopes are directly comparable. Each outcome is standardized by its own standard deviation over that outcome’s estimation rows. Rows  $m_1$  and  $m_2$  are the two moments fit in the calibration, so with two parameters the estimator is exactly identified; the held-out modern-employment slope  $m_3$  and the remaining rows are over-identifying checks read off the same solves. The structural estimate in [Section 6.2](#) was fitted to earlier estimates of  $m_1$  and  $m_2$ , +0.011 and −0.285 in  $z$ - $z$  units, on the same 133 urban centers; this column reports the current estimates. The census outcomes are estimated on all  $N = 133$  urban centers; the Susenas-2011 rows (rent, prices, wages, food share) and the PODES rice-production row are on the same set of urban centers at the coverage each source reaches, reported per row.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Figure B2: The Price Gradient in Local Agriculture Across Control Arms, Model Against Data



*Notes:* Each row is one cross-sectional regression on the same 102 urban centers with a 2002 Susenas price, run twice on the identical specification: once on the price the data observe and once on the price the model solves for, so only the outcome differs. The left panel has no controls, the headline arm of the main-text figure; the right panel adds the model-matched control set, the agro-climatic covariates the calibration's own moment operator conditions on. Cropland share is the area under all thirty crops recorded by the village census over the land area of the center, food-cropland share is its food-crop subset, and the relative price is the food price less the housing price, which removes the tendency of agricultural places to be cheap in everything. The hollow circle is the data estimate with a capped 90% interval on Conley standard errors at 200 km; each filled square is one model cell at its own optimum on the two targeted moments, across the four assumed food trade costs; the model-side operator reproduces the Stata estimate to a worst absolute gap of  $1.2 \times 10^{-13}$  on every cell. The bottom row, below the dashed rule, is the food price on agro-climatic exposure, the earlier over-identification check: its interval is wide enough to contain every model cell, which is the precision contrast that motivates the density check.

## C Identification Support

This part collects the evidence that exposure is as good as randomly assigned once the controls are in place, together with the placebo and recentering tests that support it.

### C.1 Design-Based Recentering and the Placebo Tests

*Cited in the main text. This section:* Section 4.3, p. 21; Section 5.1, p. 25. Table C1: Section 5.1, p. 25. Figure C1: Section 5.1, p. 25. Table C2: Section 4.3, p. 22; Section 5.1, p. 24. Table C3: Section 3.4, p. 16; Section 4.3, p. 22; Section 5.1, p. 24.

**Design-based recentering.** The exposure measure combines a quasi-random shock, the match between arriving settler cohorts and the sites that were ready to receive them, with components the program chose: where the sites sit, how many settlers each received, and their agro-climate. Averaging a random shock in a structure that has a non-random component means the aggregated shock itself has non-randomness in it, so a city surrounded by transmigration sites with land more similar to the Java and Bali origin mix is bound to have higher exposure regardless of how settlers are reshuffled (Section 4.3). Borusyak & Hull (2023) show that the bias in such a measure runs through a single confounder, the exposure a city would expect under counterfactual assignments of cohorts to sites, and that recentering the treatment on that expected exposure, or controlling for it, restores identification.

I construct the expected exposure directly. The sites, their locations, when each was settled, how many settlers it received, and which sites lie within each city’s travel-time catchment all stay where the program put them. Only the match moves: I reshuffle the village-level agro-climatic similarity across the sites of the same province and Repelita settlement wave, rebuild each city’s unweighted 300-minute catchment mean under each of  $S = 2,000$  reshuffles, and average them:

$$z_c = \frac{1}{|N_c|} \sum_{v \in N_c} AS_v, \quad \mu_c = \frac{1}{S} \sum_{s=1}^S z_c^{(s)}, \quad (17)$$

where  $N_c$  is the set of program villages within 300 minutes of city  $c$ ,  $AS_v$  is the agro-climatic similarity of village  $v$ , and  $z_c^{(s)}$  is the city’s exposure under draw  $s$ . I then either control for the expected exposure or subtract it from the treatment:

$$\begin{aligned} y_c &= \beta z_c + \gamma_P \mu_c^P + \gamma_{PW} \mu_c^{PW} + X_c' \delta + \varepsilon_c, \\ y_c &= \tilde{\beta} (z_c - \mu_c^{PW}) + X_c' \delta + \varepsilon_c, \end{aligned} \quad (18)$$

where  $X_c$  holds the main specification controls,  $\mu_c^{PW}$  and  $\mu_c^P$  are the expected exposures under reshuffles within province-by-wave cells and within provinces, and every treatment is standardized. Controlling for both expectations is the double-robust form of the correction, consistent if either the controls or the reshuffle model is correct. A second reshuffle holds each site’s destination agro-climate fixed and re-draws only the settler-origin composition, using the origin-destination similarity grid of Bazzi et al. (2016).

The structural-transformation results hold under the double-robust form (Table C1). The agricultural employment share falls by 0.040 and the modern employment share rises by 0.052, both statistically significant, and log modern-sector employment rises by 0.112, marginally significant; each has the headline’s sign at about three quarters of its size. Subtracting the expected exposure instead removes most of the variation in the treatment, because a city’s realized and expected exposure move closely together: they correlate at 0.75 under the whole-match reshuffle and 0.96 under the origin re-draw. The recentered estimates are closer to zero, and only the modern employment share stays statistically significant under both (Figure C1).

The same 2,000 reshuffles also give an exact test of the double-robust estimate: I re-estimate the regression under each reshuffle and ask how often a placebo  $t$ -statistic lies at least as far from the placebo average as the observed one. The test rejects the hypothesis of no effect on the modern employment share at the 10% level ( $p = 0.050$  with reshuffles

Table C1: Headline Estimates Controlling for the Expected Exposure, Double-Robust Form

|                                    | (1)              | (2)              | (3)               | (4)               | (5)                 | (6)               | (7)                 |
|------------------------------------|------------------|------------------|-------------------|-------------------|---------------------|-------------------|---------------------|
|                                    | ln pop           | ln emp           | unemp             | ln ag             | ag share            | ln modern         | modern share        |
| Agro-similarity<br>(double-robust) | 0.035<br>(0.077) | 0.024<br>(0.074) | -0.006<br>(0.006) | -0.112<br>(0.121) | -0.040**<br>(0.020) | 0.112*<br>(0.063) | 0.052***<br>(0.017) |
| Settler count                      | ✓                | ✓                | ✓                 | ✓                 | ✓                   | ✓                 | ✓                   |
| Base ln pop 1980                   | ✓                | ✓                | ✓                 | ✓                 | ✓                   | ✓                 | ✓                   |
| Base ln empl 1980                  | ✓                | ✓                | ✓                 | ✓                 | ✓                   | ✓                 | ✓                   |
| Base ln agri empl 1980             | ✓                | ✓                | ✓                 | ✓                 | ✓                   | ✓                 | ✓                   |
| Urban share 1980                   | ✓                | ✓                | ✓                 | ✓                 | ✓                   | ✓                 | ✓                   |
| Agro-climatic controls             | ✓                | ✓                | ✓                 | ✓                 | ✓                   | ✓                 | ✓                   |
| Observations                       | 134              | 134              | 134               | 134               | 134                 | 134               | 134                 |

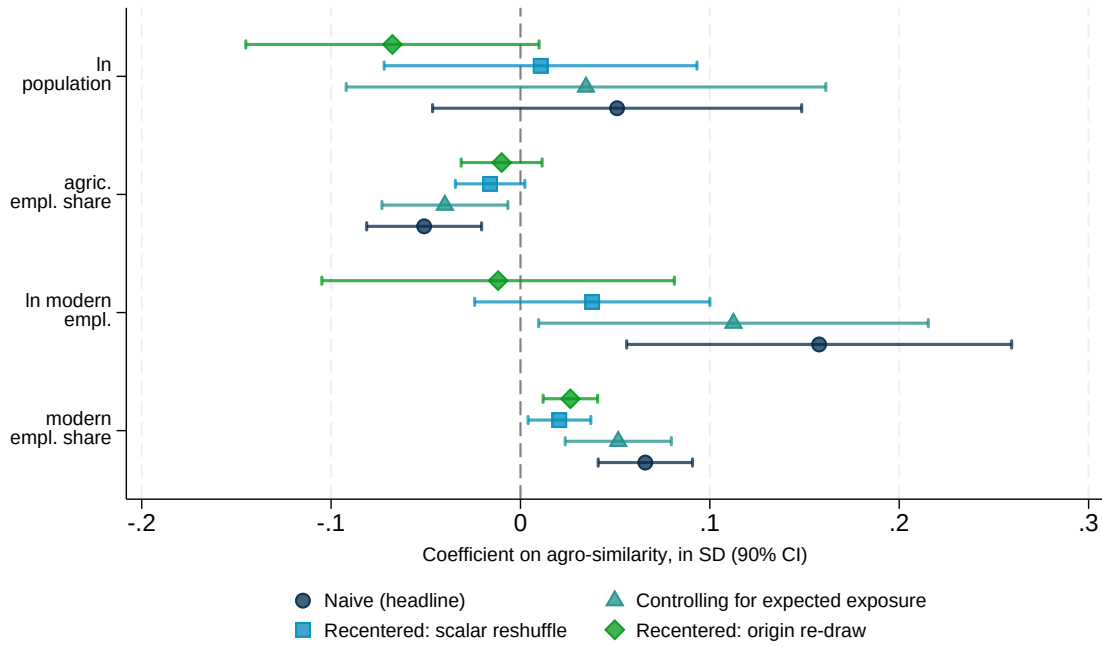
*Notes:* Each column is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Check marks show the controls. Every column also controls for the expected exposure under two random reassignments of village agro-climatic similarity, one within province and one within province and five-year plan (1979 to 1983 or 1984 to 1988), 2,000 draws each and both standardized; this is the double-robust specification of Borusyak & Hull (2023), consistent if either the controls or the reassignment model is correct. Modern-sector employment is manufacturing and services; shares and the unemployment rate are fractions. Standard errors are in parentheses. Agro-similarity (double-robust) is standardized agro-similarity with both expected exposures controlled. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

within province and wave, and 0.052 within province) and does not reject it for the agricultural share ( $p = 0.196$ ) or for log modern-sector employment ( $p = 0.181$ ). The test compares  $t$ -statistics because, once its expectation and the controls are taken out, a reshuffled exposure has about half the residual standard deviation of the realized one, so a coefficient estimated on it is spread more widely than the observed coefficient; dividing each by its own standard error puts the observed and the placebo estimates on one scale. The Online Supplement reports every randomization-inference  $p$ -value, the coefficient form included.

**Baseline placebo test, in levels.** Agro-similarity across villages does not predict the baseline outcomes of the sample cities. I regress ten 1980 baseline outcomes, urban center-level aggregates of individual 1980 census records, on agro-similarity across villages. Two constructions of the exposure inside the 300-minute identification radius are reported, the headline unweighted mean and the population $\times$ inverse-distance weighting of the exposure equation, each over all 814 transmigration villages and again over the 687 settled after the count. The right-hand block of Table C2 also rebuilds every village-derived control over those 687 villages, so nothing entering the regression could have acted on a 1980 outcome. One of the forty cells rejects zero at the 10% level, where about four would under the null.

**Controlling the 1980 birthplace composition directly.** The compositional headline does not depend on the 1980 birthplace composition of the workforce. With the 1980 birthplace shares added to the main specification, the agricultural and modern employment shares keep their size and their 1% significance, and the 2010 birthplace pattern holds, with the locally born share still falling and the same-province share still rising (Table C3).

Figure C1: Headline Estimates Under Design-Based Recentering



*Notes:* Rows are 2010 census outcomes; the modern sector is manufacturing plus services. Each marker is the coefficient on standardized agro-similarity, or on its recentered version, from one specification on the main city sample, in the main specification with the full set of controls and Conley SE within 200km; bars are 90% confidence intervals from the same standard errors. “Naive” is the headline specification. “Controlling for expected exposure” adds the expected exposure under the province and the province-by-settlement-wave reshuffles as controls (Table C1). “Scalar reshuffle” recenters the exposure against a permutation of the village match within province-by-settlement-wave cells. “Origin re-draw” recenters against a permutation that holds each site’s destination agro-climate fixed and re-draws only the settler-origin composition, using the origin-destination similarity grid of Bazzi et al. (2016). Recentering subtracts from an exposure its average over 2,000 permutations and standardizes the recentered value, following Borusyak & Hull (2023). The Online Supplement reports the full recentered estimates.

## C.2 Village-Level Analysis

*Cited in the main text.* Table C4: Section 4.3, p. 21. Figure C2: Section 4.3, p. 21. Figure C3: Section 5.5, p. 39.

Across the transmigration villages, agricultural productivity, the unitless yield index equal to one where every crop yields its benchmark, rises with the agro-climatic similarity index, winsorized or raw, over the plotted range and over the full range. The productivity measure is the total agricultural productivity of Bazzi et al. (2016), taken from the 2002 round of the village census and the 2001-2002 growing season, and it is observed for 770 of the 814 program villages, while agro-similarity is defined for all 814. Table C4 reports this village-level relevance of agro-climatic similarity for agricultural productivity, which Section 4.3 cites.

The two village curves of Figure 13 are drawn again unweighted at a bandwidth of 0.30, and with each village weighted by its 2000 population at bandwidths of 0.25 and 0.30 (Figure C3).

Table C2: Placebo Baseline Test: 1980 Outcomes Under Two Exposure Constructions and Two Village Sets

| 1980 baseline outcome      | Outcome, main city sample |      |          | All transmigration villages |                   | Villages settled after the 1980 census |                    |
|----------------------------|---------------------------|------|----------|-----------------------------|-------------------|----------------------------------------|--------------------|
|                            | Mean                      | SD   | <i>N</i> | Simple mean                 | Settler weighted  | Simple mean                            | Settler weighted   |
| ln population              | 11.093                    | 0.99 | 134      | -0.019<br>(0.122)           | -0.022<br>(0.166) | +0.017<br>(0.154)                      | -0.109<br>(0.165)  |
| ln employment              | 9.846                     | 0.96 | 134      | -0.043<br>(0.107)           | -0.042<br>(0.151) | +0.002<br>(0.140)                      | -0.118<br>(0.154)  |
| Unemployment rate          | 0.017                     | 0.02 | 134      | -0.001<br>(0.002)           | -0.002<br>(0.002) | -0.002<br>(0.002)                      | -0.003*<br>(0.002) |
| ln agricultural empl.      | 9.249                     | 1.06 | 134      | -0.129<br>(0.099)           | -0.093<br>(0.107) | -0.109<br>(0.101)                      | -0.171<br>(0.109)  |
| Agricultural empl. share   | 0.560                     | 0.22 | 134      | -0.034<br>(0.033)           | -0.021<br>(0.032) | -0.041<br>(0.026)                      | -0.020<br>(0.028)  |
| ln modern empl.            | 8.801                     | 1.31 | 134      | -0.050<br>(0.185)           | -0.052<br>(0.237) | +0.002<br>(0.218)                      | -0.149<br>(0.240)  |
| Modern empl. share         | 0.383                     | 0.20 | 134      | +0.017<br>(0.031)           | +0.007<br>(0.031) | +0.024<br>(0.027)                      | +0.005<br>(0.029)  |
| ln born same province      | 10.934                    | 1.00 | 134      | -0.001<br>(0.132)           | -0.008<br>(0.175) | +0.034<br>(0.164)                      | -0.084<br>(0.172)  |
| ln born other province     | 8.342                     | 1.88 | 134      | -0.138<br>(0.211)           | -0.125<br>(0.235) | -0.102<br>(0.225)                      | -0.266<br>(0.264)  |
| Born other province, share | 0.134                     | 0.14 | 134      | -0.014<br>(0.021)           | -0.011<br>(0.015) | -0.014<br>(0.019)                      | -0.020<br>(0.017)  |
| # significant / 10         |                           |      |          | 0                           | 0                 | 0                                      | 1                  |

Notes: Each cell is the coefficient from a separate placebo regression of a 1980 census outcome on one standardized construction of agro-similarity, with the settler count and 16 agro-climatic covariates within 300 minutes as controls, all built over the village set of the block; Conley SE within 200km in parentheses. Each column estimates on the main city sample where that construction and that outcome are measured, 134 urban centers. The ten outcomes are the seven headline outcomes at baseline (modern employment is manufacturing plus services) and three rows on province of birth, the finest level the 1980 census records. Mean, standard deviation and number of urban centers describe each outcome on the sample of the first coefficient column. In each village set, the first column is the exposure of the main specification, the simple mean of village agro-similarity within 300 travel minutes, and the second weights each village by its settler count times its inverse travel time. The left block builds every construction and control over all 814 transmigration villages; the right block rebuilds them over the 687 villages settled after the 1980 census (the other 127, placed by 1980, held 19.5% of all settlers). The last row counts the coefficients with at least one star in each column. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table C3: Headline Estimates Controlling for the 1980 Birthplace Composition

|                                                            | Main specification,<br>as reported<br>(1) | Main specification,<br>same cities as (3)<br>(2) | Adding the 1980<br>birthplace shares<br>(3) |
|------------------------------------------------------------|-------------------------------------------|--------------------------------------------------|---------------------------------------------|
| <i>Panel A. Urban growth and structural transformation</i> |                                           |                                                  |                                             |
| ln population                                              | 0.051<br>(0.059)                          | 0.051<br>(0.059)                                 | 0.053<br>(0.049)                            |
| <i>p-value</i>                                             | 0.389                                     | 0.389                                            | 0.275                                       |
| ln total employment                                        | 0.024<br>(0.057)                          | 0.024<br>(0.057)                                 | 0.027<br>(0.049)                            |
| <i>p-value</i>                                             | 0.674                                     | 0.674                                            | 0.583                                       |
| Unemployment rate                                          | 0.008<br>(0.007)                          | 0.008<br>(0.007)                                 | 0.009<br>(0.006)                            |
| <i>p-value</i>                                             | 0.216                                     | 0.216                                            | 0.159                                       |
| ln agricultural employment                                 | -0.166*<br>(0.093)                        | -0.166*<br>(0.093)                               | -0.161**<br>(0.078)                         |
| <i>p-value</i>                                             | 0.075                                     | 0.075                                            | 0.039                                       |
| Agricultural employment share                              | -0.051***<br>(0.018)                      | -0.051***<br>(0.018)                             | -0.051***<br>(0.018)                        |
| <i>p-value</i>                                             | 0.006                                     | 0.006                                            | 0.005                                       |
| ln modern employment                                       | 0.158**<br>(0.062)                        | 0.158**<br>(0.062)                               | 0.161***<br>(0.060)                         |
| <i>p-value</i>                                             | 0.011                                     | 0.011                                            | 0.007                                       |
| Modern employment share                                    | 0.066***<br>(0.015)                       | 0.066***<br>(0.015)                              | 0.066***<br>(0.014)                         |
| <i>p-value</i>                                             | 0.000                                     | 0.000                                            | 0.000                                       |
| Observations                                               | 134                                       | 134                                              | 134                                         |
| <i>Panel B. 2010 residents by birthplace</i>               |                                           |                                                  |                                             |
| Born in the same district                                  | -0.040*<br>(0.023)                        | -0.040*<br>(0.023)                               | -0.044***<br>(0.015)                        |
| <i>p-value</i>                                             | 0.080                                     | 0.080                                            | 0.003                                       |
| Born elsewhere in the same province                        | 0.042**<br>(0.018)                        | 0.042**<br>(0.018)                               | 0.043**<br>(0.018)                          |
| <i>p-value</i>                                             | 0.019                                     | 0.019                                            | 0.016                                       |
| Born in another outer-island province                      | 0.017<br>(0.021)                          | 0.017<br>(0.021)                                 | 0.015<br>(0.012)                            |
| <i>p-value</i>                                             | 0.415                                     | 0.415                                            | 0.228                                       |
| Born on Java or Bali                                       | -0.019*<br>(0.011)                        | -0.019*<br>(0.011)                               | -0.014<br>(0.009)                           |
| <i>p-value</i>                                             | 0.079                                     | 0.079                                            | 0.119                                       |
| Observations                                               | 134                                       | 134                                              | 134                                         |

*Notes:* Each cell is one regression on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Beneath each coefficient are its standard error in parentheses and the two-sided Conley *p*-value, shaded by significance level. Column (3) adds two of the three 1980 census birthplace shares of the city's population (born in another outer-island province, born on Java or Bali; born in this province is the reference), and column (2) repeats the main specification on column (3)'s cities, those with the 1980 shares measured. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table C3: Headline Estimates Controlling for the 1980 Birthplace Composition, Continued

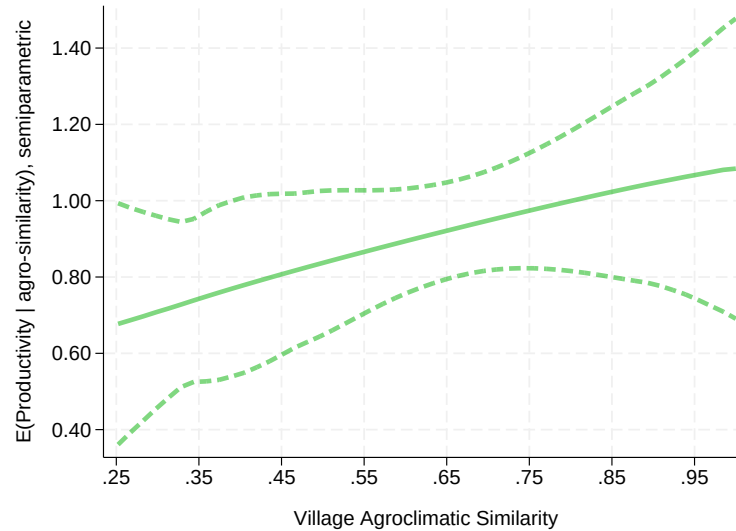
*Notes, continued:* Panel A gives the seven 2010 census outcomes of the paper's headline table and Panel B the four birthplace shares of the migration table, whose 2010 shares sum to one over each city's cohort born 1955 to 1990; shares and the unemployment rate are fractions.

Table C4: Village Agricultural Productivity on Agro-Climatic Similarity: Linear Estimates

|                          | Similarity 0.25–1 (plotted range) |                     | Full similarity range |                     |
|--------------------------|-----------------------------------|---------------------|-----------------------|---------------------|
|                          | (1)                               | (2)                 | (3)                   | (4)                 |
|                          | winsorized                        | raw                 | winsorized            | raw                 |
| Agro-climatic similarity | 0.525***<br>(0.201)               | 0.767***<br>(0.136) | 0.535***<br>(0.179)   | 0.772***<br>(0.108) |
| Winsorized               | Yes                               | No                  | Yes                   | No                  |
| Similarity range         | 0.25–1                            | 0.25–1              | All                   | All                 |
| Villages                 | 768                               | 768                 | 770                   | 770                 |

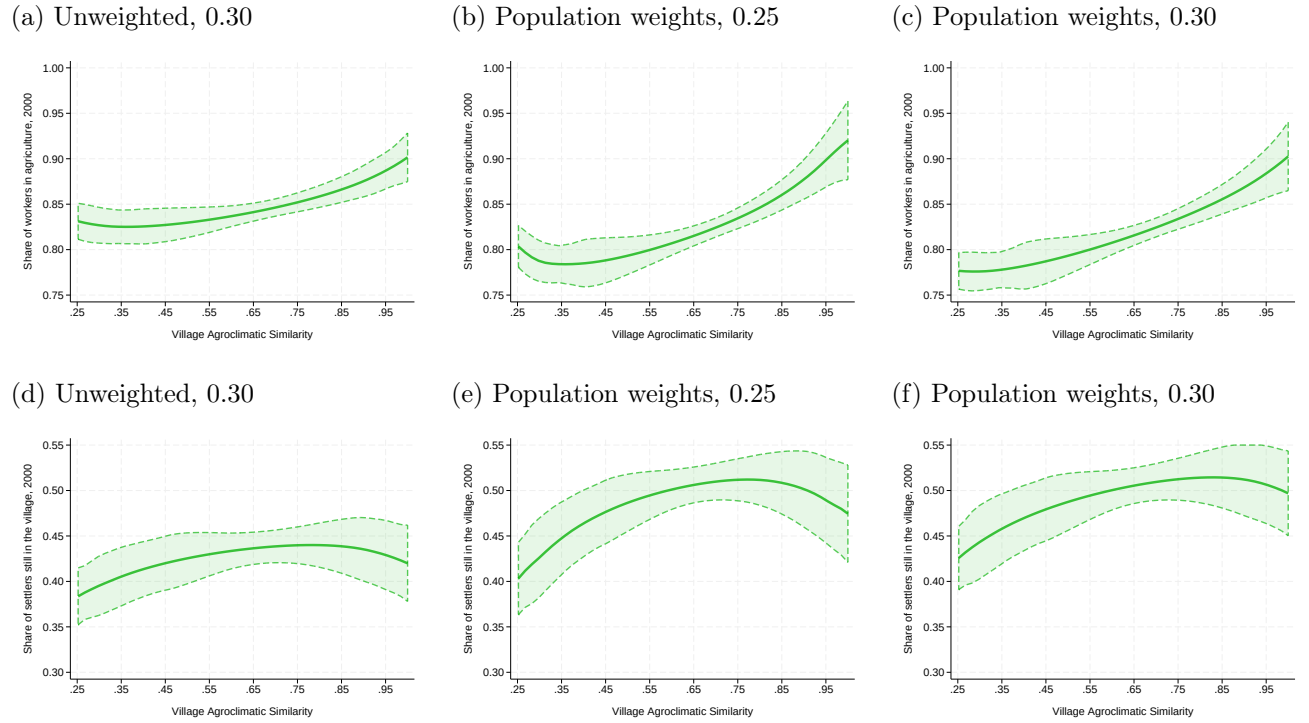
Each column regresses the village agricultural productivity index on the village agro-climatic similarity index (0 to 1 scale) across the transmigration villages with a productivity measure in the similarity range of the column; the Villages row gives their number. Conley SE within 200km in parentheses. The productivity index of [Bazzi et al. \(2016\)](#), from the 2002 village census, is the revenue-weighted geometric mean across crops of the yield per hectare relative to a benchmark yield for each crop, so it has no units. The similarity index is built from the geographic, soil, and climate characteristics of the village, so the regression has the index as its only regressor. Winsorized columns cap productivity at its 1st and 99th percentiles. Column 1 is the specification plotted in [Figure C2](#). Raw columns use the uncapped productivity index. \*\*\* 1%, \*\* 5%, \* 10%.

Figure C2: Village-Level Relevance: Agricultural Productivity Against Agro-Climatic Similarity



*Notes:* Local linear fit (Epanechnikov kernel, bandwidth 0.30) of village agricultural productivity on agro-climatic similarity, an index from 0 to 1, across the 768 transmigration villages with a productivity measure and a similarity between 0.25 and 1; the solid line is the fit and the dashed lines its 90% confidence band. Productivity is the unitless yield index of [Bazzi et al. \(2016\)](#) from the 2002 village census (PODES), equal to one where every crop yields its benchmark and winsorized at the 1st and 99th percentiles. [Table C4](#) reports the same gradient in linear form, with Conley SE within 200km.

Figure C3: The Village Agricultural Share and Settler Stay-Share at Bandwidth 0.30 and With Population Weights



*Notes:* Semiparametric local linear fits (Epanechnikov kernel) of the share of a village's workers in agriculture in the 2000 Population Census (top row) and of the settler stay-share (bottom row) on the village's agro-similarity, partialing out the village controls of Figure 13 on the same 808 transmigration villages: unweighted at a bandwidth of 0.30 (left), and with each village weighted by its 2000 population at bandwidths of 0.25 (middle) and 0.30 (right), where the weighted fits also remove the controls with the Epanechnikov kernel. Each band is a pointwise 90% confidence band that treats the villages as independent. The axes are those of Figure 13; a band that leaves its axis is cut at the axis limit, here in the population-weighted stay-share at 0.30 (band up to 0.552).

## D Robustness

*Cited in the main text. This section:* [Section 5.1, p. 26.](#)

This part re-estimates the headline results under alternative constructions of the exposure measure and alternative specifications.

### D.1 The Settlers' Size Relative to the Pre-Existing Rural Population

*Cited in the main text. Table D1:* [Section 5.1, p. 26.](#)

The settlers placed in a city's rural hinterland, meaning the rural subdistricts within 135 km, were a non-trivial share of the population that hinterland already held at the 1980 census: about 7% of it on average and roughly 5% at the median, with about a seventh of cities above 15%. That radius reproduces the 300-minute window the exposure measure uses. Cities therefore differ in how large the settlement was relative to the rural landscape it entered, and [Table D1](#) asks whether the main results change with that difference.

They do not. With the share added to the main specification, the two employment shares and the unemployment rate are unchanged; log population, log employment and log modern employment are larger by 0.02 to 0.03 log points; and log agricultural employment becomes marginally insignificant. Part of the reason is that the main specification already controls for the number of settlers placed near each city (Online Supplement), so the share's numerator is largely held and the variable moves primarily through its denominator, the size of the countryside those settlers entered.

Table D1: Headline Estimates With a Control for the Settlers' Share of the Pre-Existing Rural Population

|                                   | Population + Employment |         |            | Agric Empl |           | Modern Empl |          |
|-----------------------------------|-------------------------|---------|------------|------------|-----------|-------------|----------|
|                                   | (1)                     | (2)     | (3)        | (4)        | (5)       | (6)         | (7)      |
|                                   | ln pop                  | ln emp  | unemp rate | ln L agric | agric %   | ln L modern | modern % |
| Main specification (main text)    | 0.051                   | 0.024   | 0.008      | -0.166*    | -0.051*** | 0.158**     | 0.066*** |
|                                   | (0.059)                 | (0.057) | (0.007)    | (0.093)    | (0.018)   | (0.062)     | (0.015)  |
| + settler share of the hinterland | 0.081                   | 0.054   | 0.009      | -0.149     | -0.053*** | 0.177***    | 0.062*** |
|                                   | (0.056)                 | (0.055) | (0.007)    | (0.093)    | (0.019)   | (0.063)     | (0.015)  |
| Observations                      | 134                     | 134     | 134        | 134        | 134       | 134         | 134      |

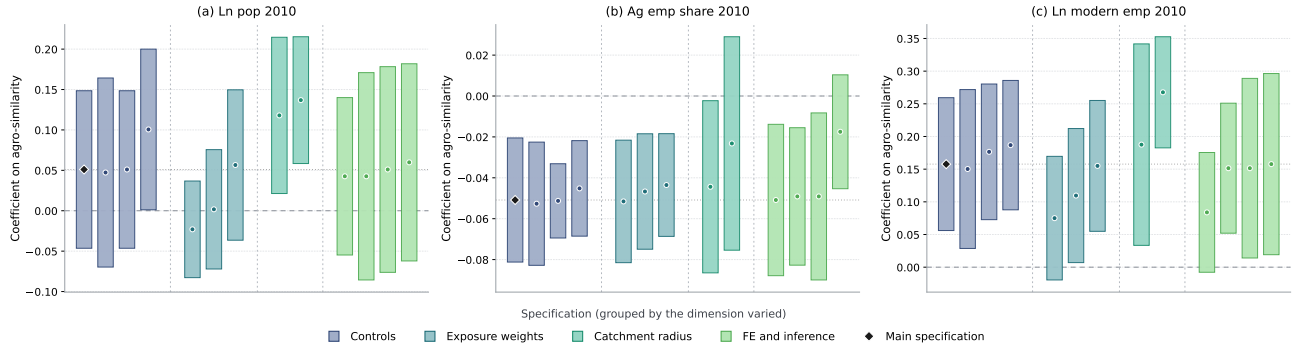
*Notes:* Each cell is the coefficient on standardized agro-similarity from one regression on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. The first row is that specification. The second row adds the settlers placed in the rural hinterland of the city as a share of the 1980 census population of that hinterland, over the rural subdistricts within 135 kilometers, the radius that reproduces the 300-minute window; with the settler count already a control, this share varies primarily through its denominator. Outcomes are the seven 2010 census outcomes, with an employment level and share for agriculture and for the modern sector (manufacturing and services), each sector pair shaded; shares and the unemployment rate are fractions. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

### D.2 The Specification Curve and Price-Measure Robustness

*Cited in the main text. This section:* [Section 3.3, p. 15.](#) Figure D2: [Section 4.2, p. 20;](#) [Section 5.1, p. 26.](#) Table D2: [Section 5.1, p. 26.](#) Figure D3: [Section 5.2, p. 31.](#)

**Specification variants on the main city sample.** The specification curve re-estimates the standardized agro-similarity coefficient across the cost-distance specification space, crossing the weight family, the travel-time cutoff, the control set, and island fixed effects on or off. For the three headline outcomes that space holds 61 specifications,

Figure D1: Specification Curve by the Dimension Varied



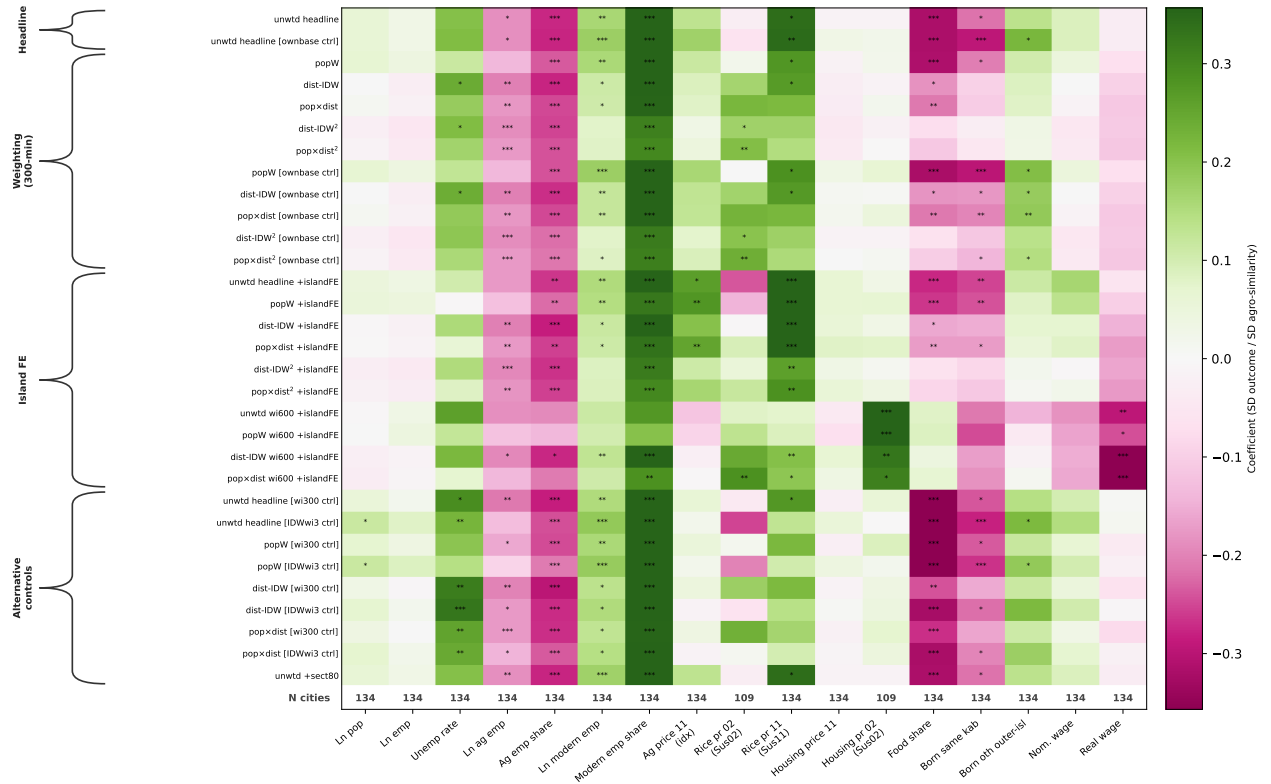
*Notes:* Each panel is one 2010 population census outcome. Each marker is the coefficient on standardized agro-similarity under one of 13 specifications on the main city sample, grouped by controls, exposure weights, catchment radius, and fixed effects and inference (Table D2). The diamond marks the main specification with the full set of controls and Conley SE within 200km; the 400- and 500-minute catchments widen the exposure and keep the controls at 300 minutes. Within a group, markers are ordered by the estimate, with the main specification first in the controls group; shaded bars are 90% confidence intervals from the same standard errors, or from standard errors clustered by province in two variants of the last group, and the dotted horizontal line marks the main estimate. For log population the own-outcome baseline coincides with the main specification, whose controls already include the 1980 population.

every one of them at the 300-minute or 600-minute cutoff and on the main city sample. Straight-line distances, province fixed effects, unrestricted catchments, and the 180-minute cutoff are separate sensitivity exercises. The 180-minute family uses a smaller sample and includes wrong-signed estimates with 90% intervals excluding zero. The displayed comparison varies weights, catchments, controls, and island fixed effects while retaining the main city sample. Within this space no specification is both wrong-signed and significant for any of the three outcomes. The weight family crosses the headline's unweighted within-300-minute mean with an inverse-distance-weighted mean, a settler-weighted mean, and the inverse-distance and settler-weighted average, and every construction inside the 300-minute window behaves like the headline specification.

The agro-similarity coefficient keeps its sign for the sectoral shares across essentially the whole specification space and is positive on average, with wider intervals, for population and prices. I extend the exercise to every outcome in one grid: each cell is the standardized agro-similarity coefficient for one outcome (columns) under one specification family (rows), built by crossing the exposure weighting, the control set, the travel-time cutoff, and island fixed effects on or off (Figure D2). Two patterns organize the grid. The sectoral columns are stable in sign: the agricultural employment share is negative and the modern-sector share positive across essentially the whole space. The population and price columns are positive on average with wider intervals; across the 61 specifications, log population clears significance on 4 against 46 for the agricultural employment share. The displayed grid is 31 specifications across 17 outcomes. Province fixed effects and their clustered-standard-error variants are excluded from the grid for the reason given above, that province effects absorb the between-province variation identifying the treatment. Island fixed effects appear as their own row group here.

**Robustness of the urban price result across measures.** The urban food-price response is small, negative and imprecise under every measure of the local price. Figure D3 re-estimates the headline main specification reduced form with the dependent variable swapped across 5 1996 price measures: the broad agricultural-goods index under national (Laspeyres), city (Paasche), and equal-weight aggregation, the equal-weight crops index, and a rice unit value. The agro-similarity coefficient on the log price runs from  $-0.028$  to  $-0.005$  across the 5 measures, and none of them is statistically significant, in logs or in index levels. The four indices are price relatives with a median near one and enter as the log of one plus the index, so their coefficients read as about half of a proportional change; the rice unit value is a price in rupiah per kilogram, and its coefficient reads as a proportional change. The measures are also internally consistent (pairwise log correlations of 0.76 to 0.94 among the national and city agricultural indices and the rice unit value, and as low as 0.42 for a pair that includes an equal-weight index) with economically sensible spatial variation. Traded food prices are

Figure D2: Specification-Curve Heatmap: Constructions and Controls



*Notes:* Each cell is the coefficient, in SD of the outcome per SD of agro-similarity, from one regression of the column's outcome under the row's specification on the main city sample, held fixed across rows; the bottom row prints each column's number of cities. Rows vary the village weights of agro-similarity, the control set and island fixed effects around the main specification with the full set of controls and Conley SE within 200km; four island fixed-effects rows measure agro-similarity within 600 minutes. Outcomes come from the 2010 population census and the 2002 and 2011 household-survey (Susenas) waves; the two 2002 columns run on 109 of those cities. Shading deepens with magnitude, one hue for positive and another for negative cells; stars use the same standard errors: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

spatially compressed (coefficient of variation  $\approx 0.11$ ) relative to the non-traded housing benchmark (0.78). Market rice prices (the World Bank's district-level market-price panel, matched to urban centers by market coordinates) do not move with agro-similarity in 2011 levels; the estimate is statistically insignificant. Households near productive transmigration villages buy 4.2% fewer kilograms of rice per person, a fall significant at the 10% level, while their spending on rice does not move, so they pay more per kilogram for less rice. Over 2002 to 2008 the rice price in the city's consumer price index does not grow faster than the index as a whole: both growth responses are statistically significant, and their difference is not.

### D.3 City Boundary Definitions and the Matched Cutoff

*Cited in the main text.* Table D3: Section 2.2, p. 8. Table D5: Section 5.1, p. 27. Figure D4: Section 5.1, p. 27. Table D4: Section 2.2, p. 10; Section 5.4, p. 37.

Three checks bound what the city definition can do to the results. Table D3 measures the same cities under four definitions, from the whole member subdistrict to the built-up core intersected with the census's own urban classification: the level of urban farming falls as the city is drawn tighter, and its 2010 mean is still about 14% under the tightest cut. Most urban centers are a combination of a few subdistricts, so I also rebuild every city from its member subdistricts whose own 1980 employment was at least one quarter, or at least one half, outside agriculture, summing the 2010 outcomes and

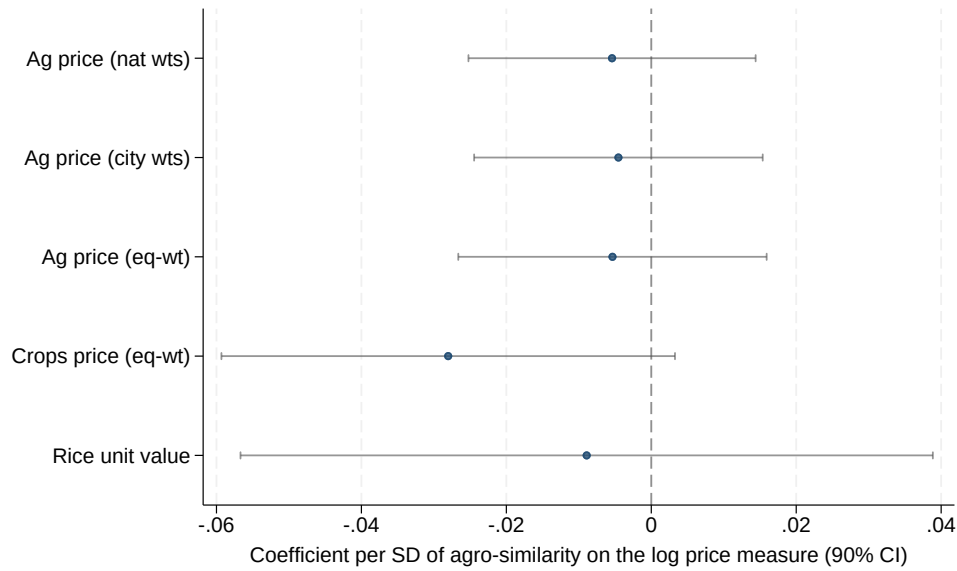
Table D2: Specification Variants in the Grouped Specification Curve

| Variant                     | What it changes relative to the main specification                                                                                                                                                                                                                                                                                                                           | <i>N</i> |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <i>Controls</i>             |                                                                                                                                                                                                                                                                                                                                                                              |          |
| Main specification          | The settler count as an inverse-travel-time-weighted sum over the transmigrant villages inside the 300-minute catchment, the four 1980 baselines (log population, log employment, the urban-population share and log agricultural employment), and the 16 agro-climatic covariates as unweighted means over the same villages. Every other row changes one thing about this. | 134      |
| Unweighted settler count    | The settler count entered as an unweighted sum over the same villages.                                                                                                                                                                                                                                                                                                       | 134      |
| Weighted covariates         | The 16 agro-climatic covariates entered as means over the same villages weighted by inverse travel time.                                                                                                                                                                                                                                                                     | 134      |
| Own-outcome base-line       | Adds the outcome's own 1980 level, or the 1980 agricultural employment share and log population density where the outcome has no 1980 level. For log population this coincides with the main specification, whose controls already include the 1980 population.                                                                                                              | 134      |
| <i>Exposure weights</i>     |                                                                                                                                                                                                                                                                                                                                                                              |          |
| Population-weighted         | Villages enter weighted by their settler count.                                                                                                                                                                                                                                                                                                                              | 134      |
| 1/d                         | Villages enter weighted by the inverse of travel time, so nearer villages count for more.                                                                                                                                                                                                                                                                                    | 134      |
| 1/d <sup>2</sup>            | The same, with the inverse of squared travel time, a steeper decay.                                                                                                                                                                                                                                                                                                          | 134      |
| <i>Catchment radius</i>     |                                                                                                                                                                                                                                                                                                                                                                              |          |
| Within 400 min              | Within 400 travel-time minutes, which every city in the estimation sample reaches.                                                                                                                                                                                                                                                                                           | 134      |
| Within 500 min              | Within 500 travel-time minutes.                                                                                                                                                                                                                                                                                                                                              | 134      |
| <i>FE and inference</i>     |                                                                                                                                                                                                                                                                                                                                                                              |          |
| Province FE                 | Adds province fixed effects. Province effects absorb the between-province variation that identifies the treatment, so this is never the headline.                                                                                                                                                                                                                            | 134      |
| Island FE                   | Adds island fixed effects.                                                                                                                                                                                                                                                                                                                                                   | 134      |
| Cluster: province           | The same point estimate, with standard errors clustered at the province level instead of Conley.                                                                                                                                                                                                                                                                             | 134      |
| Island FE, cluster province | Island fixed effects together with province-clustered standard errors.                                                                                                                                                                                                                                                                                                       | 134      |

*Notes:* Each row defines one bar in each panel of [Figure D1](#), the coefficient on standardized agro-similarity under that specification for 2010 log population, the agricultural employment share and log modern-sector (manufacturing and services) employment, from the population census. The first row is the main specification with the full set of controls and Conley SE within 200km; every other row changes exactly one thing about it, which makes a group a controlled comparison. The last row combines two of these changes, island fixed effects and province-clustered standard errors. *N* is the number of urban centers, and every row estimates on the main city sample. Standard errors are those of the main specification, except in the two rows clustered by province. In the Variant column, d is the travel time from a city to a transmigration village and FE marks fixed effects.

the four 1980 baselines over those subdistricts ([Table D4](#)). The agricultural employment share then falls by 6.2 and 7.1 percentage points per standard deviation of agro-similarity, against 5.1 for the whole city, and farm employment falls by more. Modern-sector employment rises at the quarter threshold, and its estimate loses precision at the half. The half threshold also removes old town centers, since many town subdistricts were still mostly agricultural in 1980, and 40 of the 102 cities with a 1980 urban population lose the subdistrict that held most of it. The result holds when the city is drawn wider as well. Panel D adds to each city the rural subdistricts within 50 km, and in that region the agricultural employment share falls by 5.4 percentage points per standard deviation of agro-similarity, against 5.1 for the whole city, while modern-sector employment and the modern share rise by less and population and employment stay imprecisely estimated. [Table D5](#) splits the sample at the median 1980 agricultural share, 57.9% of 1980 employment, and runs the split as an interaction, so both slopes come from one regression on all 134 centers and the difference between them has its own standard error. The two halves respond alike. The agricultural-share coefficient is  $-0.046$  in the less agricultural

Figure D3: The Effect of Agro-Similarity on Food Prices Across Price Measures, 1996



*Notes:* Each row is one regression of the log of one plus a 1996 price measure on standardized agro-similarity on the main city sample, 112 centers as the 1996 household survey (Susenas) covers each measure, in the main specification with the full set of controls and Conley SE within 200km. Bars are 90% confidence intervals from the same standard errors. Four measures are price indices built from the city's households: the survey-weighted median unit value of each raw agricultural food item over those households, relative to its median over the households of all urban centers, aggregated under national, city and equal weights, and over the rice, corn and cassava items under equal weights. These indices have a median near one, so a coefficient on their log is about half of a proportional change. The fifth measure is the city's rice unit value, the value of the rice consumed over the kilograms consumed, each summed over its households with survey weights, and its coefficient reads as a proportional change.

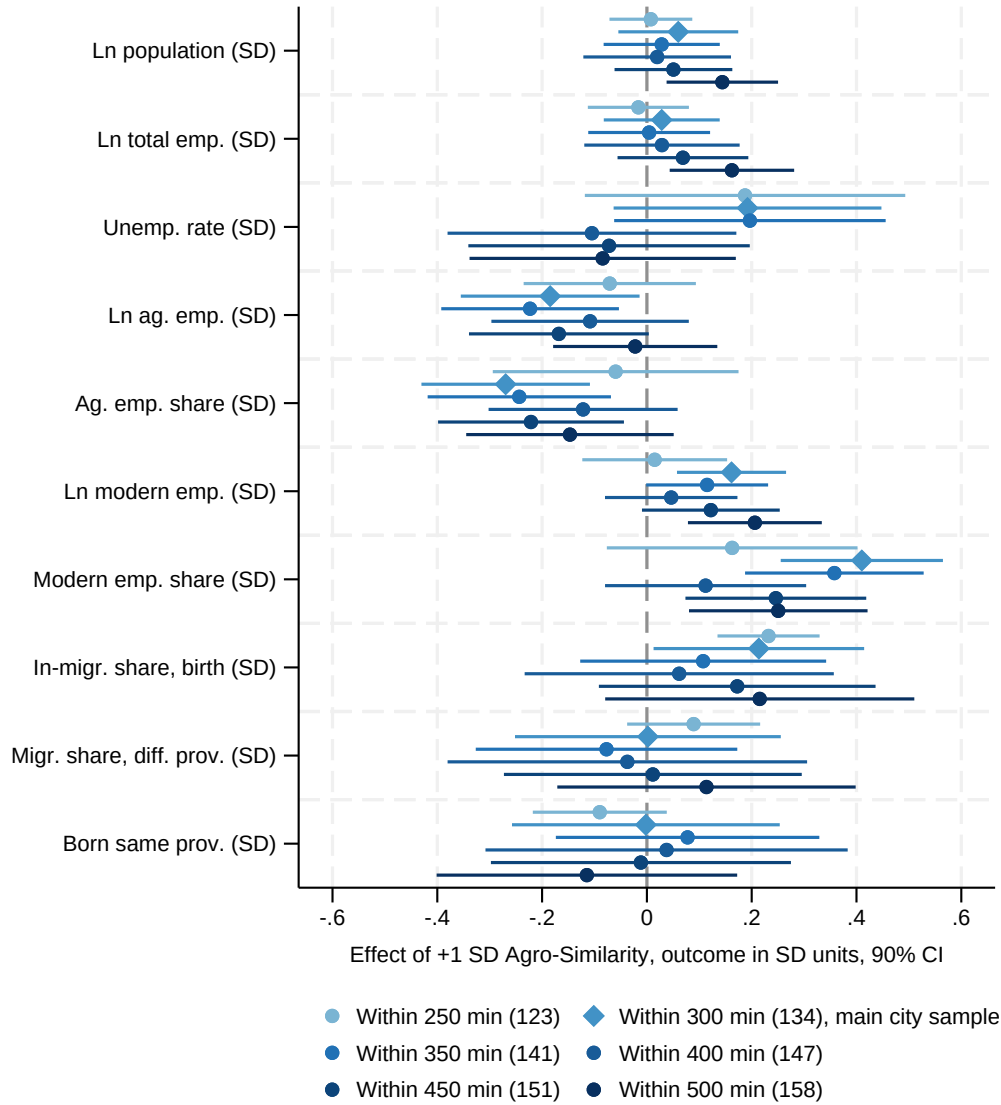
half and  $-0.056$  in the more agricultural one, the modern-employment coefficient is  $0.154$  and  $0.158$ , and no difference between the halves is distinguishable from zero, the smallest of the four  $p$ -values being  $0.422$ . The same design on 1980 population does separate the sample. Modern employment responds  $0.152$  log points more in the larger cities than in the smaller ones (s.e.  $0.055$ ) and population  $0.150$  log points more (s.e.  $0.041$ ), both distinguishable from zero at the 1% level, and the agricultural-share response is the same on both sides of that cut.

Table D3: Urban Agricultural Employment Share Under Four Definitions of the City

| Year | Definition                        | City mean | Pooled | Cities |
|------|-----------------------------------|-----------|--------|--------|
| 1980 | K whole subdistrict (as reported) | 0.594     | 0.482  | 122    |
| 1980 | U BPS-urban villages              | 0.141     | 0.086  | 96     |
| 2010 | K whole subdistrict               | 0.386     | 0.299  | 125    |
| 2010 | C built-up core                   | 0.182     | 0.112  | 124    |
| 2010 | U BPS-urban villages              | 0.158     | 0.099  | 127    |
| 2010 | CU core and BPS-urban             | 0.136     | 0.084  | 122    |

*Notes:* Agricultural employment share of the cities of the main city sample under four definitions of the city, from loosest to tightest: K sums every member subdistrict; C keeps the villages whose polygon meets the satellite footprint of the city; U keeps the villages the Indonesian census classifies as urban, a classification made independently of the satellite footprint; CU requires both. City mean averages the city shares, Pooled divides total agricultural employment by total employment, and Cities counts the cities measured; the data are the population censuses of 1980 (5% sample) and 2010.

Figure D4: The Agro-Similarity Effect Across Nested Travel-Time Windows, 250 to 500 Minutes



*Notes:* Each row is one 2010 population census outcome, standardized over all urban centers so its SD stays fixed as the sample widens: the main results table's seven outcomes (modern-sector employment is manufacturing plus services) and the shares of residents born outside the current district, born in a different province, and born in the same province. The six points in a row are coefficients from separate regressions on standardized agro-similarity over nested catchments: the urban centers whose nearest transmigration village lies within 250 to 500 travel minutes, in 50-minute steps, with a 1980 census baseline and measured 2010 outcomes (123 to 158 urban centers, as the legend reports). At each radius the exposure, the settler count and the 16 agro-climatic covariates are rebuilt over the villages within it, and the four 1980 baselines enter as in the main specification. The 300-minute point, marked with a diamond, is the main specification on the main city sample. Bars are 90% confidence intervals from Conley SE within 200km.

Table D4: Main Results for Cities Rebuilt From Their Less Agricultural Subdistricts and From Their 50 km Regions

|                                                                    | Population + Employment |                   |                  | Agric Empl           |                      | Modern Empl         |                     |
|--------------------------------------------------------------------|-------------------------|-------------------|------------------|----------------------|----------------------|---------------------|---------------------|
|                                                                    | (1)                     | (2)               | (3)              | (4)                  | (5)                  | (6)                 | (7)                 |
|                                                                    | ln pop                  | ln emp            | unemp rate       | ln L agric           | agric %              | ln L modern         | modern %            |
| <i>Panel A: Every member subdistrict</i>                           |                         |                   |                  |                      |                      |                     |                     |
| Agro-similarity                                                    | 0.051<br>(0.059)        | 0.024<br>(0.057)  | 0.008<br>(0.007) | -0.166*<br>(0.093)   | -0.051***<br>(0.018) | 0.158**<br>(0.062)  | 0.066***<br>(0.015) |
| Outcome mean                                                       | 11.842                  | 10.933            | 0.122            | 9.822                | 0.374                | 10.232              | 0.516               |
| Observations                                                       | 134                     | 134               | 134              | 134                  | 134                  | 134                 | 134                 |
| <i>Panel B: Subdistricts at least 25% non-agricultural in 1980</i> |                         |                   |                  |                      |                      |                     |                     |
| Agro-similarity                                                    | 0.042<br>(0.060)        | 0.015<br>(0.059)  | 0.008<br>(0.007) | -0.241**<br>(0.110)  | -0.062***<br>(0.018) | 0.138***<br>(0.048) | 0.066***<br>(0.015) |
| Outcome mean                                                       | 11.762                  | 10.828            | 0.128            | 9.474                | 0.298                | 10.272              | 0.582               |
| Observations                                                       | 115                     | 115               | 115              | 115                  | 115                  | 115                 | 115                 |
| <i>Panel C: Subdistricts at least 50% non-agricultural in 1980</i> |                         |                   |                  |                      |                      |                     |                     |
| Agro-similarity                                                    | -0.039<br>(0.044)       | -0.040<br>(0.045) | 0.009<br>(0.007) | -0.441***<br>(0.109) | -0.071***<br>(0.017) | 0.080<br>(0.060)    | 0.077***<br>(0.015) |
| Outcome mean                                                       | 11.827                  | 10.865            | 0.136            | 9.057                | 0.200                | 10.452              | 0.671               |
| Observations                                                       | 68                      | 68                | 68               | 68                   | 68                   | 68                  | 68                  |
| <i>Panel D: The city and its rural subdistricts within 50 km</i>   |                         |                   |                  |                      |                      |                     |                     |
| Agro-similarity                                                    | 0.041<br>(0.056)        | 0.010<br>(0.048)  | 0.006<br>(0.006) | -0.134*<br>(0.069)   | -0.054***<br>(0.020) | 0.129**<br>(0.056)  | 0.055***<br>(0.014) |
| Outcome mean                                                       | 12.724                  | 11.851            | 0.111            | 11.152               | 0.519                | 10.881              | 0.390               |
| Observations                                                       | 134                     | 134               | 134              | 134                  | 134                  | 134                 | 134                 |
| Settler count                                                      | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                   | ✓                   |
| Baseline Pop 1980                                                  | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                   | ✓                   |
| Urban share 1980                                                   | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                   | ✓                   |
| Total Empl 1980                                                    | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                   | ✓                   |
| Agri Empl 1980                                                     | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                   | ✓                   |
| Agro-climatic controls                                             | ✓                       | ✓                 | ✓                | ✓                    | ✓                    | ✓                   | ✓                   |

*Notes:* Each column is one regression of the 2010 outcome in its heading on agro-similarity, on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Panel A repeats Table 3. Panels B and C rebuild each city from its member subdistricts whose own 1980 employment was at least one quarter (Panel B) or one half (Panel C) outside agriculture, summing the 2010 outcomes and the four 1980 baselines over them; agro-similarity, the settler count and the covariates remain those of the whole city, a subdistrict two cities share is split between them by its measured weights, and the 21 member subdistricts with no 1980 employment count stay in every panel. A city with no subdistrict above the threshold leaves the panel, 19 cities in Panel B and 64 in Panel C; Panel C also loses 2 cities whose kept subdistricts have no 1980 employment count, so a 1980 baseline cannot be formed. Panel D adds to each city the rural subdistricts, those outside every city, whose nearest city it is within 50 km in a straight line, measured to the nearest member subdistrict of each city on the same island, summing the 2010 outcomes and the four 1980 baselines over that region: 709 rural subdistricts join, 6 cities do not gain a rural subdistrict, and the 56 rural subdistricts within 50 km that the 1980 census does not record are left out. Agro-similarity is standardized once, so the four panels share one scale. SE in parentheses, with stars from the same SE: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Source: 1980 and 2010 population censuses.

## D.4 Inference: The Conley Cutoff and Catchment Overlap

*Cited in the main text.* Table D6: Section 5.1, p. 24.

Table D5: Exposure Effects by the City's 1980 Agricultural Share and by Its 1980 Population

|                                                                                        | (1)                  | (2)                 | (3)                 | (4)                 |
|----------------------------------------------------------------------------------------|----------------------|---------------------|---------------------|---------------------|
|                                                                                        | Agric. share         | ln modern empl.     | ln population       | In-migrant share    |
| <i>Panel A: split at the median 1980 agricultural share, 57.9%</i>                     |                      |                     |                     |                     |
| Below median (less agricultural)                                                       | -0.046***<br>(0.017) | 0.154***<br>(0.060) | 0.055<br>(0.063)    | 0.004<br>(0.004)    |
| Above median (more agricultural)                                                       | -0.056***<br>(0.018) | 0.158**<br>(0.080)  | 0.044<br>(0.060)    | 0.007***<br>(0.002) |
| Above-median indicator (=1)                                                            | 0.089***<br>(0.022)  | -0.125**<br>(0.061) | -0.008<br>(0.056)   | -0.002<br>(0.003)   |
| Difference (above minus below)                                                         | -0.010<br>(0.014)    | 0.003<br>(0.046)    | -0.011<br>(0.039)   | 0.002<br>(0.003)    |
| <i>p</i> -value of the difference                                                      | 0.463                | 0.940               | 0.781               | 0.422               |
| <i>p</i> -value, Bartlett kernel                                                       | 0.501                | 0.946               | 0.798               | 0.366               |
| Observations                                                                           | 134                  | 134                 | 134                 | 134                 |
| <i>Panel B: split at the median 1980 population, 67,979 residents, the same design</i> |                      |                     |                     |                     |
| Below median (smaller)                                                                 | -0.047**<br>(0.022)  | 0.089<br>(0.066)    | -0.017<br>(0.081)   | 0.004<br>(0.004)    |
| Above median (larger)                                                                  | -0.057***<br>(0.018) | 0.241***<br>(0.080) | 0.133**<br>(0.065)  | 0.007***<br>(0.002) |
| Above-median indicator (=1)                                                            | -0.018<br>(0.018)    | 0.163*<br>(0.098)   | 0.150<br>(0.098)    | 0.001<br>(0.003)    |
| Difference (above minus below)                                                         | -0.010<br>(0.018)    | 0.152***<br>(0.055) | 0.150***<br>(0.041) | 0.003<br>(0.002)    |
| <i>p</i> -value of the difference                                                      | 0.580                | 0.006               | 0.000               | 0.152               |
| <i>p</i> -value, Bartlett kernel                                                       | 0.566                | 0.013               | 0.013               | 0.201               |
| Observations                                                                           | 134                  | 134                 | 134                 | 134                 |

*Notes:* Each column of each panel is one regression of a 2010 census outcome on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standardized agro-similarity enters interacted with an indicator for each side of the split, so the below-median and above-median rows are the two slopes and the indicator row is the shift in the outcome on the above-median side; the row labeled Difference is the above-median slope minus the below-median slope, from the same regression. Each split is cut at the sample median: 57.9% of 1980 employment in agriculture in Panel A, 67,979 residents in 1980 in Panel B. Standard errors (uniform kernel) are in parentheses; the Bartlett-kernel *p*-value tests the same difference, weighting a pair of cities by one minus its distance over the cutoff, and its cells are shaded by significance level. Modern employment is manufacturing plus services; the agricultural share is a fraction of 2010 employment, and the in-migrant share is the fraction of the 2010 population that lived in a different province five years earlier. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**The Conley cutoff.** The headline uses a 200-kilometer Conley cutoff. Because the cutoff enters only the variance estimator, the point estimates do not depend on it, so the choice affects only inference. At a shorter 150-kilometer radius every standard error remains well defined and two of the seven are smaller than at 200 kilometers, so the wider cutoff is not uniformly the more conservative of the two: the two agricultural and the two modern-sector columns have slightly larger standard errors at 150 kilometers (the pair for each share, from the table), and those shares are the columns the compositional result relies on. The wider direction matters more, because the exposure measure averages over village catchments that neighboring cities share. The 150 kilometer row of Table D6 reports these standard errors beside every wider radius.

**Catchment overlap and inference at wider radii.** The relevant direction to vary the cutoff is wider, and the reason is in the construction of the treatment. Exposure is an unweighted mean over the transmigration villages inside a city's 300-minute catchment, and neighboring catchments intersect, so two cities can be built partly from the same

village draws and their exposure errors are correlated at whatever distance that sharing reaches. Measured directly on the travel-time matrix the exposure itself is built from, the average city in the estimation sample draws on 30 in-window villages, 80% of those villages enter more than one city's catchment, and one village enters as many as 19. Of the 8,911 city pairs in the sample, 765 share at least one village, at a median separation of 143 kilometers and a maximum of 459; 27% of those sharing pairs sit farther apart than the 200-kilometer cutoff, which assumes independence beyond it. Grouping cities into the connected components of that sharing graph, the finest partition across which no village is shared at all, leaves 14 components over the estimation sample, and the largest holds 75 cities: roughly half the sample is linked into a single block by shared catchments.

I therefore re-estimate the headline columns across the whole ladder, from 150 out to 500 kilometers, and add a variant clustered on those components together with a wild cluster bootstrap over them (Appendix [Table D6](#)). The three composition outcomes are statistically significant under every estimator in the table except the wild bootstrap for the agricultural share, and the two size outcomes are not. The compositional standard errors are the more stable: both employment-share standard errors stay between 0.011 and 0.020 across the ladder and the clustering, and the agricultural share rejects at the 5% level at every radius ( $p = 0.010$  at 150 kilometers, 0.006 at 200 and 0.002 at 500 under the Bartlett kernel). Under the component clustering, log population returns  $p = 0.245$  with a wild cluster bootstrap  $p$  of 0.176, the agricultural share 0.014 and 0.140, and the modern share 0.000 and 0.058. Under the uniform kernel, which the headline uses, four of the table's 28 Conley cells do not return a standard error, because the estimated variance of the exposure coefficient is negative there, and all six standard errors that kernel returns at 500 kilometers are smaller than at 200 kilometers. At 200 kilometers the uniform variance matrix already has from five to six negative eigenvalues out of 23, depending on the column, so it is not positive semi-definite, and the table's Conley rows at wide radii are not a reliable guide to precision. The Bartlett kernel returns a standard error in every cell and its variance at 200 kilometers has no negative eigenvalue; from 200 to 500 kilometers its standard error falls by less than the uniform one in five of the six columns where both kernels return one, and for the unemployment rate it does not fall.

Table D6: The Headline Estimates Under a Ladder of Spatial-Inference Radii

|                                | Population + Employment |         |            | Agric Empl |           | Modern Empl |          |
|--------------------------------|-------------------------|---------|------------|------------|-----------|-------------|----------|
|                                | ln pop                  | ln emp  | unemp rate | ln L agric | agric %   | ln L modern | modern % |
| Agro-similarity                | 0.051                   | 0.024   | 0.008      | -0.166*    | -0.051*** | 0.158**     | 0.066*** |
| Conley 200 km (main), SE       | (0.059)                 | (0.057) | (0.007)    | (0.093)    | (0.018)   | (0.062)     | (0.015)  |
| Conley 200 km (main), $p$      | 0.389                   | 0.674   | 0.216      | 0.075      | 0.006     | 0.011       | 0.000    |
| Conley 150 km, SE              | (0.059)                 | (0.054) | (0.006)    | (0.098)    | (0.020)   | (0.065)     | (0.016)  |
| Conley 150 km, $p$             | 0.389                   | 0.657   | 0.188      | 0.091      | 0.010     | 0.015       | 0.000    |
| Conley 450 km, SE              | (.)                     | (.)     | (0.007)    | (0.059)    | (0.014)   | (.)         | (0.011)  |
| Conley 450 km, $p$             | .                       | .       | 0.209      | 0.005      | 0.000     | .           | 0.000    |
| Conley 500 km, SE              | (0.027)                 | (0.013) | (0.006)    | (0.054)    | (0.014)   | (.)         | (0.011)  |
| Conley 500 km, $p$             | 0.055                   | 0.071   | 0.186      | 0.002      | 0.000     | .           | 0.000    |
| Bartlett 200 km, SE            | (0.055)                 | (0.052) | (0.006)    | (0.094)    | (0.019)   | (0.059)     | (0.016)  |
| Bartlett 200 km, $p$           | 0.352                   | 0.649   | 0.153      | 0.078      | 0.006     | 0.008       | 0.000    |
| Bartlett 150 km, SE            | (0.054)                 | (0.052) | (0.006)    | (0.093)    | (0.018)   | (0.058)     | (0.015)  |
| Bartlett 150 km, $p$           | 0.345                   | 0.647   | 0.133      | 0.074      | 0.005     | 0.006       | 0.000    |
| Bartlett 450 km, SE            | (0.049)                 | (0.045) | (0.006)    | (0.081)    | (0.017)   | (0.052)     | (0.014)  |
| Bartlett 450 km, $p$           | 0.293                   | 0.599   | 0.194      | 0.041      | 0.003     | 0.003       | 0.000    |
| Bartlett 500 km, SE            | (0.046)                 | (0.043) | (0.006)    | (0.079)    | (0.017)   | (0.048)     | (0.013)  |
| Bartlett 500 km, $p$           | 0.267                   | 0.576   | 0.195      | 0.036      | 0.002     | 0.001       | 0.000    |
| Components, SE                 | (0.042)                 | (0.040) | (0.007)    | (0.060)    | (0.018)   | (0.058)     | (0.013)  |
| Components, $p$                | 0.245                   | 0.558   | 0.256      | 0.017      | 0.014     | 0.017       | 0.000    |
| Components, wild bootstrap $p$ | 0.176                   | 0.520   | 0.293      | 0.150      | 0.140     | 0.043       | 0.058    |
| Observations                   | 134                     | 134     | 134        | 134        | 134       | 134         | 134      |

*Notes:* Every column repeats the main specification of Table 3, a regression of a 2010 census outcome on standardized agro-similarity on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), with the full set of controls; only the variance estimator changes across rows, so the coefficients are identical, and their stars come from the Conley  $p$ -value at 200 kilometers. The Conley rows use a uniform kernel, which gives every pair of cities inside the cutoff the same weight, and the Bartlett rows use the Bartlett kernel, which weights a pair by one minus its distance over the cutoff. Each  $p$ -value cell takes one shade at 0.10 or above and deepening shades below 0.10, 0.05 and 0.01. The radii bracket the distance over which cities share transmigration villages: 80% of the villages within the window enter more than one city's catchment, and two cities sharing a village sit up to 459 kilometers apart. The last three rows cluster on the connected components of the village-sharing graph, the finest partition that keeps every shared village inside one group: 14 components over the 134 cities, the largest holding 75, so the clustered standard error uses few effective groups and the wild cluster bootstrap over them, with Webb weights, is the version to read; it has little power at this count. Under the uniform kernel four of the 28 cells do not return a standard error, because the estimated variance of the exposure coefficient is negative there. In each row SE is the standard error of the coefficient in the first row and  $p$  its two-sided  $p$ -value, and (main) labels the radius of the main specification. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## E Additional Results

This part reports the outcomes, decompositions, and alternative explanations that the main text states but does not tabulate.

### E.1 Additional Reduced-Form Outcome Tables

*Cited in the main text.* Table E1: Section 5.5, p. 40. Table E2: Section 5.2, p. 30. Table E3: Section 5.2, p. 31. Table E4: Section 5.1, p. 24; Section 5.3, p. 33. Table E5: Section 4.2, p. 20. Table E6: Section 5.2, p. 30. Table E7: Section 5.2, p. 30.

Table E1: All Outcomes by Location Type, Main Specification

|                                              | Cities                        | Rural, with settlement      | Rural, no settlement         |
|----------------------------------------------|-------------------------------|-----------------------------|------------------------------|
| <i>A. Size and composition</i>               |                               |                             |                              |
| ln population                                | +0.060<br>(0.069)             | +0.082<br>(0.080)           | −0.061<br>(0.074)            |
| ln employment                                | +0.028<br>(0.067)             | +0.091<br>(0.084)           | −0.090<br>(0.075)            |
| ln agricultural employment                   | −0.185*<br>(0.104)            | +0.089<br>(0.076)           | −0.214**<br>(0.094)          |
| Agricultural employment share                | −0.278***<br>(0.101)          | +0.010<br>(0.063)           | −0.319***<br>(0.083)         |
| ln modern employment                         | +0.162**<br>(0.063)           | +0.077<br>(0.076)           | +0.097*<br>(0.053)           |
| Modern employment share                      | +0.422***<br>(0.097)          | −0.025<br>(0.070)           | +0.311***<br>(0.089)         |
| Observations                                 | 134                           | 191                         | 596                          |
| <i>B. Prices, wages, and consumption</i>     |                               |                             |                              |
| Unemployment rate                            | +0.210<br>(0.170)<br>[134]    | +0.032<br>(0.090)<br>[191]  | +0.068<br>(0.091)<br>[596]   |
| Agricultural price, 2002                     | +0.126<br>(0.262)<br>[109]    | −0.225*<br>(0.115)<br>[104] | −0.039<br>(0.085)<br>[204]   |
| Housing price, 2002                          | −0.016<br>(0.141)<br>[109]    | −0.001<br>(0.085)<br>[98]   | +0.028<br>(0.077)<br>[195]   |
| ln nominal wage, 2011                        | +0.062<br>(0.101)<br>[134]    | +0.152<br>(0.178)<br>[82]   | −0.125**<br>(0.060)<br>[236] |
| ln real wage, 2011                           | +0.013<br>(0.126)<br>[134]    | +0.271**<br>(0.129)<br>[82] | −0.115<br>(0.147)<br>[236]   |
| Food spending share, 2011, stored measure    | −0.313***<br>(0.106)<br>[134] | −0.233**<br>(0.105)<br>[82] | +0.152<br>(0.119)<br>[236]   |
| ln food spending, 2011                       | +0.161<br>(0.115)<br>[134]    | −0.078<br>(0.065)<br>[188]  | +0.050<br>(0.091)<br>[576]   |
| Food spending share, 2011, pooled households | −0.313***<br>(0.106)<br>[134] | −0.063<br>(0.131)<br>[188]  | +0.120**<br>(0.061)<br>[576] |
| Rice calories per person                     | +0.451***<br>(0.172)<br>[108] | +0.172<br>(0.117)<br>[99]   | −0.010<br>(0.096)<br>[198]   |

**The response by location type, full set of outcomes.** Every outcome behind Figure 14a and Figure 14b is estimated for the three location types, each with its standard error. The size and composition outcomes, which all three areas measure on the same units, form panel A, and the prices, wages and consumption outcomes, which come from the household survey and cover a different number of units in each row, form panel B. The agricultural and housing prices are the 2002 Susenas wave, which precedes the 2004 rice import ban; the real wage is the 2011 wave, because the subdistrict level has no 2002 wage. Appendix Table E1 reports the full set. Panel C holds the eight 1996 rows; Figure 14b plots three of them, the food share, rice calories per person and the raw-food price, beside the 2011 real wage and the 2010 unemployment rate.

The goods-market outcomes behind the mechanism discussion follow. Table E3 reports the urban agricultural-goods outcomes (production, consumption, and the production-minus-consumption surplus). The non-food side (non-food spending, housing rents, and wages) is reported in Table E2, and the manufacturing/services composition of the modern sector in Table E4 below.

Table E1: All Outcomes by Location Type, Main Specification, Continued

|                                     | Cities                        | Rural, with settlement     | Rural, no settlement          |
|-------------------------------------|-------------------------------|----------------------------|-------------------------------|
| <i>C. Household survey, 1996</i>    |                               |                            |                               |
| ln total household spending, 1996   | +0.512*<br>(0.263)<br>[112]   | +0.115<br>(0.124)<br>[117] | -0.117<br>(0.101)<br>[285]    |
| ln food spending per person, 1996   | +0.460<br>(0.287)<br>[112]    | +0.117<br>(0.115)<br>[117] | -0.033<br>(0.106)<br>[285]    |
| Food spending share, 1996           | -0.462***<br>(0.177)<br>[112] | +0.058<br>(0.118)<br>[117] | -0.141*<br>(0.086)<br>[285]   |
| Agricultural price (raw food), 1996 | -0.090<br>(0.188)<br>[112]    | -0.128<br>(0.108)<br>[117] | -0.226*<br>(0.117)<br>[285]   |
| Rice unit value, 1996               | -0.050<br>(0.210)<br>[112]    | -0.198<br>(0.179)<br>[116] | -0.048<br>(0.116)<br>[283]    |
| ln calories per person, 1996        | +0.719***<br>(0.148)<br>[112] | +0.087<br>(0.178)<br>[117] | +0.041<br>(0.135)<br>[285]    |
| Rice calories per person, 1996      | +0.289***<br>(0.081)<br>[112] | +0.013<br>(0.106)<br>[117] | +0.365***<br>(0.106)<br>[285] |
| ln staple kg per person, 1996       | +0.407***<br>(0.073)<br>[112] | -0.224<br>(0.152)<br>[117] | +0.164*<br>(0.099)<br>[285]   |

*Notes:* Each coefficient is the response, in SD of the outcome, to a one-SD increase in agro-similarity, with Conley SE within 200km in parentheses. Cities are the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline); the rural subdistricts with and without a settlement lie outside every city and meet the same travel-time and baseline rule. The city column is in the main specification with the full set of controls; the two rural columns have the same controls with the settler count taken over villages within 1,000 travel minutes. Outcomes and agro-similarity are standardized once among the cities and once among both rural groups together, at exposure SDs of 0.103 over the 138 cities with a measured exposure and 0.100 over the 787 sample subdistricts, so one SD is the same amount of exposure on either side. Panel A and the unemployment rate are from the 2010 population census, the prices and rice calories from the 2002 household survey (Susenas), and the wage and food spending rows from its 2011 wave, with food spending and wages in logs. Panel C is the 1996 wave, pooled over the surveyed households of each city and subdistrict, with the agricultural price on the raw-food index and logs taken as  $\ln(x+1)$ . Panel A's observation count applies to every row of that panel; each Panel B and C row prints its own count in brackets. The stored 2011 food share exists only for the subdistricts where the 2011 survey found a wage earner living in an urban block of the statistics office, and the pooled food share averages all surveyed households of each subdistrict, as food spending does; for the cities the two are the same measure. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The main results treat the modern sector as a single object; its manufacturing and services components are reported separately here, in levels and shares, in the modern-sector layout (Appendix Table E4). Columns 5 to 7 report the shares of workers outside both sectors: construction, the remaining industry codes, and the two together. The agricultural share of Table 3, the two modern-sector shares and the share outside both sectors add up to one.

Finally, the agricultural production and food consumption/surplus responses are estimated across the crop-category hierarchy, in raw and standardized form, and the standardized versions define several of the model's calibration moments (Table B3). The raw estimates are in Table E5 and Table E6.

The earliest wave, household Susenas 1996 outcomes aggregated to the same urban centers, shows higher per-capita expenditure one to two decades after placement, while the 1996 rice price and the two food-price indices show little response (Table E7).

The cross-sectional outcomes have no 1980 baseline, so they are identified off balance in levels. Table E8 asks what that costs, by adding a 1980 variable that proxies for the unobserved baseline of the outcome and re-estimating on

Table E2: Urban Prices, Spending, and Wages

|                        | Prices & Spending   |                   | Wages            |                  |                  |
|------------------------|---------------------|-------------------|------------------|------------------|------------------|
|                        | (1)                 | (2)               | (3)              | (4)              | (5)              |
|                        | non-food %          | housing price     | ln wage nonag    | ln wage          | ln real wage     |
| Agro-similarity        | 0.016***<br>(0.006) | -0.011<br>(0.107) | 0.027<br>(0.030) | 0.019<br>(0.032) | 0.003<br>(0.029) |
| Settler count          | ✓                   | ✓                 | ✓                | ✓                | ✓                |
| Baseline Pop 1980      | ✓                   | ✓                 | ✓                | ✓                | ✓                |
| Urban share 1980       | ✓                   | ✓                 | ✓                | ✓                | ✓                |
| Total Empl 1980        | ✓                   | ✓                 | ✓                | ✓                | ✓                |
| Agri Empl 1980         | ✓                   | ✓                 | ✓                | ✓                | ✓                |
| Agro-climatic controls | ✓                   | ✓                 | ✓                | ✓                | ✓                |
| Outcome mean           | 0.420               | 1.586             | 14.062           | 13.968           | 13.749           |
| Outcome SD             | 0.052               | 0.910             | 0.273            | 0.314            | 0.234            |
| Observations           | 134                 | 134               | 134              | 134              | 134              |

Each column is one regression of a 2011 household-survey (Susenas) outcome on standardized agro-similarity on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), in the main specification with the full set of controls and Conley SE within 200km. The Observations row gives the centers the survey reaches. Standard errors are in parentheses. Non-food % is the non-food share of household expenditure, a fraction; housing price is the weighted median rent of the urban center relative to the national figure; the wage columns are the logs of the median monthly wage of non-agricultural workers, of all workers, and of that all-worker median deflated by a local food and housing price index. In the labels, ln is the natural log and nonag is non-agricultural workers; the settler count weights the settlers of each transmigration village within 300 travel minutes by the inverse of its travel time, in thousands; Baseline Pop 1980, Total Empl 1980 and Agri Empl 1980 are the logs of 1980 population, employment and agricultural employment; and SD is the standard deviation. \*\*\* 1%, \*\* 5%, \* 10%.

Table E3: Urban Agricultural Goods: Rice Production, the Rice Surplus, and Food Prices

|                        | Rice Output         |                         | Prices           |                   |
|------------------------|---------------------|-------------------------|------------------|-------------------|
|                        | (1)                 | (2)                     | (3)              | (4)               |
|                        | ln rice production  | rice surplus, tons      | rice price index | agric price index |
| Agro-similarity        | 0.735***<br>(0.252) | -3312.766<br>(3331.434) | 0.041<br>(0.025) | 0.016<br>(0.022)  |
| Settler count          | ✓                   | ✓                       | ✓                | ✓                 |
| Baseline Pop 1980      | ✓                   | ✓                       | ✓                | ✓                 |
| Urban share 1980       | ✓                   | ✓                       | ✓                | ✓                 |
| Total Empl 1980        | ✓                   | ✓                       | ✓                | ✓                 |
| Agri Empl 1980         | ✓                   | ✓                       | ✓                | ✓                 |
| Agro-climatic controls | ✓                   | ✓                       | ✓                | ✓                 |
| Outcome mean           | 8.455               | -11879.685              | 0.999            | 1.062             |
| Outcome SD             | 2.343               | 31325.719               | 0.145            | 0.121             |
| Observations           | 132                 | 108                     | 129              | 134               |

*Notes:* Each column is one regression of the outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Column 1 is the log of one plus 2003 village-census (PODES) rice production, in tons of paddy, and column 2 is the rice surplus in tons, that production in milled weight (.6402 tons of milled rice per ton of paddy, from the 2018 conversion survey of Statistics Indonesia) minus 2002 household-survey (Susenas) rice consumption. Columns 3 and 4 are the rice price index of each city and its agricultural price index, an index of raw-food prices, both from Susenas 2011. Survey outcomes are observed on the centers each source reaches, reported in the Observations row. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table E4: Modern Sector Split, and Employment Outside Agriculture and the Modern Sector

|                        | Log employment     |                    | Modern-sector shares |                     | Outside both sectors |                    |                    |
|------------------------|--------------------|--------------------|----------------------|---------------------|----------------------|--------------------|--------------------|
|                        | (1)                | (2)                | (3)                  | (4)                 | (5)                  | (6)                | (7)                |
|                        | ln L manuf         | ln L serv          | manuf empl %         | serv empl %         | constr %             | other %            | outside %          |
| Agro-similarity        | 0.249**<br>(0.120) | 0.144**<br>(0.067) | 0.010**<br>(0.005)   | 0.056***<br>(0.018) | -0.002<br>(0.004)    | -0.013*<br>(0.008) | -0.015*<br>(0.009) |
| Settler count          | ✓                  | ✓                  | ✓                    | ✓                   | ✓                    | ✓                  | ✓                  |
| Baseline Pop 1980      | ✓                  | ✓                  | ✓                    | ✓                   | ✓                    | ✓                  | ✓                  |
| Urban share 1980       | ✓                  | ✓                  | ✓                    | ✓                   | ✓                    | ✓                  | ✓                  |
| Total Empl 1980        | ✓                  | ✓                  | ✓                    | ✓                   | ✓                    | ✓                  | ✓                  |
| Agri Empl 1980         | ✓                  | ✓                  | ✓                    | ✓                   | ✓                    | ✓                  | ✓                  |
| Agro-climatic controls | ✓                  | ✓                  | ✓                    | ✓                   | ✓                    | ✓                  | ✓                  |
| Outcome mean           | 7.556              | 10.136             | 0.047                | 0.470               | 0.060                | 0.049              | 0.109              |
| 1980 sector share      | 0.063              | 0.319              | 0.063                | 0.319               | 0.035                | 0.022              | 0.057              |
| Outcome SD             | 1.356              | 0.966              | 0.042                | 0.145               | 0.027                | 0.065              | 0.078              |
| Observations           | 134                | 134                | 134                  | 134                 | 134                  | 134                | 134                |

Each column is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Columns 1 and 2 are the logs of manufacturing and of services employment, the two parts of the modern sector, and columns 3 and 4 their shares of total employment (fractions). Columns 5 to 7 are the shares of construction (census industry code 10), of the other industry codes (mining and quarrying, electricity and gas, and the census code others: codes 7, 9 and 19), and of the two together, the employment outside agriculture and the modern sector; every share divides by the workers who report an industry, so the headline table's agricultural share and columns 3 to 7 sum to one, and column 7 is the sum of columns 5 and 6. The 1980 sector share row is the mean 1980 share of the same sector across the sample. In the column labels, ln is the natural log, L and empl are employment, manuf and serv are manufacturing and services, constr is construction, other the remaining codes and outside the two together, and % is a share of total employment, a fraction; the settler count weights the settlers of each transmigration village within 300 travel minutes by the inverse of its travel time, in thousands; Baseline Pop 1980, Total Empl 1980 and Agri Empl 1980 are the logs of 1980 population, employment and agricultural employment; and SD is the standard deviation. \*\*\* 1%, \*\* 5%, \* 10%.

the same urban centers. The estimate is essentially unchanged under every proxy and remains statistically significant in every column.

Table E5: Agricultural Production by Crop Category, 2003 Village Census

|                        | Food (staples + legumes) |                     |                  | Non-food cash crops  |
|------------------------|--------------------------|---------------------|------------------|----------------------|
|                        | (1)                      | (2)                 | (3)              | (4)                  |
|                        | Ln Staples               | Ln Rice             | Ln Legumes       | Ln Coffee+Tea+Spices |
| Agro-similarity        | 0.608***<br>(0.197)      | 0.735***<br>(0.252) | 0.424<br>(0.360) | 0.103<br>(0.501)     |
| Settler count          | ✓                        | ✓                   | ✓                | ✓                    |
| Baseline Pop 1980      | ✓                        | ✓                   | ✓                | ✓                    |
| Urban share 1980       | ✓                        | ✓                   | ✓                | ✓                    |
| Total Empl 1980        | ✓                        | ✓                   | ✓                | ✓                    |
| Agri Empl 1980         | ✓                        | ✓                   | ✓                | ✓                    |
| Agro-climatic controls | ✓                        | ✓                   | ✓                | ✓                    |
| Outcome mean           | 8.953                    | 8.455               | 4.250            | 4.341                |
| Outcome SD             | 2.007                    | 2.343               | 2.220            | 2.655                |
| Observations           | 133                      | 132                 | 117              | 103                  |

*Notes:* Each column is one regression of the outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Outcomes are the log of one plus 2003 village-census (PODES) production in tons, by crop group: staples (rice, corn, cassava, sweet potato, taro, other tubers and sorghum), paddy rice, legumes (peanut, soybean and mungbean), and coffee, tea, pepper, nutmeg and clove. Coefficients are approximate semi-elasticities. Survey outcomes are observed on the centers each source reaches, reported in the Observations row. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table E6: Food Consumption per Person and the Production-Minus-Consumption Surplus, by Category

|                        | Log consumption (kg per person) |                   |                     | $\sinh^{-1}$ surplus (output minus consumption, tons) |                    |
|------------------------|---------------------------------|-------------------|---------------------|-------------------------------------------------------|--------------------|
|                        | (1)                             | (2)               | (3)                 | (4)                                                   | (5)                |
|                        | Staples, 2002                   | Staples, 2011     | Rice, 2002          | Staples Surplus                                       | Rice Surplus       |
| Agro-similarity        | 0.126***<br>(0.037)             | -0.028<br>(0.027) | 0.108***<br>(0.037) | -1.360<br>(1.194)                                     | -1.647*<br>(0.995) |
| Settler count          | ✓                               | ✓                 | ✓                   | ✓                                                     | ✓                  |
| Baseline Pop 1980      | ✓                               | ✓                 | ✓                   | ✓                                                     | ✓                  |
| Urban share 1980       | ✓                               | ✓                 | ✓                   | ✓                                                     | ✓                  |
| Total Empl 1980        | ✓                               | ✓                 | ✓                   | ✓                                                     | ✓                  |
| Agri Empl 1980         | ✓                               | ✓                 | ✓                   | ✓                                                     | ✓                  |
| Agro-climatic controls | ✓                               | ✓                 | ✓                   | ✓                                                     | ✓                  |
| Outcome mean           | 4.768                           | 4.579             | 4.653               | -3.426                                                | -6.081             |
| Outcome SD             | 0.204                           | 0.162             | 0.209               | 9.329                                                 | 7.633              |
| Observations           | 108                             | 132               | 108                 | 108                                                   | 108                |

*Notes:* Each column is one regression of the outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Columns 1 to 3 are the log of one plus annual consumption per person in kilograms from the household survey (Susenas): staples (rice, corn, cassava and tubers) in 2002 and 2011, and rice in 2002. Columns 4 and 5 are the inverse hyperbolic sine of the staples and rice surplus in tons, 2003 village-census (PODES) production minus 2002 Susenas consumption, with rice output in milled weight (.6402 tons of milled rice per ton of paddy, 2018 conversion survey). Coefficients are approximate semi-elasticities. Survey outcomes are observed on the centers each source reaches, reported in the Observations row. In the labels,  $\sinh^{-1}$  is the inverse hyperbolic sine, used because a surplus can be negative. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table E7: Household Survey Outcomes in 1996

|                 | Ln rice           | Ag price          | All-food         | House idx        | Ln rent          | Ln pcExp           | Food share           | Ag share         | Mod share         | Log wage            |
|-----------------|-------------------|-------------------|------------------|------------------|------------------|--------------------|----------------------|------------------|-------------------|---------------------|
| Agro-similarity | -0.009<br>(0.029) | -0.012<br>(0.025) | 0.007<br>(0.025) | 0.336<br>(0.238) | 0.214<br>(0.140) | 0.182**<br>(0.087) | -0.033***<br>(0.013) | 0.048<br>(0.040) | -0.033<br>(0.040) | 0.232***<br>(0.072) |
| Outcome mean    | 6.84              | 1.06              | 1.07             | 1.56             | 9.84             | 11.14              | 0.65                 | 0.44             | 0.47              | 12.43               |
| N (cities)      | 112               | 112               | 112              | 112              | 112              | 112                | 112                  | 119              | 119               | 105                 |

*Notes:* Each column is a separate urban-center regression of a 1996 outcome from the national socioeconomic household survey (Susenas) on standardized agro-similarity, on the urban centers of the main city sample that the 1996 survey reaches, counted in the bottom row, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Outcomes, in order: the log rice unit value (rupiah per kilogram); price indices of raw agricultural foods and of all foods, city prices relative to the median over the households of all urban centers; the median monthly housing rent relative to the national median, and the log of that median rent; log mean monthly spending per person; the mean household food spending share; the shares of workers in agriculture and in manufacturing and services; and log median household income per employed worker, which stands in for the wage. Ln is the natural log, and N (cities) is the number of urban centers in each regression. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table E8: Baseline Proxy Controls and the Cross-Sectional Food-Share Estimate

|                              | (1)                   | (2)                              | (3)                 | (4)                 |
|------------------------------|-----------------------|----------------------------------|---------------------|---------------------|
|                              | Population<br>density | Agricultural<br>employment share | Schooling           | All three           |
| Exposure, without the proxy  | -0.0164<br>(0.0055)   | -0.0164<br>(0.0055)              | -0.0164<br>(0.0055) | -0.0164<br>(0.0055) |
| Exposure, with the proxy     | -0.0167<br>(0.0057)   | -0.0164<br>(0.0053)              | -0.0159<br>(0.0056) | -0.0160<br>(0.0054) |
| Movement, in standard errors | -0.07                 | -0.01                            | 0.09                | 0.06                |
| Urban centers                | 134                   | 134                              | 134                 | 134                 |

*Notes:* The outcome is the 2011 household food share, the mean share of household spending on food in the national socioeconomic household survey (Susenas). Each column adds to the main specification one 1980 census proxy for the unobserved baseline of the outcome: population per square kilometer, the agricultural employment share, the number of residents who had completed secondary school, and all three together. Each regression is of the food share on standardized agro-similarity, in the main specification with the full set of controls and Conley SE within 200km. The two rows of each column use the same urban centers, those with a transmigration village within 300 travel minutes and a 1980 census baseline that have the food share and the proxy measured, counted in the bottom row. Movement is the with-proxy estimate minus the without-proxy estimate, over the without-proxy standard error. Standard errors are in parentheses.

## E.2 The In-Migration Required for the Agricultural-Share Fall

*Cited in the main text. This section:* [Section 5.2, p. 27](#). Table E11: [Section 5.2, p. 28](#).

**The identity.** A city has  $L$  workers in 2010, a share  $s$  of them in agriculture. Suppose it takes in  $M$  extra workers, a share  $a$  of whom also work in agriculture, and hold the incumbent workforce and its sector composition fixed. Its agricultural share becomes  $(sL + aM)/(L + M)$ . Writing the gross inflow rate as  $g = M/L$ , the fall in the share is  $g(s - a)/(1 + g)$ , so producing a fall of  $\Delta$  requires

$$g(a) = \frac{\Delta}{s - a - \Delta}, \quad (19)$$

and a log employment response of  $\ln(1+g(a))$ . Equation 19 has two properties the test relies on. The requirement rises steeply in  $a$ , because arrivals who farm dilute the share less. And it is infinite once  $a \geq s - \Delta$ : an inflow that agricultural cannot pull the average down that far at any size.

The base  $s$  is the comparison city's 2010 share, 0.374 on the estimation sample. Both coefficients are cross-sectional 2010 differentials, so the share the extra arrivals dilute is the share the comparison city has when they arrive. Using the 1980 share instead would attribute the common 1980-2010 structural change to the inflow as well. The 1980 base is roughly 20 percentage points higher and lowers the requirement by more than a factor of three; I report it throughout as a conservative alternative.

**Measuring the arrivals' sector mix.** No census aggregate crosses migrant status with industry. I therefore build that cross from the individual records of the 1980 and 2010 censuses, aggregating person to village, then to year-2000 subdistrict through the harmonized crosswalk, then to the reclassified urban centers. Recomputing each city's overall agricultural share from these person records reproduces the published share to machine precision for the median city (correlation 0.999).

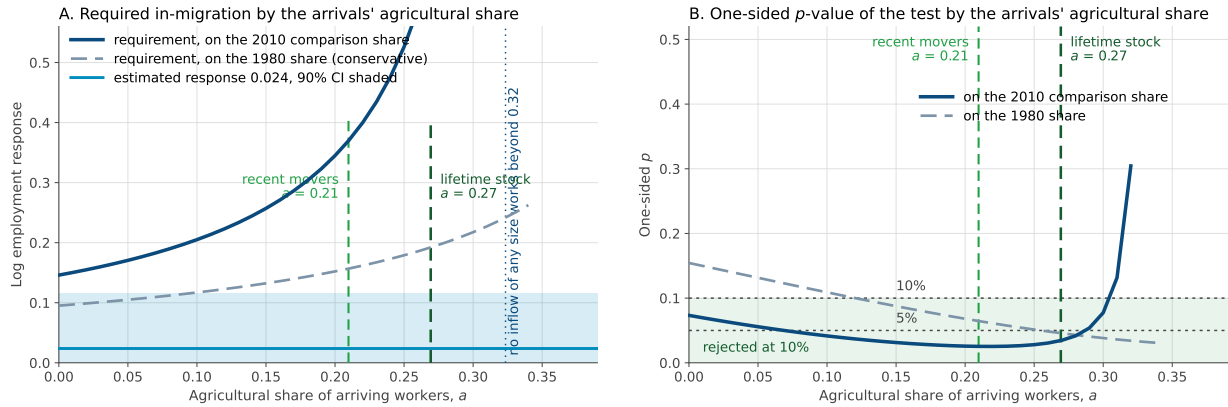
**The test.** Both  $\Delta$  and the employment response are estimated, so the threshold is itself a random variable and the test needs their joint covariance under the same spatial kernel. I therefore stack the two equations: each city contributes two rows, one for log employment and one for the agricultural share, every regressor is interacted with an equation dummy, and one Conley regression is run on the stack. The two rows of a city sit at distance zero, so the kernel returns the cross-equation covariance directly. The stacked fit is saturated and reproduces the two separate regressions exactly. The delta method then propagates the covariance through Equation 19. The resulting standard error on the gap is 0.351, against 0.057 on the employment response, so the vertical distance between the two curves in Figure E1 understates the uncertainty.

**In one city's workers.** The average city in the sample has 91,751 workers, 34,345 of them in agriculture. The high-exposure city has 2,210 more workers and 30,393 agricultural workers. If net employment growth equals arrivals, with the incumbent workforce fixed, and those arrivals farm at the measured 26.9%, they add 595 farmers, so agricultural employment would total about 34,940. It is at 30,393, a gap of roughly 4,547 workers relative to that calculation. Under those assumptions, 86,140 arrivals would be required, against net employment growth of 2,210, a shortfall of about 83,930 workers on a workforce of 91,751.

**The modern-job accounting.** Exposure adds modern-sector jobs to a city faster than it adds workers, and this subsection asks whether the workers a city gains could have filled those jobs with nobody already in the city changing sector. At the mean city, one standard deviation of exposure adds about 9,100 modern-sector jobs and 2,200 workers, the coefficients of Table 3 taken at the 2010 means. I count how many of the new jobs the workers added could fill, first if every one of them took a modern-sector job and then if they took such jobs at the rate the census measures for migrants, and I attach to each share a one-sided upper bound that uses the joint precision of the two coefficients, since both come from the same cities. The workers added fill about a quarter of the gain, 24%, and at most 82% at 95% confidence; at the migrants' rate they fill 16%, at most 54%. The main specification rejects that the net addition of workers fills the whole modern-job gain. This is an arrivals calculation with the incumbent workforce fixed and without offsetting departures or labor-force entry and exit. The checks below report how the conclusion varies with the variance estimator, job conversion, and design-based specification.

**The share of the modern-job gain.** Each coefficient of Table 3 is a proportional change, so at the mean city it is also a change in people: a coefficient  $b$  on an outcome with 2010 mean  $\bar{X}$  moves that outcome by  $\bar{X}(e^b - 1)$ . On modern-sector employment (mean 53,037) the coefficient gives 9,100 jobs; on employment (mean 90,259) it gives 2,200 workers. If nobody already working in the city changes sector, only the workers added can fill the new modern-sector jobs, and the share they fill is the workers added, times the fraction of them in modern-sector work, divided by the jobs

Figure E1: Required In-Migration by the Arrivals' Agricultural Share, and the One-Sided Test



*Notes:* Log employment and the agricultural employment share in the 2010 census are each regressed on standardized agro-similarity on the main city sample, with the main specification controls: the settler count and 16 agro-climatic covariates within 300 minutes, and four 1980 baselines; inference uses Conley SE within 200km and the joint covariance of the two coefficients. The horizontal axis is  $a$ , the agricultural share of the arriving workers; solid curves take the 2010 mean agricultural share  $s$  as the base, and dashed curves the 1980 mean. Left panel: the curves are  $\log(1 + g(a))$ , the log employment response Equation 19 requires; the flat line and shaded band are the estimated response to a one-standard-deviation rise in agro-similarity and its 90% confidence interval; the dotted line marks the asymptote  $a = s - \Delta$ , where  $\Delta$  is the estimated fall in the agricultural share per standard deviation of agro-similarity, and beyond it an inflow of any size fails to produce that fall. Right panel: the one-sided  $p$ -value at each  $a$ , with reference lines at 10% and 5%. In both panels the dashed vertical lines mark the 2010 agricultural shares of recent movers (moved district within five years) and of the lifetime stock (born outside the district). The one-sided test rejects at 10% for every  $a$  up to 0.30 and at 5% from 0.07 to 0.28.

gained:  $\phi = \mu \bar{L}(e^{b_L} - 1) / [\bar{M}(e^{b_M} - 1)]$ , with  $\bar{L}$  and  $\bar{M}$  the two means,  $b_L$  and  $b_M$  the two coefficients and  $\mu$  the fraction. With every worker added in a modern-sector job ( $\mu = 1$ ) the share is 24%; at the rate at which workers born outside their district of residence hold modern-sector jobs in the 2010 census, 65%, it is 16%. The share is uncertain because the two coefficients are, and its numerator, the employment response, is the imprecise one, so a standard error of the ratio itself (50 percentage points by the delta method) is a poor guide. I therefore test each candidate share directly, asking whether the data reject that the workers added fill that much of the gain. The two coefficients move together across cities (correlation 0.82 in one regression that stacks the two outcomes, as in the test above), and the one-sided test uses their joint covariance. The largest share the test does not reject at 5% is the one-sided 95% upper bound: 82% with every worker added in a modern-sector job and 54% at the migrants' rate. A share of 100%, the workers added filling every new job, is rejected at 5% in the first case ( $p = 0.012$ ) and at 1% in the second. The net modern-job gain exceeds net employment growth by 6,900, with a 90% interval of 1,900 to 11,900. Repeated cross-sections do not identify individual moves or occupational transitions.

**Every variance and every conversion.** The strength of the rejection varies with the standard errors and the way jobs are counted. Under the paper's uniform Conley kernel, a Bartlett kernel and errors clustered by city, every row of Table E9 that counts workers rejects a share of 100% at 5%. Province clusters, the loosest of the four, still reject in the headline row ( $p = 0.033$ , bound 93%); only the two rows that put the comparison city at the mean of the cities' shares do not reject at 5% ( $p = 0.067$  and  $0.072$ , bounds 106% and 106%), because larger cities are more modern, so the mean of the cities' modern shares is below the modern share of the mean city, and those two rows put fewer modern-sector jobs in the denominator. Counting residents aged 15 and over in place of workers weakens the comparison, because residents

Table E9: The Share of the Modern-Job Gain by Variance Estimator and Job Conversion

| Route                                                                       | Share | One-sided 95% upper bound |          |      |          | One-sided $p$ , share of 100% or more |          |       |          |
|-----------------------------------------------------------------------------|-------|---------------------------|----------|------|----------|---------------------------------------|----------|-------|----------|
|                                                                             |       | Uniform                   | Bartlett | City | Province | Uniform                               | Bartlett | City  | Province |
| Workers, every one in a modern job (headline)                               | 24%   | 82%                       | 79%      | 86%  | 93%      | 0.012                                 | 0.009    | 0.020 | 0.033    |
| Workers, modern rate of migrants, employment-weighted                       | 16%   | 54%                       | 52%      | 56%  | 61%      | <0.001                                | <0.001   | 0.001 | 0.004    |
| Workers, modern rate of migrants, unweighted                                | 15%   | 51%                       | 49%      | 54%  | 58%      | <0.001                                | <0.001   | 0.001 | 0.004    |
| All workers with an industry code, every one modern                         | 24%   | 84%                       | 81%      | 87%  | 95%      | 0.014                                 | 0.010    | 0.022 | 0.037    |
| Comparison city at the mean shares                                          | 27%   | 94%                       | 90%      | 98%  | 106%     | 0.033                                 | 0.025    | 0.045 | 0.067    |
| Share route, at the mean shares                                             | 30%   | 95%                       | 92%      | 98%  | 106%     | 0.033                                 | 0.026    | 0.043 | 0.072    |
| Residents aged 15 and over at the city rate of employment, every one modern | 52%   | 104%                      | 104%     | 110% | 118%     | 0.067                                 | 0.066    | 0.088 | 0.123    |
| Residents aged 15 and over, every one a worker in a modern job              | 92%   | 184%                      | 185%     | 194% | 210%     | 0.459                                 | 0.457    | 0.460 | 0.468    |

*Notes:* The main city sample, with the main specification controls: the settler count and 16 agro-climatic covariates within 300 minutes, and four 1980 baselines. Each row is the share of the modern-job gain that the people added could fill if nobody already working in the city changed sector, its one-sided 95% upper bound from inverting a one-sided test on the joint covariance of the two coefficients, and the one-sided  $p$ -value of a share of 100% or more. The headline row takes columns 2 and 6 of [Table 3](#) at their mean levels with every worker added in a modern job; the next two use the rate at which migrants, workers born outside their district of residence, work in manufacturing or services; the fourth counts all workers with an industry code; the fifth and sixth put the comparison city at the mean shares, the fifth with the coefficient on log modern employment and the sixth with the coefficient on the modern share; the last two count residents. Variances: Conley SE within 200km with a uniform kernel (the paper's) and a Bartlett kernel, and errors clustered by city and by province, all on one stacked regression of the main specification. Source: 1980 and 2010 population censuses.

aged 15 and over grow by more than workers and only the working residents among them fill jobs. At the city's 0.563 workers per resident aged 15 and over, the residents added fill 52% of the gain, with a bound of 104%. The accounting counts workers because a job is filled by a worker.

**Two moments.** City size and the employment shares are two moments, and one inflow has to fit both. With net employment growth equated to arrivals and the incumbent workforce fixed, arrivals imply both the agricultural share and the modern share of the more exposed city. Each implied share is a restriction that holds when the estimate is consistent with it, and the one-sided Wald test of the two restrictions together, whose null distribution is a mixture of chi-squared distributions with weights set by the correlation of the two restrictions ([Kodde & Palm 1986](#); [Wolak 1989](#)), rejects the hypothesis that arrivals with no switching produced both moments at 1% under each of the four variances, with  $p$  at most 0.002 (under province clusters). Testing the two shares jointly removes the concern that a one-moment test picks the share that happens to reject.

**The design-based specifications.** The bound needs precise coefficients, and the design-based specifications of [Appendix C.1](#) give up precision by construction, since they add the exposure a city would expect under counterfactual settler assignments as a control, or subtract it from the treatment, and either way less of the exposure's variation is left to estimate the two responses. [Table E10](#) repeats the two workers rows under all four. When every additional worker takes a modern-sector job, only the main specification rejects. With the whole-match expectation as a control the workers added could fill 35% of the gain and the bound passes 100% (133%,  $p = 0.167$ ); at the migrants' rate the test still rejects at 5% under three of the four variances ( $p = 0.021$  under the uniform kernel, bound 87%). With the recentered whole-match treatment the share is 52% with no finite upper bound ( $p = 0.359$ ). In the two origin specifications log modern-sector employment falls, so there is no gain to fill, and the table prints the coefficient in place of a share. These rows show

Table E10: The Bound in Workers Under the Design-Based Specifications

| Specification                                                          | Share                                                                    | One-sided 95% upper bound |          |      |          | One-sided $p$ , share of 100% or more |          |       |          |
|------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------|----------|------|----------|---------------------------------------|----------|-------|----------|
|                                                                        |                                                                          | Uniform                   | Bartlett | City | Province | Uniform                               | Bartlett | City  | Province |
| <i>Panel A. Every worker added takes a modern job</i>                  |                                                                          |                           |          |      |          |                                       |          |       |          |
| Main specification                                                     | 24%                                                                      | 82%                       | 79%      | 86%  | 93%      | 0.012                                 | 0.009    | 0.020 | 0.033    |
| Double-robust, whole-match expectation                                 | 35%                                                                      | 133%                      | 137%     | none | none     | 0.167                                 | 0.162    | 0.157 | 0.256    |
| Double-robust, origin expectation                                      | <i>log modern employment falls (−0.097), so there is no gain to fill</i> |                           |          |      |          |                                       |          |       |          |
| Recentered, whole-match                                                | 52%                                                                      | none                      | none     | none | none     | 0.359                                 | 0.359    | 0.353 | 0.410    |
| Recentered, origin                                                     | <i>log modern employment falls (−0.012), so there is no gain to fill</i> |                           |          |      |          |                                       |          |       |          |
| <i>Panel B. Workers added take modern jobs as often as migrants do</i> |                                                                          |                           |          |      |          |                                       |          |       |          |
| Main specification                                                     | 16%                                                                      | 54%                       | 52%      | 56%  | 61%      | <0.001                                | <0.001   | 0.001 | 0.004    |
| Double-robust, whole-match expectation                                 | 23%                                                                      | 87%                       | 89%      | none | none     | 0.021                                 | 0.029    | 0.032 | 0.088    |
| Double-robust, origin expectation                                      | <i>log modern employment falls (−0.097), so there is no gain to fill</i> |                           |          |      |          |                                       |          |       |          |
| Recentered, whole-match                                                | 34%                                                                      | none                      | none     | none | none     | 0.167                                 | 0.209    | 0.207 | 0.291    |
| Recentered, origin                                                     | <i>log modern employment falls (−0.012), so there is no gain to fill</i> |                           |          |      |          |                                       |          |       |          |

*Notes:* The main city sample, with the main specification controls: the settler count and 16 agro-climatic covariates within 300 minutes, and four 1980 baselines. Rows and columns as in Table E9, for the two routes in workers, under five specifications: the main specification; the two double-robust specifications of Appendix C.1, which add as a control the exposure a city would expect under counterfactual settler assignments, from the whole-match reshuffle or the origin re-draw; and the two recentered ones, which subtract that expected exposure from the treatment. The mean levels of employment and modern employment are the main specification's in every row. A cell that prints none has no finite upper bound. Source: 1980 and 2010 population censuses.

changes in both estimates and precision; the point estimate of the share stays below the gain in every specification that has one.

**Birthplace and age composition.** The identity treats the extra workers as people who moved in. Transmigrant placement ran through the early 1980s, so a child born to an incumbent in that decade is a prime-age worker by 2010, and a fertility response would place locally born workers in the inflow. Table E11 reports statistically insignificant responses. Splitting 2010 employment by birthplace, native-born employment does not rise detectably, and the point estimate is concentrated among workers born elsewhere, whose employment rises by 20.8% while the migrant share of the workforce rises by 3.6 percentage points; neither is distinguishable from zero. The responses of children and youths per native-born adult are also statistically insignificant. These proxies do not rule out fertility responses or labor-force entry and exit. Both ratios are computed inside the native-born population, so adult in-migrants inflate neither numerator nor denominator.

**Robustness.** I repeat the test under the assumptions that could overturn it. Two are substantive. Exposure may shift the arrivals' own sector mix, which would lower  $a$  and with it the requirement; the test rejects at the high-exposure city's own value. Exposure is also imperfectly balanced on the 1980 agricultural share, so conditioning on that share turns  $\Delta$  into a 1980-to-2010 change in the share; the test still rejects (Table E12).

**Scope of the bound.** The bound asks whether the *arrivals'* composition could produce the fall, holding the incumbent workforce fixed. The birthplace decomposition of Appendix E.3 measures a wider object, because its composition component also allows the locally born share of the workforce to fall. The two quantities are therefore not nested, and both are small: the bound puts the arrivals-only channel at 4.9% of the fall using the lifetime stock and 7.6% using five-year movers, against 16% measured directly.

Table E11: In-Migration and Births as Sources of the Extra Workers

| Outcome, 2010                                                         | Coefficient | SE     | <i>t</i> |
|-----------------------------------------------------------------------|-------------|--------|----------|
| <i>Panel A. Where the employment response comes from</i>              |             |        |          |
| Log employment, all workers                                           | 0.0248      | 0.0582 | 0.43     |
| Log employment, born in this district                                 | -0.0369     | 0.0375 | -0.99    |
| Log employment, born elsewhere                                        | 0.1890      | 0.1307 | 1.45     |
| Migrant share of employment                                           | 0.0359      | 0.0260 | 1.38     |
| <i>Panel B. Fertility, measured inside the native-born population</i> |             |        |          |
| Children 0–5 per native-born adult 16–50                              | 0.0314      | 0.0253 | 1.24     |
| Youth 5–16 per native-born adult 16–50                                | 0.0299      | 0.0282 | 1.06     |

*Notes:* Each row is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. The columns are the coefficient, its standard error and their ratio. Native-born means born in the district of residence and born elsewhere means born outside it; Panel A counts employed workers, in logs, and the migrant share is the born-elsewhere fraction of employment. Panel B's ratios are computed within the native-born population. SE is the Conley standard error and *t* the coefficient divided by it.

Table E12: The Migration Bound Under Alternative Assumptions

|                                                          | Required inflow | One-sided <i>p</i> |
|----------------------------------------------------------|-----------------|--------------------|
| Headline: 2010 base, lifetime migrant stock, $a = 0.269$ | 93.9%           | 0.034              |
| Five-year movers instead, $a = 0.210$                    | 44.7%           | 0.025              |
| $a$ at the high-exposure city's own value, 0.228         | 53.5%           | 0.026              |
| 1980 agricultural share added as a control               | 95.4%           | 0.001              |
| Both corrections together                                | 54.1%           | 0.001              |
| Conservative 1980 base, pre-treatment $a = 0.400$        | 46.5%           | 0.025              |
| Conservative 1980 base, 2010 $a = 0.210$                 | 17.0%           | 0.065              |

*Notes:* Log employment and the agricultural employment share are each regressed on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. The required inflow is the gross inflow, as a percentage of the workforce, that [Equation 19](#) needs to produce the estimated fall in the agricultural share per standard deviation of agro-similarity when arrivals farm at share  $a$  and every incumbent stays in its sector; the base is the mean agricultural share of the sample in 2010, or in 1980 in the conservative rows.  $a$  is the agricultural share of 2010 workers born outside the district (the lifetime migrant stock) or who moved district within five years (five-year movers), or of 1980 workers born outside the province; the high-exposure city is one standard deviation higher in agro-similarity, and its  $a$  adds the estimated effect of that rise. The one-sided *p*, from the same standard errors, tests the null that the log employment response is at least as large as the requirement; its cell takes one shade at 0.10 or above and deepening shades below 0.10, 0.05 and 0.01.

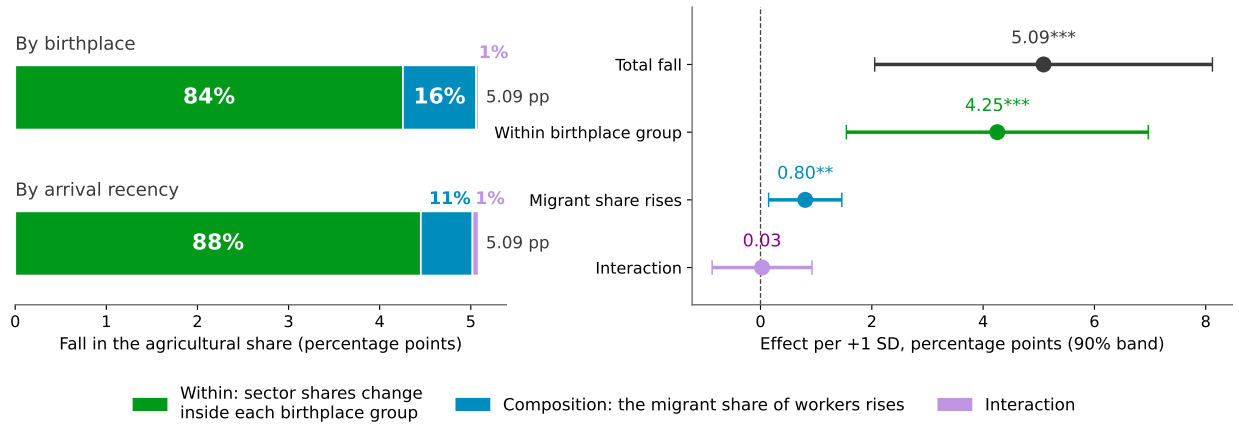
Two limitations remain. The requirement is computed on gross inflow with no offsetting outflow. A small net population response can conceal arrivals of modern-sector workers and departures of agricultural workers, producing sectoral reallocation without incumbent workers changing occupations. The native-employment coefficient in Table E11 does not identify those gross flows. And  $s$  and  $a$  enter as measured constants. They are means over the same 134 cities and are estimated far more precisely than the coefficients, but they contain sampling error that the reported standard errors omit.

### E.3 Decomposing the Agricultural-Share Fall: Construction

*Cited in the main text.* Figure E2: Section 5.2, p. 30.

This appendix gives the construction behind Figure E2: the identity the decomposition relies on, the data it needs, the calculation it replaces, and the checks at each step. The exhibits label the two components by what they measure, the migrant share of the workforce and workers switching sector; in the text I use the literature’s terms, composition and within, for the same two objects.

Figure E2: Decomposition of the Fall in the Urban Agricultural Employment Share, by Origin



*Notes:* The figure uses the main city sample in the 2010 population census; each effect is estimated by a regression across urban centers on standardized agro-similarity with the main specification controls: the settler count and 16 agro-climatic covariates within 300 minutes, and four 1980 baselines. Intervals are 90% confidence intervals from Conley SE within 200km; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Each city’s 2010 workforce is partitioned by birthplace (four groups) and, separately, by how recently a worker arrived (three groups). The stacked bars split the estimated fall in the agricultural employment share (percentage points per standard deviation, each piece labeled with its percentage of the fall) into a within component (“within birthplace group”), a composition component (“migrant share rises”) and their interaction; the right half repeats the total and the three birthplace pieces with their intervals. Because the census is a repeated cross-section, the within component includes cohort replacement.

#### The identity

Index cities by  $c$  and partition each city’s 2010 workforce into origin groups  $g$ . Let  $\omega_{gc}$  be group  $g$ ’s share of city  $c$ ’s workers and  $a_{gc}$  that group’s own agricultural employment share. The city agricultural share is then an exact accounting identity:

$$s_c = \sum_g \omega_{gc} a_{gc}. \quad (20)$$

Write each factor as its sample mean plus a deviation,  $\omega_{gc} = \bar{\omega}_g + u_{gc}$  and  $a_{gc} = \bar{a}_g + v_{gc}$ , and Equation 20 becomes

$$s_c = \underbrace{\sum_g \bar{\omega}_g \bar{a}_g}_{\text{constant}} + \underbrace{\sum_g \bar{a}_g u_{gc}}_{\text{migrant share}} + \underbrace{\sum_g \bar{\omega}_g v_{gc}}_{\text{within origin}} + \underbrace{\sum_g u_{gc} v_{gc}}_{\text{interaction}}. \quad (21)$$

The first term is the *composition* component: it moves only because the local-versus-migrant makeup of the workforce changes, holding what each group does fixed. The second is the *within* component. It moves only because groups change what they do, holding that makeup fixed. I construct the three right-hand terms as variables and regress each on the *same* main specification used for the total. Because Equation 21 is an identity and the regressions share a design matrix, the three coefficients sum to the coefficient on  $s_c$  exactly, so no residual category absorbs the difference. The adding-up holds numerically. The sum of the three minus the total is zero to ten decimal places under both partitions. Each component nonetheless has its own Conley standard error, so the split reports precision as well as magnitude.

Two partitions are used. The first is birthplace, in the four mutually exclusive groups the migration tables already use: born in the current district, born in a different district of the same province, born in a different outer-island province, and born on Java or Bali. The second is arrival recency, in three groups: never left the birth district, lifetime migrant already present five years before the census, and lifetime migrant who arrived within five years. The second is the sharper measure of the marginal inflow, since the birthplace groups pool arrivals from thirty years ago with last year's.

The interpretation comes with one caveat. The census is a repeated cross-section, so “born in this district” in 2010 is not the same list of people as in 1980. The within term measures sector composition among workers of a given origin, including whoever replaced whoever left. The within term therefore holds both sector changes inside a cohort and the replacement of one cohort by another; Budí-Ors (2026) puts cohort replacement at 68% of Indonesia's aggregate agricultural-share fall.

## The data the identity needs

Evaluating Equation 20 requires each city's workers classified by origin *and* industry in the same data. The published city file contains the birthplace shares and, quite separately, the industry counts, but nothing crossing them, so the cross is built from the 2010 individual census records for the outer islands (93.5 million persons). Three definitional choices align it with the published outcome:

- The employment base is every person with a valid industry code (40.1 million workers), which is the denominator every sectoral share in this paper uses, rather than the stricter economically active count. Using the strict base would produce shares that do not sum to the published ones
- Agriculture is industry codes 1 through 6, exactly as the census cleaning defines the sectoral dummies elsewhere in the paper
- No birth-cohort restriction is imposed. The migration tables restrict to the 1955-1990 cohort in order to match 1980 birth-district characteristics; imposing that here would place the cross on a different population than the published agricultural share and break the reconciliation below. Birthplace is recorded for 100% of the workers in the base, so no group has a thin denominator

The person-level counts are collapsed to the 2010 village, crosswalked to the year-2000 subdistrict geography the rest of the paper uses, and rolled up to urban centers over the reclassified membership. Both partitions recover the total workforce exactly in every one of the 1,999 subdistricts, and Equation 20 holds city by city to within  $2.2 \times 10^{-16}$ .

The re-measured share is checked against the published one before anything is split. Its mean across the estimation sample matches to three decimals (0.374 against 0.374), the two measures correlate at 1.000, and the main specification coefficient on the re-measured share,  $-0.0509$  (standard error 0.0184), reproduces the published  $-0.0509$  (0.0184) on the same cities.

## The required composition of arrivals

The question can be pushed some way without measuring the arrivals' sector at all. Suppose every worker the exposure adds to a city works outside agriculture, the most generous assumption available to the migration account. With the

incumbent workforce fixed and net employment growth equated to arrivals, a log employment response  $\beta_L$  implies an arrivals share  $1 - \exp(-\beta_L)$  of realized employment, and at a zero agricultural share among them the city share falls by that fraction times the baseline share. At the estimated  $\beta_L = 0.023$  and a sample mean agricultural share of 0.374 this ceiling is 0.9 percentage points, 17.1% of the estimated 5.1-point fall. An alternative version replaces the employment response with the estimated 3.5 percentage-point rise in the share of workers born outside the district and multiplies by the locally born agricultural share of 0.404, giving 1.4 points, or 28.1%.

Both point-estimate ceilings sit below the fall: under the stated assumptions, arrivals account for 28.1% of it at most. Figure E3a reports the two ceilings, the ceiling at the top of the employment response's 90% range, and the measured composition component.

The composition channel is present but small. Migrants are markedly less agricultural than the locally born, 20 to 30% against 41%, which makes a migrant-composition account plausible before it is measured. It stays small because the heavily agricultural group is two-thirds of the workforce and exposure raises the migrant share by only a few percentage points, which cannot produce the estimated 3.1-point fall (Appendix Figure E3b).

## Estimates and robustness

Dropping the five cities that draw part of their workforce from a subdistrict shared with a neighbor (below) moves the within share from 74.3% to 72.8% at  $N = 123$ . Appendix Table E13 reports every component under both partitions, and Appendix Table E14 collects the responses, baselines, ceilings, and measured components in one place.

Table E13: Decomposition of the Fall in the Urban Agricultural Employment Share, by Birthplace and Arrival Recency

|                        | By birthplace          |                       |                             |                     | By arrival recency   |                             |                     |
|------------------------|------------------------|-----------------------|-----------------------------|---------------------|----------------------|-----------------------------|---------------------|
|                        | (1)<br><b>Total</b>    | (2)<br>Migrant share  | (3)<br><b>Within origin</b> | (4)<br>Interaction  | (5)<br>Migrant share | (6)<br><b>Within origin</b> | (7)<br>Interaction  |
| Agro-similarity        | -0.0509***<br>(0.0184) | -0.0080**<br>(0.0040) | -0.0425***<br>(0.0165)      | -0.0003<br>(0.0055) | -0.0057<br>(0.0035)  | -0.0445***<br>(0.0171)      | -0.0007<br>(0.0044) |
| Settler count          | ✓                      | ✓                     | ✓                           | ✓                   | ✓                    | ✓                           | ✓                   |
| Baseline Pop 1980      | ✓                      | ✓                     | ✓                           | ✓                   | ✓                    | ✓                           | ✓                   |
| Urban share 1980       | ✓                      | ✓                     | ✓                           | ✓                   | ✓                    | ✓                           | ✓                   |
| Total Empl 1980        | ✓                      | ✓                     | ✓                           | ✓                   | ✓                    | ✓                           | ✓                   |
| Agri Empl 1980         | ✓                      | ✓                     | ✓                           | ✓                   | ✓                    | ✓                           | ✓                   |
| Agro-climatic controls | ✓                      | ✓                     | ✓                           | ✓                   | ✓                    | ✓                           | ✓                   |
| Observations           | 134                    | 134                   | 134                         | 134                 | 134                  | 134                         | 134                 |

*Notes:* Each column is one regression on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Column 1 is each city's 2010 agricultural employment share in the 2010 population census. Columns 2 to 4 split it exactly over four birthplace groups (born in the district, elsewhere in the province, in another outer-island province, and on Java or Bali): Migrant share, the composition component, lets only the birthplace makeup of the workforce vary; Within origin, the within component, lets only each group's own agricultural share vary; Interaction is the remainder. Columns 5 to 7 repeat the split over arrival recency: workers who never left their birth district, migrants present five years before the census, and migrants who arrived within five years. Because the census is a repeated cross-section, the within-origin component includes cohort replacement. Agro-climatic controls are the 16 village covariates. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

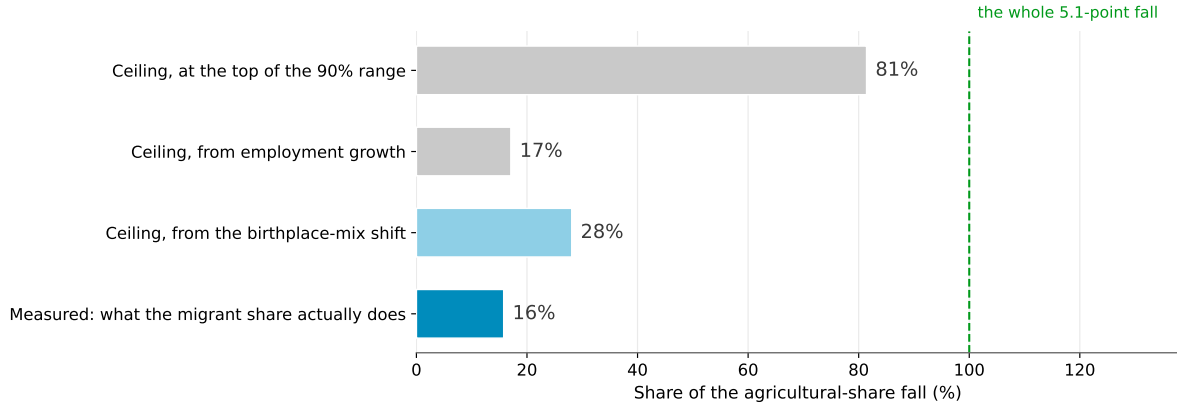
## Subdistricts claimed by two urban centers

Urban centers are satellite-drawn built-up polygons while the census reports by subdistrict, and five outer-island subdistricts are large enough to contain two separate built-up areas. The sample records both claims, and city totals built by summing constituent subdistricts therefore give the whole subdistrict to each of the two claimants, so those workers enter two cities. Across the estimation sample this affects 95,728 workers, or 0.78% of the total, and both claimants lie inside the sample in two of the five cases.

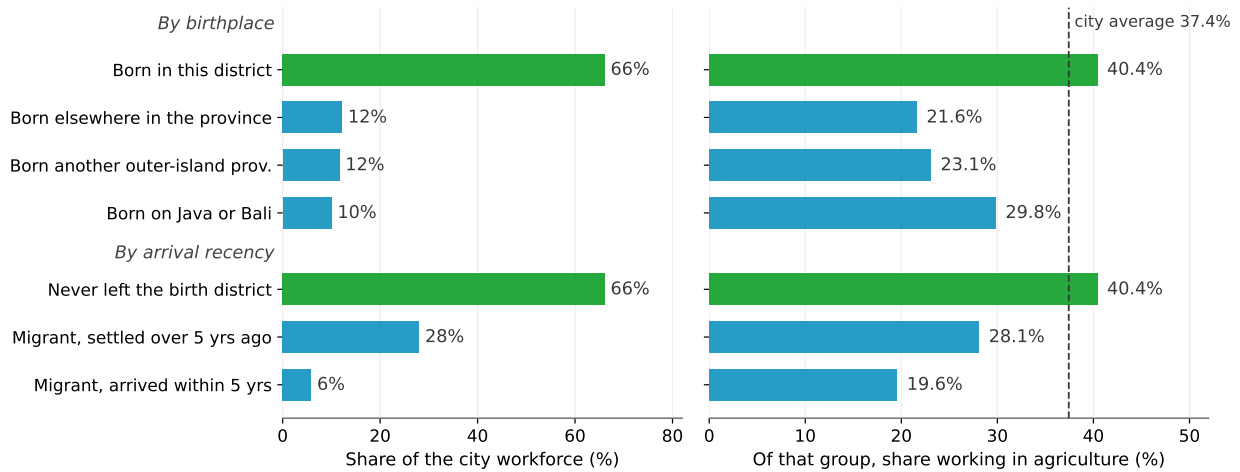
The exhibits here divide such a subdistrict. Each of its villages is assigned to the claimant whose built-up polygon covers it, or, for a village inside neither, to the nearer claimant; the subdistrict's counts are then split in proportion to the

Figure E3: The Migration Ceilings and the Size and Agricultural Share of Each Origin Group

(a) The Migration Ceilings Against the Measured Composition Component



(b) Origin Groups: Share of the Workforce and Agricultural Share



*Notes:* Both panels use the main city sample and the 2010 population census; the estimates are regressions on standardized agro-similarity with the main specification controls: the settler count and 16 agro-climatic covariates within 300 minutes, and four 1980 baselines. Panel (a): the three ceilings ask how much of the estimated agricultural-share fall migration could produce if every arrival worked outside agriculture, so they are maxima under that assumption. Each bar is a percentage of the estimated fall in the agricultural employment share per standard deviation of agro-similarity, and the dashed line at 100% is the whole fall. The employment-growth ceiling multiplies the inflow implied by the estimated log employment response by the mean 2010 agricultural share; the ceiling at the top of the 90% range uses the upper end of that response's 90% confidence interval from Conley SE within 200km; the birthplace-mix ceiling multiplies the estimated rise in the share of workers born outside the district by the agricultural share of locally born workers. The bottom bar is the measured composition component of Figure E2, which replaces that assumption with the arrivals' observed agricultural share. Panel (b): unweighted means across cities of each origin group's share of the workforce (left) and of its own agricultural employment share (right), with the unweighted mean city agricultural share as the dashed line. The two blocks are two partitions of the same workforce, so the shares within each block sum to one. Every entry is a mean across cities, so the group shares and the group agricultural shares do not multiply out to the city average; the gap is the covariance across cities between a group's share of the workforce and its agricultural share, summed over the groups.

Table E14: Migrant Composition as a Bound on the Agricultural-Share Fall

|                                                                                        | Estimate               | Share of the fall |
|----------------------------------------------------------------------------------------|------------------------|-------------------|
| <i>Panel A. Estimated responses per +1 SD of agro-similarity</i>                       |                        |                   |
| Agricultural employment share                                                          | -0.0509***<br>(0.0184) |                   |
| Conley 200 km p-value                                                                  | 0.006                  |                   |
| Log employment                                                                         | 0.0235<br>(0.0570)     |                   |
| Conley 200 km p-value                                                                  | 0.680                  |                   |
| Share of workers born outside the district                                             | 0.0354<br>(0.0257)     |                   |
| Conley 200 km p-value                                                                  | 0.170                  |                   |
| <i>Panel B. Baseline agricultural shares, sample means</i>                             |                        |                   |
| All workers, 2010                                                                      | 0.374                  |                   |
| All workers, 1980                                                                      | 0.560                  |                   |
| Workers born in the district, 2010                                                     | 0.404                  |                   |
| Workers who arrived within 5 years, 2010                                               | 0.196                  |                   |
| <i>Panel C. Maximum composition effect, arrivals assumed entirely non-agricultural</i> |                        |                   |
| Inflow based, 2010 baseline                                                            | 0.0087                 | 17.1%             |
| at the 90% upper end                                                                   | 0.0414                 | 81.4%             |
| on the 1980 baseline                                                                   | 0.0130                 | 25.6%             |
| Composition based, 2010 baseline                                                       | 0.0143                 | 28.1%             |
| at the 90% upper end                                                                   | 0.0314                 | 61.8%             |
| <i>Panel D. Measured, replacing the assumption with the observed arrival mix</i>       |                        |                   |
| Migrant share, by birthplace                                                           | -0.0080**              | 15.8%             |
| Within origin, by birthplace                                                           | -0.0425***             | 83.7%             |
| Migrant share, by arrival recency                                                      | -0.0057                | 11.1%             |
| Within origin, by arrival recency                                                      | -0.0445***             | 87.5%             |

Notes: Panels A and D report regressions on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses, with the p-value beneath each, shaded by significance level. Workers are counted in the 2010 population census, and in the 1980 census for the rows marked 1980; Panel B gives sample means. Panel C bounds the composition effect by assuming every arrival works outside agriculture: the inflow-based bound multiplies the implied employment inflow, one minus the exponential of minus the log employment response, by the baseline agricultural share; the composition-based bound multiplies the estimated rise in the share of workers born outside the district by the agricultural share of locally born workers; the upper-end rows use the upper end of the response's 90% confidence interval; the larger bound, here the composition based one, is the ceiling. Panel D replaces the assumption with the arrivals' measured agricultural share, from Table E13. Share of the fall divides each Panel C bound by the Panel A fall in the agricultural employment share, and each Panel D component by the total it decomposes. SD is one standard deviation of agro-similarity, and Log employment is the log of the city's 2010 employment. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

employment in each city's villages. Total mass is preserved by construction and checked against the village-level source, so no worker is counted twice and none is dropped. The correction is immaterial for the result, moving the within share from 74.0% to 74.2%, and the answer is equally insensitive to the splitting rule: dividing a subdistrict's rural remainder in the same proportion as its built-up cores, rather than by proximity, gives the same 74.2%.

## E.4 Public Provision as an Alternative Account

*Cited in the main text. This section: [Section 5.4](#), p. 35.*

The paper argues that agricultural productivity raised nearby city growth and reshaped urban employment through the goods market. A reader can propose a different chain. The Indonesian state settled these villages, and the same state built the roads, schools, health posts and electricity lines that serve them. If public provision followed the settlements into the places where they turned out productive, public provision could contribute to the urban response. This subsection examines road quality, village facilities, and the allocation of the INPRES program.

A facility stock is the outcome of many decisions. A grid line can follow demand that is already there, so these tests cannot tell the state's supply apart from household demand. The INPRES test below examines a particular central-government program's allocation.

**Roads.** The trade channel runs on the road network, so the sharpest form of the objection is that exposed corridors received better roads. I test this with the Indonesian Road Management System panel assembled by [Gertler et al. \(2024\)](#), which gives road-quality-implied travel speeds on a fixed network in 1990 and 2007 for Sumatra and Sulawesi, the two outer islands the source covers. Exposed rural subdistricts are more remote and slightly more slowly connected at both dates, sitting 62.8 km farther out in the road network per standard deviation of exposure (standard error 21.7) and traveling 1.02 km/h slower in 1990 (standard error 0.42) and 0.89 km/h slower in 2007 (standard error 0.49), against a base of roughly 60 to 66 km/h. The change is where the alternative would have to show itself, and it is flat. Over 1990 to 2007 the differential in road speed is 0.13 km/h with a standard error of 0.59 across 533 subdistricts, or 0.02 standard deviations with a standard error of 0.08 in fully standardized form. The estimate does not show an improvement in road quality on these surveyed corridors.

That null has two limits. Network distance is identical to the last digit across 1990 and 2007 for every Sumatran and Sulawesi district in the source, so the measure re-weights a fixed graph by road quality and cannot detect a road that was newly built. It covers Sumatra and Sulawesi. The near-baseline evidence closes part of that gap, because the 1983 village census reports local infrastructure for every outer island a few years after settlement, and it is null on each outcome in fully standardized form: roads passable by a four-wheel vehicle at 0.087 (standard error 0.067), paved roads at 0.063 (standard error 0.059), and distance to the subdistrict seat at 0.054 (standard error 0.075), beside a placebo on 1980 population density at 0.033 (standard error 0.130). These estimates do not detect an early connectivity advantage; they do not exclude smaller differences.

**Facilities.** I track what the state puts inside a village, city amenities from the 1980 village census to the 2003 and 2011 waves. Column (2) regresses the 1980 level itself on exposure, so the table says for itself whether public provision already differed before the settlements matured, and columns (3) and (4) give the change over each window ([Appendix Table E15](#)).

Two facts come out of it. The 1980 levels differ in three of the seven government-provided amenities. Senior-secondary schools and polyclinics show a 1980 gradient, at 0.0149 (standard error 0.0042) and 0.0151 (standard error 0.0051) per 1,000 people and per standard deviation of exposure, against means of 0.072 and 0.073, and the population share on a road passable by a four-wheel vehicle is higher by 5.2 percentage points (standard error 3.0), marginally significant, against a mean of 80%. So the exposed cities entered the sample period with slightly more of two facilities the state provides. And the changes are null on every one of those seven amenities, over both windows, in schooling, in health facilities and in both road-surface measures.

Adding each outcome's own 1980 level, the test Panel D reports for electricity, was run on all seven and leaves that conclusion intact. One coefficient moves. Polyclinics turn positive by 2011, at 0.0139 (standard error 0.0060) against a 1980 mean of 0.073. A facility stock does not identify its financing.

The government-provided block reports 21 coefficients, and the three that cross a conventional threshold are all 1980 levels. The 14 change coefficients are all statistically insignificant, and with the 1980 level held only the polyclinic change to 2011 is statistically significant.

**Electricity.** Grid connection is where the change regressions do move, and the two margins of it separate cleanly once the 1980 level is in the table. On the extensive margin, the share of a city's population living in a village on the state grid falls by 6.4 percentage points per standard deviation of exposure over 1980 to 2003 and by 7.2 points over 1980 to 2011, both statistically significant. That is the less exposed cities catching up. Agriculturally favored places began 10.9 points more connected, significant at the 1% level, so a change regression with no lagged outcome was measuring convergence, and adding the 1980 level, in Panel D, removes the result in both windows, leaving  $-0.1$  and  $-0.4$  points. The 1980 measure is the narrower of the two. The 1980 form records only a village's main source of electricity, so a 1980 village counts as on the grid only where the state utility is its main supplier, while a 2003 or 2011 village counts where any household takes grid electricity.

The intensive margin behaves differently and is the only electricity result that survives the 1980 level. The share of households connected also starts higher in favored places, at 5.8 percentage points per standard deviation with a standard error of 1.8, and the plain changes are small and imprecise, 1.1 points (standard error 2.4) by 2003 and 1.9 points (standard error 2.1) by 2011. With the 1980 level added the changes grow to 3.4 points by 2003, statistically insignificant, and 4.5 points by 2011, marginally significant. So the pattern is a modest deepening of connections inside places that were already connected, on a base of about 18% of households connected in 1980.

Electrification reflects both the state's supply and household demand. The grid is a state utility, while the decision to connect a household responds to demand and ability to pay. The connection estimates do not distinguish those explanations.

**The allocation rule itself.** A facility stock is the outcome of many decisions, so the sharpest available test examines a central-government program directly. Indonesia's INPRES primary-school construction program ran from 1973 to 1978 and had finished before the 1980 village census was taken (Duflo 2001). That census records INPRES-built schools, INPRES-paid teachers, INPRES-built school buildings and INPRES-built public latrines separately from everything else, so the program is identified by its own label, and the question becomes whether the center's flagship rural program had already favored agriculturally advantaged places by the start of my sample period.

Table E16 reports six versions of the test; each estimate is statistically insignificant. The INPRES share of a city's primary schools changes by  $-0.0064$  per standard deviation of exposure (standard error 0.0129) against a mean of 0.336, the INPRES share of primary-school teachers by  $-0.0047$  (standard error 0.0115) against a mean of 0.323, and the INPRES share of school buildings by  $-0.0040$  (standard error 0.0127) against a mean of 0.338, three separate questionnaire items that agree both in their null and in placing about a third of primary schooling under the program. Public latrines give the cleanest form of the contrast, since INPRES and non-INPRES latrines are counted separately within one facility type in one year and one place, and the INPRES share there is 0.0206 (standard error 0.0592) against a mean of 0.401. The placebo is null, with non-INPRES schools per 1,000 people at  $-0.0023$  (standard error 0.0658). One qualification: 13 of the 133 cities have no public latrine in 1980, so the latrine column covers 120 cities.

The road evidence covers road quality on a fixed network in Sumatra and Sulawesi, and the facility evidence covers stocks and connections. These tests do not identify their financing or rule out other public programs. The countryside is where public provision did respond (Appendix E.5). Rural public investment remains a possible explanation through food production and trade.

Table E15: Public Provision and Agricultural Productivity: City Amenities from the 1980 Baseline

|                                                                                             | 1980 baseline       |                               | Change from the 1980 baseline |                        |
|---------------------------------------------------------------------------------------------|---------------------|-------------------------------|-------------------------------|------------------------|
|                                                                                             | Mean in 1980<br>(1) | 1980 level on exposure<br>(2) | 1980 to 2003<br>(3)           | 1980 to 2011<br>(4)    |
| <i>Panel A. Government-provided: schooling and health, facilities per 1,000 people</i>      |                     |                               |                               |                        |
| Primary schools                                                                             | 1.159               | -0.0150<br>(0.0787)           | -0.0213<br>(0.0360)           | -0.0012<br>(0.0439)    |
| <i>p-value</i>                                                                              |                     | 0.849                         | 0.554                         | 0.978                  |
| Junior-secondary schools                                                                    | 0.144               | 0.0053<br>(0.0110)            | -0.0102<br>(0.0081)           | -0.0087<br>(0.0076)    |
| <i>p-value</i>                                                                              |                     | 0.628                         | 0.210                         | 0.252                  |
| Senior-secondary schools                                                                    | 0.072               | 0.0149***<br>(0.0042)         | -0.0071<br>(0.0071)           | -0.0037<br>(0.0057)    |
| <i>p-value</i>                                                                              |                     | 0.000                         | 0.314                         | 0.510                  |
| Health centers (puskesmas)                                                                  | 0.081               | -0.0001<br>(0.0111)           | -0.0001<br>(0.0105)           | 0.0014<br>(0.0091)     |
| <i>p-value</i>                                                                              |                     | 0.993                         | 0.992                         | 0.877                  |
| Polyclinics                                                                                 | 0.073               | 0.0151***<br>(0.0051)         | -0.0086<br>(0.0067)           | -0.0059<br>(0.0071)    |
| <i>p-value</i>                                                                              |                     | 0.003                         | 0.205                         | 0.405                  |
| <i>Panel B. Government-provided: road surface, share of population</i>                      |                     |                               |                               |                        |
| Asphalt main road                                                                           | 0.435               | -0.0242<br>(0.0495)           | 0.0060<br>(0.0349)            | 0.0543<br>(0.0427)     |
| <i>p-value</i>                                                                              |                     | 0.624                         | 0.863                         | 0.203                  |
| Passable by four-wheel vehicle                                                              | 0.797               | 0.0516*<br>(0.0300)           | -0.0286<br>(0.0226)           | -0.0181<br>(0.0294)    |
| <i>p-value</i>                                                                              |                     | 0.085                         | 0.205                         | 0.538                  |
| <i>Panel C. Mixed provision: grid electricity</i>                                           |                     |                               |                               |                        |
| Village connected to the grid                                                               | 0.353               | 0.1091***<br>(0.0347)         | -0.0640***<br>(0.0223)        | -0.0720***<br>(0.0224) |
| <i>p-value</i>                                                                              |                     | 0.002                         | 0.004                         | 0.001                  |
| Households connected, share of households                                                   | 0.175               | 0.0575***<br>(0.0181)         | 0.0109<br>(0.0238)            | 0.0188<br>(0.0208)     |
| <i>p-value</i>                                                                              |                     | 0.002                         | 0.648                         | 0.367                  |
| <i>Panel D. Convergence check: the same electricity changes, controlling the 1980 level</i> |                     |                               |                               |                        |
| Village connected to the grid                                                               |                     |                               | -0.0005<br>(0.0128)           | -0.0040<br>(0.0128)    |
| <i>p-value</i>                                                                              |                     |                               | 0.968                         | 0.756                  |
| Households connected, share of households                                                   |                     |                               | 0.0339<br>(0.0251)            | 0.0449*<br>(0.0233)    |
| <i>p-value</i>                                                                              |                     |                               | 0.177                         | 0.054                  |
| Observations                                                                                |                     | 133                           | 133                           | 133                    |

Notes: Column (1) is the mean 1980 level of each amenity. Each cell in columns (2) to (4) is one regression of the outcome on standardized agro-similarity on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), restricted to the centers the village census (PODES) covers in 1980 and in the endline year and counted in the Observations row, in the main specification with the full set of controls and Conley SE within 200km, except that column (2) leaves out the four 1980 baselines because its outcome is a 1980 value. Standard errors are in parentheses, with the p-value beneath, shaded by band (0.10 or more, below 0.10, below 0.05, below 0.01). \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table E15: Public Provision and Agricultural Productivity: City Amenities from the 1980 Baseline, Continued

*Notes, continued:* The outcome, from the village census, is the 1980 level in column (2) and the change from 1980 to 2003 and to 2011 in columns (3) and (4); Panel D adds the 1980 level of the outcome to the two electricity changes. Facility counts are per 1,000 people at the population of each year, and village connection to the grid is the share of city population living in villages on the state electricity grid (in 1980, villages whose main source of electricity is the grid). Puskesmas are the subdistrict public health centers.

Table E16: Central Government Programs at the 1980 Baseline

|                 | INPRES primary schooling |                                 |                          |                           | INPRES public latrines   | Placebo: the schools INPRES did not build |
|-----------------|--------------------------|---------------------------------|--------------------------|---------------------------|--------------------------|-------------------------------------------|
|                 | Share of schools<br>(1)  | Schools per 1,000 people<br>(2) | Share of teachers<br>(3) | Share of buildings<br>(4) | Share of latrines<br>(5) | Schools per 1,000 people<br>(6)           |
| Agro-similarity | -0.0064<br>(0.0129)      | -0.0098<br>(0.0243)             | -0.0047<br>(0.0115)      | -0.0040<br>(0.0127)       | 0.0206<br>(0.0592)       | -0.0023<br>(0.0658)                       |
| <i>p-value</i>  | 0.623                    | 0.687                           | 0.682                    | 0.752                     | 0.728                    | 0.972                                     |
| Mean of outcome | 0.3355                   | 0.3831                          | 0.3228                   | 0.3384                    | 0.4010                   | 0.7778                                    |
| Observations    | 133                      | 133                             | 133                      | 133                       | 120                      | 133                                       |

*Notes:* Each column is one regression of a 1980 level on standardized agro-similarity on the 133 cities of the main city sample that the 1980 village census (PODES) covers, in the main specification with the full set of controls and Conley SE within 200km, without the four 1980 census baselines, since each outcome is itself a 1980 value; the one main-sample city the 1980 village census does not cover is Pemangkat (West Kalimantan); column (5) uses the cities with a public latrine, counted in the Observations row. Standard errors are in parentheses, with the p-value beneath, its shading stepping up at each star threshold. The presidential-instruction (INPRES) primary-school construction program ran from 1973 to 1978, so the 1980 levels show where it had already built. The 1980 village census records INPRES items separately from the rest. Columns (1), (3), (4) and (5) are INPRES shares of the total, column (2) is INPRES schools per 1,000 people, and column (6), the placebo, is the schools the program did not build per 1,000 people. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## E.5 Tests of the Alternative Channels

*Cited in the main text. This section: [Section 5.4](#), p. 35.*

This subsection holds the tests behind [Section 5.4](#): public provision by area and by rural subdistrict, the state grid wave by wave, banks, new district seats, the spending of rural households, the road network, and three tests on the cities themselves, the conflict provinces, the exposure of neighboring cities and the city boundary. Every regression is the main specification with the full set of controls and Conley SE within 200km, with each outcome's own 1980 level held fixed where one exists; the Bartlett-kernel standard error prints beside the uniform one, and an estimate statistically significant at the 10% level with the uniform kernel and not with the Bartlett kernel, with province clusters, or with both prints with a dagger. The INPRES allocation test and the city facilities table are in [Appendix E.4](#), and the city-boundary test in [Appendix Table D4](#).

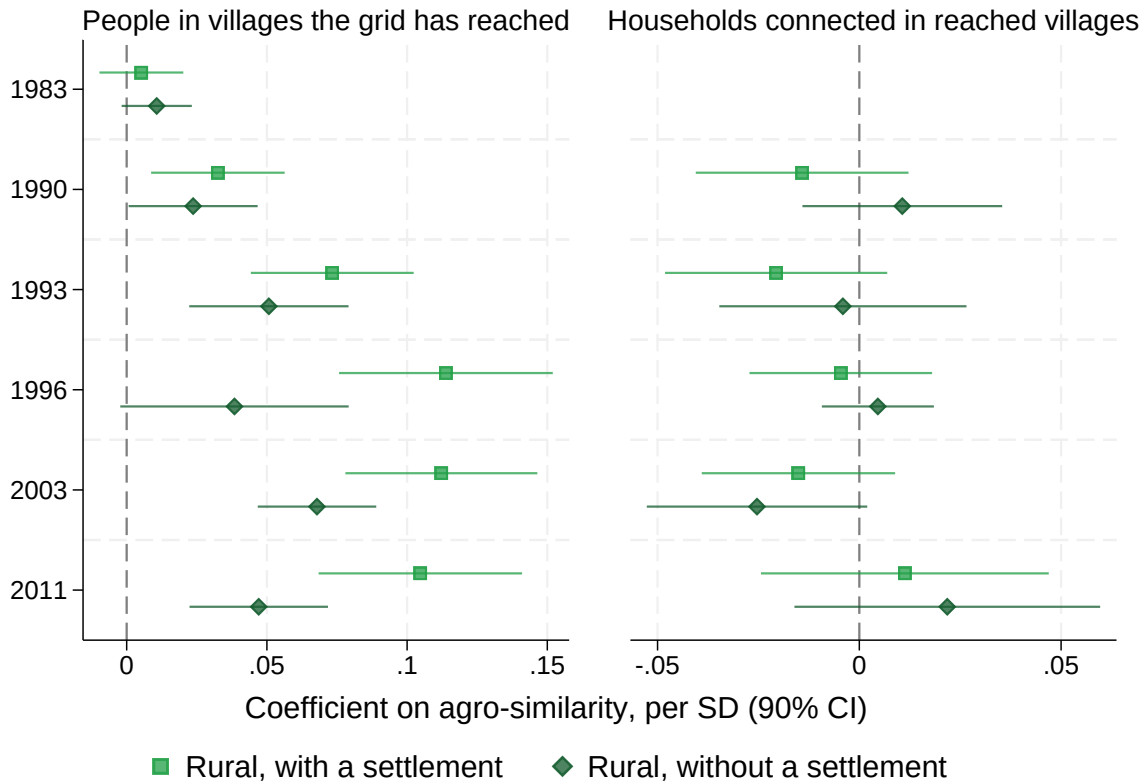
**The state grid.** The state grid was extended to more of the villages in the more exposed rural subdistricts with a settlement from 1990 on, and households in villages already on the grid did not connect at a higher rate. With the 1980 level measured on the state grid at both ends, the share of people in villages on the grid rises with exposure by 0.0325 (standard error 0.0145) in 1990 and by 0.1047 (0.0221) in 2011, while the household share within those villages changes by 0.0113 (0.0217) in 2011. Electricity from any supplier in 2011 does not move with exposure (-0.0001, standard error 0.0010), so part of the grid's rise replaced other suppliers. A grid line can follow demand that is already there, so the test does not separate the state's supply from household demand ([Figure E4](#)).

**Public provision by area.** At a given 1980 level, the city's roads do not respond to exposure, while the countryside with settlements and the countryside without settlements both gain asphalt roads by 1990 and grid connections by 2011, and senior secondary schools rise only in the countryside with settlements. The city's asphalt share changes by 0.0271 (standard error 0.0172) to 1990 and 0.0213 (0.0302) to 2011, and its four-wheel access to 1990, 0.0182 (0.0088), is statistically significant with the uniform kernel only. In the countryside with settlements the asphalt share rises by 0.0590 (0.0207) to 1990, the household grid share by 0.0553 (0.0284) to 2011 and senior secondary schools per 1,000 people by 0.0424 (0.0098); in the countryside without settlements the asphalt share rises by 0.0679 (0.0161) and the household grid share by 0.0792 (0.0194), and senior secondary schools change by 0.0030 (0.0037). Health centers per person do not move in any of the three areas ([Table E17](#)).

**Public provision by rural subdistrict.** With each rural subdistrict as a unit, asphalt roads and household electricity rose with exposure in subdistricts with and without a settlement alike, and the two slopes are statistically indistinguishable: the asphalt share to 1990 rises by 0.0743 with a settlement and 0.0868 without (difference -0.0125, standard error 0.0199), and the household grid share to 2011 by 0.0646 and 0.0520 (difference 0.0126, 0.0172). The senior-school slope is statistically insignificant in both groups, so the senior-school gain of the countryside with settlements shows up in the per-city area only. Four-wheel access to 1990 and the share of people in villages with electricity to 2011 rose more where a settlement sits, the first with the uniform kernel only. In 1980 asphalt roads were already more common in the subdistricts with a settlement, by 0.0474 (0.0184) ([Table E18](#)).

**Banks.** Banks per person and bank presence do not move with exposure in the city or in the countryside with settlements at a given 1980 level. In the city, banks per 1,000 people change by 0.0110 (standard error 0.0092) from 1980 to 2011 and the population share in villages with a bank by 0.0251 (0.0248); in the countryside with settlements, by 0.0119 (0.0072) and -0.0138 (0.0137). The countryside without settlements gains banks per 1,000 people, 0.0077 (0.0026), and its bank-presence gain is statistically significant with the uniform kernel only. With the log of the 1980 count held fixed instead, the city's gain in banks per 1,000 to 1990 is 0.0106 (0.0046) and statistically significant. With rural subdistricts as units, banks per 1,000 to 1990 rose no more where a settlement sits (difference 0.0017, 0.0021). The village censuses count banks and do not record deposits or lending ([Table E19](#)).

Figure E4: Exposure and the State Grid by Village-Census Wave



*Notes:* Each marker is the coefficient on agro-similarity (per SD over the 787 in-sample rural subdistricts) for rural subdistricts with and without a settlement, with its 90% interval from Conley SE within 200km. One regression per wave and panel: the outcome on agro-similarity times the indicators of a subdistrict with and without a settlement, the with-settlement indicator, the 1980 population shares in villages whose main source of electricity is the state grid (PLN) and another supplier (right panel: also the 1980 household shares using each), and the controls of the main specification measured at the subdistrict, with the settler count within 1,000 minutes. Sample: the 787 in-sample rural subdistricts less the 51 West Sumatra subdistricts recorded on nagari in 2011, 736 in all (191 with a settlement, 545 without), less the 71 of West Kalimantan from 1993 on, when the province merged its villages; the 41 in-sample subdistricts that belong to a reclassified city are left out. Left: the share of the subdistrict's people in villages with a household on PLN. Right: households on PLN over all households of those villages, from 1990. Electricity from any supplier in 2011, with a settlement: -0.0001 (standard error 0.0010). Source: village censuses (PODES) 1980 to 2011 on year-2000 subdistricts; population census 1980.

**New district seats.** Exposure does not predict the seat of a new district. The probability of hosting the seat of a district created 1999 to 2010 changes by -0.006 (standard error 0.052) per standard deviation of exposure, and that of hosting a new kota, the urban type of district, by 0.057 (0.030), statistically significant with the uniform kernel only. Without the 33 cities that host a new seat, the agricultural employment share changes by -0.053 (0.017) and log modern-sector employment by 0.172 (0.064), as on the main city sample. Log modern-sector employment without education, health, social and government services rises by 0.165 (0.071), statistically significant under both Conley kernels and marginally insignificant with province clusters; the industry codes left out hold public and private employers alike, so the cut does not separate public from private employment (Table E20).

**Spending of rural households.** Rural households around more exposed cities do not spend more per person, and the non-food share of their spending does not move. In 2011, log spending per person changes by -0.0306 (standard error 0.0273) in the whole hinterland and by -0.0225 (0.0393) in the countryside with settlements, and the non-food share

Table E17: Public Provision in the City and Its Countryside

| 1980 control:                                      | 1990<br>value<br>Level<br>(1)     | 2011<br>value<br>Level<br>(2)     | 1980<br>value<br>None<br>(3)      | 1990<br>value<br>Level<br>(4)     | 2011<br>value<br>Level<br>(5)     | 1980<br>value<br>None<br>(6)      |
|----------------------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| <i>Panel A: The city</i>                           |                                   |                                   |                                   |                                   |                                   |                                   |
| Asphalt main road<br>(share)                       | 0.0271<br>(0.0172)<br>[0.0280]    | 0.0213<br>(0.0302)<br>[0.0295]    | -0.0526<br>(0.0459)<br>[0.0519]   |                                   |                                   |                                   |
| Mean / cities                                      | 0.688 / 131                       | 0.817 / 122                       | 0.436 / 122                       |                                   |                                   |                                   |
| Main road passable by car<br>all year (share)      | 0.0182**†<br>(0.0088)<br>[0.0159] | 0.0193<br>(0.0127)<br>[0.0136]    | 0.0723**†<br>(0.0359)<br>[0.0333] |                                   |                                   |                                   |
| Mean / cities                                      | 0.932 / 131                       | 0.971 / 122                       | 0.808 / 122                       |                                   |                                   |                                   |
| <i>Panel B: The countryside</i>                    |                                   |                                   |                                   |                                   |                                   |                                   |
|                                                    | With settlements                  |                                   |                                   | Without settlements               |                                   |                                   |
| Households connected to<br>electricity (share)     | 0.0108**<br>(0.0043)<br>[0.0048]  | 0.0553*†<br>(0.0284)<br>[0.0262]  | -0.0048<br>(0.0050)<br>[0.0044]   | 0.0061<br>(0.0123)<br>[0.0114]    | 0.0792***<br>(0.0194)<br>[0.0198] | 0.0020<br>(0.0042)<br>[0.0037]    |
| Mean / cities                                      | 0.061 / 131                       | 0.512 / 131                       | 0.012 / 131                       | 0.163 / 125                       | 0.629 / 125                       | 0.026 / 125                       |
| Population in a village<br>with grid power (share) | 0.0460***<br>(0.0137)<br>[0.0139] | 0.0719***<br>(0.0216)<br>[0.0215] | -0.0272*<br>(0.0158)<br>[0.0130]  | 0.0460***<br>(0.0165)<br>[0.0166] | 0.0552***<br>(0.0168)<br>[0.0168] | -0.0058<br>(0.0128)<br>[0.0125]   |
| Mean / cities                                      | 0.192 / 131                       | 0.813 / 123                       | 0.037 / 123                       | 0.432 / 125                       | 0.915 / 119                       | 0.081 / 119                       |
| Asphalt main road<br>(share)                       | 0.0590***<br>(0.0207)<br>[0.0218] | -0.0220<br>(0.0215)<br>[0.0191]   | -0.0303<br>(0.0258)<br>[0.0220]   | 0.0679***<br>(0.0161)<br>[0.0175] | -0.0055<br>(0.0254)<br>[0.0227]   | 0.0095<br>(0.0302)<br>[0.0309]    |
| Mean / cities                                      | 0.349 / 129                       | 0.513 / 121                       | 0.184 / 121                       | 0.525 / 123                       | 0.696 / 116                       | 0.282 / 116                       |
| Main road passable by car<br>all year (share)      | 0.0310<br>(0.0192)<br>[0.0198]    | 0.0201*<br>(0.0106)<br>[0.0091]   | 0.0396<br>(0.0387)<br>[0.0397]    | 0.0270***<br>(0.0101)<br>[0.0107] | 0.0286***<br>(0.0084)<br>[0.0083] | 0.0609<br>(0.0371)<br>[0.0390]    |
| Mean / cities                                      | 0.848 / 129                       | 0.926 / 121                       | 0.591 / 121                       | 0.877 / 123                       | 0.943 / 116                       | 0.714 / 116                       |
| Primary schools<br>per 1,000                       | 0.0147<br>(0.0519)<br>[0.0438]    | 0.0521<br>(0.0511)<br>[0.0465]    | 0.0470<br>(0.0976)<br>[0.0989]    | -0.0131<br>(0.0340)<br>[0.0357]   | 0.0492**<br>(0.0226)<br>[0.0206]  | -0.0665<br>(0.0712)<br>[0.0700]   |
| Mean / cities                                      | 1.285 / 131                       | 0.902 / 131                       | 1.454 / 131                       | 1.352 / 125                       | 0.962 / 125                       | 1.363 / 125                       |
| Junior secondary schools<br>per 1,000              | -0.0028<br>(0.0120)<br>[0.0104]   | 0.0191*<br>(0.0104)<br>[0.0094]   | 0.0187<br>(0.0120)<br>[0.0106]    | -0.0072<br>(0.0087)<br>[0.0077]   | -0.0044<br>(0.0112)<br>[0.0100]   | 0.0176***<br>(0.0065)<br>[0.0064] |
| Mean / cities                                      | 0.176 / 131                       | 0.277 / 131                       | 0.103 / 131                       | 0.201 / 125                       | 0.264 / 125                       | 0.114 / 125                       |
| Senior secondary schools<br>per 1,000              | 0.0059<br>(0.0064)<br>[0.0059]    | 0.0424***<br>(0.0098)<br>[0.0120] | 0.0113**<br>(0.0050)<br>[0.0042]  | 0.0071<br>(0.0045)<br>[0.0041]    | 0.0030<br>(0.0037)<br>[0.0043]    | 0.0065<br>(0.0051)<br>[0.0048]    |
| Mean / cities                                      | 0.044 / 131                       | 0.132 / 131                       | 0.015 / 131                       | 0.075 / 125                       | 0.127 / 125                       | 0.029 / 125                       |
| Health centers<br>per 1,000                        | -0.0251**<br>(0.0127)<br>[0.0145] | 0.0072<br>(0.0186)<br>[0.0174]    | -0.0039<br>(0.0107)<br>[0.0104]   | -0.0140<br>(0.0148)<br>[0.0149]   | -0.0066<br>(0.0155)<br>[0.0146]   | 0.0074<br>(0.0086)<br>[0.0075]    |
| Mean / cities                                      | 0.290 / 131                       | 0.274 / 131                       | 0.112 / 131                       | 0.210 / 125                       | 0.255 / 125                       | 0.090 / 125                       |

*Notes:* Each cell is one regression on the main city sample (134 cities), with the outcome measured over the area's subdistricts observed on the item in 1980 and in the end year. Columns 1 and 4 regress the 1990 value and columns 2 and 5 the 2011 value on exposure, each holding the outcome's own 1980 value over the same subdistricts fixed; columns 3 and 6 regress the 1980 value itself, over the subdistricts of columns 2 and 5, on exposure, without the four 1980 census baselines, which are not controls for a 1980 outcome. Every regression is in the main specification with the full set of controls and Conley SE within 200km, adding the 1980 control its column names. Below each estimate: its SE with the uniform kernel in parentheses and with the Bartlett kernel in brackets, then the mean of the outcome and the cities in the column; a city leaves a column when its area has no subdistrict observed on the item in both years. The stars use the uniform-kernel SE: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with province clusters, or with both.

by -0.0060 (0.0054) and -0.0022 (0.0078); the 2002 wave and the spending on durables and services are in the table. The household survey (Susenas) starts after 1980, so these regressions do not add a 1980 level (Table E21).

**The road network.** The road network around more exposed cities did not improve more than its earlier level predicts. Road density on the current map, with the 1950 density held fixed, changes by -0.0764 (standard error 0.0894), and the roughness of the surveyed corridors over 1990 to 2007, with its 1990 level held fixed, by 0.0574 (0.3016). Roads were already denser around more exposed cities on the 1950 map, drawn before the program, by 0.244 (0.065) in log density, and four-wheel access in the catchment was better in 1983, by 0.093 (0.024), a difference that reflects where the settlements were sited. The table is the exhibit of the roads paragraph of [Appendix E.4](#) ([Table E22](#)).

**The conflict provinces.** The effect does not differ measurably in the provinces of the conflicts of around 2000. With exposure interacted with an indicator of Aceh, West Kalimantan, Central Kalimantan and Central Sulawesi on the main city sample, the interaction term is  $-0.027$  (standard error 0.032) on the agricultural employment share and  $-0.015$  (0.084) on log modern-sector employment, and it does not reach the 10% level on any of the seven outcomes under any of the three variances; the slope outside stays within a fifth of the headline, at  $-0.042$  (0.021) and  $0.170$  (0.081). Leaving the 26 cities of the four provinces out instead gives  $-0.024$  (0.022) on the farm share and  $0.080$  (0.092) on log modern-sector employment, about half their size on the main city sample, with every standard error larger. The conflict-province cities are about as exposed as the others; the halving comes from estimating the control coefficients on the 108 remaining cities, and the standard deviation of exposure left after the controls falls from 0.46 to 0.30 ([Table E23](#)).

**The exposure of neighboring cities.** The exposure of neighboring cities leaves the shift in sector shares in place, and its estimates on city size are mixed. With the inverse-distance-weighted exposure of the 137 other measured cities added beside a city's own, the farm share and the modern share do not move with it (0.007, standard error 0.075;  $-0.015$ , 0.075), and the own-exposure estimates stay at the headline,  $-0.052$  against  $-0.051$  on the farm share. Log population, log employment and log modern-sector employment fall with the exposure of neighbors, by  $-0.603$  (0.281),  $-0.597$  (0.272) and  $-0.612$  (0.306); the weighted measure varies little once own exposure and the controls are known (its standard deviation is 0.29 against 1.00 for own exposure, and the two explain 0.87 of it), and the size estimate loses the 5% level without three pairs of twin cities within 12 km of each other or with province effects. The plain mean over the other measured cities within 100 km, on the 125 cities with such a neighbor, has the same negative sign on the three size outcomes and is statistically insignificant on every outcome with the uniform kernel and with province clusters, with log population at  $-0.133$  (0.085) ([Table E24](#)).

Table E17: Public Provision in the City and Its Countryside, Continued

|                                                                                                                      | 1990<br>value<br>Log<br>(1)                  | 2011<br>value<br>Log<br>(2)                   | 1990<br>value<br>Log<br>(4)                   | 2011<br>value<br>Log<br>(5)                   |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| <i>Panel C: The 1980 control as <math>\ln(1 + \text{the 1980 numerator})</math>, on the cities of Panels A and B</i> |                                              |                                               |                                               |                                               |
| <i>The city</i>                                                                                                      |                                              |                                               |                                               |                                               |
| Asphalt main road<br>(share)                                                                                         | 0.0137<br>(0.0176)<br>[0.0322]               | 0.0151<br>(0.0316)<br>[0.0300]                |                                               |                                               |
| Main road passable by car<br>all year (share)                                                                        | 0.0211* <sup>†</sup><br>(0.0119)<br>[0.0180] | 0.0239** <sup>†</sup><br>(0.0109)<br>[0.0132] |                                               |                                               |
| <i>The countryside, also holding <math>\ln(1 + \text{its 1980 village-census population})</math> fixed</i>           |                                              |                                               |                                               |                                               |
|                                                                                                                      | With settlements                             |                                               | Without settlements                           |                                               |
| Households connected to<br>electricity (share)                                                                       | 0.0107* <sup>†</sup><br>(0.0056)<br>[0.0056] | 0.0709**<br>(0.0310)<br>[0.0283]              | 0.0064<br>(0.0118)<br>[0.0112]                | 0.0524***<br>(0.0158)<br>[0.0158]             |
| Population in a village<br>with grid power (share)                                                                   | 0.0527***<br>(0.0146)<br>[0.0144]            | 0.0864***<br>(0.0195)<br>[0.0210]             | 0.0359** <sup>†</sup><br>(0.0182)<br>[0.0192] | 0.0431** <sup>†</sup><br>(0.0186)<br>[0.0177] |
| Asphalt main road<br>(share)                                                                                         | 0.0170<br>(0.0254)<br>[0.0285]               | -0.0596**<br>(0.0257)<br>[0.0237]             | 0.0392<br>(0.0244)<br>[0.0252]                | -0.0048<br>(0.0263)<br>[0.0244]               |
| Main road passable by car<br>all year (share)                                                                        | 0.0036<br>(0.0156)<br>[0.0217]               | -0.0009<br>(0.0092)<br>[0.0092]               | 0.0034<br>(0.0101)<br>[0.0114]                | 0.0359***<br>(0.0101)<br>[0.0107]             |
| Primary schools<br>per 1,000                                                                                         | -0.0104<br>(0.0567)<br>[0.0501]              | 0.0281<br>(0.0581)<br>[0.0514]                | -0.0296<br>(0.0373)<br>[0.0386]               | 0.0227<br>(0.0231)<br>[0.0213]                |
| Junior secondary schools<br>per 1,000                                                                                | -0.0035<br>(0.0136)<br>[0.0116]              | 0.0172<br>(0.0105)<br>[0.0091]                | -0.0060<br>(0.0082)<br>[0.0078]               | 0.0012<br>(0.0104)<br>[0.0098]                |
| Senior secondary schools<br>per 1,000                                                                                | 0.0060<br>(0.0071)<br>[0.0061]               | 0.0354***<br>(0.0103)<br>[0.0118]             | 0.0098* <sup>†</sup><br>(0.0050)<br>[0.0052]  | 0.0046<br>(0.0036)<br>[0.0042]                |
| Health centers<br>per 1,000                                                                                          | -0.0165<br>(0.0134)<br>[0.0146]              | -0.0192<br>(0.0195)<br>[0.0204]               | -0.0017<br>(0.0152)<br>[0.0159]               | 0.0045<br>(0.0165)<br>[0.0162]                |

*Notes, continued:* The city is its member subdistricts, each weighted by population times the share of the subdistrict assigned to the city; the countryside with settlements is the rural subdistricts (in no city) with a program village within 300 travel minutes of the city (3 cities have none), the city's own subdistricts left out; the countryside without settlements is the other rural subdistricts within those 300 minutes, with every subdistrict that holds a program village removed (8 cities have none); both are pooled by population. Road shares are of the population in villages reached by land. Electricity is measured on the state grid (PLN) in every year: in 1980 the household share is the share of households using electricity in villages whose main source is the grid, and a grid village has that main source; in 1990 and 2011 the household share counts households on the grid, and a grid village has at least one of them. Schools are public plus private (1980: the four primary types and the secondary totals). Health centers count health and auxiliary centers, without polyclinics, in 1990 and 2011; the 1980 form has one line for health centers (puskesmas) and no separate line for auxiliary centers (pustu), so it does not show whether the 1980 count includes them. A count is left missing in 1980 where the item is missing for any village of the subdistrict (34 outer-island subdistricts for primary schools, 32 for health centers) and in 1990 where it is missing for every village of the subdistrict (4 subdistricts). Rates per 1,000 use each year's village-census population. A presence share rises with the size of its unit. Columns 2, 3, 5 and 6 of the two road shares leave out the West Sumatra subdistricts whose 2011 record includes a nagari, a larger unit than the 1980 village (13 member subdistricts of the cities, 0 in the countryside with settlements and 65 without; 4, 0 and 2 cities with no other subdistrict leave those columns), and those columns of the two road shares and of the grid village in Panel B also leave out the 8 cities of West Kalimantan, whose villages were merged into larger units between the 1990 and 1993 village censuses; the 2011 grid-village share keeps the nagari subdistricts, measured on hamlets of about the size of the 1980 village (1,010 people on average against 948). Without the 62 of those subdistricts that hold a primary-school value in both years, the primary-school estimate of column 5 of Panel B is 0.0359 (0.0240) [0.0223] on 123 cities. The 1980 numerator is the 1980 count for a rate and, for a share, the 1980 share times 1980 population (road shares are of land-reached villages, the connection share of households). Source: village census (PODES) 1980, 1990 and 2011. The stars use the uniform-kernel SE: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . <sup>†</sup> significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with province clusters, or with both.

Table E18: Public Provision in Rural Subdistricts With and Without a Settlement

|                                                  | 1980 to 1990                      |                                             |                                    | 1980 to 2011                      |                                             |                                  | 1980                                        |
|--------------------------------------------------|-----------------------------------|---------------------------------------------|------------------------------------|-----------------------------------|---------------------------------------------|----------------------------------|---------------------------------------------|
|                                                  | With a<br>settlement<br>(1)       | Without a<br>settlement<br>(2)              | Difference<br>(3)                  | With a<br>settlement<br>(4)       | Without a<br>settlement<br>(5)              | Difference<br>(6)                | level<br>Difference<br>(7)                  |
| Asphalt main road<br>(population share)          | 0.0743***<br>(0.0126)<br>[0.0202] | 0.0868***<br>(0.0191)<br>[0.0186]<br>0.4397 | -0.0125<br>(0.0199)<br>[0.0250]    | 0.0148<br>(0.0251)<br>[0.0246]    | 0.0413*<br>(0.0242)<br>[0.0249]<br>0.6291   | -0.0265<br>(0.0236)<br>[0.0245]  | 0.0474***<br>(0.0184)<br>[0.0196]<br>0.2043 |
| Mean<br>Subdistricts                             |                                   | 182 / 563                                   |                                    |                                   | 168 / 476                                   |                                  | 186 / 578                                   |
| Four-wheel access all year<br>(population share) | 0.0803***<br>(0.0230)<br>[0.0279] | 0.0371**<br>(0.0151)<br>[0.0184]<br>0.8347  | 0.0433***†<br>(0.0218)<br>[0.0257] | 0.0409**<br>(0.0170)<br>[0.0171]  | 0.0506***<br>(0.0114)<br>[0.0121]<br>0.9179 | -0.0097<br>(0.0130)<br>[0.0151]  | 0.0508**<br>(0.0217)<br>[0.0247]<br>0.6009  |
| Mean<br>Subdistricts                             |                                   | 182 / 563                                   |                                    |                                   | 168 / 476                                   |                                  | 186 / 578                                   |
| Households connected<br>(household share)        | 0.0114<br>(0.0071)<br>[0.0073]    | 0.0170**<br>(0.0073)<br>[0.0086]<br>0.1301  | -0.0057<br>(0.0065)<br>[0.0076]    | 0.0646***<br>(0.0212)<br>[0.0200] | 0.0520**<br>(0.0207)<br>[0.0185]<br>0.5832  | 0.0126<br>(0.0172)<br>[0.0176]   | 0.0002<br>(0.0043)<br>[0.0045]<br>0.0239    |
| Mean<br>Subdistricts                             |                                   | 191 / 586                                   |                                    |                                   | 191 / 589                                   |                                  | 191 / 589                                   |
| Village with electricity<br>(population share)   | 0.0381**<br>(0.0185)<br>[0.0175]  | 0.0283<br>(0.0173)<br>[0.0188]<br>0.3439    | 0.0097<br>(0.0162)<br>[0.0169]     | 0.1056***<br>(0.0214)<br>[0.0186] | 0.0478***<br>(0.0130)<br>[0.0119]<br>0.8803 | 0.0578**<br>(0.0249)<br>[0.0202] | 0.0047<br>(0.0103)<br>[0.0111]<br>0.0677    |
| Mean<br>Subdistricts                             |                                   | 191 / 586                                   |                                    |                                   | 173 / 539                                   |                                  | 191 / 589                                   |
| Primary schools<br>per 1,000 people              | 0.0013<br>(0.0291)<br>[0.0323]    | -0.0316<br>(0.0338)<br>[0.0335]<br>1.4847   | 0.0329<br>(0.0310)<br>[0.0320]     | 0.0194<br>(0.0327)<br>[0.0309]    | -0.0010<br>(0.0236)<br>[0.0236]<br>1.0802   | 0.0204<br>(0.0261)<br>[0.0261]   | 0.0639<br>(0.0499)<br>[0.0531]<br>1.4662    |
| Mean<br>Subdistricts                             |                                   | 185 / 570                                   |                                    |                                   | 185 / 572                                   |                                  | 185 / 572                                   |
| Junior secondary schools<br>per 1,000 people     | 0.0025<br>(0.0101)<br>[0.0086]    | -0.0047<br>(0.0102)<br>[0.0084]<br>0.2053   | 0.0073<br>(0.0128)<br>[0.0103]     | -0.0066<br>(0.0083)<br>[0.0087]   | -0.0102<br>(0.0097)<br>[0.0088]<br>0.2903   | 0.0036<br>(0.0091)<br>[0.0079]   | -0.0069<br>(0.0092)<br>[0.0095]<br>0.1182   |
| Mean<br>Subdistricts                             |                                   | 191 / 586                                   |                                    |                                   | 191 / 589                                   |                                  | 191 / 589                                   |
| Senior secondary schools<br>per 1,000 people     | 0.0037<br>(0.0044)<br>[0.0045]    | 0.0005<br>(0.0050)<br>[0.0044]<br>0.0669    | 0.0032<br>(0.0034)<br>[0.0039]     | 0.0076<br>(0.0075)<br>[0.0070]    | 0.0051<br>(0.0048)<br>[0.0048]<br>0.1293    | 0.0025<br>(0.0063)<br>[0.0059]   | -0.0013<br>(0.0029)<br>[0.0025]<br>0.0225   |
| Mean<br>Subdistricts                             |                                   | 191 / 586                                   |                                    |                                   | 191 / 589                                   |                                  | 191 / 589                                   |
| Puskesmas and pustu<br>per 1,000 people          | 0.0033<br>(0.0123)<br>[0.0111]    | -0.0062<br>(0.0104)<br>[0.0098]<br>0.2721   | 0.0094<br>(0.0086)<br>[0.0098]     | -0.0200<br>(0.0155)<br>[0.0146]   | -0.0099<br>(0.0172)<br>[0.0155]<br>0.3092   | -0.0101<br>(0.0108)<br>[0.0114]  | 0.0067<br>(0.0061)<br>[0.0063]<br>0.1116    |
| Mean<br>Subdistricts                             |                                   | 187 / 574                                   |                                    |                                   | 187 / 577                                   |                                  | 187 / 577                                   |

*Notes:* Each row is one outcome: columns 1 to 3 come from one regression of the 1990 value and columns 4 to 6 from one regression of the 2011 value, on exposure interacted with the indicators of a subdistrict with and without a settlement, the with-settlement indicator, the outcome's own 1980 value and the controls; columns 3 and 6 subtract the slope without a settlement from the slope with one, and column 7 does the same in one regression of the 1980 value itself, without the four 1980 census baselines, which are not controls for a 1980 outcome. Every regression is in the main specification with the full set of controls and Conley SE within 200km, with the settler count taken over villages within 1,000 travel minutes. Below each estimate: its SE with the uniform kernel in parentheses and with the Bartlett kernel in brackets, then the mean of the outcome and the subdistricts with and without a settlement in the regression. The stars use the uniform-kernel SE: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with least squares clustered by province (20 provinces,  $t$  with 19 degrees of freedom; the 1980 to 2011 presence shares, which leave West Kalimantan out, on 19 provinces), or with both.

Table E18: Public Provision in Rural Subdistricts With and Without a Settlement, Continued

*Notes, continued:* Sample: the 787 in-sample rural subdistricts (191 with a settlement, holding at least one of the 814 settlement villages, and 596 without); the 41 in-sample subdistricts that belong to a reclassified city are left out, and a regression also needs the end-year value and the 1980 value of its outcome. Exposure is agro-similarity within 300 travel minutes, standardized over the 787 (one SD is 0.100). Road shares are of the population in villages reached by land, and missing where every village is reached by water only (2 subdistricts in 2011); the 1993 wave is not used, since its item asks about most village roads. Electricity is measured on the state grid (PLN) in every year: households connected is the share of households using electricity in villages whose main source is the grid in 1980, and PLN households over all households in 1990 and 2011; a village with electricity has the grid as its main source in 1980 and any PLN household in 1990 and 2011, so its change mixes the change of definition with the change in the outcome. The only 1980 control of the two electricity rows is the 1980 state-grid level, while the estimates of the state grid wave by wave from 1983 to 2011 (Figure E4) hold the 1980 levels of the grid and of other suppliers fixed separately, so the two do not form one series. Schools are public plus private in every wave (1980: the four primary types and the secondary totals). Health centers count puskesmas plus pustu, without polyclinics, in 1990 and 2011; the 1980 form has one line for health centers (puskesmas) and no separate line for auxiliary centers (pustu), so it does not show whether the 1980 count includes them. A 1980 count is left missing in a subdistrict that takes any share of a 1980 subdistrict where the item is missing for any village (34 outer-island subdistricts for primary schools, 32 for health centers). Shares run from 0 to 1; rates per 1,000 use each wave's own village-census population. The 1980 to 2011 columns of the two road shares leave out the 51 West Sumatra subdistricts whose 2011 record includes a nagari, a larger unit than the village of 1980; the 2011 village-with-electricity share measures them on hamlets (jorong), units of about the size of the 1980 village (a mean of 936 people per hamlet in 2011 and 855 per village in 1980, over the 51 nagari subdistricts), and keeps them; the 1980 to 2011 columns of the three presence shares (asphalt, four-wheel access and village with electricity) also leave out the 71 subdistricts of West Kalimantan (18 with a settlement), whose villages were merged into larger units between the 1990 and 1993 village censuses. Column 7 does not apply either rule. Source: village census (PODES) 1980, 1990 and 2011 on year-2000 subdistricts; population census 1980.

Table E19: Banks in the City and Its Countryside

|                                                                  | Bank presence                                 |                                              | Banks per 1,000 people            |                                              |
|------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------|-----------------------------------|----------------------------------------------|
|                                                                  | 1980 to 1990                                  | 1980 to 2011                                 | 1980 to 1990                      | 1980 to 2011                                 |
| <i>Panel A. The city</i>                                         |                                               |                                              |                                   |                                              |
| Agro-similarity                                                  | -0.0194<br>(0.0161)<br>[0.0168]               | 0.0251<br>(0.0248)<br>[0.0222]               | 0.0055<br>(0.0039)<br>[0.0040]    | 0.0110<br>(0.0092)<br>[0.0093]               |
| Agro-similarity, 1980 log count                                  | -0.0075<br>(0.0194)<br>[0.0186]               | 0.0337<br>(0.0306)<br>[0.0283]               | 0.0106**<br>(0.0046)<br>[0.0051]  | 0.0154<br>(0.0098)<br>[0.0097]               |
| Mean of outcome                                                  | 0.0533                                        | 0.1047                                       | 0.0119                            | 0.0648                                       |
| Cities                                                           | 131                                           | 122                                          | 131                               | 133                                          |
| <i>Panel B. The countryside with settlements</i>                 |                                               |                                              |                                   |                                              |
| Agro-similarity                                                  | -0.0085<br>(0.0143)<br>[0.0115]               | -0.0138<br>(0.0137)<br>[0.0130]              | 0.0037<br>(0.0025)<br>[0.0024]    | 0.0119* <sup>†</sup><br>(0.0072)<br>[0.0060] |
| Agro-similarity, 1980 log count                                  | -0.0151* <sup>†</sup><br>(0.0085)<br>[0.0080] | -0.0102<br>(0.0182)<br>[0.0169]              | -0.0007<br>(0.0028)<br>[0.0028]   | 0.0041<br>(0.0062)<br>[0.0055]               |
| Mean of outcome                                                  | 0.0108                                        | 0.0527                                       | 0.0045                            | 0.0304                                       |
| Cities                                                           | 131                                           | 123                                          | 131                               | 131                                          |
| <i>Panel C. The countryside without settlements</i>              |                                               |                                              |                                   |                                              |
| Agro-similarity                                                  | -0.0050<br>(0.0083)<br>[0.0079]               | 0.0185* <sup>†</sup><br>(0.0112)<br>[0.0118] | 0.0016<br>(0.0024)<br>[0.0023]    | 0.0077***<br>(0.0026)<br>[0.0034]            |
| Mean of outcome                                                  | 0.0083                                        | 0.0311                                       | 0.0005                            | 0.0265                                       |
| Cities                                                           | 125                                           | 117                                          | 124                               | 124                                          |
| Agro-similarity, 1980 log count                                  | 0.0009<br>(0.0082)<br>[0.0081]                | 0.0114<br>(0.0084)<br>[0.0101]               | 0.0002<br>(0.0022)<br>[0.0022]    | 0.0069***<br>(0.0023)<br>[0.0034]            |
| Cities                                                           | 124                                           | 116                                          | 124                               | 124                                          |
| <i>Panel D. Rural subdistricts with and without a settlement</i> |                                               |                                              |                                   |                                              |
| Agro-similarity, with a settlement                               | 0.0088<br>(0.0070)<br>[0.0078]                | -0.0140<br>(0.0124)<br>[0.0123]              | 0.0069***<br>(0.0017)<br>[0.0022] | 0.0024<br>(0.0036)<br>[0.0041]               |
| Agro-similarity, without a settlement                            | -0.0041<br>(0.0060)<br>[0.0054]               | -0.0038<br>(0.0085)<br>[0.0090]              | 0.0052*<br>(0.0028)<br>[0.0025]   | 0.0050**<br>(0.0024)<br>[0.0029]             |
| Difference                                                       | 0.0130<br>(0.0083)<br>[0.0082]                | -0.0102<br>(0.0141)<br>[0.0124]              | 0.0017<br>(0.0021)<br>[0.0020]    | -0.0026<br>(0.0039)<br>[0.0036]              |
| Mean of outcome                                                  | -0.0021                                       | 0.0202                                       | -0.0016                           | 0.0213                                       |
| Subdistricts                                                     | 777                                           | 661                                          | 768                               | 771                                          |
| with a settlement                                                | 191                                           | 173                                          | 186                               | 186                                          |
| Difference, 1980 log count                                       | 0.0079<br>(0.0067)<br>[0.0076]                | -0.0143<br>(0.0128)<br>[0.0118]              | 0.0025<br>(0.0024)<br>[0.0026]    | -0.0016<br>(0.0044)<br>[0.0041]              |
| Subdistricts                                                     | 768                                           | 652                                          | 768                               | 771                                          |
| with a settlement                                                | 186                                           | 168                                          | 186                               | 186                                          |

*Notes:* Each column regresses the change in the area's bank measure from 1980 to the year in the column header on standardized agro-similarity, with the outcome's own 1980 level held fixed. Panels A to C use the main city sample (134 cities) and the main specification with the full set of controls and Conley SE within 200km; a row whose label ends in 1980 log count holds fixed instead the log of the area's 1980 banks plus one and, in Panels B and C, the log of its 1980 village-census population plus one. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . <sup>†</sup> significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test the table does not print), or with both.

Table E19: Banks in the City and Its Countryside, Continued

*Notes, continued:* Panel A is the city's member subdistricts, weighted by membership share; Panel B the countryside with settlements, the rural subdistricts with a settlement village within 300 travel minutes of the city, its own left out; Panel C the countryside without settlements, the other rural subdistricts in that 300-minute window with no settlement village. Panel D uses the 787 rural subdistricts within 300 travel minutes of a settlement village, measured in the 1980 and 2010 censuses, that do not belong to a reclassified city (191 with a settlement village, 596 without), with agro-similarity, the mean over the settlement villages within 300 minutes, interacted with an indicator for each group, the first indicator, the settler count weighted by inverse travel time within 1,000 minutes, the four 1980 census baselines and the sixteen agro-climatic means within 300 minutes; Difference is the first slope less the second. A subdistrict with a missing 1980 bank count for any of its villages leaves the per-1,000 columns and the area's 1980 log count. Bank presence is the population share in villages with a bank, and banks per 1,000 is banks over the village-census population; the village censuses count banks and do not record deposits or lending. 1980 counts banks on one line, 2011 commercial and rural banks, and the 1990 line the commercial banks and some of the rural banks. The 1980 to 2011 presence columns leave out West Kalimantan, whose villages were merged between the 1990 and 1993 village censuses, and the West Sumatra subdistricts recorded as nagari in 2011, a larger unit than the 1980 village. Sources: village censuses (PODES) of 1980, 1990 and 2011 on year-2000 subdistricts; the 1980 population census; the settlement-village records.

Table E20: New District Seats and Modern-Sector Employment Without Government Services

|                                                                            |                                 |                                            |                                           |                                            |                                           |                                             |
|----------------------------------------------------------------------------|---------------------------------|--------------------------------------------|-------------------------------------------|--------------------------------------------|-------------------------------------------|---------------------------------------------|
| <i>Panel A. Hosting the seat of a new district, linear probability</i>     |                                 |                                            |                                           |                                            |                                           |                                             |
|                                                                            | New seat,<br>1999 to 2010       | New seat,<br>2000 to 2010                  | New kota seat,<br>1999 to 2010            |                                            |                                           |                                             |
| Agro-similarity                                                            | -0.006<br>(0.052)<br>[0.056]    | -0.065<br>(0.045)<br>[0.055]               | 0.057* <sup>†</sup><br>(0.030)<br>[0.040] |                                            |                                           |                                             |
| District seat, end of 1998                                                 | -0.281***<br>(0.088)<br>[0.090] | -0.151* <sup>†</sup><br>(0.091)<br>[0.086] | 0.079<br>(0.079)<br>[0.074]               |                                            |                                           |                                             |
| Mean of outcome<br>Cities                                                  | 0.246<br>134                    | 0.179<br>134                               | 0.104<br>134                              |                                            |                                           |                                             |
| <i>Panel B. Main outcomes without the cities hosting a new seat</i>        |                                 |                                            |                                           |                                            |                                           |                                             |
|                                                                            | Log population                  |                                            | Agricultural share                        | Log modern employment                      |                                           |                                             |
|                                                                            | Member                          | Town                                       | Member                                    | Town                                       | Member                                    | Town                                        |
| Agro-similarity                                                            | 0.065<br>(0.066)<br>[0.069]     | 0.056<br>(0.066)<br>[0.070]                | -0.053***<br>(0.017)<br>[0.018]           | -0.045***<br>(0.015)<br>[0.018]            | 0.172***<br>(0.064)<br>[0.071]            | 0.137*** <sup>†</sup><br>(0.052)<br>[0.066] |
| Mean of outcome<br>Cities                                                  | 11.856<br>101                   | 11.886<br>98                               | 0.375<br>101                              | 0.364<br>98                                | 10.257<br>101                             | 10.312<br>98                                |
| <i>Panel C. Modern-sector employment without government services, 2010</i> |                                 |                                            |                                           |                                            |                                           |                                             |
|                                                                            | All modern                      | Without codes 14 to 18                     |                                           | Without codes 16 to 18, strict             |                                           |                                             |
|                                                                            | Log                             | Log                                        | Share                                     | Log                                        | Share                                     |                                             |
| Agro-similarity                                                            | 0.177***<br>(0.063)<br>[0.060]  | 0.164** <sup>†</sup><br>(0.072)<br>[0.066] | 0.038***<br>(0.005)<br>[0.007]            | 0.165** <sup>†</sup><br>(0.071)<br>[0.066] | 0.039***<br>(0.005)<br>[0.007]            |                                             |
| Own 1980 numerator, log                                                    | 0.199**<br>(0.091)<br>[0.097]   | 0.055<br>(0.080)<br>[0.078]                | 0.021* <sup>†</sup><br>(0.012)<br>[0.013] | 0.058<br>(0.078)<br>[0.077]                | 0.020* <sup>†</sup><br>(0.012)<br>[0.013] |                                             |
| Mean of outcome<br>Cities                                                  | 10.232<br>134                   | 9.634<br>134                               | 0.287<br>134                              | 9.675<br>134                               | 0.305<br>134                              |                                             |

*Notes:* Each column is one regression on the main city sample (134 cities), in the main specification with the full set of controls and Conley SE within 200km; Panel A adds the city's district-seat status at the end of 1998, and Panel C adds the outcome's own 1980 numerator as the log of the count plus one. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets. Panel A: linear probability models of hosting the seat of a district created 1999 to 2010 (33 cities), created 2000 to 2010 (24 cities), or of a new kota, the urban type of district, created 1999 to 2010 (14 cities). A city hosts a seat when one of its member subdistricts holds the regent's or mayor's office, located in the village censuses of 2011 and 2014, and a new seat is held in both waves; the status at the end of 1998 is read from the 2011 and 2014 seats of districts established by 1998 and equals 1 for 94 cities. Panel B: log population, the agricultural employment share and log modern-sector employment, 2010, whose estimates on all 134 cities are in Table 3, without the 33 cities hosting a new seat in a member subdistrict (Member, 101 cities), and also without the 3 cities whose town is the capital of such a district although the seat subdistrict is not a member (Town, 98 cities). Panel C: modern-sector employment, 2010, in logs and as a share of employment; column 1 is the headline sector (census industry codes 8 and 11 to 18), columns 2 and 3 keep codes 8 and 11 to 13 (manufacturing, trade, hotels and restaurants, and transport), and columns 4 and 5 leave out codes 16 to 18 (education, health, and social, government and private services) on the strict employment definition, the employed with a recorded industry. Codes 16 to 18 hold public and private employers alike, so columns 4 and 5 do not separate public from private employment. Sources: population censuses of 1980 and 2010; village censuses of 2011 and 2014; World Bank district proliferation crosswalk. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . <sup>†</sup> significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test the table does not print), or with both.

Table E21: Spending of Rural Households Around the City

|                                                  | 2002                         |                             |                              |                              |                               | 2011                         |                              |                                |                                |                              |
|--------------------------------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|-------------------------------|------------------------------|------------------------------|--------------------------------|--------------------------------|------------------------------|
|                                                  | Spending                     | Non-food                    | Durables                     | Any durable                  | Services                      | Spending                     | Non-food                     | Durables                       | Any durable                    | Services                     |
| <i>Panel A. The whole hinterland</i>             |                              |                             |                              |                              |                               |                              |                              |                                |                                |                              |
| Agro-similarity                                  | 0.001<br>(0.028)<br>[0.025]  | 0.001<br>(0.003)<br>[0.004] | -0.133<br>(0.170)<br>[0.162] | -0.017<br>(0.019)<br>[0.018] | 0.082**<br>(0.041)<br>[0.038] | -0.031<br>(0.027)<br>[0.026] | -0.006<br>(0.005)<br>[0.005] | -0.317**<br>(0.138)<br>[0.136] | -0.033**<br>(0.016)<br>[0.015] | -0.022<br>(0.037)<br>[0.038] |
| Mean of outcome                                  | 11.898                       | 0.268                       | 5.226                        | 0.674                        | 8.981                         | 13.057                       | 0.369                        | 4.342                          | 0.475                          | 10.155                       |
| Households per city                              | 2248                         | 2248                        | 2248                         | 2248                         | 2248                          | 2814                         | 2112                         | 2112                           | 2112                           | 2112                         |
| Cities                                           | 123                          | 123                         | 123                          | 123                          | 123                           | 128                          | 128                          | 128                            | 128                            | 128                          |
| <i>Panel B. The countryside with settlements</i> |                              |                             |                              |                              |                               |                              |                              |                                |                                |                              |
| Agro-similarity                                  | -0.002<br>(0.052)<br>[0.049] | 0.006<br>(0.009)<br>[0.009] | 0.479<br>(0.368)<br>[0.331]  | 0.040<br>(0.035)<br>[0.034]  | 0.175<br>(0.116)<br>[0.098]   | -0.023<br>(0.039)<br>[0.035] | -0.002<br>(0.008)<br>[0.008] | 0.153<br>(0.314)<br>[0.312]    | 0.015<br>(0.030)<br>[0.031]    | 0.039<br>(0.060)<br>[0.065]  |
| Mean of outcome                                  | 11.896                       | 0.263                       | 5.370                        | 0.689                        | 8.952                         | 13.112                       | 0.387                        | 4.979                          | 0.529                          | 10.271                       |
| Households per city                              | 541                          | 541                         | 541                          | 541                          | 541                           | 673                          | 508                          | 508                            | 508                            | 508                          |
| Cities                                           | 118                          | 118                         | 118                          | 118                          | 118                           | 131                          | 131                          | 131                            | 131                            | 131                          |

*Notes:* Each column regresses one outcome, a mean over the rural households of the area the panel names, on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Panel A: the whole hinterland, every rural subdistrict within 300 travel minutes of the city, the city's own subdistricts left out. Panel B: the countryside with settlements, the rural subdistricts that hold the settlement villages within 300 travel minutes of the city, a subdistrict of an urban center left out. Means are weighted by the household survey weights. Spending is  $\ln(x+1)$  of monthly spending per person, in rupiah; non-food the non-food share of monthly spending; durables  $\ln(x+1)$  of monthly spending per person on durable goods (furniture, appliances, electronics, vehicles, jewelry), recalled over 12 months in 2002 and over 3 months in 2011, and any durable the share of households reporting a durable purchase in that period; services  $\ln(x+1)$  of monthly spending per person on each survey's service items, which differ between 2002 and 2011, so the two service columns are not comparable across years. Households per city is the mean number of surveyed rural households behind a city's means; in 2011 every column but spending uses the households answering the consumption modules, and a city leaves a column when its area has no surveyed rural household. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets. Sources: Susenas 2002 and 2011, the core survey and the consumption modules; population censuses of 1980 and 2010. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table E22: Road Density and Road Condition Around the City

|                                                                    | Exposure<br>(1)                 | Earlier level<br>(2)                         |
|--------------------------------------------------------------------|---------------------------------|----------------------------------------------|
| <i>Current density, catchment</i>                                  |                                 |                                              |
| 1950 density held fixed                                            | -0.0764<br>(0.0894)<br>[0.0991] | 0.3263* <sup>†</sup><br>(0.1735)<br>[0.1655] |
| Mean / cities                                                      | 1.664 / 131                     |                                              |
| 1992 density held fixed                                            | -0.0449<br>(0.0933)<br>[0.1004] | 0.1970<br>(0.1236)<br>[0.1223]               |
| Mean / cities                                                      | 1.664 / 131                     |                                              |
| <i>Current density, within 5 km of a program village</i>           |                                 |                                              |
| 1950 density held fixed                                            | -0.1668<br>(0.1992)<br>[0.1750] | 0.4125***<br>(0.1543)<br>[0.1527]            |
| Mean / cities                                                      | 1.802 / 131                     |                                              |
| 1992 density held fixed                                            | -0.0422<br>(0.1674)<br>[0.1574] | 0.5451***<br>(0.1469)<br>[0.1459]            |
| Mean / cities                                                      | 1.802 / 131                     |                                              |
| Corridor roughness, change 1990 to 2007, 1990 level held fixed     | 0.0574<br>(0.3016)<br>[0.3024]  | -0.4182***<br>(0.0967)<br>[0.0862]           |
| Mean / cities                                                      | -0.992 / 132                    |                                              |
| Corridor unpaved share, change 1990 to 2007, 1990 level held fixed | 0.0146<br>(0.0202)<br>[0.0238]  | -0.9318***<br>(0.0364)<br>[0.0424]           |
| Mean / cities                                                      | -0.466 / 125                    |                                              |

*Notes:* Each row is one regression on the cities of the main city sample (134 cities) that hold the row's measures, in the main specification with the full set of controls and Conley SE within 200km, adding the row's earlier level. The four density rows hold 131 cities (3 cities have no subdistrict hosting their catchment villages once the subdistricts that belong to a city are removed), the roughness row 132 (2 have no corridor survey post with roughness measured in both years) and the unpaved row 125 (9 have no post with the surface recorded in both years). Road density is  $\ln(1 + \text{km of road per 100 square km})$  on the current national and provincial road map, over the union of the subdistricts hosting the city's catchment villages less every subdistrict that belongs to a city, or over the part of that union within 5 km of a catchment village; the level held fixed is the same density on the 1950 atlas (main and secondary roads) or on the 1992 Digital Chart of the World, and column 2 reports its coefficient. Corridors are the road-condition survey posts within 10 km of the straight line from the city to each catchment village, measured in both 1990 and 2007, with roughness in meters per km (International Roughness Index) and unpaved as a share of posts. Below each estimate: its standard error with the uniform kernel in parentheses and with the Bartlett kernel in brackets, then the mean of the outcome and the cities. The stars use the uniform-kernel standard errors: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . <sup>†</sup> significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with province clusters, or with both. Source: road maps and the road-condition survey of the public-provision appendix.

Table E23: Main Results Inside and Outside the Conflict Provinces

|                                                                                          | Population                   | Employment                   | Unemployment                 | Agricultural employment      |                                 | Modern employment              |                                |
|------------------------------------------------------------------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|---------------------------------|--------------------------------|--------------------------------|
|                                                                                          | Log                          | Log                          | Rate                         | Log                          | Share                           | Log                            | Share                          |
| <i>Panel A. Main city sample, agro-similarity interacted with the conflict provinces</i> |                              |                              |                              |                              |                                 |                                |                                |
| Agro-similarity                                                                          | 0.056<br>(0.084)<br>[0.071]  | 0.045<br>(0.079)<br>[0.067]  | 0.007<br>(0.007)<br>[0.006]  | -0.122<br>(0.119)<br>[0.111] | -0.042**†<br>(0.021)<br>[0.022] | 0.170**†<br>(0.081)<br>[0.072] | 0.063***<br>(0.020)<br>[0.020] |
| × conflict province                                                                      | -0.021<br>(0.098)<br>[0.091] | -0.066<br>(0.089)<br>[0.087] | 0.009<br>(0.008)<br>[0.008]  | -0.125<br>(0.167)<br>[0.145] | -0.027<br>(0.032)<br>[0.033]    | -0.015<br>(0.084)<br>[0.096]   | 0.015<br>(0.035)<br>[0.032]    |
| Mean of outcome                                                                          | 11.842                       | 10.933                       | 0.122                        | 9.822                        | 0.374                           | 10.232                         | 0.516                          |
| Cities                                                                                   | 134                          | 134                          | 134                          | 134                          | 134                             | 134                            | 134                            |
| <i>Panel B. Without the conflict provinces</i>                                           |                              |                              |                              |                              |                                 |                                |                                |
| Agro-similarity                                                                          | 0.059<br>(0.099)<br>[0.102]  | 0.040<br>(0.097)<br>[0.098]  | -0.003<br>(0.009)<br>[0.009] | -0.030<br>(0.154)<br>[0.150] | -0.024<br>(0.022)<br>[0.028]    | 0.080<br>(0.092)<br>[0.099]    | 0.037<br>(0.026)<br>[0.027]    |
| Mean of outcome                                                                          | 11.896                       | 10.981                       | 0.122                        | 9.855                        | 0.371                           | 10.279                         | 0.518                          |
| Cities                                                                                   | 108                          | 108                          | 108                          | 108                          | 108                             | 108                            | 108                            |

*Notes:* Each column is one regression in the main specification with the full set of controls and Conley SE within 200km. Panel A keeps the main city sample (134 cities in 17 provinces) and adds agro-similarity times an indicator of the four conflict provinces, and the indicator itself: its first row is the slope outside the conflict provinces and its second the difference inside them, with every control coefficient common to both groups. Panel B leaves out the 26 cities of the provinces where communal or separatist violence displaced people around 2000: Aceh, on Sumatra (8 cities; the separatist conflict), West Kalimantan (8) and Central Kalimantan (5) (the Dayak-Madurese violence), and Central Sulawesi (5; the Poso conflict), so 108 cities in 13 provinces remain and every control coefficient is estimated again on them; Maluku and North Maluku, the provinces of the Maluku conflict, hold no city of the main city sample. Panel B's estimates for the agricultural employment share, log modern-sector employment and the modern-sector share are 47% to 56% of their size in Table 3, while in Panel A the slope outside the conflict provinces stays within a fifth of that table's estimate on these three outcomes; without the 26 cities the SD of agro-similarity left after the controls falls from 0.46 to 0.30, and every SE in Panel B is larger than in that table. Agro-similarity is standardized once over every city with a measured exposure, so both panels share one scale. Outcomes are measured in 2010; shares and the unemployment rate are fractions. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets. Source: population censuses of 1980 and 2010. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test the table does not print), or with both.

Table E24: Main Results With the Agro-Similarity of Neighboring Cities

|                                                                         | Population                                | Employment                     | Unemployment                               | Agricultural employment      |                                 | Modern employment                           |                                |
|-------------------------------------------------------------------------|-------------------------------------------|--------------------------------|--------------------------------------------|------------------------------|---------------------------------|---------------------------------------------|--------------------------------|
|                                                                         | Log                                       | Log                            | Rate                                       | Log                          | Share                           | Log                                         | Share                          |
| <i>Panel A. Every other measured city, weighted by inverse distance</i> |                                           |                                |                                            |                              |                                 |                                             |                                |
| Agro-similarity                                                         | 0.129* <sup>†</sup><br>(0.068)<br>[0.067] | 0.100<br>(0.068)<br>[0.065]    | 0.004<br>(0.006)<br>[0.006]                | -0.100<br>(0.107)<br>[0.105] | -0.052***<br>(0.019)<br>[0.020] | 0.236***<br>(0.074)<br>[0.072]              | 0.068***<br>(0.019)<br>[0.018] |
| Agro-similarity of neighbors                                            | -0.603**<br>(0.281)<br>[0.289]            | -0.597**<br>(0.272)<br>[0.282] | 0.035** <sup>†</sup><br>(0.015)<br>[0.023] | -0.513<br>(0.384)<br>[0.363] | 0.007<br>(0.075)<br>[0.073]     | -0.612** <sup>†</sup><br>(0.306)<br>[0.319] | -0.015<br>(0.075)<br>[0.072]   |
| Mean of outcome<br>Cities                                               | 11.842<br>134                             | 10.933<br>134                  | 0.122<br>134                               | 9.822<br>134                 | 0.374<br>134                    | 10.232<br>134                               | 0.516<br>134                   |
| <i>Panel B. Measured cities within 100 km, plain mean</i>               |                                           |                                |                                            |                              |                                 |                                             |                                |
| Agro-similarity                                                         | 0.100<br>(0.085)<br>[0.073]               | 0.058<br>(0.085)<br>[0.073]    | 0.014* <sup>†</sup><br>(0.008)<br>[0.008]  | -0.164<br>(0.121)<br>[0.119] | -0.059**<br>(0.024)<br>[0.024]  | 0.210**<br>(0.101)<br>[0.088]               | 0.072***<br>(0.021)<br>[0.020] |
| Agro-similarity of neighbors                                            | -0.133<br>(0.085)<br>[0.074]              | -0.107<br>(0.084)<br>[0.077]   | -0.006<br>(0.006)<br>[0.008]               | -0.111<br>(0.132)<br>[0.110] | -0.014<br>(0.027)<br>[0.025]    | -0.063<br>(0.074)<br>[0.074]                | 0.012<br>(0.024)<br>[0.023]    |
| Mean of outcome<br>Cities                                               | 11.853<br>125                             | 10.944<br>125                  | 0.121<br>125                               | 9.848<br>125                 | 0.381<br>125                    | 10.228<br>125                               | 0.510<br>125                   |

*Notes:* Each column is one regression in the main specification with the full set of controls and Conley SE within 200km, with the agro-similarity of neighboring cities added beside the city's own. Panel A: the mean of the standardized agro-similarity of every other city with a measured exposure (137 cities for each city), each weighted by 1 over its great-circle distance in km, defined for all 134 cities of the main city sample. Panel B: the plain mean over the other measured cities within 100 km, which leaves out the 9 cities with no measured city that close, so 125 remain. City centers are the coordinates the Conley SE use. Agro-similarity is standardized once over the 138 cities with a measured exposure, and both neighbor measures are means of it, so each coefficient is the change in the outcome per SD of agro-similarity; on the main city sample the Panel A measure has an SD of 0.29 and the Panel B measure an SD of 0.83. Outcomes are measured in 2010; shares and the unemployment rate are fractions. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets. Source: population censuses of 1980 and 2010. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . <sup>†</sup> significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test the table does not print), or with both.

## F The Model

This part derives the spatial equilibrium the main text summarizes, and reports its calibration and its channel decomposition.

This appendix derives the two-sector Stone-Geary spatial equilibrium exactly as it is used in [Section 6](#): the von Thünen land split with the built-up land endowment, the closed land-rent rebate, the measured 1980 amenity, and a location choice that ranks Stone-Geary indirect utility. The main text displays the model's core equations; this appendix derives the full system behind them. The two sectors are agriculture ( $A$ ), which produces a tradable food good using land and labor, and a modern non-agricultural composite ( $S$ ), which produces a non-traded good using housing land and labor.

### F.1 Model Details: Notation, Technology, and Choice Probabilities

*Cited in the main text. This section: [Section 6.1](#), p. 44.*

This subsection states the parts of the model that [Section 6.1](#) summarizes in words: the notation, the expenditure shares, the production technology with the land market, and the two stages of the worker's choice.

**Expenditure shares.** With supernumerary income  $\tilde{w}_{js} \equiv w_{js} + R_j - p_A^j \bar{a}$  defined as in [Section F.2](#), the implied expenditure shares are

$$s_A^j = \frac{p_A^j c_A}{w_{js} + R_j} = \frac{p_A^j \bar{a} + \psi_A \tilde{w}_{js}}{w_{js} + R_j} \quad (22)$$

$$s_S^j = \frac{\psi_S \tilde{w}_{js}}{w_{js} + R_j} \quad (23)$$

**Technology.** Total labor  $\bar{L}$  is fixed and each worker supplies one unit inelastically. Each location has a fixed land endowment  $\bar{T}_j$ , split between farming ( $T_{jA}$ ) and housing ( $T_{jH} = \bar{T}_j - T_{jA}$ ).

- **Agriculture:**

$$Y_{jA} = (A_j^A T_{jA})^{1-\beta} (L_{jA})^\beta, \quad 0 < \beta < 1 \quad (24)$$

Constant returns in  $(T_{jA}, L_{jA})$  jointly, diminishing returns to labor holding land fixed, so land scarcity is the congestion force in farming.  $A_j^A$  varies across space and is where the agro-climatic productivity shock from the transmigration program enters the model.

- **Modern / Services:**

$$Y_{jS} = \bar{A}_j^S N_j^\eta (T_{jH})^{1-\gamma_S} (L_{jS})^{\gamma_S}, \quad 0 < \gamma_S < 1 \quad (25)$$

where  $\bar{A}_j^S$  is the exogenous modern productivity scale and the external term  $N_j^\eta$ , entering (25) exactly once, is the agglomeration externality. The agglomeration term enters only modern production, and it does not improve the model's overall fit; a sensitivity arm at  $\eta = 0.10$  modestly helps one targeted moment ([Section F.6](#)). The headline sets  $\eta = 0$ , and the flat modern wage (**F12**) rests on the price and land terms.

**Wages, land, and the rebate.** Competitive factor markets price both inputs. Each sector's wage equals its marginal revenue product of labor ([Section F.3](#), [Section F.5](#)), and under Cobb-Douglas, factor payments are revenue shares, so labor collects  $\beta$  of agricultural revenue and  $\gamma_S$  of modern revenue while land collects the rest. A single rental market allocates  $\bar{T}_j$  across farming and housing by equalizing the per-hectare bids, so land shares follow value shares. Land goes to whichever use bids more per hectare, so farming's share of the land equals its share of the location's land value (the von Thünen split, [Section F.7](#)), and a booming city's agriculture can shrink through land, not only through labor. The endowment is measured as the PODES planted crop area plus the GHS built-up footprint, the housing land the

Table F1: Key Notation

| Symbol                   | Interpretation                                                                                                                                                                                                                                                                                                   |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $J$                      | number of locations (calibration panel $J = 300$ : the 128 urban centers of the calibration sample and 172 rural district remainders; Maluku and North Maluku excluded)                                                                                                                                          |
| $\mathcal{S} = \{A, S\}$ | sector set: Agriculture, Modern non-agriculture / Services (manufacturing folded into $S$ )                                                                                                                                                                                                                      |
| $L_{js}$                 | labor employed in location $j$ sector $s$                                                                                                                                                                                                                                                                        |
| $N_j = \sum_s L_{js}$    | total population in location $j$                                                                                                                                                                                                                                                                                 |
| $\bar{T}_j$              | total land endowment in location $j$ (crop area + built-up footprint)                                                                                                                                                                                                                                            |
| $T_{jA}, T_{jH}$         | land allocated to farming and to housing ( $T_{jA} + T_{jH} = \bar{T}_j$ )                                                                                                                                                                                                                                       |
| $A_j^A, \bar{A}_j^S$     | agricultural TFP; exogenous modern productivity scale (total modern TFP $\bar{A}_j^S N_j^\eta$ )                                                                                                                                                                                                                 |
| $\bar{a}$                | subsistence food requirement (Stone-Geary), $\bar{a} = 0.108$                                                                                                                                                                                                                                                    |
| $\psi_A, \psi_S$         | expenditure shares on discretionary consumption ( $\psi_A + \psi_S = 1$ ; $\psi_A = 0.550$ )                                                                                                                                                                                                                     |
| $\beta$                  | labor share in agriculture ( $\beta = 0.5$ ; land gets $1 - \beta$ )                                                                                                                                                                                                                                             |
| $\gamma_S$               | labor share in services ( $\gamma_S = 0.7$ ; housing land gets $1 - \gamma_S$ )                                                                                                                                                                                                                                  |
| $\eta$                   | agglomeration elasticity in services ( $\eta = 0$ headline)                                                                                                                                                                                                                                                      |
| $\tau_{kj}$              | iceberg cost of shipping food from $k$ to $j$ , $\tau_{kj} = \max(d_{kj}, 1)^\delta$                                                                                                                                                                                                                             |
| $\delta$                 | iceberg cost-distance elasticity (reported over the band $\delta \in [0.5, 0.8]$ )                                                                                                                                                                                                                               |
| $\sigma$                 | CES / Armington substitution elasticity across food sources ( $\sigma = 4$ )                                                                                                                                                                                                                                     |
| $R_j$                    | per-capita land-rent rebate to residents of $j$                                                                                                                                                                                                                                                                  |
| $\tilde{w}_{js}$         | supernumerary income (income net of the subsistence food bill)                                                                                                                                                                                                                                                   |
| $\Phi_L^j$               | measured 1980 location amenity (built-up, density, infrastructure)                                                                                                                                                                                                                                               |
| $\varepsilon_{js}(i)$    | idiosyncratic sector efficiency shock (scales the worker's effective wage), Fréchet with shape $\theta_S$                                                                                                                                                                                                        |
| $v_j(i)$                 | idiosyncratic location-preference shock, Fréchet with shape $\theta_L$                                                                                                                                                                                                                                           |
| $\theta_S, \theta_L$     | Fréchet shape parameters (sector choice; location choice)                                                                                                                                                                                                                                                        |
| $\psi_M$                 | expenditure share on the catchment-traded modern good, two-tradable-sector variant only ( $\psi_M = 0.082$ , the Susenas-2011 traded-services share; see <a href="#">Section 6.3</a> )                                                                                                                           |
| $\delta_S$               | distance decay on the catchment-traded modern good, two-tradable-sector variant only (measured at $\delta_S = 1.091$ , s.e. 0.033, from IFLS service-flow gravity in the Online Supplement; solved at $\delta_S = 1.0$ , the highest decay below the measured estimate at which the traded-service block exists) |
| $P_j$                    | ideal price index in location $j$                                                                                                                                                                                                                                                                                |
| $W_j$                    | sector-inclusive value (option value of living in $j$ )                                                                                                                                                                                                                                                          |
| $\pi_j, \pi_{s j}$       | choice probabilities (location; sector conditional on location)                                                                                                                                                                                                                                                  |

satellite observes. Agricultural productivity is recovered on the crop base only, winsorized at the 1st and 99th percentiles and normalized by its median, while the split operates on the full  $\bar{T}_j$ . Using the built-up endowment removes a crop-area floor artifact at cities (about 40% of cities previously sat at the minimum-farmland floor, and none do under the built-up

total) without diluting the agricultural base. This built-up  $\bar{T}_j$  is the headline land input. A pure-crop-area endowment is retained as robustness (Section F.11).

**Stage 2, sector choice within location  $j$ .** A worker draws a sector-specific efficiency shock  $\varepsilon_{js}$  from Fréchet with shape  $\theta_S$ . With scales equal to the sectoral wages, the sector-choice probability is

$$\pi_{A|j} = \frac{w_{jA}^{\theta_S}}{w_{jA}^{\theta_S} + w_{jS}^{\theta_S}} \quad (26)$$

so location agricultural employment is  $L_{jA} = \pi_{A|j} N_j$  (and  $L_{jS} = (1 - \pi_{A|j}) N_j$ ). Ranking wages is utility-consistent here because the rebate and the price level are common across sectors within a location, so the wage-maximizing sector is also the utility-maximizing one (Section F.8). The sector-inclusive value  $W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}$  is the option value of living in  $j$  before sector shocks are drawn; its level normalization is stated in the same appendix.

**Stage 1, location choice.** A worker draws a location taste shock  $v_j$  from Fréchet with shape  $\theta_L$ . The location stage ranks the Stone-Geary *indirect utility* (Section F.2) evaluated at the location's expected income and scaled by the measured amenity. Writing the location value as  $v_j = \Phi_L^j \tilde{w}_j / P_j$  with  $\tilde{w}_j = W_j + R_j - p_A^j \bar{a}$ , the location probability is

$$\pi_j = \frac{v_j^{\theta_L}}{\sum_{k=1}^J v_k^{\theta_L}}, \quad v_j = \Phi_L^j \frac{\tilde{w}_j}{P_j}, \quad (27)$$

where  $\Phi_L^j = \exp(\kappa z_j^{\text{amen}}) / \text{median}$  is the measured 1980 amenity (standardized built-up, population density, and PODES-1983 infrastructure, with  $\kappa = 0.5$ ), and population and sectoral labor follow as  $N_j = \bar{L} \pi_j$  and  $L_{js} = N_j \pi_{s|j}$ . The amenity anchors where people live (the cross-sectional level fit).

**Trade costs.** In the limit of large  $\sigma$  the CES aggregate (39) converges to the cheapest-supplier rule  $p_A^j = \min_k \{\tau_{kj} \text{mc}_k\}$ . The cost-distance elasticity  $\delta$  (how fast  $\tau$  rises with travel time) is distinct from the trade elasticity  $\sigma - 1 = 3$ ; the outer-island setting supports the upper half of the band  $\delta \in [0.5, 0.8]$  on independent grounds, since inter-island shipping over weakly connected routes plausibly has a higher per-minute trade cost than a within-continent road network. What the posted-cost convention simplifies and what it omits is stated in Section F.4.

## F.2 Preferences, Demand, and Indirect Utility

*Cited in the main text. This section: Section 6.1, p. 45.*

A worker living in location  $j$  and working in sector  $s$  consumes food  $c_A$  and the modern good  $c_S$  with generalized Stone-Geary (Kongsamut-Rebello-Xie) preferences:

$$U(c_A, c_S) = (c_A - \bar{a})^{\psi_A} c_S^{\psi_S}, \quad \psi_A + \psi_S = 1, \quad c_A \geq \bar{a}, \quad (28)$$

where  $\bar{a} > 0$  is the subsistence food requirement (the single non-homotheticity) and  $\psi_A, \psi_S$  are the discretionary-expenditure weights. The first  $\bar{a}$  units of food buy survival, not utility, so a poorer household spends a larger share on food and a richer one a larger share on the modern good.

**Step 1: the budget.** The worker earns the sectoral wage  $w_{js}$  and receives the per-capita land-rent rebate  $R_j$  (derived in Section F.7):

$$p_A^j c_A + p_S^j c_S = w_{js} + R_j, \quad (29)$$

with location-specific prices  $p_A^j$  (food) and  $p_S^j$  (modern good).

**Step 2: the Lagrangian.** Since  $\ln$  is monotone, maximizing (28) is the same as maximizing  $\psi_A \ln(c_A - \bar{a}) + \psi_S \ln c_S$ . With multiplier  $\lambda$  on (29),

$$\mathcal{L} = \psi_A \ln(c_A - \bar{a}) + \psi_S \ln c_S - \lambda(p_A^j c_A + p_S^j c_S - w_{js} - R_j).$$

**Step 3: first-order conditions.**

$$\frac{\partial \mathcal{L}}{\partial c_A} = \frac{\psi_A}{c_A - \bar{a}} - \lambda p_A^j = 0, \quad \frac{\partial \mathcal{L}}{\partial c_S} = \frac{\psi_S}{c_S} - \lambda p_S^j = 0.$$

**Step 4: solve.** The two FOCs give  $p_A^j(c_A - \bar{a}) = \psi_A/\lambda$  and  $p_S^j c_S = \psi_S/\lambda$ . Adding them and using the budget yields  $1/\lambda = w_{js} + R_j - p_A^j \bar{a}$ . Define this object, the income left after the subsistence food bill, as the **supernumerary income**

$$\boxed{\tilde{w}_{js} \equiv w_{js} + R_j - p_A^j \bar{a}} \quad (30)$$

and substitute back to obtain the Marshallian demands

$$c_A^* = \bar{a} + \frac{\psi_A}{p_A^j} \tilde{w}_{js}, \quad c_S^* = \frac{\psi_S}{p_S^j} \tilde{w}_{js}. \quad (31)$$

The household first buys the subsistence bundle  $\bar{a}$ , then splits the remaining income  $\tilde{w}_{js}$  in fixed shares  $\psi_A : \psi_S$ . As income grows, the food expenditure share falls toward  $\psi_A$  from above (the Engel curve), which is the demand-side engine of structural transformation.

**Step 5: indirect utility and the price index.** Substituting (31) back into (28),

$$V_{js} = \underbrace{\psi_A^{\psi_A} \psi_S^{\psi_S}}_{\text{const.}} \times \frac{\tilde{w}_{js}}{P_j}, \quad P_j = (p_A^j)^{\psi_A} (p_S^j)^{\psi_S}. \quad (32)$$

The Cobb-Douglas constant multiplies every location's value equally and is dropped. So a location is attractive when supernumerary income is high relative to the price index over the supernumerary basket. Evaluated at the location-level expected income  $\tilde{w}_j$  (Section F.8), this indirect utility is the object the location choice ranks: not the nominal wage, but  $\tilde{w}_j/P_j$ .

### F.3 Agriculture: Wage and the Return to Land

Farming at  $j$  combines effective land  $A_j^A T_{jA}$  (quality-adjusted hectares allocated to farming) with labor:

$$Y_{jA} = (A_j^A T_{jA})^{1-\beta} L_{jA}^\beta, \quad 0 < \beta < 1. \quad (33)$$

The function has constant returns in  $(T_{jA}, L_{jA})$  jointly and diminishing returns to labor holding land fixed, so land scarcity is the congestion force in farming.

**The wage (labor FOC).** A competitive farmer takes  $p_A^j$  as given and hires labor until the value of the marginal product equals the wage:

$$w_{jA} = p_A^j \frac{\partial Y_{jA}}{\partial L_{jA}} = p_A^j \beta (A_j^A T_{jA})^{1-\beta} L_{jA}^{\beta-1}. \quad (34)$$

**The land payment.** Land collects the value of its marginal product times land used. With Cobb-Douglas, factor payments are revenue shares:

$$v_{jA} \equiv T_{jA} p_A^j \frac{\partial Y_{jA}}{\partial T_{jA}} = (1 - \beta) p_A^j Y_{jA}. \quad (35)$$

**Product exhaustion.** Because (33) has constant returns, the two payments exactly exhaust revenue (Euler's theorem):

$$w_{jA} L_{jA} + v_{jA} = p_A^j Y_{jA}. \quad (36)$$

Nothing is left on the table, which lets the land-market accounting in Section F.7 close each location's budget.

## F.4 Trade in Food: Cost-Distance Enters Here

**The iceberg cost.** Shipping one unit of food from  $k$  to  $j$  melts part of it away, at an iceberg factor that is a power law in the route-aware cost-distance  $d_{kj}$  (minutes):

$$\tau_{kj} = \max(d_{kj}, 1)^\delta, \quad \delta > 0. \quad (37)$$

The floor at one minute makes the own-location cost exactly  $\tau_{jj} = 1$  (the diagonal of  $d$  is zero); the smallest off-diagonal distance in the sample is about five minutes, so the floor binds only own-location shipping. This is the within-location  $k = j$ ,  $\tau_{jj} \geq 1$  normalization that closes the system of food prices: every location can always source food from itself at unit trade cost, so  $p_A^j$  is well defined at each  $j$  even when the location imports nothing.

**The farm-gate supply price.** Each origin  $k$  posts a farm-gate marginal cost obtained by inverting the labor marginal product in (33):

$$\text{mc}_k = \frac{1}{\partial Y_{kA} / \partial L_{kA}} = \frac{L_{kA}^{1-\beta}}{\beta (A_k^A T_{kA})^{1-\beta}}. \quad (38)$$

**Remark (the omitted wage factor).** A textbook marginal cost would be  $w_{kA} \times 1/\text{MPL}_k$ , the wage bill of the marginal unit. The supply price in (38) includes only  $1/\text{MPL}_k$ . In the self-sourcing limit the omission is a normalization. With  $\tau_{jj} = 1$  and all food sourced at home,  $p_A^j = \text{mc}_j$ , so substituting (38) into the wage first-order condition (34) gives  $w_{jA} = 1$ . Away from that limit the suppressed factor is an endogenous, location-specific wage, one number per origin rather than a single numeraire. Combining (38) with (34) gives  $\text{mc}_k = p_A^k / w_{kA}$ , so the supply price falls in the origin's own agricultural wage at a given local food price index. General-equilibrium wage responses to the shock therefore do not reach export prices. The farm-gate price is a pure supply-side object, and cross-location differences in it are differences in inverse productivity: a productivity gain at  $k$  lowers  $\text{mc}_k$ , and with it the food price at every destination within reach. The food block is then a price-transmission system, and it does not pass the origin's wage into the Armington cost of traded food. The traded-services block of Section 6.3 does, pricing at the mill cost  $\text{mc}_k^M = w_{kS} / A_k^M$ . The solved food blocks do sit close to the self-sourcing limit in expenditure shares, which is where the convention is exact (see the openness paragraph below).

**The food price at each location (CES over origins).** The food price at  $j$  aggregates the costs of food from all origins, trade costs included, with elasticity  $\sigma$ :

$$p_A^j = \left[ \sum_k (\tau_{kj} \text{mc}_k)^{1-\sigma} \right]^{1/(1-\sigma)}, \quad \sigma > 1, \quad (39)$$

with import shares  $\lambda_{kj} \propto (\tau_{kj} \text{mc}_k)^{1-\sigma}$ . Two limits: as  $\sigma \rightarrow \infty$  the CES aggregate converges to the cheapest-supplier rule  $p_A^j = \min_k \{\tau_{kj} \text{mc}_k\}$ ; and with a single origin the food price at  $j$  is exactly  $\tau \cdot \text{mc}$ . Equation (39) is the smooth “buy from

the cheapest place, shipping included” rule the solver needs for differentiability. This is the channel that transmits the agro-similarity shock: a productivity gain at transmigration-village hosts lowers their  $mc_k$ , lowering  $p_A^j$  at every location within shipping reach. The modern good is non-traded, so  $p_S^j$  is set by local clearing (Section F.9).

**On the choice of the trade-cost band.** The estimation reports  $\delta$  over the band  $[0.5, 0.8]$ , spanning the range used in the quantitative-trade literature for cost-distance-based iceberg costs (Allen & Arkolakis 2014; Tombe & Zhu 2019); the estimated elasticities barely move across the band (Section 6.2), so no headline result turns on the point chosen. Raising  $\delta$  toward 0.6 and above also moves the untargeted price and agricultural-output moments toward the data, which the outer-island setting supports on independent grounds: inter-island shipping over weakly connected routes plausibly has a higher per-minute trade cost than a within-continent road network.

**Implied trade-cost levels and the openness of the solved equilibrium.** With the diagonal pinned at  $\tau_{jj} = 1$  and  $d$  measured in minutes, the level of the off-diagonal costs is set jointly by  $\delta$  and by the unit-of-time convention: measuring  $d$  in hours instead would divide every off-diagonal distance by 60 and so rescale every  $\tau_{kj}$  by  $60^{-\delta}$  (about a factor of 26 at  $\delta = 0.8$ ) relative to the fixed own-supply cost. The borrowed band  $\delta \in [0.5, 0.8]$  therefore disciplines how fast the cost rises with travel time, not its level against the self-supply option. The implied levels are large. At the median in-window pair distance of 187 minutes,  $\tau$  is 13.7 at  $\delta = 0.5$  and 65.6 at  $\delta = 0.8$ , and at the 300-minute window edge, 17.3 and 95.9. The solved equilibrium is correspondingly close to food self-sufficiency in expenditure shares. At the featured  $\delta = 0.8$  solves, under either land-market closure, the median city buys 99.9% of its food spending from its own location (mean 99.5%, 10th percentile 99.1%), the median rural unit 99.99%, and the expenditure-weighted trade cost cities actually pay is about 1.06. The trade channel of Section 6.3 accordingly operates through the local food price, the response of  $p_A^j$  to nearby marginal costs, rather than through large traded quantities.

## F.5 Services: Housing Land, Agglomeration, Wage

Services at  $j$  combine labor with housing land  $T_{jH}$  (the land the city occupies) and an agglomeration external effect, with  $\bar{A}_j^S$  the exogenous modern productivity scale and  $N_j^\eta$  the external term, entering once:

$$Y_{jS} = \bar{A}_j^S N_j^\eta T_{jH}^{1-\gamma_S} L_{jS}^{\gamma_S}, \quad 0 < \gamma_S < 1, \quad \eta \geq 0. \quad (40)$$

The  $N_j^\eta$  term is external: an individual firm takes city size as given, so its private labor FOC contains  $\gamma_S$  only, and the land bid is the residual revenue share:

$$w_{jS} = p_S^j \gamma_S \bar{A}_j^S N_j^\eta T_{jH}^{1-\gamma_S} L_{jS}^{\gamma_S-1}, \quad v_{jS} = (1 - \gamma_S) p_S^j Y_{jS}, \quad (41)$$

and the private payments again exhaust revenue,  $w_{jS} L_{jS} + v_{jS} = p_S^j Y_{jS}$ . The service good is non-traded: it must be produced where it is consumed, which ties  $p_S^j$  to local demand in Section F.9.

## F.6 The Flat Modern Wage With and Without Agglomeration

*Cited in the main text. This section: Section 5.2, p. 30.*

Decomposing the modern wage FOC (41) in city size, with the price and land responses explicit, gives

$$\frac{\partial \ln w_{jS}}{\partial \ln N_j} = \underbrace{\eta}_{\geq 0, \text{ agglom.}} + \underbrace{(\gamma_S - 1)}_{< 0, \text{ dim. ret.}} \cdot \underbrace{\frac{\partial \ln L_{jS}}{\partial \ln N_j}}_{\approx 1} + \underbrace{(1 - \gamma_S)}_{> 0} \cdot \underbrace{\frac{\partial \ln T_{jH}}{\partial \ln N_j}}_{> 0, \text{ land}} + \underbrace{\frac{\partial \ln p_S^j}{\partial \ln N_j}}_{> 0, \text{ price}} \quad (42)$$

The rising-price term and the housing-land term (a growing city bids land into housing through the von Thünen split, raising the marginal product of modern labor) push the wage up while diminishing returns push it down. At the headline  $\eta = 0$  the rising-price and housing-land terms are the only offsetting forces; at  $\eta > 0$  agglomeration adds a further

offsetting force. In the featured solves the offset holds in levels: the cross-city standard deviation of the shock's modern-wage response is about 0.0003 log points, flat for any practical purpose, in line with the data's flat non-agricultural wage (**F12**: 0.027, s.e. 0.030, [Table E2](#) col 3). In the standardized units the calibration reports its moments in, the model's modern-wage response slope is +0.16 under the closed rebate and +0.18 under the open one, against +0.02 in the data ([Table B3](#)), so the tiny wage movements the model does produce are more aligned with exposure than the data's are. The  $\eta = 0.10$  sensitivity arm modestly improves the modern-employment moment.  $\eta$  is fixed outside the estimation.

## F.7 The Land Market: von Thünen Split, Rebate, Built-up Endowment

*Cited in the main text. This section: [Section 6.1](#), p. 44.*

**One land market.** The location's endowment  $\bar{T}_j$  is allocated between farming ( $T_{jA}$ ) and housing ( $T_{jH} = \bar{T}_j - T_{jA}$ ) by a single rental market. Each use bids its average land value per hectare: farming bids  $v_{jA}/T_{jA}$  from (35), housing bids  $v_{jS}/T_{jH}$  from (41).

**Rents equalize.** If farming paid more per hectare, land would flow to farming, and vice versa. Setting the two bids equal,  $v_{jA}/T_{jA} = v_{jS}/(\bar{T}_j - T_{jA})$ , and solving,

$$T_{jA} = \bar{T}_j \frac{v_{jA}}{v_{jA} + v_{jS}}, \quad T_{jH} = \bar{T}_j - T_{jA}. \quad (43)$$

Land shares follow value shares: the sector generating the larger land value occupies the larger share of the location. This single equation is the von Thünen mechanism, and it is why a booming city's agriculture can shrink through land, not just through labor.

**The closed rebate.** Land income must accrue to someone. The headline (closed) variant gives each location's residents equal shares of their own location's land income:

$$R_j = \frac{v_{jA} + v_{jS}}{N_j}. \quad (44)$$

This closes the budget (all local value added becomes local income). Substituting  $L_{jA} = \pi_{A|j} N_j$  into (35) shows the agricultural part of the rebate,  $v_{jA}/N_j$ , scales as  $N_j^{\beta-1}$  at given prices, land, and sector shares, falling in city size, so the rebate is also a dispersion force. The open (channel-fix) variant replaces (44) with a national pool  $r = \sum_j (v_{jA} + v_{jS}) / \sum_j N_j$ .

**The built-up land endowment.** The endowment is measured as

$$\bar{T}_j = T_j^{\text{crop}} + T_j^{\text{built}}, \quad (45)$$

the PODES planted crop area plus the GHS built-up footprint (the housing land the satellite observes). Two rules keep the economics straight: agricultural productivity is recovered on the crop base only,  $A_j^A = (L_{jA,0}/T_j^{\text{crop}})^\beta$ , winsorized at the 1st and 99th percentiles across the  $J$  locations and normalized by its median (built-up is housing, not farmland), and the split (43) operates on the full  $\bar{T}_j$ . This built-up endowment removes the crop-area floor artifact at cities without diluting the agricultural base; a pure-crop-area endowment is the land robustness.

## F.8 Worker Choice: the Two Fréchet Stages

**A Fréchet primer.** A random variable  $X$  with scale  $u$  is Fréchet with shape  $\theta$  if  $\Pr(X \leq x) = \exp(-(x/u)^{-\theta})$ . Two properties make it the workhorse of discrete choice: it is max-stable, and the probability that draw  $s$  is the largest is  $u_s^\theta / \sum_{s'} u_{s'}^\theta$ . A larger  $\theta$  means less idiosyncratic dispersion, so choices chase the scales  $u_s$  more aggressively.

## Stage 2: the sector choice

After settling at  $j$ , a worker draws sector tastes and picks the sector with the higher taste-scaled wage. With scales  $u_s = w_{js}$ , direct integration of the Fréchet maximum gives

$$\pi_{A|j} = \frac{w_{jA}^{\theta_S}}{w_{jA}^{\theta_S} + w_{jS}^{\theta_S}}, \quad L_{jA} = \pi_{A|j} N_j, \quad L_{jS} = (1 - \pi_{A|j}) N_j. \quad (46)$$

**The sector stage ranks utility, not income.** The choice (46) uses wages, which is utility-consistent under a productivity-shock interpretation of the Fréchet draw: if the idiosyncratic term is labor efficiency, a worker's income is  $\varepsilon_s w_{js} + R_j$  and utility is  $(\varepsilon_s w_{js} + R_j - p_A^j \bar{a})/P_j$ ; the additive rebate and the price level are common across sectors *within* a location, so the utility-maximizing sector is the wage-maximizing sector. The wage-ranked sector stage is therefore utility maximization with efficiency shocks, not income maximization.

The expected income of a worker at  $j$ , before sector tastes are drawn, is the Fréchet expected maximum

$$\mathbb{E}[\max_s w_{js} \varepsilon_s] = \Gamma\left(1 - \frac{1}{\theta_S}\right) \underbrace{(w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}}_{\equiv W_j}. \quad (47)$$

**Remark (the Gamma factor).** The model uses the inclusive value  $W_j$  without the  $\Gamma(1 - 1/\theta_S)$  factor. The factor is constant across locations, so the sector shares (46) are unaffected; the location shares are not exactly invariant to it, because  $W_j$  enters  $\tilde{w}_j = W_j + R_j - p_A^j \bar{a}$  additively. Omitting it is a normalization of the expected-income level. Separately from this constant factor,  $W_j$  itself is not the head-count average wage. The income concept the solver deploys, and the location-varying wedge it opens against the wage bill, are stated in [Section F.9](#). The factor diverges as  $\theta_S \downarrow 1$ . At the estimated  $\theta_S = 1.043 > 1$  the expected maximum exists, but the estimate sits close enough to that boundary that the omitted level is large,  $\Gamma(1 - 1/1.043) \approx 24$ , so the normalization is quantitatively consequential. The nested-*logit* version (Gumbel taste shocks in log utilities, which accommodate any  $\theta_S > 0$  and give the same sector shares) is therefore retained as a robustness reframe. The shares (46) are well defined at any  $\theta_S > 0$ .

## Stage 1: the location choice

The location stage ranks the indirect utility (32) evaluated at the location's expected income and scaled by the measured amenity: the location value is

$$v_j = \Phi_L^j \frac{\tilde{w}_j}{P_j}, \quad \tilde{w}_j = W_j + R_j - p_A^j \bar{a}, \quad \Phi_L^j = \frac{\exp(\kappa z_j^{\text{amen}})}{\text{median}}, \quad (48)$$

where  $z^{\text{amen}}$  is the standardized 1980 amenity index (GHS built-up, population density, PODES-1983 infrastructure) and  $\kappa$  scales its dispersion. The location choice therefore ranks Stone-Geary indirect utility  $\tilde{w}_j/P_j$ , not the nominal wage.

A worker draws a location taste shock  $v_j \sim \text{Gumbel}(0, 1/\theta_L)$  (equivalently a Fréchet location value), so realized location utility is  $\ln \tilde{U}_j = \ln v_j + v_j$ . The probability of choosing  $j$  is  $\pi_j = \Pr[\ln \tilde{U}_j \geq \ln \tilde{U}_k \forall k]$ . Conditioning on  $v_j = u$  and integrating,

$$\pi_j = \int_{-\infty}^{\infty} \prod_{k \neq j} F_v(u + \ln v_j - \ln v_k) f_v(u) du, \quad (49)$$

with  $F_v(x) = \exp[-e^{-\theta_L x}]$  and  $f_v(x) = \theta_L e^{-\theta_L x} \exp[-e^{-\theta_L x}]$ . Defining  $S_j \equiv \sum_{k \neq j} (v_k/v_j)^{\theta_L}$ , the integrand collapses to  $\exp[-e^{-\theta_L u}(1 + S_j)] \cdot \theta_L e^{-\theta_L u}$ . The change of variables  $z = e^{-\theta_L u}$  (so  $dz = -\theta_L e^{-\theta_L u} du$ ) turns (49) into an elementary integral,

$$\pi_j = \int_0^\infty e^{-z(1+S_j)} dz = \frac{1}{1+S_j} = \frac{v_j^{\theta_L}}{\sum_k v_k^{\theta_L}},$$

so the location probability is

$$\pi_j = \frac{v_j^{\theta_L}}{\sum_{k=1}^J v_k^{\theta_L}}, \quad N_j = \bar{L} \pi_j. \quad (50)$$

**The costly-migration generalization.** The migration-cost extension replaces (50) with an origin-destination allocation over the 1980 origin population,  $\pi_{oj} \propto (v_j d_{oj}^{-\nu})^{\theta_L}$ , with the composite  $\theta_L \nu = 1.865$  pinned by the IFLS migration gravity. At  $\nu = 0$  it collapses exactly to (50). It is a gated robustness, not part of the headline.

## F.9 Market Clearing, the Value Balance, and Equilibrium

**Modern-market clearing.** The modern good is non-traded, so local demand equals local supply. Evaluating the demand (31) at the location-level income  $\tilde{w}_j$  for each of the  $N_j$  residents,

$$p_S^j Y_{jS} = N_j \psi_S \tilde{w}_j \quad \Longleftrightarrow \quad p_S^j = \frac{N_j \psi_S \tilde{w}_j}{Y_{jS}}. \quad (51)$$

The right-hand form is the solver's price update. Modern prices are high where many people with high supernumerary income chase little modern output: the cost-of-living force that disperses population.

**The income concept, and the value balance it does and does not satisfy.** Every demand-side equation above spends the same income: per-worker income in the modern-clearing condition (51), in the location value (48), and in the subsistence bill is  $W_j + R_j$ , the sector-inclusive value plus the per-head rebate, while production and the mapping to census employment use head counts  $L_{js} = \pi_{s|j} N_j$ . These are two different aggregates of the same wages. Factor payments add up to the head-count wage bill: with  $\bar{w}_j \equiv \pi_{A|j} w_{jA} + (1 - \pi_{A|j}) w_{jS}$  the average wage per head,  $N_j \bar{w}_j + v_{jA} + v_{jS} = p_A^j Y_{jA} + p_S^j Y_{jS}$ , and wages and land payments stay local under the closed rebate. Under that wage-bill income the food account closes exactly. Spending  $\bar{w}_j + R_j$  per worker and subtracting modern clearing leaves

$$p_A^j Y_{jA} = p_A^j C_{jA}, \quad (52)$$

the value of food produced equal to the value of food consumed at every location. This system does not satisfy (52), because it spends  $W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}$  per worker, and by the power-mean inequality  $W_j > \bar{w}_j$  whenever both sectors are active, at any  $\theta_S > 0$ , with a gap that varies across locations with the sectoral mix. Measured food spending therefore exceeds the value of local food output by exactly the option-value wedge  $N_j(W_j - \bar{w}_j)$ , and the wedge is large at the featured solves: the median ratio  $W_j/\bar{w}_j$  is 1.91 under the closed rebate and 1.60 under the open one, so in aggregate the wedge is 89% of the head-count wage bill closed and 59% open. Consistently with this, the equilibrium conditions are exactly the list below, the food block determines the food price index (39) from origin marginal costs, and food quantities are demand-determined at that index. What the channel argument actually uses is weaker and survives. Under the closed rebate every argument of location  $j$ 's demand, the wages inside  $W_j$  and the rents inside  $R_j$ , is generated at  $j$ , so no location holds a claim on another location's output with which to finance a persistent net food purchase. Solved, that local-financing property binds. The baseline two-sector model's trade-only leg on the agricultural share is wrong-signed and cities move out of agriculture through migration, and the resolutions in Section 6.3 work precisely by creating a cross-location claim, modern exports under the second tradable sector, or a national land-income claim under the open rebate.

**Equilibrium.** Given fundamentals  $\{A_j^A, \bar{A}_j^S, \Phi_L^j, \bar{T}_j, d\}$  and parameters  $\{\beta, \gamma_S, \eta, \psi_A, \bar{a}, \delta, \sigma, \kappa, \theta_L, \theta_S\}$ , an equilibrium is a vector  $\{N_j, L_{jA}, L_{jS}, T_{jA}, p_A^j, p_S^j, w_{jA}, w_{jS}, R_j\}$  such that: (1) wages satisfy (34) and (41); (2) food prices satisfy (39) with costs (37) and (38); (3) the land split (43) and rebate (44) hold; (4) sector employment satisfies (46) with  $L_{jA} + L_{jS} = N_j$ ; (5) modern markets clear location by location, (51); (6) population satisfies (50) with  $\sum_j N_j = \bar{L}$ ; and (7) the numeraire is the reference city's modern wage,  $w_{j^*S} = 1$  ( $j^*$  the median-population urban unit). Conditions 1–4 solve as an inner fixed point at given  $(N_j, p_S^j)$ ; conditions 5–6 are the outer fixed point in  $(\ln N, \ln p_S)$ .

**The first-order comparative static behind the main text.** Differentiating the sector share (46) in the log wage ratio gives  $d\pi_{A|j} = \theta_S \pi_{A|j} (1 - \pi_{A|j}) d\ln(w_{jA}/w_{jS})$ . At fixed local inputs the agricultural wage (34) is proportional to the local food price, so  $d\ln(w_{jA}/w_{jS}) = d\ln p_A^j$ ; the CES price (39) gives  $\partial \ln p_A^j / \partial \ln mc_v = s_{vj}$ , the expenditure share of origin  $v$  in city  $j$ 's sourcing; and the marginal cost (38) gives  $\partial \ln mc_v / \partial \ln A_v^A = -(1 - \beta)$ . Chaining the three yields the main text's (14). The expression holds the city's own labor, land, and modern price fixed, so it is the impact response. The solved general equilibrium adds the adjustment of all three.

## F.10 From Equilibrium Conditions to Estimating Equations

The equilibrium conditions map directly to the main empirical specifications.<sup>26</sup>

**Sectoral employment share.** From the sector-choice condition (46) and the equilibrium wages, the agricultural employment share falls with a lower local food price, motivating

$$\underbrace{\text{empl\_ag\_pct}_j}_{\approx L_{jA}/N_j} = \beta_0 + \beta_1 \underbrace{\text{AgSim}_j}_{\propto A_j^{A, \text{nearby}}} + \beta_2 \text{TM\_count}_j + \beta_3 X_j + \varepsilon_j \quad (53)$$

The model predicts  $\beta_1 < 0$ : higher agro-similarity in nearby villages lowers  $p_A^j$  (through trade), lowers  $w_{jA}/w_{jS}$ , and lowers  $L_{jA}/L_{jS}$  (matching **F3**).

**Population.** From the location-choice condition (50),  $\ln \text{pop}_j = \beta_0 + \beta_1 \text{AgSim}_j + \beta_2 \text{TM\_count}_j + \beta_3 X_j + \varepsilon_j$ . The coefficient  $\beta_1$  captures how agro-similarity raises the city's real-income index  $v_j$ ; the model predicts it positive but modest when  $\theta_L$  is low (high migration frictions), matching **F1**.

**Modern-good price.** From the modern-clearing condition (51),  $\ln \text{housing\_rent}_j = \beta_0 + \beta_1 \text{AgSim}_j + \beta_2 \text{TM\_count}_j + \beta_3 X_j + \varepsilon_j$ , with  $\beta_1 > 0$ : a prediction the 1996 rents confirm and the 2011 rent response, indistinguishable from zero, leaves untested at endline (**F9**).

**Modern wage.** From (41), the effect of agro-similarity on  $w_{jS}$  is ambiguous:  $p_S^j$  rises (up),  $L_{jS}$  rises with  $\gamma_S < 1$  (down, diminishing returns), housing land  $T_{jH}$  expands (up, the land term), and at  $\eta > 0$  agglomeration adds a further offset. If these roughly cancel, the modern-wage effect is small and insignificant, matching **F12** (Section F.6 decomposes the offset).

## F.11 Calibration Detail: Targets, Headline Estimate, Robustness, and Fit

*Cited in the main text. This section: Section 6.2, p. 47.*

This subsection holds the calibration detail behind Section 6.2: the recovered fundamentals, the estimator's mechanics, the targets, the headline estimate with its robustness, the fit on the untargeted moments, and the 1980-baseline forward solve.

<sup>26</sup>The calibration targets are the standardized ( $z$ -on- $z$ ) agro-similarity-response slopes reported in Table B3; the model's predicted responses are computed in the same  $z$ -on- $z$  units, so model and data slopes are directly comparable.

**City fundamentals.** A household evaluates the amenity for any given location relative to the other locations, once wages are taken into account. Modern-sector productivity is recovered from the census modern wage bill, agricultural productivity from agricultural labor per hectare of crop land, and the location amenity from the measured 1980 index of built-up area, population density, and PODES infrastructure. A location missing any of these inputs is excluded.

**The estimator.** Starting from an initial  $\theta$ , I re-solve the model, recompute the two moments, and update  $\theta$  until the model's moments match the data's. The estimator solves the model without the shock, solves it with a five-log-point agricultural TFP shock at every transmigration-village host, runs the same specification, with the same controls, on the resulting model panel of the same cities, and chooses  $\theta$  to minimize the standard-error-weighted distance between the model's standardized ( $z$ -on- $z$ ) slopes and the coefficients estimated in the data. The model panel itself contains the calibration sample's urban centers as its urban units and each remaining outer-island district's rural remainder as one rural location, dropping Maluku and North Maluku, which the 1980 crosswalk and the PODES-based inputs do not cover:  $J = 300$  locations, 128 urban and 172 rural, the count in Table F1. **The targets.** The targets are the two slopes the reduced form features. Population growth pins down the location elasticity, because as people are migrating that shapes the population allocation across places, and the agricultural-share-decline moment pins down the sectoral elasticity, people choosing sectors within each location; the modern-employment slope, nested by those two, is held out as an untargeted moment. With two parameters and two targeted moments the estimator is exactly identified, and fit is judged on the untargeted moments.

**Headline estimate.** On the built-up  $\bar{T}_j$  endowment, under the closed land-rent rebate, the traded-services specification estimated to the two  $z$ - $z$  targets gives

$$\theta = (\theta_L, \theta_S) = (0.372, 1.043)$$

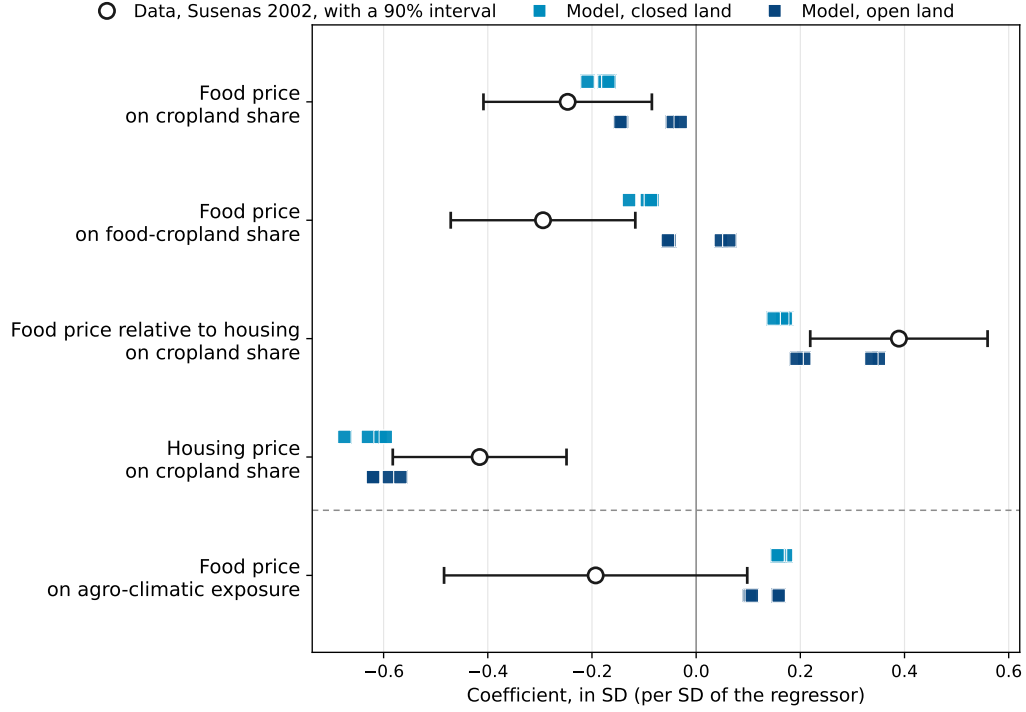
at  $\delta = 0.8$ . The estimate is fitted to the two coefficients as estimated before a correction to the 1980 census crosswalk (Online Supplement) that moved the 1980 baseline controls of nine urban centers, and it matches both exactly: the model's agricultural-employment-share slope is  $-0.178$  against a target of  $-0.178$ , and its population slope  $+0.112$  against  $+0.112$ . Table B3 reports the coefficients after that correction,  $-0.168$  and  $+0.100$ , on the same 128 urban locations of the calibration panel. Across the trade-cost band  $\delta \in [0.5, 0.8]$  the estimate is tight,  $\theta_L$  between 0.339 and 0.385 and  $\theta_S$  between 1.02 and 1.04, so the food trade cost is reported as a band and no result below depends on it. The single-tradable-good specification cannot reach these targets at any  $\theta$ . Its best attainable miss on the agricultural-share moment is 1.61 standard errors, and under exact identification a specification that cannot reach the targets at any  $\theta$  is rejected as a specification, which is the model-selection argument for the traded-services specification. The nest holds ( $\theta_S > \theta_L$ ). Workers respond more strongly to within-location sectoral wage gaps than to across-location real-income gaps, the ordering under which structural transformation is strong and population reallocation modest. Appendix Section F.12 sets out the traded-services specification.

**Robustness on the land specification and closure.** A pure-crop-area endowment flips the nest and rests on the floor imposed on the crop-area endowment, so the built-up endowment is the baseline specification. Opening the land-rent rebate to a national pool (the von Thünen open closure) is estimated alongside the closed rebate, and the channel decomposition is reported under both closures (Section F.12).

**Fit on the untargeted moments.** On the modern-employment slope, the model gives  $+0.135$  against the  $+0.149$  it was scored against, inside that interval and close to the  $+0.136$  of the current reduced form. The two figures below are the model-fit summary and the untargeted price test that Section 6.2 discusses.

**The 1980-baseline forward solve.** Solving the model forward from the recovered pre-program (1980) baseline, at an earlier calibration, the same shock moves the two economy-wide average prices identically ( $\Delta \ln p_A = +0.082$  and  $\Delta \ln p_S = +0.082$ , in units of the numeraire, the reference city's modern wage). Because the two responses are equal to three decimals, the relative price  $p_A/p_S$  is unchanged in that solve, and since the reallocation channel operates entirely through relative prices in the sector choice, the exercise shows that the recovered baseline supports a forward solve but

Figure F1: The Food Price Falls in the Agricultural Density of a City's Surroundings, in the Data and in the Model



*Notes:* The untargeted price test implied by the von Thünen logic: a city near abundant cropland faces a cheaper food price. Each row is one cross-sectional regression on the same 102 urban centers with a 2002 Susenas price, with the price and the agricultural density standardized and no controls; controls are left out because the agro-climatic covariates are themselves determinants of how much land is cropped, and the same regressions with the reduced form's controls are in Figure B2. The hollow circle is the data estimate with a 90% interval on Conley standard errors at a 200-kilometer cutoff; each filled square is the model at its own estimate for one of the four assumed food trade costs, with the model side handed the same measured density so the two sides run the identical regression. Neither side is a targeted moment: the estimator never saw this regression. The closed-rebate specification sits inside the data's 90% interval at all four food trade costs; the open-rebate specification misses at the two trade costs at which  $\theta_L$  is at or near the upper bound of the search, which Table B3 reports as a bound. Because the model's land base is all crops, the gradient it responds to is the all-crops one.

cannot test the productivity transmission. The price side of the transmission is visible in the end-period solves instead: exposure lowers the food price in the city (a standardized cross-sectional response of  $-0.066$  under the closed rebate,  $-0.093$  on the trade-only counterfactual) while the local modern price rises ( $+0.159$ ), and the open-rebate solve shows the same signs, so the relative price of food falls with exposure as the mechanism requires.

## F.12 The Channel Decomposition at the Estimated Parameters

*Cited in the main text. This section: Section 6.1, p. 45.*

**Local financing in place of a value balance.** The demand system spends the inclusive value. Per-worker income in the location choice, the modern-market clearing, and the subsistence bill is  $W_j + R_j$ , while factor payments add up to the head-count wage bill  $\pi_{A|j}w_{jA} + (1 - \pi_{A|j})w_{jS}$ , which is strictly smaller whenever both sectors are active. An exact food value balance,  $p_A^j Y_{jA} = p_A^j C_{jA}$ , therefore holds under the wage-bill income and not under the  $W_j$  income the solver uses; the appendix states the income concept the solver uses and quantifies the resulting wedge at the featured solves (Section F.9). The closed rebate does guarantee local financing. Every argument of a location's demand, the wages inside  $W_j$  and the rents inside  $R_j$ , is generated at that location, so no city holds a claim on another location's output with

which to finance a persistent net food purchase, and, solved, the baseline model's trade-only response on the agricultural share is wrong-signed. Cities move out of agriculture through migration. That local-financing property is the constraint the channel decomposition in [Section 6.3](#) relaxes.

**Decomposition & the Tradable Sector.** The local-financing property implies that with a single tradable good, a closed land-rent rebate, and non-traded services, a city holds no claim on outside output with which to finance a persistent net food purchase, so no city can move out of agriculture through trade and nothing else. In the baseline closed model, the city's declining agricultural share is therefore driven mostly by migration, and shutting migration down leaves the trade-only response wrong-signed. This is the Matsuyama tension. The mechanism this paper studies, a city importing food and reallocating labor to the modern sector, requires a way for the city to run a net food deficit. A second tradable sector resolves it, because urban services are reached by travel rather than shipped: the modern composite is split into a catchment-traded component (census trade, transport, and skilled private services, plus manufacturing) and a purely local component, and the traded part takes an Armington structure at a steep distance decay. The block has two parameters. The traded expenditure share is  $\psi_M = 0.082$ , the Susenas-2011 traded-services share of household spending, so the expenditure side is calibrated services-only while the employment side of the traded bundle also includes manufacturing. The distance decay is measured directly from IFLS household service-flow gravity at  $\delta_S = 1.091$ , with a standard error of 0.033 (Online Supplement); the model solves at  $\delta_S = 1.0$ , the highest decay below the measured estimate at which the traded-service economy exists. A city can then export modern output to pay for net food imports, which is the open-economy case of the Matsuyama mechanism. Each urban region with its rural hinterland is the closed economy of [Matsuyama \(1992\)](#), and within it the villages specialize in food while the city, connected through cost-distance to the villages that feed it, imports food and moves its own workers into the modern sector.

For the agricultural employment share all three counterfactuals are informative, because a city can change what it does whether or not anyone moves. For city size the response with migration shut is zero by construction, since shutting migration means fixing populations, so the informative comparison there runs between the full response and the response with trade shut.

**The decomposition at the estimated  $\theta$ .** In the world of trade only, switching migration off and re-solving leaves a trade-only agricultural-share response of  $-0.145$  against the full-model  $-0.178$  under the closed land-rent rebate, and  $-0.181$  against  $-0.182$  when the rebate opens to a national pool (so land income, not modern exports, finances the city's net food purchases). The trade-only counterfactual is sufficient to reproduce 81% to 100% of the model's full response, stable across the two closures. Even with migration shut down, the response to the agricultural productivity gain runs almost entirely through trade. The migration-only response is large under the closed rebate ( $-0.215$ ) and modest under the open one ( $-0.083$ ), and the interaction absorbs the difference ( $+0.182$  and  $+0.082$ ), so how the two channels share the remainder depends on the land-market closure while the trade result does not. For city size the pattern reverses. Its trade-only response is zero by construction, so the informative report is the other two counterfactuals. The migration-only counterfactual reproduces the full population response under the closed rebate ( $+0.115$  against the full-model  $+0.112$ , with a negligible interaction) but produces  $+0.043$  of the full  $+0.110$  under the open one, where the interaction accounts for the remaining  $+0.067$ , so under the open rebate most of the size response arises from migration operating alongside trade rather than from migration without trade. This two-closure bracket is the reported decomposition, and the shares describe the calibrated model's response to the productivity shock. Consistent with the friction estimated in [Section F.13](#), holding population fixed leaves most of the structural transformation intact in the traded-services specification, while in the closed single-tradable baseline the same exercise leaves the trade-only response wrong-signed, which motivates the second tradable sector.

## F.13 Migration Friction: The Gravity Estimate and the Costly-Migration Variant

*Cited in the main text. This section: [Section 6.1](#), p. 45.*

The frictionless location choice (27) is generalized to an origin-destination allocation over the 1980 origin population,  $\pi_{oj} \propto (v_j d_{oj}^{-\nu})^{\theta_L}$ , where the composite  $\theta_L \nu$  is pinned by the subdistrict-to-subdistrict migration gravity estimated

in the IFLS:

$$\theta_L \nu = 1.865$$

At the estimated  $\theta_L = 0.372$  the structural friction elasticity is  $\nu \approx 5.0$ , against Bryan and Morten's effective  $\tilde{\nu} \approx 3.18$ ; the choice-probability distance term  $d_{oj}^{-1.865}$  does not depend on  $\theta_L$ , so only the reported  $\nu$  scales with it.

The costly-migration model, in which both the baseline and the shock bear the moving cost from 1980 origins, is a robustness check. At the  $\theta$  estimated on the built-up endowment it is near-immobile. The rural productivity boom holds people near the transmigration-village hosts, and city growth does not absorb them (a stay-home effect), so the population and modern moments turn wrong-signed and the level fit falls, while the channel diagnosis is unchanged. The closed land market blocks trade from producing the contraction. The migration cost is a friction that dampens over-mobility. A dynamic migration model, motivated by the gap between the per-decade gravity estimate and the 30-year stay share, is left for future work.